

Probing the Dark Universe

Machine Learning and Statistical Approaches to Gamma-Ray
Dark Matter Searches

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ABSTRACT

This thesis explores...



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INTRODUCTION

The introduction

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THE DARK MATTER PROBLEM

The nature of dark matter is an open question at the heart of fundamental physics. Over the past century, independent observations have established that the visible matter we ~~can~~ detect — stars, gas, dust — accounts for only a small fraction of the total matter content of the Universe. Flat rotation curves in spiral galaxies [1, 2], the dynamics and gravitational lensing of galaxy clusters [3], and the detailed pattern of temperature anisotropies in the Cosmic Microwave Background [4] all point consistently toward a dominant, non-luminous matter component. These observations, spanning scales from individual galaxies to the observable Universe as a whole, converge on the Λ CDM ~~concordance~~ model, in which approximately 85% of the matter density is composed of cold, non-baryonic dark matter [4]. The non-baryonic character of this component is independently confirmed by Big Bang Nucleosynthesis, which constrains the baryon density to $\Omega_b h^2 \approx 0.022$ — far below the total matter density $\Omega_m h^2 \approx 0.14$ measured by the CMB [5].

If dark matter is not made of ordinary ~~atoms~~, then what is it? Proposed candidates span nearly ninety orders of magnitude in mass, from ultralight axions to primordial black holes, and none is singled out by the data alone [5]. Among these, the Weakly Interacting Massive Particle (WIMP) has long occupied a privileged position because of a ~~quantitative coincidence~~ known as the “WIMP miracle”: a stable particle with mass and coupling strength at the electroweak scale, produced thermally in the early Universe, naturally yields a relic abundance consistent with the observed dark matter density [5, 6]. The same annihilation process that sets the relic abundance also predicts that WIMPs should ~~continue~~ to annihilate today in regions of high dark matter density, producing Standard Model particles — gamma rays, neutrinos, positrons, and antiprotons — that can in principle be detected by astrophysical observatories [5, 7]. This annihilation signature is what makes ~~the WIMP~~ a natural target for the gamma-ray searches developed in this thesis, and it is why we adopt it as our working hypothesis: not because it is the only viable candidate, but because its predicted Standard Model ~~yields~~ are what gamma-ray observatories can probe most directly.

This chapter lays the groundwork for the searches presented in the rest of the thesis. We begin by reviewing the multi-scale evidence for dark matter, from galactic rotation curves to the cosmic microwave background (Section 1.1). We then introduce the WIMP paradigm, deriving the thermal freeze-out mechanism and the resulting relic abundance calculation that defines

the target cross-section for experimental searches (Section 1.2). Section 1.3 surveys the three complementary experimental strategies — direct, collider, and indirect detection — and argues that indirect detection via gamma-rays provides the most direct test of the WIMP hypothesis. Finally, Section 1.4 develops the indirect detection formalism in detail — annihilation channels, spectral signatures, density profiles, and the J -factor/ D -factor framework — that connects particle physics parameters to observable gamma-ray fluxes. The observational targets and the current status of indirect searches, reviewed at the end of this chapter, provide the direct motivation for the advanced statistical and machine learning approaches that constitute the core of this thesis.

1.1 Evidence for Dark Matter

The case for dark matter rests on a convergence of independent lines of evidence spanning a wide range of astrophysical scales. We follow the hierarchical classification of Cirelli et al. [5], organizing the evidence from galactic (“mini”) to cluster (“midi”) to cosmological (“maxi”) scales. The internal consistency of these measurements — obtained with fundamentally different techniques, probing different physical environments — constitutes one of the strongest arguments ~~that dark matter is not an artifact of any single observational systematic~~ in favor of a DM interpretation in terms of new physics.

1.1.1 Galactic Scale

Among the earliest and most compelling evidence for dark matter are the internal kinematics of galaxies. In a spiral galaxy, the circular velocity $v_c(r)$ of stars and gas at distance r from the center is determined by the enclosed mass through $v_c(r) = \sqrt{GM(r)/r}$. If the visible mass — concentrated in a central bulge and an exponential disk — were the only source of gravity, the rotation curve should decline as $v_c \propto r^{-1/2}$ beyond the luminous extent, following the standard Keplerian falloff [5].

Observations tell a different story. The first hints of anomalous galactic dynamics appeared with Babcock’s 1939 study of M31, though the measurements lacked the radial extent and precision to be conclusive [8]. The modern picture took shape in 1970, when Rubin and Ford [1] obtained the first high-precision optical rotation curve of M31, tracing approximately 70 H II regions out to ~ 22 kpc and finding no evidence of the expected Keplerian decline. In the same year, Freeman identified similar discrepancies in the disk dynamics of several spiral galaxies [9]. Over the following decades, radio observations of the 21-cm hydrogen line extended rotation curves well beyond the optical disk, establishing that $v_c(r)$ remains approximately constant — or even rises slightly — out to the largest measurable radii in hundreds of systems [10, 11]. To maintain a constant circular velocity, the enclosed mass must grow linearly with radius, $M(r) \propto r$, which requires an extended, roughly spherical halo of invisible matter with density $\rho(r) \propto r^{-2}$ at large radii. The comprehensive analysis of Persic, Salucci, and Stel [2] synthesized rotation curves across a wide range of galaxy luminosities into a “Universal Rotation Curve”, demonstrating that the dark-to-luminous mass ratio increases systematically with decreasing galaxy luminosity.

Modern data have only strengthened this picture. The Spitzer Photometry and Accurate Rotation Curves (SPARC) compilation [12] provides high-quality mass models for 175 disk galaxies spanning a broad range of surface brightnesses and morphologies. These data confirm the ubiquity of flat rotation curves and reveal tight empirical scaling relations — such as the baryonic Tully-Fisher relation linking asymptotic rotation velocity to total baryonic mass — that any successful dark matter model must reproduce. Figure 1.1 illustrates this evidence: the left panel shows the original optical rotation curve of M31 from Rubin and Ford [1], while the right

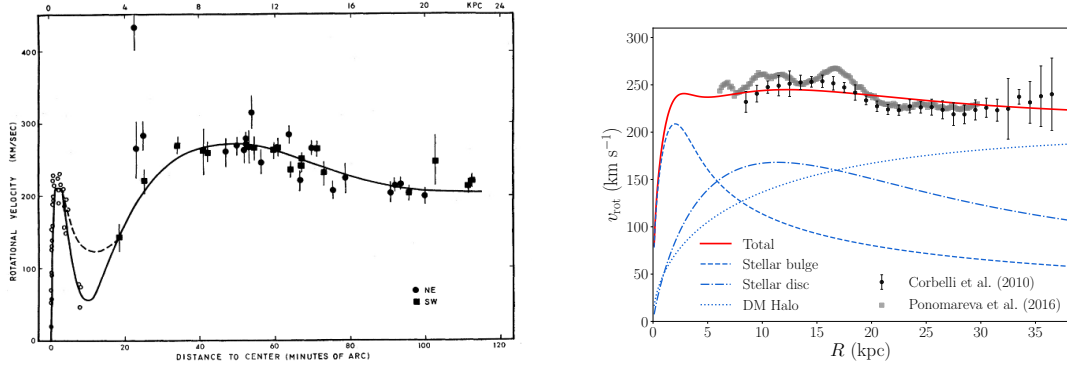


Fig. 1.1. Rotation curves and mass decomposition of M31. *Left:* the original optical rotation curve measured by Rubin and Ford [1], one of the first datasets to show that the observed velocities do not follow the expected Keplerian decline. *Right:* a modern mass decomposition from Afruni et al. [13], showing the contributions of the stellar disk, gas, and dark matter halo to the total rotation curve; the dark matter halo dominates at large radii.

panel presents a modern mass decomposition demonstrating the dominant contribution of the dark matter halo at large radii.

An independent line of evidence comes from the satellite galaxies of the Milky Way. Dwarf spheroidal galaxies (dSphs) — small, gas-poor stellar systems such as Draco, Sculptor, and Segue 1 — orbit within the gravitational potential of our Galaxy and represent some of the smallest and faintest galaxies known. Unlike the disk galaxies discussed above, dSphs **lack organized rotation**; their stars move on essentially random orbits, making these systems **pressure-supported** rather than rotationally supported. The simple relation $v_c(r) = \sqrt{GM(r)/r}$ that governs rotating disks is therefore not directly applicable, and the enclosed mass **must instead be** inferred from the line-of-sight velocity dispersion σ_* of the stellar population. Wolf et al. [14] showed that the mass enclosed within the three-dimensional half-light radius is robustly estimated by $M(< r_{1/2}) \approx 3\sigma_*^2 r_{1/2}/G$, a result that is largely insensitive to the unknown velocity anisotropy. Applying this estimator reveals extreme mass-to-light ratios: from $M/L \sim 10 M_\odot/L_\odot$ in the classical dwarfs, to several hundred in Draco ($M/L \sim 300$), to well over a thousand in ultra-faint systems like Segue 1 [5, 14, 15]. These ratios cannot be explained by baryonic mechanisms alone, establishing dSphs as the most dark-matter-dominated objects known. Their low astrophysical backgrounds also make them prime targets for indirect dark matter searches, a point to which we return in Section 1.3.

1.1.2 Galaxy Cluster Scale

Moving to larger scales, the dynamics of galaxy clusters provide a second, **independent** class of evidence for dark matter. The earliest quantitative argument **dates** to 1933, when Zwicky studied the radial velocities of galaxies in the Coma Cluster [16]. Applying the virial theorem — which relates the time-averaged kinetic energy of a gravitationally bound system to its potential energy, $2\langle K \rangle + \langle U \rangle = 0$ — he estimated the cluster’s total mass and found a mass-to-light ratio of ~ 400 , far exceeding what could be accounted for by the luminous galaxies alone. Zwicky concluded that the cluster must contain a large amount of unseen matter, which **he termed** “**dunkle Materie**” — dark matter.

Modern X-ray observations have placed these early dynamical estimates on much firmer

ground. Galaxy clusters contain massive reservoirs of ionized gas — the intracluster medium (ICM) — heated to temperatures of 10^7 – 10^8 K as it falls into the cluster potential well. This gas radiates thermally through bremsstrahlung emission, observed in the X-ray band [17]. Under the assumption of hydrostatic equilibrium, the outward gas pressure gradient balances the inward gravitational pull, so the total gravitating mass can be reconstructed from the observed gas temperature and density profiles. These analyses consistently show that the visible galaxies plus the X-ray-emitting gas account for only $\sim 15\%$ of the total cluster mass [5], with the rest attributable to a dark, non-luminous component.

The Sunyaev-Zel'dovich (SZ) effect [18] offers a complementary probe. When cosmic microwave background (CMB) photons traverse a cluster, they undergo inverse-Compton scattering off the hot ICM electrons, gaining energy and producing a characteristic spectral distortion [5, 19]. The amplitude of this distortion is proportional to the integrated electron pressure along the line of sight and, unlike X-ray surface brightness, does not diminish with redshift. Combined with independent constraints on the baryon density from Big Bang Nucleosynthesis (BBN) and the CMB (where $\Omega_b \approx 0.05$), SZ measurements of the cluster baryon fraction $f_b \approx \Omega_b/\Omega_m$ yield a total matter density $\Omega_m \approx 0.3$, well above the baryonic contribution [4, 5].

Gravitational lensing provides a direct method for mapping the total mass distribution of clusters, independent of assumptions about dynamical equilibrium. Strong lensing in the dense cluster core produces highly distorted arcs and multiple images of background galaxies, while weak lensing measures the subtle statistical shearing of background galaxy shapes over larger angular scales [5, 19]. Both regimes reconstruct the total mass distribution — visible and dark — from the geometry of light paths alone.

The most striking application of gravitational lensing is the Bullet Cluster (1E 0657-558), analyzed by Clowe et al. [3]. This system consists of two galaxy clusters that passed through each other roughly 150 million years ago. X-ray observations reveal that the collisional baryonic gas from the two clusters interacted strongly, producing a bow shock and lagging behind the galaxies. Weak-lensing mass maps, however, show that the bulk of the total mass is spatially offset from the gas, coinciding instead with the collisionless galaxies [3]. This spatial separation is visible in Figure 1.2: the green mass contours reconstructed through weak lensing are displaced from the X-ray-emitting gas, demonstrating that the dominant mass component is collisionless. The observed separation matches the behavior expected of collisionless dark matter and contradicts modified gravity theories such as MOND [20], in which the lensing signal should trace the baryonic gas [5]. Harvey et al. [21] extended this analysis to 72 colliding clusters, confirming the separation of dark matter from baryonic gas at greater than 7σ significance and setting stringent upper limits on the dark matter self-interaction cross-section [5].

1.1.3 Cosmological Scale

At the largest scales, precision cosmological observations provide the tightest constraints on dark matter and its non-baryonic nature.

The cosmic microwave background (CMB) radiation is the thermal relic of the hot, dense early Universe. Roughly 380,000 years after the Big Bang, the expanding cosmos cooled sufficiently for protons and electrons to combine into neutral hydrogen — an epoch known as *recombination*. Before this transition, baryonic matter existed as a hot plasma tightly coupled to photons through Thomson scattering; after recombination, photons streamed freely, and this “last scattering surface” is what we observe today as the CMB. Its temperature is remarkably uniform at ~ 2.725 K, but carries tiny anisotropies at the level of $\sim 10^{-5}$ that encode detailed information about the matter-energy content of the cosmos.

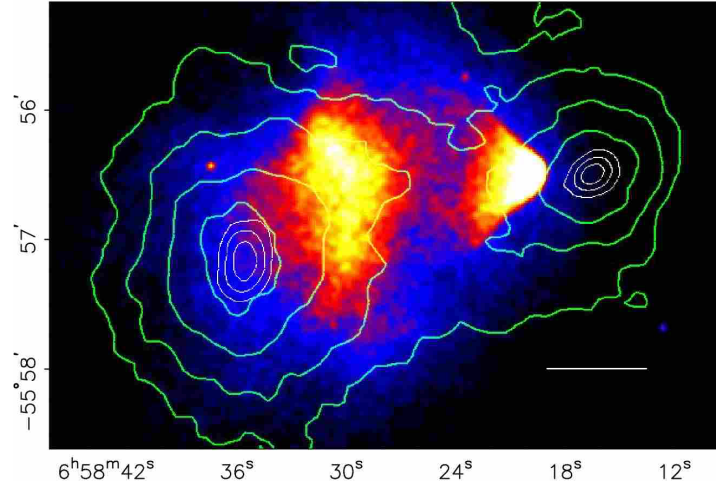


Fig. 1.2. The Bullet Cluster (1E 0657-558). The color map shows the X-ray surface brightness of the hot intracluster gas from *Chandra* observations, while the green contours trace the total mass distribution reconstructed through weak gravitational lensing. The clear spatial offset between the two demonstrates that the dominant mass component is collisionless, as expected for dark matter. The white bar corresponds to 200 kpc. Credit: Cirelli et al. [5], after Clowe et al. [3].

These anisotropies are conventionally decomposed into angular scales via the temperature power spectrum C_ℓ , which quantifies the variance of temperature fluctuations at each angular multipole ℓ — with larger ℓ corresponding to smaller angular separations on the sky. Peaks in C_ℓ correspond to sound waves in the pre-recombination baryon-photon plasma that were caught at maximum compression or rarefaction at the moment of decoupling; the positions and heights of these peaks encode the composition and geometry of the Universe [4, 5]. Baryons load the oscillating fluid, enhancing the compressional (odd-numbered) peaks relative to the rarefaction (even-numbered) ones, so that the ratio of the first to the second peak directly constrains the physical baryon density. Dark matter, which does not couple to the photon pressure, provides the gravitational scaffolding within which the oscillations occur: its abundance sets the overall depth of the potential wells and modulates the relative heights of successive peaks. In particular, a universe without cold dark matter would produce a strongly suppressed third peak (cf. Figure 1.3), in clear contradiction with the observed spectrum [5]. The Planck 2018 analysis yields $\Omega_c h^2 = 0.1200 \pm 0.0012$ for the cold dark matter density and $\Omega_b h^2 = 0.02237 \pm 0.00015$ for the baryon density [4], confirming that dark matter is roughly five times more abundant than ordinary matter. Figure 1.3 illustrates the contrast between the observed spectrum and the prediction for a baryon-only universe.

The observed large-scale structure of the Universe delivers a complementary argument for dark matter. In a purely baryonic universe, structure formation would have been severely hindered: before recombination, baryons were tightly coupled to photons through Thomson scattering, and their density perturbations were erased on small scales by photon diffusion — a process known as *Silk damping* [22]. Cold dark matter, which does not interact electromagnetically, is immune to this damping and began to cluster gravitationally during the matter-dominated era, well before recombination [23]. The potential wells established by dark matter then acted as seeds into which baryons fell after decoupling, seeding the formation of galaxies and clusters (see Section 5.1 for a detailed treatment of dark matter halo formation and substructure). Without

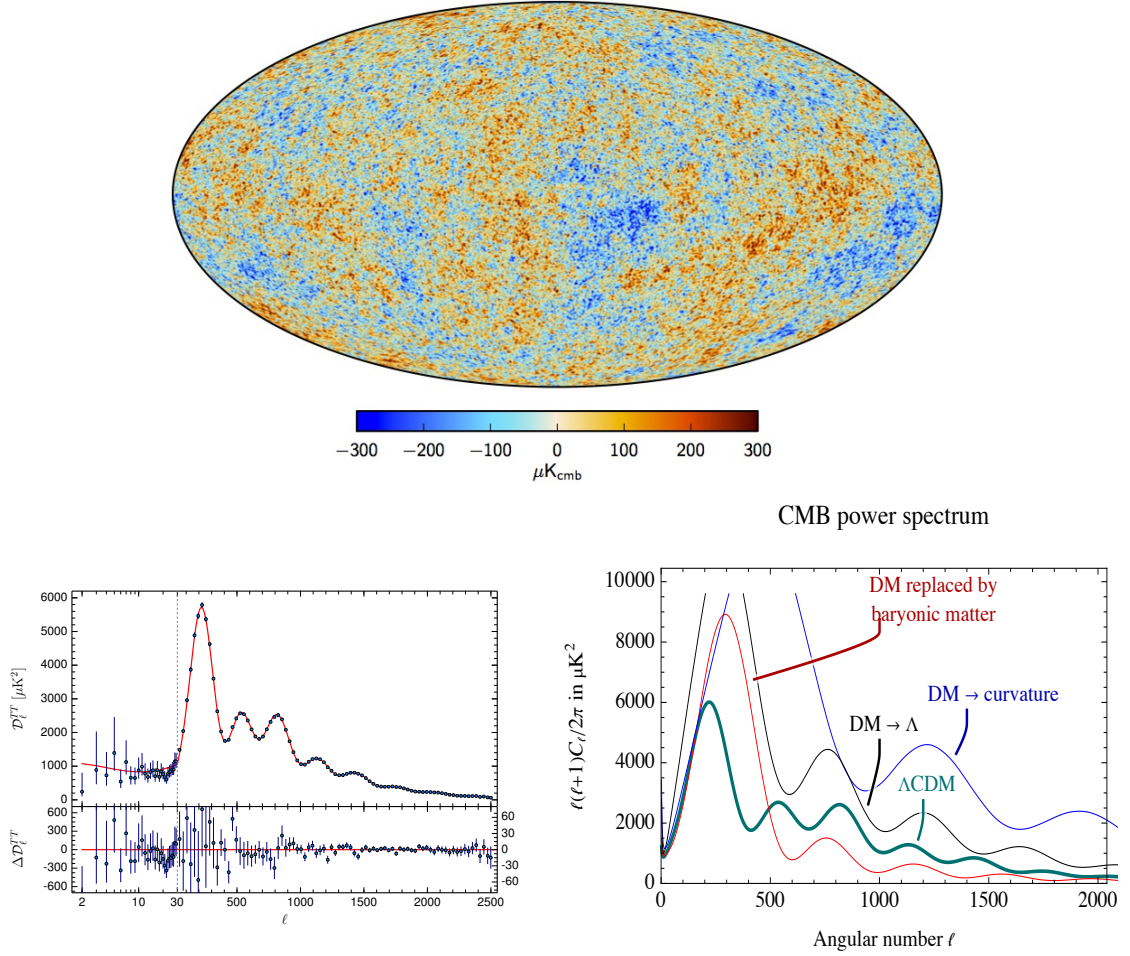


Fig. 1.3. The imprint of dark matter on the cosmic microwave background. *Top:* the Planck CMB temperature anisotropy map [4], showing the $\sim 10^{-5}$ fluctuations in the otherwise uniform 2.725 K radiation field. *Bottom left:* the observed angular power spectrum C_ℓ^{TT} of the CMB temperature anisotropies, measured by the Planck satellite [4]. *Bottom right:* the same quantity computed in a universe without cold dark matter, showing the dramatic suppression of the third acoustic peak. Adapted from Cirelli et al. [5].

this pre-existing dark matter framework, the matter power spectrum $P(k)$ cannot be reproduced; Figure 1.4 illustrates the striking difference in predicted large-scale structure between a universe with and without cold dark matter.

Beyond the overall shape of the matter power spectrum, the primordial sound waves left a second, sharper imprint on the distribution of matter. Baryon acoustic oscillations (BAO) preserve this imprint in the large-scale galaxy distribution. The same pressure waves that produced the acoustic peaks in the CMB power spectrum were also imprinted on the baryonic matter: when the baryon-photon plasma decoupled at recombination, the outward-propagating sound shells froze in place, leaving a characteristic comoving scale of ~ 150 Mpc in the spatial correlations of galaxies. This scale appears as a well-defined bump in the two-point correlation function and provides a “standard ruler” for measuring the expansion history of the Universe. Its first detection in the galaxy distribution by Eisenstein et al. [24] delivered independent confirmation of the ΛCDM

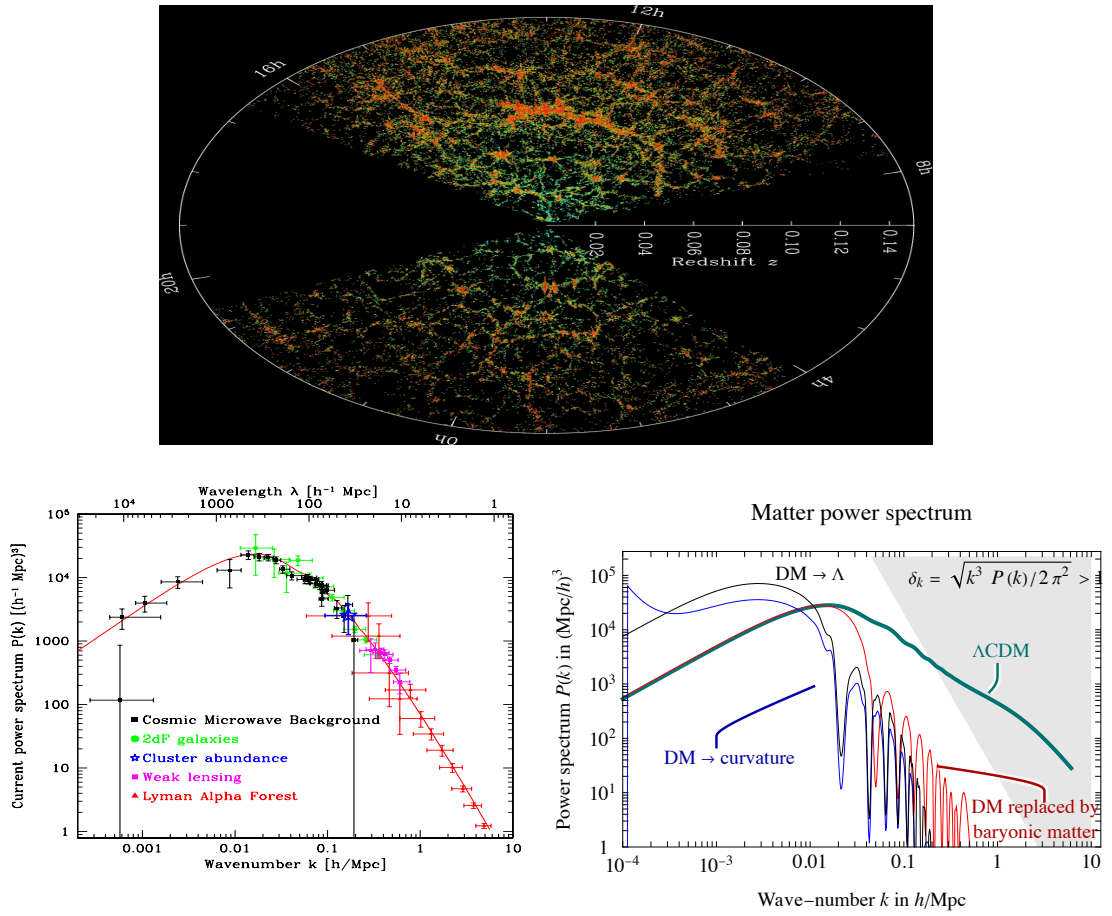


Fig. 1.4. The imprint of dark matter on large-scale structure. *Top:* the large-scale distribution of galaxies from the Sloan Digital Sky Survey (SDSS). *Bottom left:* the observed matter power spectrum $P(k)$. *Bottom right:* the same quantity computed in a universe without cold dark matter, showing the absence of small-scale structure. Adapted from Cirelli et al. [5].

parameters. BAO measurements of the expansion rate, combined with distance measurements from Type Ia supernovae — which established the accelerated expansion of the Universe [25, 26] — and CMB data, constrain the dark energy density to $\Omega_\Lambda \approx 0.7$ and the total matter density to $\Omega_m \approx 0.3$ [4, 27].

The evidence reviewed so far establishes that the total matter density is $\Omega_m h^2 \approx 0.14$, while the CMB independently constrains the baryon density to $\Omega_b h^2 \approx 0.022$ [4] — nearly seven times smaller. Could this missing matter simply be a form of ordinary matter that happens to be dark: cold molecular gas clouds, compact stellar remnants, or some as-yet-undetected baryonic component hiding in the intergalactic medium? Big Bang Nucleosynthesis (BBN) decisively rules out this possibility. The production of light elements during the first few minutes after the Big Bang is highly sensitive to the baryon-to-photon ratio, and the observed primordial abundances of deuterium, ^4He , and ^7Li independently constrain the baryon density to $\Omega_b h^2 \approx 0.022$ [28, 29] — in agreement with the CMB, but through entirely different physics. Crucially, this constraint applies to the *total* baryon density, not just the visible fraction: any baryonic matter, however dark or diffuse, would have participated in nucleosynthesis and altered the primordial element

yields. The observed abundances leave no room for a hidden baryonic reservoir. Whatever dark matter is, it must therefore be *non-baryonic* — a new form of matter beyond the Standard Model — a conclusion that motivates the particle physics discussion of the following section.

1.2 The WIMP Paradigm

[AA: done, it's in a good state, need a few more tweaks]

The evidence assembled in the previous section suggests that dark matter constitutes the dominant form of matter in the Universe and that it cannot be made of Standard Model particles. This conclusion opens a large parameter space of possible candidates: viable masses span roughly ninety orders of magnitude, from ultra-light bosonic fields with $m \sim 10^{-22}$ eV that behave as classical waves on galactic scales [30] up to macroscopic primordial black holes, with interaction strengths ranging from purely gravitational to just below current experimental sensitivity [5]. In this section, we narrow our focus to the candidate class which is most relevant to Chapters 4 through 8 of this thesis: *weakly interacting massive particles* (WIMPs). In fact, the so-called *WIMP miracle* — the observation that a thermal relic with electroweak-scale mass and coupling naturally reproduces the observed dark matter abundance — has made WIMPs one of the most explored paradigms over the past two decades.

Section 1.2.1 surveys the candidate landscape — the main classes (WIMPs, axions, sterile neutrinos, primordial black holes, FIMPs) together with modifications of the cold collisionless paradigm such as warm and self-interacting dark matter — to place the WIMP focus in context. Section 1.2.2 describes the thermal freeze-out mechanism, presents the Boltzmann equation governing the comoving abundance, and identifies the thermally averaged annihilation cross section $\langle\sigma v_{\text{rel}}\rangle$ as the central observable linking early-universe cosmology to present-day signals; the canonical benchmark $\langle\sigma v\rangle_{\text{cosmo}} \approx 2.2 \times 10^{-26} \text{ cm}^3/\text{s}$ emerges from this analysis. Section 1.2.3 reviews the theoretical bounds that bracket the allowed WIMP mass range, with the Lee–Weinberg argument setting a lower edge near a few GeV and partial-wave unitarity capping the upper end near 100 TeV, jointly defining the GeV–100 TeV thermal window targeted by gamma-ray experiments. Section 1.2.4 discusses the refinements that complicate the mapping between the freeze-out cross section and the present-day annihilation rate: velocity dependence, Sommerfeld enhancement, co-annihilation, and resonances. The thermal-relic cross section, established by the freeze-out calculation, serves as the primary benchmark for the indirect-detection analyses throughout this thesis.

1.2.1 The Landscape of Candidates

Having ruled out every Standard Model particle as the dominant matter component — baryons fall a factor of six short of the total matter density (Section 1.1.3), while hot relics such as Standard Model neutrinos free-stream out of the forming potential wells that observations require [19] — the search turns to new physics. As Figure 1.5 illustrates, the range of possibilities is enormous: candidate masses span roughly ninety orders of magnitude, from ultra-light bosonic fields near 10^{-22} eV, below which the de Broglie wavelength would exceed the size of observed dwarf galaxies [30], up to macroscopic primordial black holes of tens of solar masses, and interaction strengths range from purely gravitational to just below current direct-detection sensitivity. A separate, quantum-mechanical floor on fermionic candidates comes from the *Tremaine–Gunn bound* [31, 32]: Pauli exclusion limits the phase-space density that a fermionic species can achieve, requiring $m \gtrsim 0.4\text{--}1$ keV to match the dense inner profiles observed in dwarf-spheroidal halos.

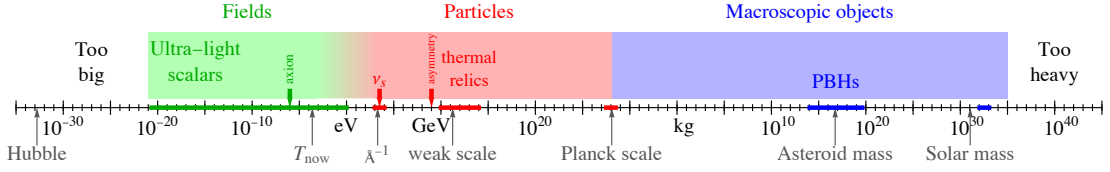


Fig. 1.5. The dark-matter mass landscape: viable candidate masses span roughly ninety orders of magnitude, from ultra-light bosonic fields to macroscopic primordial black holes. Each class is characterized by a distinct production mechanism and probed by a complementary set of experiments. Credit: Cirelli et al. [5]. [AA: consider replacing with a custom figure for the final version]

Dark matter must therefore be a new, non-baryonic, cold or at most warm form of matter; the open question is which microphysical realization Nature has chosen.

Weakly interacting massive particles occupy the GeV–TeV mass range and interact with Standard Model particles through weak-scale couplings. They are produced through thermal freeze-out in the early Universe, a mechanism we describe in detail in Section 1.2.2. The observation that a particle with weak-scale mass and coupling naturally yields the observed relic abundance — the WIMP miracle — has made WIMPs one of the dominant paradigms in the dark matter field for over two decades. Concrete realizations include the lightest neutralino of supersymmetric extensions of the Standard Model [33] and *minimal dark matter* scenarios in which a single electroweak multiplet, stabilized by an accidental symmetry, yields the correct relic density without additional model-building [34]. For this thesis, the appeal of the WIMP framework is more immediate: the same annihilation cross section that sets the relic density also fixes the rate at which WIMPs annihilate into gamma rays in present-day halos, so that the indirect-detection signals studied in Chapters 4–8 probe directly the parameter that cosmology constrains.

Axions are among the most popular alternatives to WIMPs, and the tightening of direct-detection limits on weakly interacting particles has only increased their relative weight in the community. The axion was originally proposed by Peccei and Quinn [35] as the pseudo-Nambu–Goldstone boson of a spontaneously broken global symmetry introduced to solve the strong CP problem of QCD [36, 37]. It acquires a small mass at the QCD phase transition, and an initial misalignment between its field value and the potential minimum sources a coherent oscillating condensate that today behaves as cold dark matter. This *misalignment mechanism* [38–40] is generic to any light scalar with a late-developing potential and naturally explains why such a feebly coupled particle can saturate the observed relic density. Because axions are bosons, they evade the Tremaine–Gunn bound and can sit at occupation numbers $N \gg 1$, behaving as a classical field rather than as a gas of quanta. The phenomenologically viable range of axion masses is broad, spanning roughly 10^{-12} to 10^7 eV once astrophysical, cosmological, and laboratory bounds are combined [41, 42].

A separate keV-scale window is occupied by sterile (right-handed) neutrinos, which do not couple to the Standard Model weak interaction but mix slightly with active neutrinos. Two production mechanisms are commonly considered: the Dodelson–Widrow scenario of non-resonant oscillations in the early Universe [43], and the Shi–Fuller mechanism in which a primordial lepton asymmetry drives resonant oscillations and shifts the produced spectrum to colder momenta [44]. Resonant production is the more flexible of the two because it evades the

most stringent free-streaming bounds from Lyman- α forest data and Milky Way satellite counts, which otherwise push the allowed mass above 20–30 keV [32]. Sterile neutrinos are quasi-stable but decay radiatively as $N \rightarrow \nu\gamma$, yielding a monochromatic X-ray line that provides a distinctive observational signature; the canonical reviews include those of Boyarsky et al. [32] and the keV white paper of Drewes et al. [45].

Primordial black holes (PBHs) offer a qualitatively different option in which the dark matter is not a new particle at all but a population of macroscopic objects formed from the collapse of large density perturbations before Big Bang Nucleosynthesis. Because PBHs are not made of baryons after BBN, they bypass the baryon-budget constraint of Section 1.1.3. Microlensing surveys, dynamical bounds on wide stellar binaries and ultra-faint dwarfs, and CMB spectral-distortion limits together exclude PBHs as the dominant dark-matter component across most of the mass range [46, 47]. The single surviving window for a monochromatic PBH population to constitute all of the dark matter is the asteroid-mass range, $10^{-16} M_{\odot} \lesssim M \lesssim 3 \times 10^{-12} M_{\odot}$, where Hawking-evaporation bounds from below and microlensing limits from above leave a narrow gap that next-generation experiments aim to close. [AA: check again this number, the plot is small]

An alternative production mechanism is *freeze-in*, in which feebly interacting massive particles (FIMPs) couple so weakly to the thermal bath that they never reach equilibrium [48]. Their abundance is built up gradually through rare decays and $2 \rightarrow 2$ scatterings, and the final relic density grows with the coupling rather than shrinking — the inverse of the freeze-out logic described in Section 1.2.2. The viable FIMP mass range is correspondingly broad, and the extreme smallness of the couplings places FIMPs out of reach of direct-detection and collider searches, shifting attention to cosmological and astrophysical probes.

Beyond these candidate-based classifications, dark matter may also depart from the cold, collisionless idealization on which Λ CDM is built. Several persistent tensions at sub-galactic scales motivate this possibility: the cusp/core problem in dwarf galaxies, the diversity of rotation curves at fixed maximum circular velocity, and the too-big-to-fail puzzle of the Milky Way satellites all point to fewer dense, cuspy halos than collisionless N -body simulations predict [19]. One response is *warm dark matter* (WDM), a kinematic classification encompassing any candidate with a free-streaming length intermediate between hot and cold relics; the resulting suppression of small-scale power softens these tensions while leaving the large-scale successes of Λ CDM intact. Resonantly produced keV sterile neutrinos are one concrete realization, but WDM as a classification is broader, and the astrophysical tests that constrain it are reviewed in [49]. A second, orthogonal direction is *self-interacting dark matter* (SIDM), originally proposed by Spergel and Steinhardt [50] to thermalize the inner regions of halos through elastic scattering and so produce cored rather than cuspy density profiles. The review of Tulin and Yu [51] summarizes the status of the program, and the ETHOS framework [52] provides a systematic way to integrate non-standard dark-matter interactions into cosmological hydrodynamical simulations for direct comparison with dwarf-galaxy data.

This catalogue of candidates makes clear how little the existing evidence constrains the microphysics of the dark sector. We narrow our attention now to the WIMP paradigm — the candidate class probed by the gamma-ray indirect-detection analyses developed in the later chapters of this thesis. The appeal of this framework is that its phenomenology is governed by a single quantity, the velocity-averaged annihilation cross section $\langle\sigma v\rangle$, which simultaneously fixes the cosmological relic abundance, the elastic-scattering rate on nuclei, the production cross section at colliders, and the present-day annihilation rate in galactic halos. The following subsection reviews the freeze-out mechanism that explains how $\langle\sigma v\rangle$ governs the relic density, and why the resulting target value defines the natural benchmark for the experimental program that follows.

1.2.2 Thermodynamics of the Early Universe

[AA: is this section too qualitative?]

Consider a hypothetical species χ in chemical equilibrium with the Standard Model plasma of the early Universe, sustained by pair-annihilation reactions $\chi\chi \leftrightarrow \text{SMSM}$ that proceed in both directions at equal rates. As the Universe expands and cools, two processes run in competition: the cosmological expansion, which dilutes everything and lowers the temperature, and the annihilation rate, which depends on the density of χ particles still available to find each other. Eventually the expansion wins: the reactions become too slow to maintain equilibrium, the comoving number density stops tracking its equilibrium value, and whatever population remains is locked in as the relic abundance we observe today. The purpose of this subsection is to make that picture quantitative enough to identify what kind of particle reproduces $\Omega_\chi h^2 \approx 0.12$, with a result that may seem surprising: the observable that controls the answer is not the mass of χ but its velocity-averaged annihilation cross section.

A particle that annihilates strongly stays in chemical equilibrium for a long time: it follows the equilibrium population as the latter drops exponentially, and only departs from it when the equilibrium abundance has already become small. Such a particle leaves behind few relics. A particle that annihilates weakly, by contrast, loses contact with the bath earlier, while its equilibrium abundance is still relatively large, and therefore freezes out at a higher residual density. In the thermal-relic context, the weakest particle wins. The dark-matter abundance is set by how little a candidate annihilates, not by how massive it is; the mass enters the final answer only logarithmically, through the temperature at which decoupling occurs, while the velocity-averaged annihilation rate $\langle\sigma v_{\text{rel}}\rangle$ sets the relic abundance to first approximation.

Figure 1.6 captures this picture in a plot of the comoving abundance $Y \equiv n_\chi/s$ against the dimensionless inverse temperature $x \equiv M_\chi/T$. At high temperatures ($x \ll 1$), χ is effectively relativistic and present in numbers comparable to the photons; the actual abundance hugs the equilibrium curve Y_{eq} . Once the temperature drops below the rest mass ($x \gtrsim 1$), producing a χ pair costs an exponential Boltzmann penalty, and Y_{eq} falls off as e^{-x} . The actual abundance tracks this decline only as long as the annihilation rate $\Gamma = n_\chi \langle\sigma v_{\text{rel}}\rangle$ exceeds the Hubble rate H ; once Γ drops below H , annihilations can no longer keep up with expansion, Y peels away from Y_{eq} , and the comoving abundance plateaus at its asymptotic value Y_∞ . For a weakly interacting candidate at the electroweak scale, this freeze-out happens at $T_{\text{fo}} \sim M_\chi/(20\text{--}25)$ — already non-relativistic, but well before the present epoch.

A quantitative description of this process follows the comoving number density through the Boltzmann equation [5, 53],

$$\dot{n}_\chi + 3Hn_\chi = -\langle\sigma v_{\text{rel}}\rangle(n_\chi^2 - n_{\text{eq}}^2), \quad (1.2.1)$$

where H is the Hubble expansion rate, n_{eq} the equilibrium number density, and n_χ the dark matter number density. The $3Hn_\chi$ piece on the left is the dilution due to Hubble expansion; the $-\langle\sigma v_{\text{rel}}\rangle n_\chi^2$ piece on the right is the depletion of χ by annihilation into Standard Model final states; and the $+\langle\sigma v_{\text{rel}}\rangle n_{\text{eq}}^2$ piece is the inverse process, in which χ pairs are recreated from the thermal bath at the rate dictated by detailed balance. It is convenient to factor out the trivial expansion dilution by passing from n_χ to the comoving abundance $Y \equiv n_\chi/s$, where s is the entropy density. Because the total entropy in a comoving volume is conserved during adiabatic expansion, Y remains constant whenever the dark-matter number per comoving volume is also conserved; variations of Y thus track only annihilation and creation, rather than the cosmological dilution shared by every species.

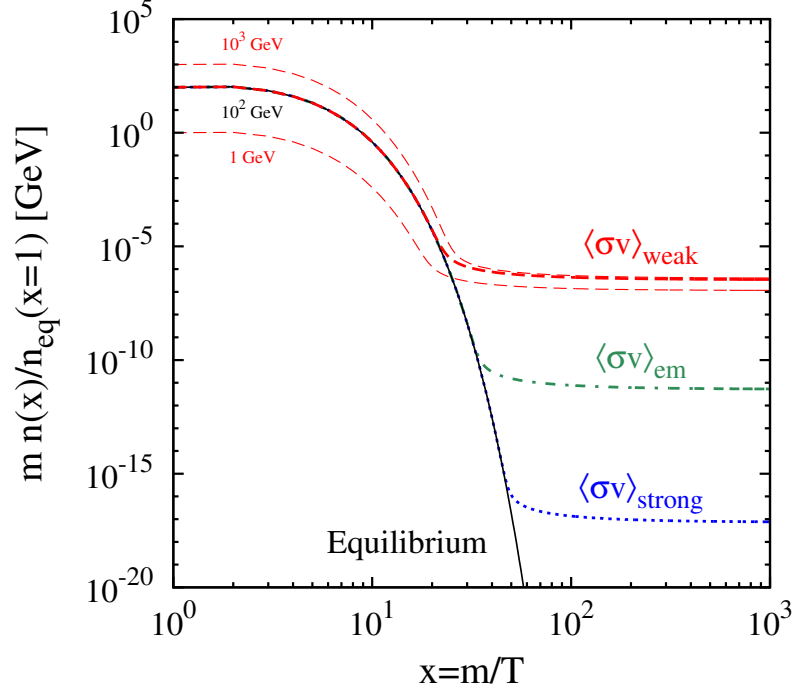


Fig. 1.6. Comoving abundance $Y \equiv n_\chi/s$ as a function of the dimensionless inverse temperature $x \equiv M_\chi/T$. The dashed curve traces the equilibrium abundance Y_{eq} , exponentially Boltzmann-suppressed once T falls below M_χ . The solid curves show the actual abundance for three values of the velocity-averaged annihilation cross section, illustrating that stronger annihilation tracks equilibrium longer and freezes out at a smaller residual abundance. Credit: Steigman et al. [6].

A subtlety worth flagging is the meaning of $\langle\sigma v_{\text{rel}}\rangle$, since the dark-matter literature uses the term “cross section” for two physically distinct objects. The σv_{rel} that enters Eq. (1.2.1) is the cross section for the process $\chi\chi \rightarrow \text{SMSM}$ multiplied by the relative velocity of the incoming pair, then averaged over a Maxwell–Boltzmann distribution at temperature T ; it carries units of cm^3/s . The cross section quoted in direct-detection experiments, by contrast, is the elastic cross section $\sigma_{\chi N}$ for the recoil process $\chi + N \rightarrow \chi + N$ on a target nucleon, measured in cm^2 and involving no thermal average. The two thus refer to different reactions, different kinematics, and different epochs in the dark-matter history. Throughout this thesis the symbol $\langle\sigma v\rangle$ in cm^3/s always denotes the annihilation case. One further subtlety is that the thermal average entering the Boltzmann equation is evaluated at freeze-out velocities $v_{\text{rel}} \sim 0.2c$, whereas dark-matter particles in present-day galactic halos move at velocities set by the gravitational potential, $v_{\text{rel}} \sim 10^{-3}c$ — roughly two orders of magnitude smaller. Whether $\langle\sigma v\rangle$ depends on the relative velocity therefore controls the mapping between the cosmological cross section that fixes the relic abundance and the annihilation rate accessible to indirect detection; we explore this issue in Section 1.2.4.

Solving Eq. (1.2.1) through the freeze-out transition, in the limit of s -wave annihilation and weak-scale parameters, yields a compact result [5, 6]:

$$\frac{\Omega_\chi h^2}{0.12} \approx \frac{2.2 \times 10^{-26} \text{ cm}^3/\text{s}}{\langle\sigma v_{\text{rel}}\rangle}, \quad (1.2.2)$$

in which the relic abundance is essentially the inverse of the annihilation cross section, with the mass entering only through logarithmic corrections to the prefactor. The numerator is the canonical *thermal cross section*, $\langle\sigma v\rangle_{\text{cosmo}} \approx 2.2 \times 10^{-26} \text{ cm}^3/\text{s}$: any dark-matter candidate that annihilates at this rate reproduces, to within a few percent, the observed cosmological abundance. For an annihilation amplitude characterized by a coupling g , with $\alpha \equiv g^2/(4\pi)$, the natural size of the cross section is $\langle\sigma v\rangle \sim \alpha^2/M_\chi^2$. Plugging in the weak-interaction value $\alpha \sim \alpha_{\text{weak}} \sim 0.03$ and demanding a match to $\langle\sigma v\rangle_{\text{cosmo}}$ returns a mass M_χ of order a few hundred GeV to a few TeV — the electroweak scale, recovered without any input from particle-physics model-building. That coincidence is the *WIMP miracle*: a stable particle with mass and coupling strength at the electroweak scale, produced thermally in the early Universe, naturally yields a relic abundance consistent with the observed dark matter density [5, 6]. Equally important for this thesis, Eq. (1.2.2) fixes the same $\langle\sigma v\rangle$ that controls the annihilation rate of dark-matter particles in present-day astrophysical environments, providing the link between cosmological abundance and indirect-detection signals that makes the gamma-ray searches of Chapters 4–8 a direct test of the thermal freeze-out hypothesis.

1.2.3 Theoretical Bounds on the WIMP Mass

The relic-abundance formula of the previous subsection, combined with some additional physical requirements, brackets the viable WIMP mass from both below and above. Consider first a particle that interacts with the Standard Model only through the weak interaction, with no dark mediator of its own. At freeze-out energies the annihilation cross section for such a particle, dominated by exchange of W and Z bosons, scales as $\sigma_0 \approx G_F^2 M_\chi^2/2\pi$, where G_F is the Fermi constant [54]. This is an *s-wave* process, meaning the cross section is independent of the relative velocity — a feature we return to repeatedly, since it controls whether the same $\langle\sigma v\rangle$ that fixes the relic abundance also drives present-day annihilations in galactic halos. Because σ_0 scales as M_χ^2 , lowering the candidate mass reduces the annihilation cross section quadratically; combined with the inverse dependence of the relic abundance on $\langle\sigma v\rangle$ in Eq. (1.2.2), this means a sufficiently light particle annihilates too weakly to thin itself out and ends up overproducing dark matter. The original Lee–Weinberg analysis [55], contemporaneous with the closely related arguments of Hut [56] and Vysotsky, Dolgov, and Zeldovich [57], obtained $M_\chi \gtrsim 2 \text{ GeV}$ for a heavy Dirac neutrino; modern evaluations with updated effective-degree-of-freedom counts give [5, 54]

$$M_\chi \gtrsim 3 \text{ GeV} \quad (\text{Lee–Weinberg bound}), \quad (1.2.3)$$

below which a thermal relic coupled to the Standard Model weak interaction is cosmologically forbidden. This GeV-scale floor is specific to weak-mediated annihilation; when that assumption is dropped, the much weaker BBN bound discussed below takes over, extending the allowed lower edge into the MeV regime.

The qualifier “coupled to the Standard Model weak interaction” is essential. The Lee–Weinberg bound says nothing about models in which a new mediator — a dark photon, a light scalar, or a hidden-sector gauge boson — carries the annihilation. With such ingredients the relic abundance can be reproduced down to the sub-GeV and even keV regimes, which is precisely why a parallel low-mass direct-detection program (SENSEI, DAMIC, and related semiconductor experiments) has grown alongside the conventional GeV–TeV searches; we mention it here only to delimit the scope of the bounds derived in the rest of this subsection, which assume Standard Model mediation.

The upper end of the mass window has a more universal origin, which is often referred to as the *unitarity bound*. Expanding the amplitude in partial waves of definite orbital angular

momentum ℓ , the contribution of each partial wave to a two-body cross section obeys $\sigma_\ell \leq 4\pi(2\ell + 1)/k^2$, with k the center-of-mass momentum [58]. For non-relativistic dark-matter annihilation, $k = M_\chi v/2$, so the s -wave ($\ell = 0$) ceiling on the velocity-averaged cross section scales as $\langle\sigma v\rangle_{\max} \propto 1/M_\chi^2$. This is a ceiling that any theory must respect, regardless of the coupling structure. As the candidate mass increases, the unitarity ceiling drops; at some mass it sinks below the thermal benchmark $\langle\sigma v\rangle_{\text{cosmo}}$, and beyond that point even a maximally coupled candidate must be overabundant. The classic estimate of Griest and Kamionkowski places this crossover at [58]

$$M_\chi \lesssim 100 \text{ TeV} \quad (\text{unitarity bound}), \quad (1.2.4)$$

above which thermal-relic dark matter is no longer viable as a single-particle, two-body annihilation problem and requires either bound-state formation, multi-step annihilation cascades, or a non-thermal cosmological history.

An independent lower bound, insensitive to which mediator carries the annihilation, comes from Big Bang Nucleosynthesis [59]. A dark-matter particle still in chemical contact with the plasma at $T \sim 1 \text{ MeV}$ would contribute to the radiation energy density during nucleosynthesis and inject electromagnetic or hadronic energy through residual annihilations, both of which spoil the otherwise excellent agreement between predicted and observed primordial light-element abundances. Requiring freeze-out to complete before BBN enforces $M_\chi \gtrsim 3 \text{ MeV}$, a floor that holds for any thermal relic regardless of its couplings.

Taken together, these arguments delineate a mass window

$$3 \text{ GeV} \lesssim M_\chi \lesssim 100 \text{ TeV} \quad (1.2.5)$$

for WIMPs that annihilate through Standard Model weak interactions, with the BBN floor extending the lower edge into the MeV regime for candidates with non-Standard-Model mediators. Within this window the thermal benchmark $\langle\sigma v\rangle_{\text{cosmo}} \approx 2.2 \times 10^{-26} \text{ cm}^3/\text{s}$ is the natural target for indirect-detection experiments: any instrument that reaches this sensitivity directly tests the thermal freeze-out hypothesis. The Fermi Large Area Telescope already reaches the benchmark in the $b\bar{b}$ channel up to masses of several tens of GeV looking at dwarf spheroidal galaxies, an analysis we revisit in Chapter 4 when studying dark matter subhalos. The Cherenkov Telescope Array Observatory, with its larger collection area and lower energy threshold relative to current imaging atmospheric Cherenkov arrays, will extend indirect-detection sensitivity into the TeV domain, which is the subject of Chapter 8. We close with a caveat: the bounds derived here assume the simplest freeze-out scenario — a single stable species annihilating through a velocity-independent s -wave amplitude — and several physically motivated effects can shift the mapping between $\langle\sigma v\rangle_{\text{cosmo}}$ and the present-day indirect-detection signal in either direction. We briefly survey alternative freeze-out scenarios in the next subsection.

1.2.4 Beyond Standard Freeze-out

Velocity-dependent cross sections. The bounds derived in Section 1.2.3 assumed the simplest possible freeze-out scenario: a single stable species, two-body s -wave annihilation with velocity-independent cross section, and weak-scale perturbative couplings. Several physically motivated effects modify this picture and, more importantly, alter the mapping between the cosmological cross section $\langle\sigma v\rangle_{\text{cosmo}}$ that fixes the relic abundance and the present-day annihilation rate probed by indirect detection. Non-relativistic two-body annihilation admits the partial-wave expansion

$$\sigma v_{\text{rel}} \simeq \sigma_0 + \sigma_1 v_{\text{rel}}^2 + \mathcal{O}(v_{\text{rel}}^4), \quad (1.2.6)$$

where σ_0 is the s -wave ($\ell = 0$) contribution and $\sigma_1 v_{\text{rel}}^2$ the p -wave ($\ell = 1$) contribution [5]. When the s -wave piece is non-vanishing, the cross section is approximately constant from freeze-out velocities ($v_{\text{rel}} \sim 0.3c$) down to typical halo velocities ($v_{\text{rel}} \sim 10^{-3}c$), and $\langle\sigma v\rangle_{\text{cosmo}}$ remains a meaningful benchmark for indirect-detection searches. When the leading contribution is p -wave, in contrast, the present-day rate is suppressed by $v_{\text{rel}}^2 \sim 10^{-6}$ relative to its value at freeze-out, and the channel becomes effectively invisible to instruments that rely on annihilation in cold galactic halos. Selection rules — for instance, specific initial-state quantum numbers or symmetries of the underlying interaction — can cause the s -wave amplitude to vanish identically in particular models, leaving p -wave as the leading contribution and rendering the channel undetectable with gamma-ray telescopes.

Sommerfeld enhancement. A second class of corrections arises when dark matter couples to a mediator with mass $M_V \ll M_\chi$. The long-range attractive force distorts the incoming-pair wave function and enhances the annihilation probability at low velocities, a non-perturbative effect known as the *Sommerfeld enhancement* [60, 61]. The enhancement factor S scales as $S \propto 1/v_{\text{rel}}$ in the Coulomb regime $v_{\text{rel}} \ll \pi\alpha_g$, saturates at $v_{\text{rel}} \sim M_V/M_\chi$ for Yukawa mediators, and can exhibit resonant peaks at specific mass ratios where a zero-energy bound state appears in the dark-sector potential. Halo velocities $v_{\text{rel}} \sim 10^{-3}c$ therefore probe a regime in which S can reach $\mathcal{O}(10^2\text{--}10^3)$ [60], while freeze-out velocities $v_{\text{rel}} \sim 0.2c$ see only moderate enhancement. [AA: check if we didn't cut this part from section 1.4] The Sommerfeld effect is essential for heavy electroweak WIMPs — the pure wino at $M_\chi \sim 3\text{TeV}$, the higgsino near $M_\chi \sim 1.1\text{TeV}$, and Minimal Dark Matter candidates more generally — where it shifts the thermal-relic mass from hundreds of GeV to multi-TeV scales and dominates the present-day indirect-detection phenomenology; we return to this point in Section 1.4 when discussing J -factors and the loop-suppressed channels of Minimal Dark Matter. For the pure wino in particular, the enhanced halo rate at $v_{\text{rel}} \sim 10^{-3}c$ is large enough that current gamma-ray constraints effectively rule the candidate out for cuspy halo profiles. For the GeV-scale candidates relevant to the Galactic Center excess analyzed in Chapter 4, by contrast, the standard thermal cross section fits the observed signal without invoking Sommerfeld enhancement, as Daylan et al. [62] showed explicitly for a $\sim 30\text{--}50\text{GeV}$ WIMP annihilating at $\langle\sigma v\rangle \approx (1\text{--}3) \times 10^{-26} \text{cm}^3/\text{s}$.

Co-annihilation and resonance. Two further refinements deserve mention because they directly affect the relic-abundance–indirect-detection connection. The first is *co-annihilation*: if the dark sector contains additional particles whose masses satisfy $M' - M_\chi \lesssim \text{a few} \times T_{\text{fo}}$, the heavier partners remain Boltzmann-abundant during freeze-out and contribute to depleting the dark-matter density through both their own annihilations and χ -partner co-annihilation processes [33, 63]. After freeze-out the heavier states decay into χ , so that only the lightest species remains in the present-day halo; its annihilation cross section is then smaller than the effective freeze-out value, often by a substantial margin when the co-annihilating partners carry stronger couplings, and the link between $\Omega_\chi h^2$ and the indirect-detection signal is correspondingly weakened. The second is *resonant annihilation*: when dark matter annihilates through an s -channel mediator whose mass lies near the threshold $m_{\text{med}} \approx 2M_\chi$, the cross section follows a Breit–Wigner form with sharp velocity dependence [53, 63]. Whether the present-day rate is enhanced or suppressed relative to the freeze-out value depends on whether the resonance is accessed at thermal velocities, at halo velocities, or sits between them. Both effects introduce model dependence into the mapping from $\langle\sigma v\rangle_{\text{cosmo}}$ to the observable annihilation rate, and conclusions drawn from either freeze-out or indirect detection alone must be interpreted accordingly.

These refinements lead to several distinct theoretical scenarios, of which the standard case — s -wave annihilation, no Sommerfeld enhancement, no co-annihilation, no resonance — is the simplest and the **least model-dependent**. In that case, the freeze-out cross section also fixes the galactic-halo annihilation rate, providing the cleanest target for indirect-detection experiments. In richer scenarios the mapping remains calculable but model-dependent: Sommerfeld enhancement can boost halo rates by orders of magnitude, p -wave suppression or off-resonance Breit–Wigner effects can lower them, and co-annihilation can shift the relic-abundance target relative to the halo signal. Throughout this thesis we adopt the standard thermal-relic value of $\langle\sigma v\rangle$ as the primary benchmark when quoting cross-section bounds, while flagging explicitly the regimes in which these refinements would shift the interpretation. With this theoretical setting in place, we turn to the **three complementary** experimental strategies for testing the WIMP hypothesis.

1.3 Searching for Dark Matter

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~~If dark matter interacts with ordinary matter through any non-gravitational force, then~~ the same coupling that governs annihilation in the early universe also mediates scattering in underground laboratories and production at particle colliders. **These three approaches** — direct detection, collider production, and indirect detection — form what is commonly known as the “detection triangle”.

Each vertex of this triangle probes a different kinematic regime ~~of the same underlying interaction~~. Direct and indirect detection both observe dark matter in its natural habitat: the galactic halo, where particles drift at non-relativistic velocities of $v \sim 10^{-3}c$. Colliders, by contrast, collide particles at energies **far exceeding the dark matter mass**, accessing a fully relativistic regime. The same dark matter–Standard Model coupling can yield very different cross-sections depending on the kinematic regime: velocity-dependent or momentum-dependent suppressions that are negligible at collider energies can reduce the non-relativistic rate probed by direct and indirect detection by orders of magnitude. An interaction that produces a strong signal in one channel can therefore be entirely invisible in another. A signal in one corner combined with a non-detection in another would narrow the type of interaction at play, and together they carry complementary information that no single channel could yield. **As such, only a signal in all three would constitute an unambiguous identification.** [AA: is this phrase too strong? in the end, even if we found a new particle at colliders or in direct detection surveys, this would not mean they are the same dark matter particles responsible for the LSS of the universe.]

This section reviews direct detection and collider searches ~~in turn, with emphasis on the physical principles rather than experimental details~~. We defer a more detailed discussion of indirect detection — which is particularly relevant for this thesis — to Section 1.4.

1.3.1 Direct Detection

Direct detection experiments search for the faint recoil produced when a dark matter particle from the galactic halo scatters off an atomic nucleus inside a ~~carefully shielded~~ underground detector. We restrict the discussion to WIMP–nucleus scattering, being the channel most relevant to this thesis. A WIMP with mass M_χ and velocity $v \sim 10^{-3}c$ strikes a target nucleus of mass m_A essentially at rest, transferring a recoil energy $E_R \sim \mu_A^2 v^2 / m_A$, where $\mu_A = M_\chi m_A / (M_\chi + m_A)$ is the reduced mass of the system [64]. For a WIMP and a xenon nucleus of comparable mass ($M_\chi \sim m_A \sim 100$ GeV), this yields $E_R \sim 20$ keV — small enough that detecting it requires highly sensitive instruments, but large enough to be within reach of modern technology. In addition to

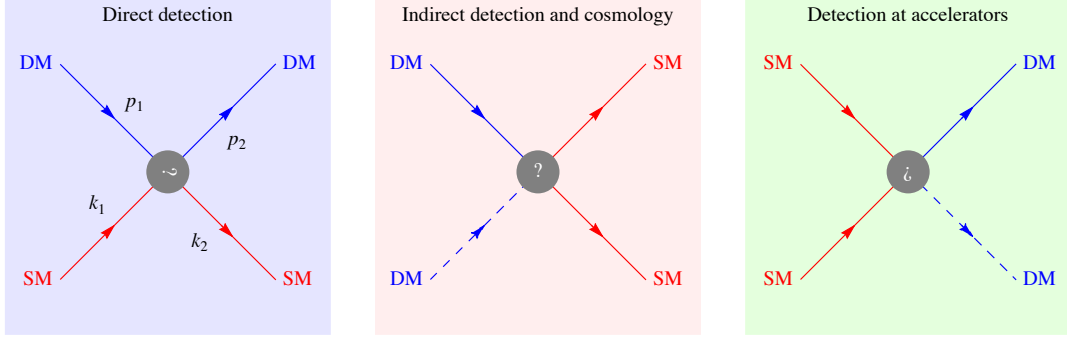


Fig. 1.7. The detection triangle: three complementary strategies for detecting dark matter, each corresponding to a different reading of the same underlying interaction vertex. *Left*: direct detection — dark matter scatters off a Standard Model target in a laboratory. *Center*: indirect detection — dark matter annihilates into Standard Model particles in astrophysical environments. *Right*: collider production — Standard Model particles produce dark matter at accelerators. Figure from Cirelli et al. [5].

the recoil energy spectrum, the Earth’s orbital motion around the Sun modulates the WIMP flux by $\sim 5\%$ annually (peaking around June 2), providing a distinctive time-dependent signature [65]. The DAMA/LIBRA experiment has observed a persistent annual modulation signal in NaI(Tl) crystals over more than two decades [66], but its interpretation as WIMPs is in strong tension with null results from other experiments using different target materials, and the claimed signal remains controversial [67].

The physics of the scattering divides into two categories determined by the non-relativistic limit of the interaction [5, 68]. In **spin-independent (SI)** scattering, the dark matter couples to the nucleon number density without reference to nuclear spin. In the limit of small momentum transfer — where the de Broglie wavelength of the dark matter particle exceeds the nuclear radius — the amplitudes from individual nucleons add coherently, producing a total cross-section that scales as $\sigma_{\text{SI}} \propto A^2$, where A is the atomic mass number [5, 68]. This quadratic enhancement is the primary reason why the most sensitive experiments use heavy target elements, particularly xenon ($A \approx 131$). **Spin-dependent (SD)** scattering, by contrast, couples to the nuclear spin J and receives no coherent enhancement, leading to bounds that are typically three to five orders of magnitude weaker than the SI case [5].

The current generation of direct detection experiments is dominated by dual-phase liquid xenon time projection chambers (TPCs), which detect both the scintillation light (S1 signal) and the ionization charge (S2 signal) from a nuclear recoil event [5]. The ratio S2/S1 provides powerful discrimination between nuclear recoils (the expected dark matter signal) and electron recoils (the dominant background). Three experiments lead the field: **LZ** (LUX-ZEPLIN) [69] in the United States, **XENONnT** [70] at the Gran Sasso laboratory in Italy, and **PandaX-4T** [71] in China. Together, they have pushed the upper limit on the SI WIMP–nucleon cross-section down to $\sigma_{\text{SI}} \lesssim 10^{-47} \text{ cm}^2$ near the minimum of the exclusion curve at $M_\chi \sim 30 \text{ GeV}$ [69–71]. At higher masses ($M_\chi \sim 100 \text{ GeV}$), the limits relax to $\sim 10^{-46} \text{ cm}^2$ as the local number density of dark matter particles decreases [72]. Beyond these noble liquid TPCs, a complementary program of cryogenic bolometers (SuperCDMS, CRESST-III) and semiconductor-based detectors (SENSEI, DAMIC) extends sensitivity to sub-GeV dark matter by searching for electron recoils rather than

nuclear recoils [5].

Standard Model neutrinos — solar, atmospheric, and from the diffuse supernova background — scatter coherently off nuclei and produce recoils kinematically indistinguishable from a WIMP signal; this irreducible background is the **neutrino fog** (or neutrino floor). The question of whether it would ever show up in data was settled in 2021, when XENON1T recorded the first observation of solar ^8B coherent elastic neutrino-nucleus scattering (CEvNS) in a dark matter detector [73]. PandaX-4T and XENONnT have since reported independent indications at roughly 2.6σ and 2.7σ , consistent with the expected solar rate [74, 75]; XENONnT also ran the first dedicated dark matter search inside the fog [75], and LZ set the world-best spin-independent WIMP bound while reporting its own ^8B indications [72]. What this means practically is that the fog is already a feature of current data: sensitivity gains slow from here, scaling as the inverse square root of accumulated exposure rather than inversely [76], and the next generation of detectors — notably the DARWIN/XLZD program [77] — will reach multi-ton xenon exposures designed to operate within the fog rather than to avoid it. Pushing substantially beyond it will likely require directional detectors capable of resolving the apparent “dark matter wind” from the Cygnus constellation against the isotropic neutrino background.

Direct detection excels in the mass range where the WIMP mass is comparable to the target nucleus mass — roughly tens of GeV for xenon targets [5]. At lower masses, the recoil energy drops below experimental thresholds; at higher masses, the local number density decreases as $n \propto 1/M_\chi$, reducing the event rate. These mass-dependent limitations, combined with our entry into the neutrino fog, motivate a complementary search strategy. Direct detection constrains the WIMP–nucleon scattering cross-section but is insensitive to the annihilation cross-section $\langle\sigma v_{\text{rel}}\rangle$ that determines the thermal relic abundance — the observable probed by indirect detection.

For comprehensive recent treatments of direct detection covering experimental techniques, current bounds, and the path beyond the neutrino fog, we refer the reader to the reviews by Schumann [78] and Cirelli et al. [5], and to Hooper [68].

1.3.2 Collider Searches

If dark matter couples to quarks or gluons, it can be pair-produced in high-energy collisions at the Large Hadron Collider. The challenge is detection: dark matter particles carry no electric or color charge and escape the detector without leaving a trace, just as neutrinos do [5, 68]. The only unavoidable signature is **missing transverse energy** (E_T^{miss}), inferred from the momentum imbalance of the visible particles in the event [5].

Pure pair-production of dark matter ($pp \rightarrow \chi\chi$) produces no visible particles and goes entirely unnoticed. A detectable signal requires that an additional Standard Model particle — a jet, a photon, or a W/Z boson — is radiated from the initial state, recoiling against the invisible dark matter pair. These are the **mono-X searches**: mono-jet, mono-photon, mono- Z , and mono-Higgs, each paying a cross-section penalty of roughly $g^2/(4\pi)^2 \sim 10^{-3}$ for the additional emission [5]. A complementary channel probes **invisible Higgs decays**: if the dark matter is lighter than half the Higgs boson mass ($M_\chi < M_h/2$), the Higgs can decay directly into a dark matter pair. Current LHC measurements constrain the invisible branching ratio to $\text{BR}_{\text{inv}} \lesssim 0.11$ at 95% C.L., essentially excluding Higgs-portal thermal relic dark matter below $M_h/2 \approx 63$ GeV [5, 79]. These results are interpreted within **simplified models** that include an explicit mediator and its couplings to quarks and dark matter, enabling direct comparison with direct and indirect detection bounds [80]. Current data exclude s -channel mediators in roughly the 0.1–4 TeV range and push the lower bound on colored mediators (squarks, gluinos) to 1–2 TeV [5, 68], probing interaction strengths that non-relativistic direct detection cannot reach.

Despite their power, collider experiments face a fundamental limitation: even if a new invisible particle is produced and characterized, the LHC cannot establish that it is cosmologically stable or that it constitutes the dark matter observed in gravitational systems. ~~Confirming the cosmological connection — that the particle froze out in the early universe with the correct relic abundance — requires measuring the annihilation cross-section directly. This is the domain of indirect detection.~~

For a comprehensive ~~recent~~ treatment of dark matter searches at colliders, including the ~~simplified model framework and the LHC mono-X programme~~, we refer the reader to the review by Boveia and Doglioni [80], and to the broader reviews by Cirelli et al. [5] and Hooper [68].

Among the three vertices of the detection triangle, direct detection and collider searches — reviewed above — constrain the scattering cross-section ~~and the production mechanism, respectively~~, but neither directly measures the annihilation cross-section $\langle\sigma v_{\text{rel}}\rangle$ that determines the thermal relic abundance. ~~Indirect detection is the only strategy that probes this quantity, and gamma-rays are its most informative messenger. Because indirect detection is the approach pursued throughout this thesis, we devote the next section entirely to it.~~

1.4 Indirect Detection via Gamma-Rays

~~Section 1.3 established that indirect detection via gamma-rays directly probes the annihilation cross-section — the quantity that governs the thermal relic abundance and the one quantity that neither underground laboratories nor colliders can access. This section develops the underlying formalism. We follow the signal from particle physics — the annihilation and decay rates, the spectra they produce — through astrophysics — the density profiles that determine where the dark matter sits — to derive the equations that predict the gamma-ray flux from any given target. These expressions, and the assumptions that enter them, provide the quantitative foundation for every analysis presented in this thesis.~~

~~The development begins with the microphysics of annihilation and decay and the gamma-ray spectra they produce (§1.4.1–§1.4.2), then introduces the dark matter density profiles that determine how signal strength varies across the sky (§1.4.3). These ingredients combine into the J -factor and D -factor formalism — a factorization of the predicted flux into a particle physics term and an astrophysical term (§1.4.4) — before we survey the observational targets that define the thesis programme (§1.4.5–§1.4.7).~~

1.4.1 Annihilation and Decay Physics

In the Λ CDM framework, structure forms hierarchically from cold, non-relativistic dark matter, and the particles that populate present-day galactic halos retain typical velocities of order $v \sim 10^{-3} c$ [68]. They therefore annihilate or decay essentially at rest. This kinematic fact has two immediate consequences. First, the total center-of-mass energy available in an annihilation event is simply $\sqrt{s} = 2m_\chi$, so the most energetic photon that can be produced carries energy $E_\gamma = m_\chi$. For decay, the available energy is instead m_χ , and momentum conservation among at least two ~~daughter~~ particles limits the maximum photon energy to $E_\gamma = m_\chi/2$. These hard kinematic endpoints are among the most distinctive features of a dark matter signal, since known astrophysical gamma-ray sources do not generically produce spectra with such a sharp cutoff tied to a single mass scale.

The second consequence is more profound and concerns the spatial morphology of the expected signal. Annihilation is a two-body process ~~— two dark matter particles must find each~~

~~other~~ — and its rate per unit volume is therefore proportional to the square of the local dark matter density:

$$\frac{d\Gamma_{\text{ann}}}{dV} = \frac{\langle\sigma v_{\text{rel}}\rangle}{2} \frac{\rho^2}{m_\chi^2}, \quad (1.4.1)$$

where $\langle\sigma v_{\text{rel}}\rangle$ is the thermally averaged annihilation cross-section and the factor of 1/2 avoids double-counting for self-conjugate particles [68]. For Dirac dark matter, where particle and antiparticle are distinct, the factor becomes 1/4, but the thermal cross-section required for the correct relic abundance is correspondingly larger, leaving the observable rate unchanged [68]. Decay, by contrast, is a single-particle process with a rate proportional to the first power of the density:

$$\frac{d\Gamma_{\text{dec}}}{dV} = \frac{\rho}{m_\chi \tau}, \quad (1.4.2)$$

where τ is the dark matter lifetime [68]. Current gamma-ray observations constrain $\tau \gtrsim 10^{28}$ s [81, 82], far exceeding the age of the universe — making decay an intrinsically rare process whose detection requires enormous integrated exposures. The ρ^2 versus ρ distinction has direct observational consequences: annihilation signals are sharply concentrated in the densest regions of a halo — the inner kiloparsec of the Galactic Center, or the cores of dwarf galaxies — while decay signals are spread diffusely across the entire halo volume, producing a much flatter angular profile. This morphological difference provides a powerful diagnostic: an excess that tracks the ρ^2 profile of the inner Galaxy but vanishes at high latitudes points toward annihilation, while a persistent, broadly distributed signal favours decay.

The stable particles emerging from an annihilation or decay event — photons, neutrinos, electrons, positrons, protons, antiprotons — are determined by the ~~primary~~ channel into which the dark matter annihilates. In a model-independent analysis, one parametrizes the annihilation as $\chi\chi \rightarrow f\bar{f}$ for a specific Standard Model final state f (quarks, leptons, or gauge bosons), and the total gamma-ray spectrum is given by a sum over channels weighted by branching ratios [5, 83, 84]:

$$\frac{dN_\gamma}{dE} = \sum_f \text{BR}_f \frac{dN_\gamma^{(f)}}{dE}. \quad (1.4.3)$$

In practice, limits are derived for one channel at a time — setting $\text{BR}_f = 1$ for a chosen final state — so each "benchmark channel" maps to a distinctive spectral shape. The spectra $dN_\gamma^{(f)}/dE$ for each channel are computed using Monte Carlo event generators such as Pythia, with the results tabulated in standard libraries such as PPPC 4 DM ID [83] or the more recent CosmiXs [84]. These spectra exhibit qualitatively different shapes depending on f . In hadronic channels such as $b\bar{b}$ or $t\bar{t}$, the quarks hadronize into jets containing many neutral pions, which decay almost instantly via $\pi^0 \rightarrow \gamma\gamma$. The resulting gamma-ray spectrum is broad and soft, peaking at $E_\gamma \sim m_\chi/20$ in $E^2 dN/dE$ units [5, 68]. The $b\bar{b}$ channel has become the standard benchmark in Fermi-LAT analyses for two complementary reasons: the resulting gamma-ray spectrum depends dominantly on neutral pion decay in the hadronization cascade and is therefore representative of nearly all hadronic annihilation channels; and for typical WIMP masses (~ 30 – 60 GeV), the $b\bar{b}$ spectrum peaks at ~ 1 – 3 GeV — the energy range where Fermi-LAT exhibits its greatest sensitivity [68].

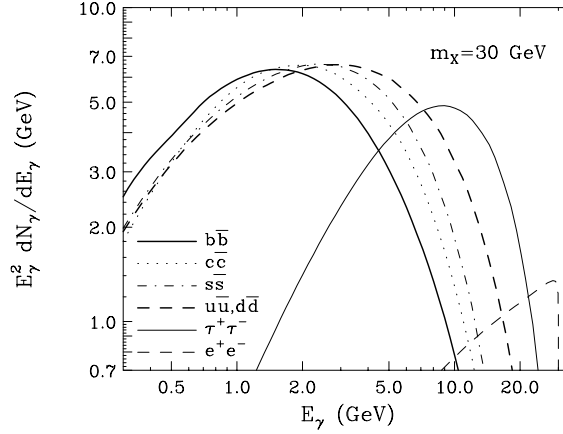


Fig. 1.8. Prompt gamma-ray spectrum dN_γ/dE_γ produced per dark matter annihilation for a 30 GeV WIMP, shown for a variety of Standard Model final states. Hadronic channels ($b\bar{b}$, $c\bar{c}$) produce soft, broad spectra peaked at $E_\gamma \sim m_\chi/20$, while leptonic channels ($\tau^+\tau^-$, $\mu^+\mu^-$) yield harder spectra concentrated near the kinematic endpoint. Figure from Daylan et al. [62].

Leptonic channels behave differently. The $\tau^+\tau^-$ final state undergoes hadronic decays that produce fewer but more energetic pions, yielding a harder spectrum peaking at $E_\gamma \sim m_\chi/3$ [68]. Lighter leptons (e^+e^- , $\mu^+\mu^-$) do not produce pions at tree level — their gamma-ray spectrum is dominated by final-state radiation and carries far less total flux, concentrated near the kinematic endpoint [5]. Gauge boson channels (W^+W^- , ZZ) produce a combination of both behaviours: the hadronic decays of the W and Z generate a broad low-energy component similar to $b\bar{b}$, while leptonic decays and final-state radiation add a harder tail extending toward $E_\gamma \sim m_\chi$ [5, 68].

The more hadrons produced in the cascade, the softer the spectrum, because the available energy is shared among a larger multiplicity of pions. Channels that produce fewer, more energetic particles yield harder spectra with sharper features. In practice, the distinction between the $b\bar{b}$ and $\tau^+\tau^-$ benchmarks brackets the range of spectral shapes relevant for most thermal relic models [68]. These qualitative differences are illustrated in Figure 1.8.

1.4.2 Spectral Features and Signatures

The continuum spectra described above are broad, featureless apart from a cutoff at m_χ , and not unique to dark matter — astrophysical gamma-ray sources produce similar power-law-like spectra through conventional particle acceleration (cf. Chapter 2). The discriminating power of gamma-ray searches therefore lies in spectral features that have no plausible astrophysical counterpart; and unlike charged cosmic rays, whose spectral structures are smeared by propagation through galactic magnetic fields, sharp features in the gamma-ray spectrum propagate intact from source to detector and can be searched for directly [85].

Monochromatic lines. The cleanest such feature is a gamma-ray line at $E_\gamma = m_\chi$, produced when two dark matter particles annihilate directly into a pair of photons, $\chi\chi \rightarrow \gamma\gamma$ [5]. Because dark matter carries no electric charge, this process cannot occur at tree level; it must proceed through quantum loops of charged particles, suppressing the cross-section by a factor of order α^2 relative to annihilation into Standard Model final states [5, 7]. The expected flux is correspondingly small, yet no known astrophysical mechanism produces a sharp line superimposed on a smooth continuum — the monochromatic signal has long been regarded as a “smoking gun” for dark matter [7, 86]. Related processes such as $\chi\chi \rightarrow \gamma Z$ or γh produce secondary lines

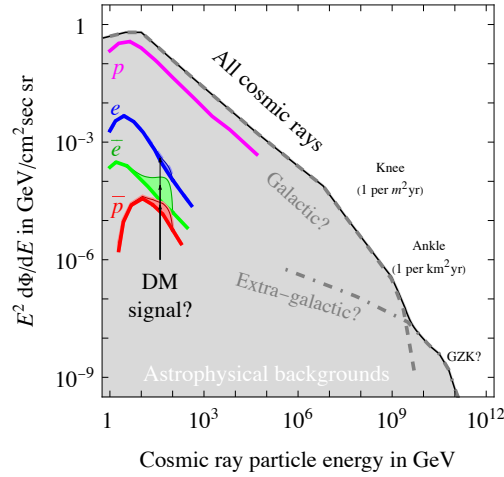


Fig. 1.9. Notable sharp gamma-ray spectral features from dark matter annihilation. From left to right: the monochromatic line from $\chi\chi \rightarrow \gamma\gamma$, the “saw-tooth” shape of virtual internal bremsstrahlung (VIB), the “top-hat” (box-shaped) spectrum from cascade decays $\chi\chi \rightarrow XX \rightarrow \gamma\gamma\gamma\gamma$, and the single saw-tooth from $\chi\chi \rightarrow XX \rightarrow \gamma\nu\gamma\nu$. Unlike charged cosmic-ray features, these sharp structures propagate intact to the detector and can be searched for directly. Figure from Cirelli et al. [5].

displaced to lower energies, since part of the available centre-of-mass energy goes into the mass of the Z or Higgs boson; detecting multiple lines at predictable separations would provide a strong consistency check [5]. An exception to the generic loop suppression arises in models with non-perturbative Sommerfeld enhancement, where the line cross-section can be boosted to a level comparable with the continuum contribution for a range of dark matter masses [5].

Virtual internal bremsstrahlung (VIB). When a photon is radiated from a charged particle mediating the annihilation, the resulting spectrum acquires a characteristic *saw-tooth* shape: rising steeply toward a sharp cutoff at $E_\gamma \lesssim m_\chi$ [5, 87]. In models where two-body annihilation into light fermions is suppressed, this radiative three-body channel can become the dominant process, making the saw-tooth feature the leading spectral signature [5, 87].

These spectral features are illustrated schematically in Figure 1.9.

Electroweak corrections. For TeV-scale dark matter ($m_\chi \gg M_W$), radiation of W and Z bosons from the annihilation products becomes significant and modifies the predicted spectra in two ways [5, 88]. First, sharp peaks such as monochromatic lines are smeared, and a new continuum of lower-energy gamma-rays appears from the hadronic and leptonic decays of the radiated gauge bosons. Second, the emitted weak bosons mix the messenger channels: purely leptonic final states ($\chi\chi \rightarrow e^+e^-, \mu^+\mu^-$) acquire an antiproton yield that would be entirely absent without electroweak radiation, altering multi-messenger predictions [5].

Prompt versus secondary emission. The gamma-ray production mechanisms discussed so far — pion decay, final-state radiation, lines, VIB — all occur directly in the annihilation or decay event and are collectively termed *prompt emission* [5, 89]. Prompt photons propagate freely through the Galaxy and arrive at the detector with their spectral shape intact, because the Galaxy is effectively transparent to gamma-rays below a few hundred TeV [68].

A qualitatively different contribution arises when the energetic electrons and positrons produced by dark matter propagate through the interstellar medium. These leptons upscatter

ambient low-energy photons to gamma-ray energies via *inverse Compton scattering* (ICS) [68, 90]. The target photon field — the interstellar radiation field (ISRF) — consists of three components: the cosmic microwave background, optical starlight, and dust-reprocessed infrared light [91]. As a representative example, a 1 TeV electron upscatters a CMB photon to ~ 1.5 GeV, or a UV starlight photon to ~ 150 GeV [5]. The propagating $e^\pm$ also emit *bremsstrahlung* radiation when deflected by the electric fields of interstellar gas nuclei, a process that dominates their energy losses in the 1–10 GeV range near the Galactic plane where gas densities are highest [85, 90]. These processes produce *secondary* gamma-ray emission that is spatially extended and spectrally softer than the prompt component, and whose morphology depends on the magnetic field structure, gas distribution, and radiation fields of the host galaxy [68, 91]. Secondary emission therefore depends not only on the dark matter distribution but also on the astrophysical environment — magnetic fields, radiation fields, gas densities — that reprocesses the primary electrons and positrons. Despite this additional layer of modelling uncertainty, secondary emission plays a central role in the interpretation of the Galactic Center excess discussed in Chapter 4, where the identification of the signal relies on subtracting the diffuse gamma-ray background dominated by cosmic-ray ICS and bremsstrahlung in the dense inner Galaxy [68].

1.4.3 How Dark Matter is Distributed: Density Profiles

The annihilation and decay rates derived in Section 1.4.1 depend directly on the dark matter density $\rho(r)$ — and for annihilation, on its square. The predicted gamma-ray flux from any astrophysical target is therefore directly determined by the assumed density profile. The same measurement that constrains the particle physics simultaneously probes the dark matter distribution, but the result is only as reliable as the profile assumed — a tension that runs through every chapter of this thesis.

The standard profiles used in the field come from cosmological N -body simulations — numerical experiments that follow the gravitational collapse of billions of dark matter particles from primordial density perturbations to the present-day cosmic web [5]. These simulations show that the spherically averaged density $\rho(r)$ of collapsed halos follows a universal logarithmic slope $s(r) \equiv d \ln \rho / d \ln r$ that transitions smoothly from $s \approx -1$ in the inner region to $s \approx -3$ in the outskirts [5]. The results can be captured by several fitting functions, each with a characteristic density ρ_s and a scale radius r_s .

NFW (Navarro–Frenk–White). The most widely adopted benchmark has a density that diverges as $\rho \propto r^{-1}$ toward the center (a “cusp”) and falls off as $\rho \propto r^{-3}$ in the outer halo [92]:

$$\rho_{\text{NFW}}(r) = \frac{\rho_s}{\frac{r}{r_s} \left(1 + \frac{r}{r_s}\right)^2}, \quad (1.4.4)$$

A common generalization replaces the fixed inner slope with a free parameter γ , giving the “generalized NFW” (gNFW) profile $\rho \propto (r/r_s)^{-\gamma} (1 + r/r_s)^{\gamma-3}$ [5]. The standard NFW corresponds to $\gamma = 1$; steeper values ($\gamma > 1$) describe adiabatically contracted halos, while $\gamma < 1$ accommodates shallower cusps. This freedom is directly relevant for the Galactic Center excess analysis in Chapter 4, where the morphology of the signal constrains γ .

Einasto. Originally proposed by Einasto (1965) [93] and later identified as a fitting function for simulated halos [5], this profile has a logarithmic slope that varies continuously with radius, without converging to a fixed power law at the center. It is controlled by a shape parameter α_{Ein} :

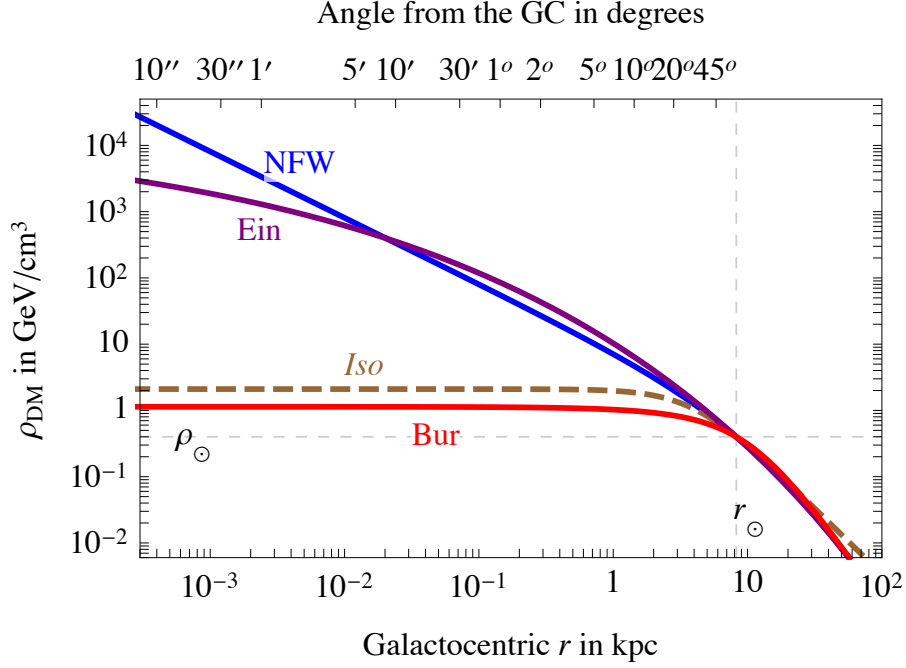


Fig. 1.10. Dark matter density profiles for the Milky Way: NFW (red), Einasto (blue), Burkert (green), and Isothermal (purple) as a function of galactocentric distance. The profiles are normalized to agree at $r \sim r_\odot$ but diverge by orders of magnitude in the inner kiloparsec — the region most relevant for indirect detection from the Galactic Center. Figure from Cirelli et al. [5].

$$\rho_{\text{Ein}}(r) = \rho_s \exp \left\{ -\frac{2}{\alpha_{\text{Ein}}} \left[\left(\frac{r}{r_s} \right)^{\alpha_{\text{Ein}}} - 1 \right] \right\}. \quad (1.4.5)$$

High-resolution simulations increasingly favour Einasto over NFW as a fitting function, with $\alpha_{\text{Ein}} \approx 0.17$ providing a good average fit across a range of halo masses [5]. At the specific radii probed by indirect detection, however, the two profiles yield similar predictions, and both are used as standard benchmarks.

Burkert (cored). A profile with a constant-density core at small radii, motivated by observed rotation curves of dwarf and low-surface-brightness galaxies rather than N -body simulations [19, 94]:

$$\rho_{\text{Bur}}(r) = \frac{\rho_s}{\left(1 + \frac{r}{r_s} \right) \left(1 + \left(\frac{r}{r_s} \right)^2 \right)}. \quad (1.4.6)$$

These three profiles are compared in Figure 1.10, which shows both the radial density dependence and the fitted parameters for the Milky Way. At large radii (around $r_\odot$) all profiles agree, but they diverge by orders of magnitude in the inner kiloparsec — the region most relevant for indirect detection from the Galactic Center.

When fit to the Milky Way, the parameters of these profiles are fixed by requiring consistency with the total Galactic mass ($\sim 4.7 \times 10^{11} M_\odot$) and the local dark matter density at the Sun's position ($r_\odot \approx 8.12$ kpc) [5, 89]. The most recent determinations place the local density at $\rho_\odot \approx 0.40 \pm 0.10$ GeV/cm³ [5], though many analyses continue to use the historical benchmark

of 0.3 GeV/cm^3 ~~for consistency with existing limits~~. Characteristic Milky Way parameters are $r_s \approx 14.5 \text{ kpc}$ and $\rho_s \approx 0.57 \text{ GeV/cm}^3$ for NFW, and $r_s \approx 13.6 \text{ kpc}$ with $\rho_s \approx 0.15 \text{ GeV/cm}^3$ for Einasto [5].

At the solar radius, all three profiles agree by construction. They differ sharply in the inner kiloparsec, where no direct observational data constrain the dark matter profile and the resolution of current simulations reaches its limit (roughly a fraction of a kiloparsec for a Milky Way-sized halo) [5, 19, 92]. The NFW and Einasto profiles predict densities at $r \sim 100 \text{ pc}$ that exceed the Burkert core prediction by one to two orders of magnitude [5]. This discrepancy propagates directly into the ~~J factor: the predicted gamma-ray flux from the Galactic Center can vary by orders of magnitude depending on whether one assumes a cusp or a core.~~

The disagreement between cuspy simulations and cored observations **is the cusp-core problem** [5, 19]. Dark matter-only N -body simulations consistently produce cusps with inner slopes $\gamma \approx 1\text{--}1.5$, while rotation curve measurements — particularly in dwarf galaxies — often favour constant-density cores ($\gamma \approx 0\text{--}0.5$) [19]. Several **resolutions** have been proposed. From the particle physics side, self-interacting dark matter (with $\sigma/m \approx 0.5\text{--}10 \text{ cm}^2/\text{g}$) transfers heat to halo centers and produces cores of the observed size [19, 50, 51], while ultralight “fuzzy” dark matter generates solitonic cores through quantum pressure on scales comparable to the de Broglie wavelength [5, 30]. From the astrophysical side, modern hydrodynamical simulations show that repeated bursts of supernova-driven outflows can rapidly alter the central gravitational potential, pushing dark matter outward and flattening a primordial cusp into a core [5, 19].

The effectiveness of baryonic feedback depends non-monotonically on the stellar-to-halo mass ratio $M_\star/M_{\text{vir}}$: core formation peaks at intermediate ratios ($M_\star/M_{\text{vir}} \approx 5 \times 10^{-3}$, the regime of bright dwarfs), where supernovae are frequent enough to redistribute dark matter, but is negligible in ultrafaint dwarfs (too few stars) and in massive galaxies where adiabatic contraction recreates a cusp [5, 19]. For the Milky Way ($M_\star/M_{\text{vir}} \approx 0.08$), this places the expected inner profile in the cuspy regime — **though the argument rests on the assumption that feedback and contraction have been correctly modelled, which remains uncertain.**

Substructure and boost factors. Real dark matter halos are not smooth. N -body simulations reveal a rich hierarchy of gravitationally bound subhalos embedded within the main halo, with a mass function $dn/dM \propto M^{-2}$ extending from cluster masses down to the resolution limit [5, 95, 96]. Because annihilation scales as ρ^2 , the clumpy density field produces a higher total signal than a smooth halo of the same average density. This follows from Jensen’s inequality applied to the convex function ρ^2 : for any non-uniform density distribution, the volume-averaged squared density satisfies $\langle \rho^2 \rangle > \langle \rho \rangle^2$, with equality only for a perfectly homogeneous halo. This enhancement is parametrized by a multiplicative **boost factor** $B \equiv \langle \rho^2 \rangle / \langle \rho \rangle^2$ [5].

Early estimates of the boost factor ranged from modest ($B \sim 10$) to extreme ($B \sim 10^9$), but **modern analyses find much smaller values.** The survival of subhalos is governed by tidal stripping from the host halo’s gravitational field, gravitational “disk shocking” as subhalos cross the dense baryonic disk, and encounters with individual stars — all of which preferentially destroy subhalos in the inner Galaxy [5, 97]. When these effects are included, the boost factor for the Galaxy as a whole is $B \lesssim \mathcal{O}(10)\text{--}\mathcal{O}(30)$ [5, 97, 98]. Toward the Galactic Center, where subhalos are most heavily depleted by tidal forces, the boost is expected to be negligible [5, 97]. The outskirts of the halo, by contrast, retain a richer substructure population, **and the individual subhalos that survive there are the targets of the searches described in Chapter 5.**

The subhalo hierarchy does not extend to arbitrarily small masses. It is truncated at a minimum mass M_{min} set by the kinetic decoupling of the dark matter particle from the thermal plasma in the early universe [5]. After kinetic decoupling, dark matter particles free-stream over

a characteristic length λ_{fs} , erasing density perturbations below the corresponding mass scale $M_{\text{min}} \simeq (4\pi/3)\rho_{\text{DM}}\lambda_{\text{fs}}^3$ [5]. For a 100 GeV WIMP with weak-scale couplings, $M_{\text{min}} \sim 10^{-6}M_{\odot}$ — roughly an Earth mass [5]. This scale lies far below the resolution of any current simulation and represents a fundamental uncertainty in the predicted total annihilation rate, since the integral over the subhalo mass function is dominated by the smallest masses.

The density profiles and substructure properties introduced here feed directly into the J -factor and D -factor formalism of Section 1.4.4, which combines them with the particle physics of Section 1.4.1 to produce the governing equations for gamma-ray indirect detection.

1.4.4 The Flux Factorization: J -Factor and D -Factor

Combining the particle physics of Section 1.4.1 with the density profiles of Section 1.4.3, we can now write the differential gamma-ray flux from dark matter annihilation observed from a direction ψ relative to the Galactic Center [5, 68]:

$$\frac{d\Phi_{\gamma}}{dE d\Omega} = \underbrace{\frac{1}{4\pi} \frac{\langle\sigma v_{\text{rel}}\rangle}{2m_{\chi}^2} \frac{dN_{\gamma}}{dE}}_{\text{Particle Physics}} \times \underbrace{J(\psi)}_{\text{Astrophysics}}. \quad (1.4.7)$$

The prefactor $1/(4\pi)$ reflects the isotropic emission of photons over the full sphere, and the factor of $1/2$ avoids double-counting for self-conjugate (Majorana) particles, as discussed in Section 1.4.1. The J -factor is the line-of-sight integral of the density squared:

$$J(\psi) = \int_{\text{l.o.s.}} \rho^2(r(s, \psi)) ds, \quad (1.4.8)$$

where s parametrizes distance along the line of sight and r is the galactocentric radius, related to s and the observation angle ψ by $r = (r_{\odot}^2 + s^2 - 2r_{\odot}s \cos \psi)^{1/2}$, with $r_{\odot} \simeq 8.33$ kpc the Sun–Galactic Center distance [5]. In this dimensionful convention, J carries units of $\text{GeV}^2 \text{cm}^{-5}$; an alternative dimensionless definition, in which ρ and s are normalized to $\rho_{\odot}$ and $r_{\odot}$ respectively, is also common in the literature [5].

For decaying dark matter, the analogous expression is [5, 89]:

$$\frac{d\Phi_{\gamma}}{dE d\Omega} = \frac{1}{4\pi} \frac{\Gamma}{m_{\chi}} \frac{dN_{\gamma}}{dE} \times D(\psi), \quad (1.4.9)$$

where $\Gamma = 1/\tau$ is the decay rate. No symmetry factor appears, since decay is a single-particle process. The D -factor integrates the density linearly:

$$D(\psi) = \int_{\text{l.o.s.}} \rho(r(s, \psi)) ds, \quad (1.4.10)$$

and carries units of GeV cm^{-2} . Because D depends on ρ rather than ρ^2 , it is far less sensitive to the assumed density profile than the J -factor, and substructure boost factors are entirely absent for decay [5]. This robustness makes decay searches less model-dependent than their annihilation counterparts, though at the cost of weaker expected signals.

The power of this factorization is that it cleanly separates the unknowns. The particle physics factor — containing m_{χ} , $\langle\sigma v_{\text{rel}}\rangle$ (or τ), and the spectral shape dN_{γ}/dE — encodes the dark matter model we wish to test. The astrophysical factor — J or D — encodes the distribution of dark matter along the line of sight, determined by the density profile of the host halo and constrained independently through kinematic measurements. A non-detection in a target with a

~~well-determined J -factor~~ translates directly into an upper limit on $\langle\sigma v_{\text{rel}}\rangle$; for decay, an upper limit on Γ (equivalently, a lower limit on τ).

When applied to an extended region of the sky, the integrated flux is obtained by integrating J (or D) over the solid angle of interest. For comparing different targets, one typically quotes the J -factor integrated over a cone of half-angle $\theta_\star$ centered on the target:

$$J_{\text{int}}(\theta_\star) = \int_0^{\theta_\star} J(\psi) 2\pi \sin \psi \, d\psi. \quad (1.4.11)$$

For dwarf spheroidal galaxies observed by Fermi-LAT, typical integrated J -factors (within $\theta_\star = 0.5^\circ$) range from $\sim 10^{16}$ to $\sim 10^{19.5}$ $\text{GeV}^2 \text{cm}^{-5}$, depending on the system's mass, distance, and the quality of the kinematic data used to reconstruct its density profile [5, 68, 99]. The choice of integration angle reflects the angular extent of the target: dSphs are resolved by Fermi-LAT and extended emission can be recovered within $\sim 0.5^\circ$, while dark matter subhalos at typical Galactic distances are unresolved and effectively point-like, so their J -factors are integrated within $\theta_\star \sim 0.1^\circ$, matching the instrument's point-spread function. These values are orders of magnitude smaller than the Galactic Center J -factor, but the absence of astrophysical backgrounds in dwarfs compensates for the reduced signal.

Several caveats affect the simple factorization above. First, it assumes a *velocity-independent* annihilation cross-section. If $\langle\sigma v_{\text{rel}}\rangle$ depends on the dark matter velocity — as it does for p -wave annihilation ($\sigma v \propto v^2$) or Sommerfeld-enhanced models ($\sigma v \propto 1/v$) — the cross-section can no longer be factored out of the line-of-sight integral [5]. In such cases, one must compute an *effective J -factor* that folds the full, position-dependent velocity distribution into the integral, and the clean particle-physics/astrophysics separation is lost.

Second, even for velocity-independent cross-sections, the J -factor depends quadratically on the density profile, making it extremely sensitive to the uncertainties discussed in Section 1.4.3. For the Galactic Center, the J -factor computed with an NFW profile can exceed the Burkert value by orders of magnitude [5]; even within the NFW family, varying the inner slope from $\gamma = 0.7$ to $\gamma = 1.3$ changes the predicted flux by more than an order of magnitude [68]. The astrophysical factor is therefore the dominant systematic uncertainty in annihilation searches.

1.4.5 Observational Targets for Gamma-Ray Searches

~~The formalism of the preceding sections produces a clear prescription:~~ to maximise the gamma-ray signal from dark matter, one should observe targets with large J -factors (for annihilation) or D -factors (for decay) and low astrophysical backgrounds. No single target satisfies both criteria simultaneously, and the field has consequently developed a multi-target observation programme in which the complementary strengths of different targets compensate each other's weaknesses. We now review the principal targets, noting their advantages, limitations, and connection to the analyses developed in this thesis.

The Galactic Center. The center of the Milky Way offers the largest J -factor of any astronomical source, owing to its proximity (~ 8 kpc) and the expected steep rise of the dark matter density at small radii [68]. For an NFW profile with $\gamma = 1$ and a local density $\rho_\odot = 0.4 \text{ GeV/cm}^3$, the predicted flux from the inner 10° exceeds that from any individual dwarf galaxy by more than three orders of magnitude [68]. If a steeper inner slope ($\gamma \approx 1.3$) is assumed, the flux increases by an additional factor of ~ 5 ; for a shallower slope ($\gamma \approx 0.7$) it decreases by ~ 3 [68]. In either case, the Galactic Center remains the single brightest expected source of dark matter annihilation products on the sky.

The advantage is offset by severe astrophysical contamination. The inner few degrees of the Galaxy host bright diffuse emission from cosmic-ray protons and electrons scattering off interstellar gas (through pion production and bremsstrahlung) and ambient radiation fields (through inverse Compton scattering), together with numerous point sources — supernova remnants, pulsars, and the central supermassive black hole [5, 68]. Models of this diffuse foreground are constructed from gas maps and the locally measured cosmic-ray spectrum, and while they capture the broad features of the observed emission, they cannot account for its detailed spectral and morphological structure [68]. This imperfect background knowledge is the primary obstacle to extracting a dark matter signal from the Galactic Center.

Despite these difficulties, the Fermi-LAT has detected a statistically significant excess of gamma-rays from the inner Galaxy at energies between ~ 0.5 and 5 GeV — the so-called Galactic Center excess (GCE) [5, 62, 100, 101]. The spectrum and approximately spherical morphology of this excess are consistent with the predictions for a ~ 40 – 70 GeV dark matter particle annihilating into $b\bar{b}$ quarks at a rate close to the thermal relic value [5, 62]. The dark matter interpretation is compelling, but an unresolved population of millisecond pulsars in the Galactic bulge provides an equally plausible astrophysical explanation [5, 89]. Multiple statistical analyses have investigated the point-source versus diffuse origin of the excess, reaching contradictory conclusions [5]. The morphological debate — whether the excess traces a spherical dark matter halo or the stellar distribution of the Galactic bulge — remains unresolved [5]. This persistent ambiguity motivates the dedicated analysis in Chapter 4, which examines the millisecond pulsar hypothesis by constraining the pulsar luminosity function from globular cluster observations; the residual maps and spectrum of the excess are presented there, where its origin is examined in detail.

Dwarf spheroidal galaxies. The Milky Way’s satellite dwarf galaxies occupy the opposite end of the signal-versus-background trade-off. Dwarf spheroidals are among the most dark matter-dominated objects known: their shallow gravitational potential wells allowed early supernova feedback to expel most of the baryonic gas, halting star formation and leaving behind systems with mass-to-light ratios ranging from tens of $M_\odot/L_\odot$ for classical dwarfs like Sculptor to thousands for ultrafaint dwarfs such as Segue 1 [5]. The scarcity of stars, gas, and other active astrophysical objects means that their gamma-ray backgrounds are subdominant [5]. The expected dark matter flux from any individual dwarf is far smaller than from the Galactic Center — roughly three orders of magnitude lower for a typical system [68] — but the absence of competing emission makes them the cleanest targets available.

Eight “classical” dwarf spheroidals (Sculptor, Fornax, Leo I and II, Ursa Minor, Draco, Carina, and Sextans) have been known since before the 1990s [5]. The advent of digital surveys — SDSS, DES, and Gaia — has brought the total count to nearly fifty, with many of the recent discoveries classified as ultrafaint dwarfs [5]. This growing census is expected to continue with the Rubin Observatory.

By combining observations of dozens of these systems in a stacked analysis, the Fermi-LAT collaboration has derived some of the tightest constraints on the annihilation cross-section to date [68, 102, 103]. The original combined analysis of 15 kinematically confirmed dSphs excluded the thermal relic cross-section ($\langle\sigma v_{\text{rel}}\rangle \simeq 2.2 \times 10^{-26} \text{ cm}^3/\text{s}$) for dark matter masses below ~ 100 GeV in the $b\bar{b}$ channel [102]; the most recent legacy analysis extends this to 50 dSphs with 14 years of data, yielding slightly tighter constraints in the 20–100 GeV mass range while confirming the earlier exclusion [103]. The dominant systematic uncertainty lies in the J -factors themselves, which must be inferred from the kinematics of the dwarfs’ constituent stars. For ultrafaint systems, the number of kinematic tracers can be extremely small, and removing or adding a single star from the sample can shift the reconstructed J -factor by orders of magnitude [5].

Additional systematics arise from the cusp–core ambiguity in the inner density profile and from the choice of integration angle $\theta_\star$ [5]. Despite these challenges, dwarfs remain the gold standard for model-independent constraints. Chapter 5 of this thesis searches for dark matter subhalos — which may include unassociated dwarf satellites — among the unidentified Fermi-LAT sources.

Dark subhalos. Λ CDM simulations predict far more gravitationally bound subhalos within the Milky Way’s halo than the ~ 50 known satellite galaxies can account for [5]. Most of these subhalos are expected to be entirely dark — devoid of stars and gas — yet sufficiently massive to produce a detectable gamma-ray flux if the annihilation cross-section is near the thermal relic value [5]. Such dark subhalos would appear in the Fermi-LAT data as unassociated point sources with no counterpart at other wavelengths. Their positions are unknown a priori, turning the search into a classification problem: distinguishing rare dark matter subhalos from the much larger population of unidentified astrophysical sources (primarily pulsars and faint blazars). This challenge is addressed in Chapter 5, where machine learning methods are applied to classify unassociated Fermi-LAT sources in the presence of dataset shift between simulations and real data.

The unresolved gamma-ray background. The cumulative, nearly isotropic gamma-ray flux that remains after subtracting resolved point sources and Galactic diffuse emission constitutes the unresolved gamma-ray background (UGRB) [104]. Chapters 6 and 7 develop methods to characterize the unresolved source populations that compose the UGRB, though without restricting to the dark matter component. This diffuse glow receives guaranteed contributions from unresolved blazars (BL Lac objects and flat-spectrum radio quasars), misaligned active galactic nuclei, star-forming galaxies, and millisecond pulsars [104]. Unresolved BL Lac objects alone can account for the majority of the UGRB above ~ 100 GeV, while at lower energies other populations are required [104]. A dark matter component — originating from annihilation or decay in all halos and subhalos across cosmic history — may also be present, but it is expected to be subdominant in the total intensity spectrum [5, 104].

Because the dark matter signal is buried beneath dominant astrophysical foregrounds, its extraction requires methods that go beyond the energy spectrum alone. The angular anisotropies of the UGRB carry additional information: different source classes imprint different patterns of spatial fluctuations, reflecting their distinct clustering properties, redshift distributions, and luminosity functions [104]. Cross-correlating these anisotropies with independent gravitational tracers of the large-scale structure — galaxy catalogs, cosmic shear from weak lensing surveys, or 21cm intensity mapping of neutral hydrogen — can isolate the component that traces the dark matter distribution [5, 89, 104]. The approach is particularly powerful because dark matter annihilation scales as ρ^2 and has a redshift evolution distinct from that of astrophysical gamma-ray emitters, providing a spectral and tomographic handle for separation [89]. This cross-correlation technique forms the basis of the analysis in Chapter 8, where the sensitivity of next-generation instruments to the dark matter cross-correlation signal is forecast.

The complementary strengths of these targets — ~~the Galactic Center’s raw flux, the dwarfs’ clean backgrounds, the dark subhalos’ model independence, and the UGRB’s cosmological reach~~ — motivate the multi-pronged programme developed in Parts II through IV of this thesis. Each target exposes different aspects of the dark matter parameter space and is sensitive to different systematic uncertainties. ~~Progress requires pursuing all of them simultaneously, using the formalism established in this section as the common foundation.~~

1.4.6 Multi-Messenger Context

Dark matter annihilation and decay produce all stable Standard Model particles, not only gamma-rays. Each additional messenger opens a complementary window onto the dark sector, but none offers the same combination of directional, spectral, and morphological information that makes gamma-rays the preferred channel for the analyses in this thesis [5]. A brief survey of the alternatives clarifies why.

Neutrinos share the key advantage of photons — electrical neutrality — and therefore propagate in straight lines, preserving directional information [5, 68]. They can also escape environments opaque to photons: dark matter captured in the solar core, for instance, annihilates into neutrinos that are the only annihilation products to reach the surface, making high-energy solar neutrinos a probe of the dark matter–nucleon scattering cross-section [5]. Searches by IceCube, ANTARES, and Super-Kamiokande have found no signal, but for dark matter masses below a few hundred GeV they set bounds on the spin-dependent cross-section roughly one order of magnitude stronger than direct detection experiments [5, 105]. For galactic targets, however, neutrino constraints remain generally weaker than the corresponding gamma-ray bounds, becoming competitive only above roughly 1–10 TeV depending on the annihilation channel [5].

Charged cosmic rays — positrons, antiprotons, and heavier anti-nuclei — suffer from a fundamental limitation: galactic magnetic fields randomize their arrival directions, erasing all spatial information and leaving only the energy spectrum as a diagnostic [5, 89]. The resulting searches have produced suggestive anomalies. The positron fraction measured by PAMELA and AMS-02 rises steeply above 10 GeV [5, 106], but fitting it with dark matter requires a cross-section three orders of magnitude above the thermal relic value, in severe tension with gamma-ray and CMB constraints; the excess is now widely attributed to nearby pulsars [5, 68]. AMS-02 antiproton data show a tentative excess at 10–20 GeV compatible with an ~ 80 GeV dark matter particle annihilating into $b\bar{b}$ at roughly the thermal cross-section [5, 107]. These parameters overlap with those invoked for the Galactic Center excess (Chapter 4), but the significance drops below $\sim 2\sigma$ once systematic uncertainties in the propagation model and production cross-sections are included [5]. Looking ahead, the detection of low-energy anti-deuterons or anti-helium — species with nearly vanishing astrophysical backgrounds — would provide a qualitatively different signature; AMS-02 and the upcoming GAPS experiment may reach the sensitivity needed to probe the thermal relic cross-section [5, 68].

An independent and increasingly powerful constraint comes from the cosmic microwave background: dark matter annihilation during and after recombination injects energy into the photon–baryon plasma, distorting the CMB temperature and polarisation anisotropies in a way that is tightly bounded by Planck data [5, 108]. For s -wave annihilation, these bounds exclude the thermal relic cross-section for masses below ~ 44 GeV in the e^+e^- channel and below ~ 16 GeV in the $b\bar{b}$ channel [108] — limits that are largely independent of astrophysical modelling and therefore highly complementary to gamma-ray searches.

~~This thesis focuses exclusively on gamma-rays — the messenger that provides both spatial and spectral information while relying on the fewest assumptions about particle propagation. Multi-messenger results are noted where they bear on the gamma-ray analyses, but the searches in Parts II–V rely entirely on the gamma-ray formalism developed above.~~

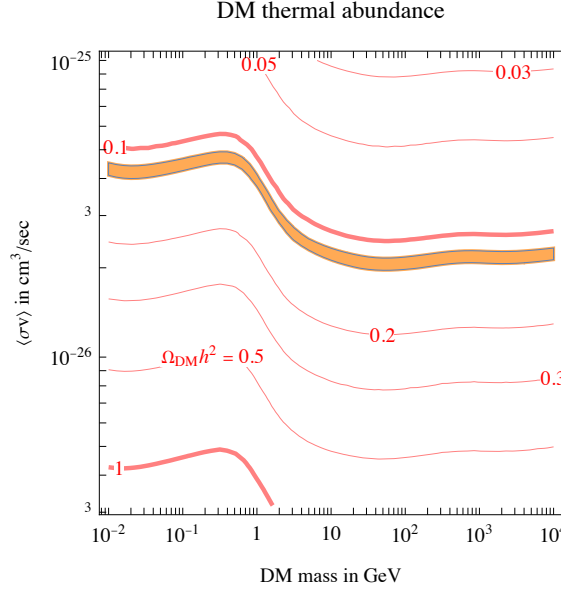


Fig. 1.11. Dark matter freeze-out abundance $\Omega_{\text{DM}} h^2$ as a function of DM mass and annihilation cross-section $\langle\sigma v\rangle$ for Majorana dark matter. The measured cosmological abundance $\Omega_{\text{DM}} h^2 = 0.120 \pm 0.001$ is reproduced within the orange band at 3σ . For smaller (larger) annihilation cross-sections — i.e. in the lower (upper) part of the plot — the freeze-out happens too early (too late) and therefore the DM relic abundance is too high (too low). This cross-section is the target for indirect detection searches. Figure from Cirelli et al. [5].

1.4.7 The Status of the Field

Despite more than a decade of increasingly sensitive searches across all three detection strategies, no unambiguous signal of dark matter has been found [5]. The current status is best summarized as a set of progressively tightening upper bounds on $\langle\sigma v_{\text{rel}}\rangle$ and τ , punctuated by anomalies whose dark matter interpretations have been challenged by plausible astrophysical alternatives.

Gamma-ray constraints. The tightest indirect detection limits come from the Fermi-LAT stacked analysis of Milky Way dwarf spheroidal galaxies. The original combined analysis of 15 dSphs with six years of data excluded the thermal relic cross-section for dark matter masses in the range ~ 10 – 100 GeV in the $b\bar{b}$ channel [102]; below ~ 10 GeV the Fermi-LAT effective area drops steeply and the constraints weaken. The most recent legacy analysis extends the sample to 50 dSphs and 14 years of Fermi-LAT exposure, confirming and slightly strengthening these constraints [68, 103]; no significant excess is found in the combined likelihood analysis, though a marginal (2 – 3σ local) excess near $M_\chi \approx 150$ – 230 GeV in the $b\bar{b}$ channel has motivated projections for future observations [103]. For a generic annihilation channel, marginalizing over branching ratios yields a model-independent lower limit of $m_\chi \gtrsim 20$ GeV [5]. Constraints on the dark matter lifetime, derived primarily from the isotropic gamma-ray background and Milky Way halo observations, reach $\tau \gtrsim 10^{27}$ – 10^{28} seconds depending on the decay channel and dark matter mass — some ten orders of magnitude longer than the age of the universe [5, 81, 82]. The central target for these searches — the thermal relic cross-section — is illustrated in Figure 1.11, which shows how the freeze-out abundance varies with the dark matter mass and annihilation cross-section.

Direct detection and collider constraints. The tightening of parameter space is not

~~confined to indirect searches.~~ The current generation of dual-phase liquid xenon detectors — LZ, XENONnT, and PandaX-4T — has pushed the spin-independent WIMP–nucleon cross-section below 10^{-46} cm^2 for masses near the electroweak scale [5, 68]. At the LHC, mono-jet and mono-photon searches have found no evidence for new physics, excluding strongly interacting superpartners (squarks, gluinos) up to masses of 1–2 TeV and charged superpartners up to several hundred GeV [5, 68]. The theoretical basis of the most popular WIMP candidate — the supersymmetric neutralino — has been substantially weakened by the absence of supersymmetry at collider energies [5].

Unresolved anomalies. ~~Against this backdrop of null results,~~ several persistent anomalies keep the possibility of discovery alive. The Galactic Center gamma-ray excess — a signal first reported in 2009 and consistent with a $\sim 40\text{--}70$ GeV dark matter particle annihilating into $b\bar{b}$ at the thermal cross-section — remains ~~the most debated~~ [5, 62, 68]. Its leading astrophysical competitor, an unresolved population of millisecond pulsars in the Galactic bulge, ~~has been alternately supported and challenged by photon-count statistics, morphological analyses, and luminosity function arguments~~ [5]. Chapter 4 of this thesis contributes directly to this debate. The charged cosmic-ray anomalies discussed in Section 1.4.6 — the AMS-02 antiproton excess and the positron fraction rise — remain consistent with astrophysical explanations once systematic uncertainties are fully accounted for [5, 68].

The methodological case for this thesis. Taken together, these results define the landscape in which this thesis is situated. The viable parameter space for the simplest thermal WIMP models has narrowed substantially — what Cirelli et al. describe as a field “at a turning point” [5]. The standard approach to indirect detection — searching for a bright, spectrally distinctive excess from a single target above the astrophysical background — has not delivered a ~~discovery~~. Yet the thermal relic benchmark is not the prediction of a specific model but a cosmological ~~target~~: if dark matter is a thermal relic, the annihilation cross-section must have the value that produces the observed abundance, regardless of the underlying theory. Current exclusions cover only a portion of the mass range and a subset of annihilation channels; heavier candidates, ~~leptonic final states~~, and velocity-dependent cross-sections remain largely unexplored. Extracting these signals demands methods that go beyond simple signal-over-background thresholding.

~~The analyses of this thesis pursue that transition.~~ The Galactic Center excess demands careful modelling of the interstellar emission and the unresolved point-source population — distinguishing dark matter annihilation from millisecond pulsars requires statistical tools, not a single spectral fit (Part II). Individual dark matter subhalos among the Fermi-LAT unassociated sources cannot be identified by classification alone; a fully probabilistic model that accounts for both covariate and prior probability shifts between associated and unassociated sources is needed to derive statistically rigorous constraints (Part II). Below the point-source detection threshold, deep learning applied to gamma-ray sky maps can recover the source-count distribution of faint populations down to fluxes a factor of 50 below the Fermi-LAT sensitivity limit, while probabilistic catalogues extend spatial source information into the sub-threshold regime (Part III). At the largest scales, the dark matter contribution to the unresolved gamma-ray background can potentially be isolated through its cross-correlation with independent tracers of the large-scale structure (Part IV).

~~The mathematical framework developed in this section — the flux factorization, the J -factor and D -factor formalism, the density profiles and their uncertainties — underpins every one of these analyses.~~

1.5 Summary

This chapter has established the theoretical and observational foundation for the dark matter searches that follow. Multiple independent lines of evidence — flat rotation curves in spiral galaxies, the dynamics and gravitational lensing of galaxy clusters, and the precision cosmology of the cosmic microwave background — converge on the Λ CDM concordance model, in which roughly 85% of the matter density is composed of cold, non-baryonic dark matter. Big Bang Nucleosynthesis independently confirms the non-baryonic character of this component, requiring that dark matter consist of a new form of matter beyond the Standard Model.

Among the candidates proposed to fill this role, the Weakly Interacting Massive Particle stands out because of the “WIMP miracle”: a stable particle with electroweak-scale mass and coupling, produced thermally in the early universe, naturally yields a relic abundance consistent with observation. The thermal freeze-out calculation fixes the annihilation cross-section at $\langle\sigma v_{\text{rel}}\rangle_{\text{cosmo}} \approx 2.2 \times 10^{-26} \text{ cm}^3/\text{s}$, providing a concrete, parameter-free experimental target in the GeV–TeV mass window bracketed by the Lee–Weinberg and unitarity bounds. The same interaction that sets the relic density can be probed from three complementary directions — scattering in underground detectors (direct detection), production at colliders, and annihilation in astrophysical environments (indirect detection). Among these, indirect detection via gamma-rays occupies a unique position: it directly probes the annihilation cross-section, the quantity that determines the cosmological abundance.

The J -factor and D -factor formalism developed in this chapter factorizes the predicted gamma-ray flux into a particle physics term and an astrophysical term, providing the mathematical framework that underpins every analysis in this thesis. The current null results of WIMP searches — with Fermi-LAT constraints from dwarf galaxies already excluding the thermal relic cross-section for masses below $\sim 100 \text{ GeV}$ in the $b\bar{b}$ channel — demonstrate that the era of simple signal-over-background searches is reaching its limits. Extracting weaker signals from noise-dominated data demands the statistical and machine learning methods developed in the remainder of this thesis. The next chapter introduces the instrument that collects the data: the Fermi Large Area Telescope and the astrophysical gamma-ray sky that constitutes both the observational setting and the primary source of systematic uncertainty for the analyses that follow.

THE GAMMA-RAY SKY AND FERMI-LAT

The gamma-ray sky is rich and complex, dominated by diffuse emission that traces the interstellar gas of the Milky Way, punctuated by thousands of individual point sources, and underlaid by a faint, isotropic extragalactic glow. For indirect searches for dark matter, these astrophysical gamma rays constitute the primary background. Extracting a faint dark matter signal—whether from annihilation at the Galactic Center or from the collective emission of unresolved subhalos—requires us to understand where these background photons originate and how our instruments record them.

Standard methodologies successfully classify the brightest objects and map the prominent diffuse components, but they struggle with overlapping populations and instrumental constraints such as finite spatial resolution and detection thresholds. These challenges obscure the faintest dark matter signatures and the broader population of unresolved sources. Distinctly isolating a dark matter signal from this background therefore requires novel statistical and machine learning techniques capable of modeling complex systematic uncertainties.

In this chapter we establish the astrophysical and instrumental framework that underpins the analyses in the remainder of this thesis. Section 2.1 reviews the fundamental production mechanisms of high-energy gamma rays, detailing the hadronic and leptonic processes that generate the Galactic foreground. Section 2.2 surveys the dominant source populations, progressing from Galactic entities such as millisecond pulsars to extragalactic classes like blazars, and formalizes the source-count distribution and its connection to the isotropic background. Finally, Section 2.3 describes the *Fermi* Large Area Telescope, focusing on the instrumental response functions, detection thresholds, and data products that define the practical limits of current observations. The physical complexity and instrumental limitations introduced here constitute the primary systematic uncertainties that we address using probabilistic methods and simulation-based inference in the following chapters.

2.1 Gamma-Ray Production Mechanisms

Astrophysical gamma rays from the Milky Way constitute the primary background in the search for dark matter signals. Before cataloging individual point sources or attempting to extract

information from the faint extragalactic emission, one must understand the physical processes that generate this intense foreground.

This section reviews the standard gamma-ray production mechanisms. These processes fall into two broad categories: hadronic interactions involving cosmic-ray nucleons colliding with interstellar gas, and leptonic interactions involving energetic electrons and positrons scattering off radiation fields and gas. Operating in different energy regimes and tracing distinct target environments, they combine to produce the macroscopic Galactic Diffuse Emission that dominates the observed gamma-ray sky.

2.1.1 Hadronic Emission

High-energy cosmic-ray protons and heavier nuclei interact with the interstellar medium through the strong nuclear force, primarily colliding with ambient hydrogen gas. These inelastic proton-proton collisions produce secondary particles, most notably pions:

$$p + p \rightarrow \pi^0 + \pi^\pm + X. \quad (2.1.1)$$

The reaction is subject to a strict kinematic threshold. A cosmic-ray proton colliding with a stationary target proton must have a total energy of at least $E_p > m_p + 2m_\pi + m_\pi^2/2m_p \approx 1.2$ GeV in order to produce one or more pions [68]. Once this threshold is crossed, the total inelastic cross section increases only mildly with energy, rising from $\sigma_{\text{inel}}^{pp} \approx 30\text{--}40$ mb at $E_p \lesssim 10$ GeV to $\sigma_{\text{inel}}^{pp} \approx 100$ mb near $E_p \sim 1$ EeV [68]. Because a significant fraction of the proton's kinetic energy is transferred in each collision (inelasticity $K_{pp} \sim 0.5$), only a few successive interactions suffice to extract most of the cosmic ray's original energy [85].

While charged pions ($\pi^\pm$) decay into muons and neutrinos, neutral pions decay electromagnetically with a mean lifetime of only $\tau_{\pi^0} \approx 8.4 \times 10^{-17}$ s:

$$\pi^0 \rightarrow \gamma\gamma. \quad (2.1.2)$$

In the rest frame of the π^0 , this decay produces two photons of equal energy $E_\gamma = m_{\pi^0}/2 \approx 67.5$ MeV. In the laboratory frame, however, a pion moving with energy E_π generates a gamma-ray spectrum that is flat between the kinematic bounds $(E_\pi - \sqrt{E_\pi^2 - m_\pi^2})/2$ and $(E_\pi + \sqrt{E_\pi^2 - m_\pi^2})/2$, and vanishes outside this interval [68]. This flat distribution has a remarkable consequence: regardless of the spectral shape of the parent pion population, the resulting gamma-ray spectrum—when plotted as $E^2 dN/dE$ —always peaks at $m_{\pi^0}/2 \approx 67.5$ MeV. This universal spectral feature, known as the *pion bump*, is the definitive spectral signature of hadronic cosmic-ray interactions with gas and has been conclusively observed in both supernova remnants and the Galactic Plane emission [110]. Figure 2.1 shows the inclusive π^0 production cross section as a function of proton energy across the primary cosmic-ray regime; the parametrization of Kafexhiu et al. [109] is widely adopted in current gamma-ray analyses.

Hadronic emission dominates the Galactic Diffuse Emission along the Galactic Plane, where the column density of interstellar gas is highest. As a result, it constitutes the primary astrophysical background for any dark matter search targeting the inner Galaxy. Mismodeling the pion-decay component introduces systematic errors that directly impact the extraction of the Galactic Center Excess and the identification of dark matter subhalos. While hadronic emission dominates wherever gas is abundant, leptonic processes—inverse Compton scattering, bremsstrahlung, and synchrotron radiation—provide the complementary diffuse components, particularly at high Galactic latitudes.

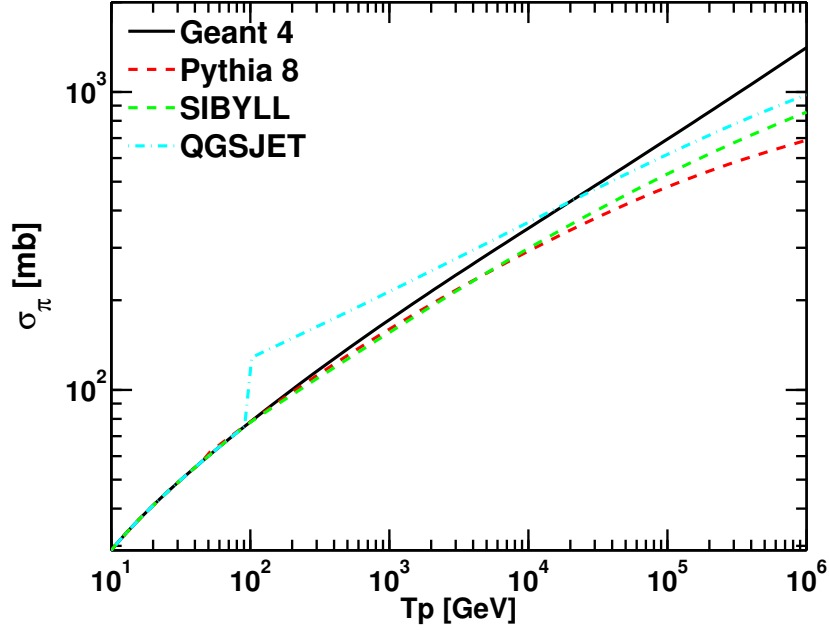


Fig. 2.1. Inclusive π^0 production cross section for inelastic proton-proton collisions as a function of the incident proton energy. Different theoretical models (Geant4, Pythia8, QGSJET, SIBYLL) diverge at extremely high energies but provide consistent predictions across the primary cosmic-ray regime. Credit: Figure 13 from Kafexhiu et al. (2014) [109]. [AA: replace this with a pp cross section to show the pion bump, maybe]

2.1.2 Leptonic Emission

Relativistic electrons and positrons produce high-energy photons through three primary mechanisms: inverse Compton scattering, bremsstrahlung, and synchrotron radiation. These processes operate across different energy regimes and target environments, and their relative importance depends on the local astrophysical conditions along each line of sight.

Inverse Compton scattering. Relativistic electrons can up-scatter low-energy ambient photons—the Cosmic Microwave Background (CMB), infrared dust emission, and starlight—into the X-ray and gamma-ray bands [90]. The behavior of this process depends critically on the ratio of the electron energy to the target photon energy ϵ . In the Thomson regime ($E_e \ll m_e^2/\epsilon$), a given electron up-scatters photons to a maximum energy of $E_\gamma^{\max} \approx 4\epsilon E_e^2/m_e^2$. The quadratic dependence on E_e implies that higher-energy electrons are vastly more efficient radiators, which steepens the resulting photon spectrum relative to the parent electron population. Quantitatively, a power-law electron spectrum with index α ($dN_e/dE_e \propto E_e^{-\alpha}$) produces a gamma-ray spectrum with a softer index $-(\alpha + 1)/2$ [68]. In this regime, the ICS energy loss rate takes a form mathematically identical to that of synchrotron radiation:

$$\left. \frac{dE_e}{dt} \right|_{\text{ICS}}^{\text{Thomson}} = -\frac{4}{3} \sigma_T \rho_{\text{rad}} \gamma_e^2 c, \quad (2.1.3)$$

where ρ_{rad} is the energy density of the target photon field [68].

At highly relativistic energies ($E_e \gg m_e^2/\epsilon$), the interaction transitions into the Klein-Nishina regime [68]. Here the scattering cross section drops below the Thomson value, and the energy loss

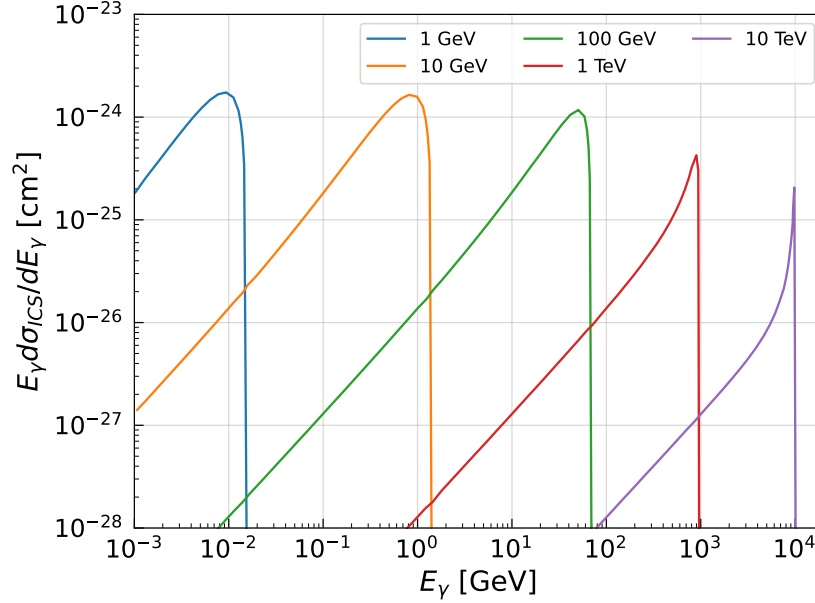


Fig. 2.2. The differential cross section for inverse Compton scattering of a high-energy electron with a 1 eV target photon, for several values of E_e . At high energies ($E_e \gg m_e^2/c$), the process enters the Klein-Nishina regime: the cross section is severely suppressed and the scattered photon carries nearly all the electron’s energy.

rate acquires an additional suppression factor of $\sim m_e^4/(E_e^2 T^2)$ relative to the Thomson prediction, where T is the effective temperature of the target radiation field. Consequently, the scattered gamma rays carry away nearly all the incoming electron’s energy ($E_\gamma \approx E_e$), and the differential cross section peaks sharply just below E_e rather than being spread across a broad range. This transition from the Thomson to the Klein-Nishina regime is illustrated in Figure 2.2.

In practice, cosmic-ray electrons scatter off starlight, infrared dust emission, and the CMB. At higher electron energies, Klein-Nishina suppression progressively reduces the efficiency of scattering off higher-frequency photons, shifting the dominant ICS target toward the CMB [68].

Bremsstrahlung. In gas-rich regions of the interstellar medium, electrons undergo bremsstrahlung as they decelerate in the Coulomb fields of neutral and ionized atomic gas. The bremsstrahlung cross section scales with the square of the target nuclear charge ($\propto Z^2$), making hydrogen and helium the dominant targets in the ISM. A distinctive feature of bremsstrahlung is that its energy loss rate scales linearly with the electron energy at leading order ($b_{\text{brem}} \propto E_e$) [89], resulting in an energy loss timescale that is effectively independent of E_e . As a consequence, the gamma-ray spectrum produced through bremsstrahlung closely mirrors the spectral shape of the parent electron population [68]. This process serves as the dominant cooling mechanism for cosmic-ray electrons in the MeV to GeV range in gas-dense environments such as the Galactic Plane, though it is broadly subdominant to hadronic emission in the same regions.

Synchrotron radiation. In the presence of the Galactic magnetic field ($B \sim \text{few } \mu\text{G}$), ultra-relativistic electrons gyrate around magnetic field lines and emit synchrotron radiation. This process primarily generates photons in the radio and microwave bands, but it traces the same underlying electron populations that produce gamma rays via ICS and bremsstrahlung. The synchrotron energy loss rate scales as $b_{\text{syn}} \propto B^2 E_e^2$ [68], mirroring the E_e^2 dependence of

Thomson-regime ICS. Which of the two mechanisms dominates depends on the local ratio of magnetic field energy density to radiation field energy density ($\rho_{\text{mag}}/\rho_{\text{rad}}$). At the highest electron energies, Klein-Nishina suppression of ICS breaks this symmetry: the ICS loss rate falls below its Thomson-regime prediction, leaving synchrotron radiation as the primary cooling channel for the most energetic cosmic-ray electrons [68].

These three leptonic processes, together with the hadronic pion-decay mechanism described in the preceding section, constitute the main set of fundamental interactions that generate the diffuse gamma-ray sky. The following section describes how these individual components are combined into a global model of the Galactic Diffuse Emission.

2.1.3 The Galactic Diffuse Emission Model

The individual production mechanisms described in the preceding sections combine to produce the Galactic Diffuse Emission (GDE)—the brightest feature in the gamma-ray sky, accounting for roughly 80% of all photons detected by the *Fermi*-LAT [104, 111]. This emission arises from the integrated interactions of propagating cosmic rays with the interstellar gas and radiation fields of the Milky Way.

To model this complex, spatially structured foreground, astrophysicists rely on numerical cosmic-ray propagation codes, of which GALPROP is the most established [91]. Starting from a set of assumptions about cosmic-ray source distributions (primarily supernova remnants), injection spectra, and diffusion parameters, GALPROP solves the steady-state diffusion-loss equation for all relevant cosmic-ray species as they propagate through the Galactic magnetic field and interstellar medium. The output is a set of sky maps predicting the gamma-ray intensity from each emission component: a π^0 -decay map that traces the gas column density (dominated along the Galactic Plane), an inverse Compton map that traces the interstellar radiation field (producing an extended halo at high latitudes), and a bremsstrahlung map that follows the gas distribution but with a different spectral shape.

The resulting spatial and spectral morphology is highly structured and energy-dependent. Along the Galactic Plane, where atomic and molecular gas is dense, hadronic pion decay dominates the high-energy emission, while bremsstrahlung and ionization govern the cooling of low-energy cosmic-ray electrons. At high Galactic latitudes, the gas density drops while the interstellar radiation field remains significant. There, inverse Compton scattering overtakes bremsstrahlung as the dominant energy-loss mechanism for electrons above a few tens of MeV, producing an extended, diffuse gamma-ray glow [89]. The energy- and position-dependent balance between these loss channels is shown in Figure 2.3.

When analyzing *Fermi*-LAT data, the standard procedure is to model the observed sky through template decomposition [111]. The total photon counts in each spatial and energy bin are fit to a linear combination of: the GALPROP-derived π^0 -decay, ICS, and bremsstrahlung templates; a catalog of individually resolved point sources; spatial templates for large-scale structures such as Loop I and the Fermi Bubbles; and a spatially uniform isotropic component representing the combined extragalactic background and residual instrumental contamination. The normalizations of these Galactic templates are typically left as free parameters and fitted independently in each energy bin, allowing the data to correct for imperfections in the GALPROP predictions. The Fermi collaboration applies precisely this procedure to produce the official Galactic interstellar emission model `gll_iem_v07`, which serves as the standard foreground template in *Fermi*-LAT analyses. Despite the physical sophistication of GALPROP and the flexibility of the template-fitting procedure, the GDE cannot be modeled perfectly. Imperfect modeling of the interstellar emission remains the largest single source of systematic uncertainty

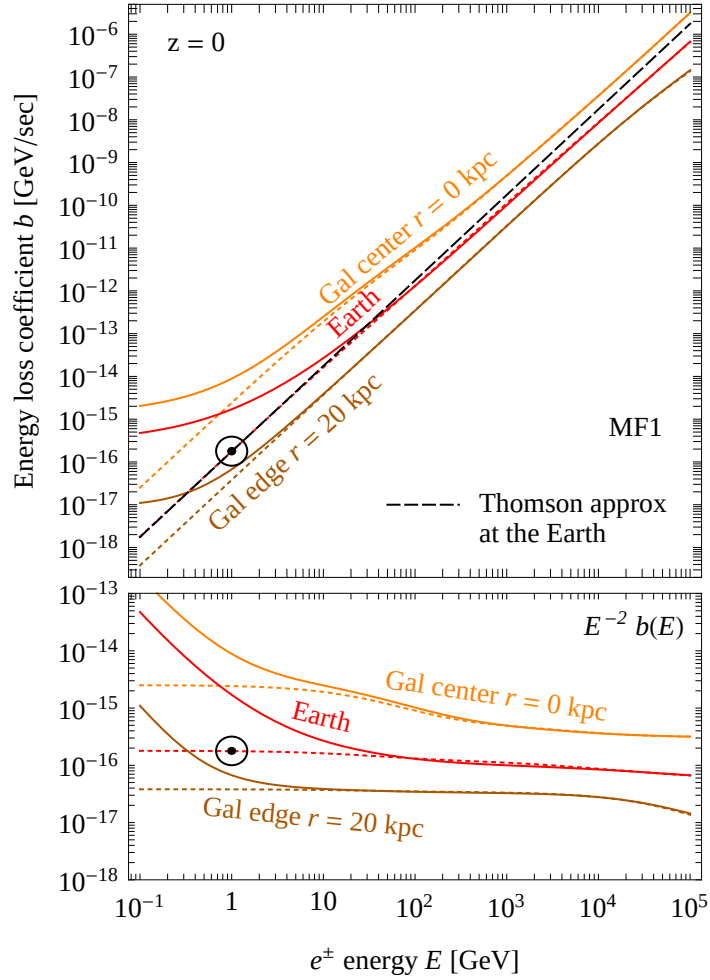


Fig. 2.3. Energy loss coefficient function for electrons and positrons in the Milky Way’s galactic disk at various radial distances from the Galactic Center. In the gas-rich Plane, ionization and bremsstrahlung dominate at low energies; at higher energies, inverse Compton scattering and synchrotron radiation become the primary loss channels. The transition to the Klein-Nishina regime suppresses the ICS efficiency at the highest energies, causing deviations from the Thomson approximation. Credit: Figure 7.1 from Cirelli et al. (2024) [5].

in gamma-ray astrophysics, introducing errors between 15% and 30% depending on the energy range [104, 111]. Uncertainties in the gas column densities (particularly the conversion from CO emission to molecular hydrogen), the interstellar radiation field, and the cosmic-ray propagation parameters all contribute. Handling the Galactic foreground properly is a necessary condition for any analysis of the gamma-ray sky: without adequate foreground control, both spectral characterization and source detection are biased. This challenge recurs throughout this thesis—in the extraction of the Galactic Center Excess (Chapter 4), the search for dark matter subhalos (Chapter 5), and the reconstruction of the extragalactic source-count distribution (Chapter 6). The probabilistic and simulation-based methods developed in later chapters are designed to operate robustly in the presence of these foreground uncertainties, rather than to eliminate them.

2.2 The Astrophysical Gamma-Ray Sky

The high-energy sky observed by the Fermi Large Area Telescope (Fermi-LAT) is a complex superposition of processes spanning our local Galactic neighborhood and the broader universe. Phenomenologically, this emission falls into two domains. The Galactic sky is dominated by the diffuse foreground of cosmic-ray interactions and populated by Galactic gamma-ray sources, most notably pulsars. The extragalactic sky, in contrast, is characterized by an isotropic background arising primarily from unresolved active galactic nuclei and star-forming galaxies. A detailed understanding of the astrophysical processes behind these conventional emitters is necessary for dark matter indirect detection, where every ordinary astrophysical signal is a potential background.

2.2.1 Galactic Sources

The gamma-ray emission from the Milky Way comprises two broad components: a diffuse foreground and a diverse population of individual point sources. The Galactic Diffuse Emission dominates the observed sky and, as discussed in Section 2.1.3, its template-based modeling remains the largest source of systematic uncertainty in Fermi-LAT analyses. Alongside this diffuse foreground, the Galactic sky is populated by individual energetic sources, among which pulsars are the most prominent gamma-ray emitters.

Young pulsars are rapidly spinning neutron stars with large surface magnetic fields (10^{11} – 10^{13} G), emitting gamma rays primarily through curvature radiation in their magnetospheric gaps. This emission typically continues over millions of years as the pulsar gradually decelerates (spins down). Over longer timescales, neutron stars in binary systems can undergo a recycling process: the older neutron star accretes matter and angular momentum from its companion, spinning back up to periods of 2–10 milliseconds [112]. These millisecond pulsars (MSPs) possess weaker magnetic fields (10^8 – 10^9 G) but maintain their rapid rotation for gigayears. While young pulsars remain tightly clustered near their birthplaces in the Galactic disk, MSPs have orbited the Galaxy numerous times. Their natal kicks and prolonged dynamical history broaden their spatial distribution, causing MSP populations to extend to higher Galactic latitudes and to densely populate the Galactic bulge [113].

Their concentration in the Galactic bulge places MSPs squarely in the regions most targeted by dark matter searches, but the challenge extends beyond spatial overlap: curvature radiation produces a characteristic power-law spectrum with an exponential cutoff at a few GeV. This spectral shape is essentially indistinguishable from the expected photon yield of a ~ 50 GeV WIMP annihilating into $b\bar{b}$ quarks [5] — a topic we will extensively explore in Chapters 4 and 5.

Among the remaining Galactic gamma-ray sources, supernova remnants and pulsar wind nebulae are routinely detected by the Fermi-LAT but represent subdominant backgrounds for the dark matter searches undertaken in this thesis.

2.2.2 Extragalactic Sources

The high-latitude sky is dominated by the Extragalactic Gamma-Ray Background (EGB), which encompasses the total gamma-ray emission originating outside our Galaxy. The EGB combines both individually resolved extragalactic sources and the unresolved glow of fainter objects [104, 114]. The Unresolved Gamma-Ray Background (UGRB) is the residual, almost isotropic emission that remains after subtracting the Galactic Diffuse Emission and all cataloged point sources [104, 111]. It is composed by the cumulative emission of all the currently unresolved gamma-ray sources, plus a residual truly diffuse isotropic background. The Fermi-LAT has

measured the UGRB from 100 MeV to 820 GeV, and characterizing its composition is essential for identifying the origins of high-energy cosmic radiation and for constraining dark matter annihilation.

The dominant contributors to the extragalactic sky are blazars, a subclass of active galactic nuclei (AGNs) with relativistic jets oriented along the observer’s line of sight. Accretion onto a supermassive black hole powers these jets, producing a bimodal Spectral Energy Distribution (SED) with a low-frequency synchrotron peak and a high-frequency inverse-Compton peak [104]. Blazars are classified into two populations: Flat-Spectrum Radio Quasars (FSRQs) and BL Lacertae objects (BL Lacs). FSRQs exhibit broad optical emission lines and typically have lower-frequency synchrotron peaks, resulting in softer gamma-ray spectra that contribute most to the background below 1 GeV. BL Lacs lack strong emission lines, have harder energy spectra, and frequently exhibit high-frequency synchrotron peaks. As a result, nearby, low-luminosity BL Lacs produce the majority of the EGB emission above 100 GeV [114]. At these energies, gamma rays interact with the Extragalactic Background Light (EBL) to produce electron-positron pairs, an attenuation effect that naturally explains the exponential cutoff observed in the EGB spectrum above roughly 250 GeV. As a unified class, resolved and unresolved blazars account for approximately $50^{+12}_{-11}\%$ of the total EGB photons [114]. Because they define both the resolved catalogs and the upper end of the unresolved background, the classification and modeling of blazars are central to the research presented in Chapters 5, 6, and 7.

The remainder of the extragalactic gamma-ray emission is primarily due to misaligned AGNs and star-forming galaxies (SFGs). Misaligned AGNs, including Fanaroff-Riley radio galaxies, possess jets oriented away from our line of sight [104]. Without strong Doppler boosting, they are intrinsically fainter than blazars but far more abundant, contributing an estimated 20% to 25% of the UGRB [104, 115]. Star-forming galaxies, on the other hand, produce gamma rays through hadronic interactions as described in Section 2.1.1: cosmic-ray protons accelerated by supernovae collide with interstellar gas to produce neutral pions that decay into gamma rays. While their individual luminosities are low, SFGs are ubiquitous and their aggregate emission closely tracks the cosmic star-formation history. Current models suggest they account for 10% to 30% of the EGB [114, 116], potentially dominating among the faintest source populations. Detailed population models demonstrate that the combined emission from blazars, misaligned AGNs, and SFGs can account for the entire amplitude and spectral shape of the measured EGB (Figure 2.4), establishing tight constraints on any dark matter contribution [114].

The connection between resolved source catalogs and the unresolved background is formalized by the source-count distribution, dN/dS , which describes the differential number density of sources per unit solid angle as a function of their gamma-ray flux S . Integrating the dN/dS distribution from zero flux up to the telescope’s point-source detection threshold yields the intensity of the UGRB. While the bright end of the dN/dS is directly anchored by catalog measurements, probing the distribution below the detection threshold requires statistical techniques that extract population-level information from the collective imprint of unresolved sources on the photon-count map. The formal definition of dN/dS , its connection to the underlying luminosity function, and the methods developed to reconstruct it in the unresolved regime are the subject of Chapter 6.

2.3 The Fermi Large Area Telescope

The overlapping source populations, diffuse foregrounds, and the faint unresolved emission described in the previous section place specific demands on both the detecting instrument and the data analysis pipeline. To interpret the astrophysical gamma-ray sky, we must first understand

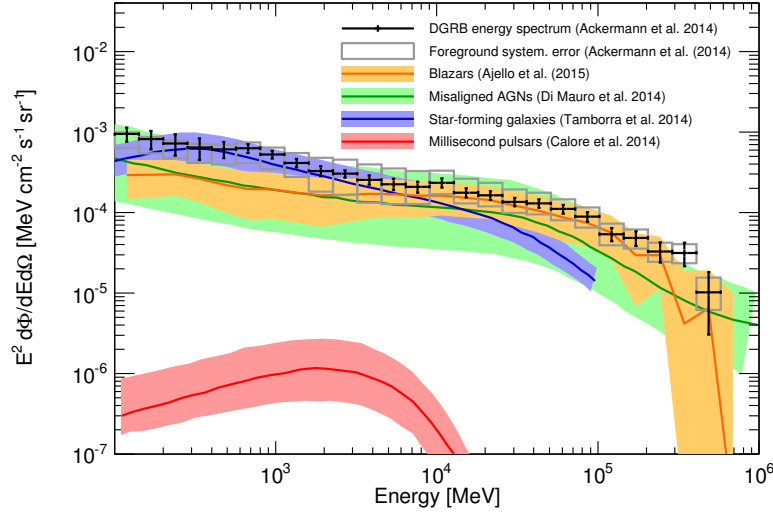


Fig. 2.4. The energy spectrum of the Diffuse Gamma-Ray Background and its expected composition. The black points represent the Fermi-LAT measurement, with gray bands indicating systematic uncertainty from the Galactic diffuse foreground subtraction. The colored lines and bands illustrate the predicted contributions from unresolved blazars (orange), misaligned AGNs (green), star-forming galaxies (blue), and millisecond pulsars (red). Credit: Figure 9 from [104]. [AA: try to replace later]

the telescope that observes it. Launched in 2008, the *Fermi* Large Area Telescope (LAT) has provided the most sensitive all-sky survey in the GeV domain to date, operating continuously for over fifteen years. While the detailed pipelines for data extraction and analysis are discussed directly within the subsequent chapters, it is necessary to establish the basic instrumental parameters and data products that define the practical limits of current observations. In this section, we review the fundamental design of the LAT, the energy-dependent instrumental response functions that govern its spatial and energy resolution, and the resulting catalogs of point sources that serve as the foundation for both individual target searches and statistical population studies.

2.3.1 Instrument Overview

The *Fermi* Large Area Telescope (LAT) is a pair-conversion telescope designed to measure cosmic gamma rays. Because high-energy gamma rays cannot be reflected or refracted, they must be detected indirectly through their interaction with matter. Fermi-LAT is therefore structured around four primary subsystems that convert incident photons into charged particles, track their trajectories, measure their energy, and reject the overwhelming background of cosmic rays [117]. Figure 2.5 illustrates the instrument components.

The conversion process takes place in the precision tracker, which consists of a 4×4 array of 16 modules. Each module contains 18 x, y tracking planes of single-sided silicon microstrip detectors, interleaved with high- Z tungsten conversion foils. An incoming gamma ray preferentially undergoes pair production within the tungsten, yielding an electron-positron (e^+e^-) pair. The silicon strip detectors then record the passage of these charged particles through successive

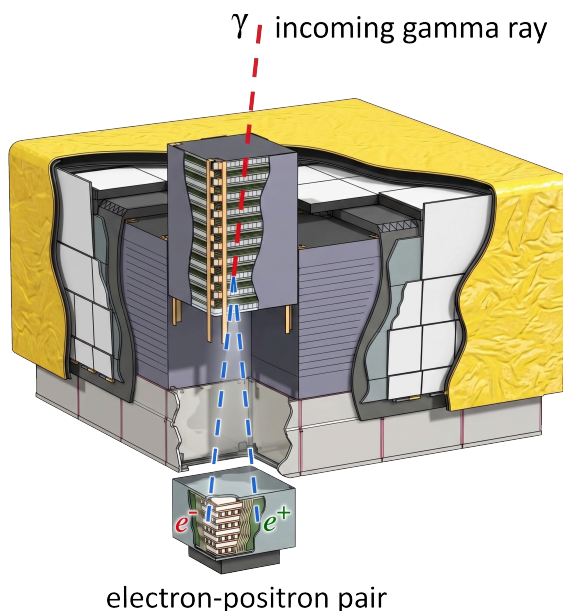


Fig. 2.5. Schematic diagram of the Large Area Telescope, showing the tracker modules, calorimeter, and the surrounding Anticoincidence Detector. The telescope’s dimensions are $1.8 \text{ m} \times 1.8 \text{ m} \times 0.72 \text{ m}$. Adapted from Figure 1 from [117].

layers. By precisely tracing the particle tracks back to their common vertex, the instrument reconstructs the arrival direction of the original photon.

Below the tracker lies the calorimeter, composed of 96 cesium iodide (CsI(Tl)) scintillating crystals per module, arranged in a configuration of 8 layers with alternating orientation. The calorimeter measures the total energy deposited by the electromagnetic shower and images its longitudinal and transverse development profile. This three-dimensional shower imaging is essential not only for energy reconstruction—allowing measurements to extend beyond the calorimeter’s physical depth up to TeV energies—but also for background rejection, as it separates the narrow electromagnetic showers from photon-induced events from the broader hadronic cascades produced by cosmic rays.

The entire tracker array is covered by the Anticoincidence Detector (ACD), a segmented shield of 89 plastic scintillator tiles read out by wavelength-shifting fibers and photomultiplier tubes. The ACD provides a charged-particle detection efficiency exceeding 0.9997 averaged over its area, enabling rejection of the charged cosmic-ray background that outnumber gamma rays by several orders of magnitude [117]. The segmentation of the ACD is necessary to suppress the “backsplash” effect: at high energies, secondary particles from electromagnetic showers in the calorimeter can scatter back into the shield, creating false veto signals. By considering only the ACD segment nearest the candidate photon’s entry point, the instrument avoids falsely rejecting valid high-energy gamma-ray events.

This hardware configuration enables the LAT to achieve substantial improvements over previous-generation instruments. The LAT operates across an energy range spanning from 20 MeV to over 300 GeV. Its compact aspect ratio grants a large field of view of approximately 2.4 sr at 1 GeV, ensuring that nearly all pair-conversion events initiated in the tracker pass into the calorimeter for energy measurement. In its primary survey mode, the spacecraft rocks alternately by $\pm 35^\circ$ from the zenith on consecutive orbits, achieving nearly uniform exposure

of the entire sky every two orbits—approximately three hours. Combined with a peak effective area of $\sim 9,500 \text{ cm}^2$ at normal incidence and a dead time of only $26.5 \mu\text{s}$ per event readout, these capabilities result in a sensitivity approximately 25 times greater than EGRET, enabling the detection and characterization of thousands of faint source populations [117].

2.3.2 Instrument Response Functions

The physical characteristics of the LAT—its tracking efficiency, multiple scattering within the conversion foils, and incomplete shower containment—are encapsulated in the Instrument Response Functions (IRFs). The IRFs map the true energy, direction, and conversion location of incident gamma rays to their reconstructed quantities. Three IRF components are particularly relevant for the analyses in this thesis: the Point Spread Function, the energy dispersion, and the effective area.

The Point Spread Function (PSF) describes the angular resolution of the telescope, quantifying the probability distribution of reconstructed directions around the true source position. At energies below a few GeV, the resolution is dominated by the multiple Coulomb scattering of the newly produced electron-positron pair as it traverses the tungsten conversion foils. Consequently, the PSF is strongly energy-dependent, as shown in Figure 2.6: at 100 MeV, the 68% containment radius spans approximately 3.5° to 5° ; it tightens to roughly 0.6° at 1 GeV and reaches an asymptotic limit of less than 0.15° above 10 GeV [117]. Off-axis, the containment radius degrades by up to a factor of 1.7 at 55° incidence. The broad PSF at lower energies creates a severe source confusion limit: because the spatial profiles of nearby point sources overlap with each other and with the intense Galactic diffuse emission, it becomes statistically difficult to disentangle individual faint objects. This confusion directly limits the depth of point-source catalogs and motivates the use of statistical population methods—such as those developed in Chapters 6 and 7—rather than standard source-by-source extraction.

The energy dispersion quantifies the accuracy of the reconstructed photon energy. It is determined by the calorimeter’s ability to contain and measure the electromagnetic shower. For on-axis photons, the equivalent Gaussian 1σ energy resolution ranges from 9% to 15% in the 100 MeV–1 GeV band, improves to 8%–9% between 1 and 10 GeV, and then degrades to $\sim 18\%$ near 300 GeV as shower leakage from the finite calorimeter depth becomes significant [117]. At large off-axis angles ($> 60^\circ$), where the effective calorimeter depth increases, the resolution can improve to $\sim 6\%$ above 10 GeV. For most analyses in this thesis, the energy resolution is sufficiently precise that energy dispersion is not a dominant systematic.

Not every gamma ray that enters the LAT produces a usable event: some pass through without converting, others are rejected during background filtering. The effective area captures this by expressing the instrument’s photon-collecting power as the area of an ideal, perfectly efficient detector that would record the same number of events. At normal incidence, it rises steeply from $\sim 100 \text{ cm}^2$ at 20 MeV to a broad plateau of $\sim 8,000$ – $9,500 \text{ cm}^2$ between 1 GeV and several hundred GeV, before declining at the highest energies [117]. For comparison, the LAT’s overall footprint is $1.8 \times 1.8 \text{ m}^2$; the peak effective area of $\sim 9,500 \text{ cm}^2$ —roughly a third of the geometric area—reflects the compounded probabilities of pair conversion in the tungsten, successful track reconstruction, and survival of background-rejection cuts. For a survey instrument, however, the relevant figure of merit is not the on-axis effective area alone, but the *acceptance*—the effective area integrated over all incident directions. While the effective area quantifies the detection efficiency for photons arriving from a single direction, the acceptance captures the instrument’s total photon-collecting power across the full field of view simultaneously. The LAT’s acceptance peaks at $\sim 2.5 \text{ m}^2 \text{ sr}$ above 1 GeV [117], a direct consequence of the large 2.4 sr field of view enabled by the tracker’s

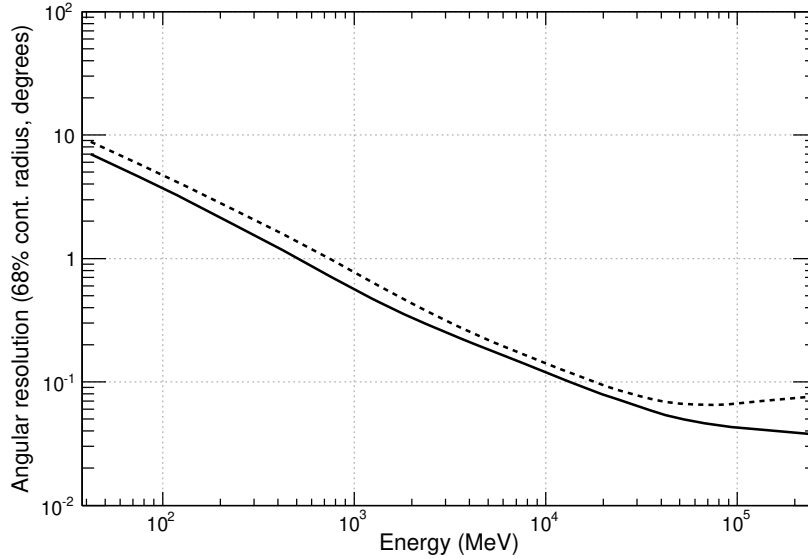


Fig. 2.6. The 68% containment radius of the Point Spread Function as a function of energy at normal incidence (solid) and at 60° off-axis (dashed), for photons converting in the thin *front* section of the tracker. Photons converting in the thick *back* section have a PSF approximately twice as wide at 1 GeV. Credit: Figure 17 from [117].

compact aspect ratio. Combined with the scanning survey mode described in Section 2.3.1, this large acceptance allows the LAT to achieve nearly uniform full-sky exposure every three hours, operating as a continuous all-sky monitor rather than a pointed observatory.

The accuracy of the IRFs relies directly on the underlying event reconstruction algorithms. The Pass 8 analysis framework represents a comprehensive overhaul of the LAT event-level reconstruction, designed to maximize the scientific return of the data [118]. A major improvement in Pass 8 was the mitigation of “ghost signals”—instrumental pile-up and out-of-time residual hits from previous events that previously degraded the reconstruction. This enabled a significant expansion of the effective area, particularly below 100 MeV and above 1 TeV, while simultaneously reducing background contamination. Pass 8 also introduced event subclassifications that partition data by the quality of the reconstructed direction (PSF event types: PSF0–PSF3, from broadest to narrowest) and the quality of the energy measurement (EDISP event types). This partitioning allows analyses to select event classes optimized for their specific scientific goals—angular resolution for morphological studies, or photon statistics for spectral analyses.

Subsequent iterations, most notably the P8R3 release [119], further refined the residual background rejection, producing a spatially uniform and clean dataset that serves as the baseline for all modern *Fermi*-LAT analyses, including the work presented in this thesis.

2.3.3 Data Products and Catalogs

With the IRFs established, we now turn to how the LAT data are translated into astrophysical catalogs. Converting raw photon events into measurements of individual sources requires disentangling point-like objects from the structured diffuse backgrounds. The *Fermi*-LAT collaboration achieves this through a maximum likelihood framework [120, 121]. The observed sky is modeled as a linear combination of spatial and spectral templates: the highly structured Galactic diffuse

emission, a spatially uniform isotropic component capturing the extragalactic background and residual instrumental noise, and individual point sources each parametrized by a position and spectral model. The normalizations and spectral parameters of all components are simultaneously adjusted to maximize the Poisson likelihood function against the observed photon counts across the sky.

To evaluate the significance of a candidate point source, the framework computes the Test Statistic (TS), defined as

$$TS = 2 \ln \frac{\mathcal{L}_1}{\mathcal{L}_0},$$

where $\mathcal{L}_1$ is the maximum likelihood of the model with the candidate source included and $\mathcal{L}_0$ is the maximum likelihood of the null hypothesis without it [120]. By Wilks' theorem, TS is approximately distributed as χ^2 with a number of degrees of freedom equal to the additional free parameters of the source model. For a point source with four free parameters (two spatial and two spectral), the catalog detection threshold of $TS > 25$ corresponds to a significance of just over 4σ [121, 122]. As a practical rule of thumb, for an isolated source with known position (one spectral degree of freedom), the significance is well approximated by $\sqrt{TS} \sigma$. This threshold defines the boundary between what is considered a “resolved” source entering the catalog and what remains part of the unresolved diffuse background.

The results of these analyses have been systematically compiled into a series of catalogs of increasing depth. The progression from the First *Fermi*-LAT Catalog (1FGL, based on 11 months of data) through the Third Catalog (3FGL, 4 years, 3,033 sources) to the Fourth Catalog (4FGL, 8 years, 5,064 sources) [121, 122] reflects both the accumulation of exposure and successive improvements in event reconstruction and background modeling. The most recent published incremental release, 4FGL-DR4, extends the dataset to 14 years and contains 7,194 sources [123], with more recent data releases continuing to grow the catalog. Throughout this evolution, the detection threshold has remained at $TS > 25$, so the growing source count is driven primarily by longer exposure, improvements in the galactic foreground modeling, and improved sensitivity from Pass 8 event reconstruction.

Each catalog entry reports the source localization (with 95% confidence ellipse), the integrated energy flux in multiple bands, the best-fit spectral model, a variability index, and multi-wavelength associations. The spectral models employed include the simple power law, the log-parabola (commonly preferred for blazars), and the power law with an exponential cut-off (standard for pulsars). These spectral characterizations—in particular the flux, spectral index, and curvature from the log-parabola fits—form the feature space used in Chapter 5 for a population-level study of unassociated sources.

The association procedure matches each gamma-ray source against catalogs at radio, infrared, optical, and X-ray wavelengths using Bayesian and likelihood-ratio methods. Sources receiving a high-confidence match are labeled as “associated”; those with firm identification through periodicity, correlated variability, or resolved angular extent are labeled as “identified.” In the 4FGL-DR4, 4021 sources are identified or associated with extragalactic objects and 593 with Galactic source classes. However, a substantial fraction—2577 sources—lacks any plausible multi-wavelength counterpart and remains unassociated [123]. These unassociated sources are concentrated along the Galactic plane, where the dense background complicates counterpart searches, and at high latitudes, where their average spectral properties resemble those of blazars of unknown type.

This large population of unassociated sources is scientifically important for multiple reasons. Beyond its likely composition of faint blazars and pulsars below the sensitivity of current multi-

wavelength surveys, it also constitutes the primary search pool for exotic gamma-ray emitters such as dark matter subhalos. As discussed in Chapter 1, Λ CDM cosmology predicts the existence of thousands of Galactic subhalos below $\sim 10^7 M_\odot$ —too low in mass to retain gas or form stars, and therefore invisible at all wavelengths except gamma rays, where WIMP annihilation could produce a faint, spectrally soft signal closely resembling an unassociated *Fermi*-LAT point source [5, 95]. Separating such candidates from the astrophysical contaminants within the 4FGL catalog is the central challenge addressed in Chapter 5, where machine learning techniques are used to search for dark matter subhalo candidates among the unassociated sources.

2.4 Summary

This chapter has established the astrophysical and instrumental context for the dark matter searches developed in the remainder of this thesis. The gamma-ray sky observed by the *Fermi*-LAT is a superposition of hadronic and leptonic diffuse emission from cosmic-ray interactions in the Milky Way, thousands of individually resolved point sources—predominantly pulsars and blazars—and a faint, nearly isotropic extragalactic background composed of unresolved active galactic nuclei and star-forming galaxies. Each of these components is relevant to the analyses that follow: the Galactic Diffuse Emission is the dominant foreground whose imperfect modeling introduces systematic uncertainties in every *Fermi*-LAT measurement; the spectral degeneracy between millisecond pulsars and dark matter annihilation is at the heart of the Galactic Center Excess debate; and the source-count distribution of unresolved extragalactic populations defines the observable that connects resolved catalogs to the diffuse background.

The *Fermi*-LAT itself imposes fundamental limitations on what can be extracted from the data. The energy-dependent Point Spread Function creates a source confusion limit that prevents standard detection methods from resolving faint objects at low energies. The $TS > 25$ detection threshold partitions the sky into resolved sources entering the catalog and an unresolved component that may contain both astrophysical emitters and dark matter signals. The large population of unassociated sources—about one third of the 4FGL catalog—lacks clear multi-wavelength counterparts and constitutes the primary search pool for exotic emitters such as dark matter subhalos.

These challenges—complex, spatially structured backgrounds; spectral degeneracies between dark matter and conventional sources; instrumental limitations that leave a large fraction of the source population unresolved or unclassified—cannot be addressed by traditional signal-over-background analyses alone. They motivate the statistical and machine learning methods introduced in the next chapter: simulation-based inference to recover population properties below the detection threshold, neural posterior estimation to bypass intractable likelihoods, and domain-adaptive models to operate reliably when training datasets do not perfectly match the observed data.

STATISTICAL METHODS FOR NOISE-DOMINATED REGIMES

The gamma-ray sky described in the previous chapter presents a formidable data analysis challenge. Faint signals from potential dark matter annihilation or decay are embedded in bright, spatially structured astrophysical backgrounds; source populations overlap in both position and spectral shape; and instrumental effects—finite angular resolution, energy-dependent sensitivity, detection thresholds—leave a large fraction of the emitting population unresolved or unclassified. Extracting new physics from this environment requires moving beyond standard signal-over-background methods toward a more comprehensive statistical toolkit.

[AA: This has nice momentum, but maybe there’s a way to rewrite it to have shorter sentences?] The analyses presented in this thesis draw on techniques spanning classical statistics, modern machine learning, and domain-specific tools developed for astrophysical inference. Each method addresses a different aspect of the extraction problem: frequentist and Bayesian approaches provide the language for parameter estimation and hypothesis testing; simulation-based inference bypasses intractable likelihoods that arise in complex forward models; neural networks learn directly from high-dimensional data when explicit physical models are unavailable; domain adaptation corrects for the systematic mismatches between simulated training data and real observations; and angular cross-correlations exploit the connection between gamma-ray emission and the large-scale distribution of matter to access collective signals invisible to single-instrument analyses.

This chapter introduces the mathematical framework used throughout the thesis. Each method is presented at a conceptual level, with forward references to the research papers and chapters where the full technical details appear. We begin with the foundations of frequentist and Bayesian inference in Section 3.1, establishing the likelihood function, maximum likelihood estimation, Bayesian posteriors, and hypothesis testing. Section 3.2 introduces the simulation-based inference paradigm for problems where the likelihood cannot be evaluated analytically. Section 3.3 covers the machine learning tools—convolutional neural networks, density estimation, and mixture models—used for regression and classification tasks on gamma-ray data. Section 3.4 formalizes the dataset shift problem that arises when training and target distributions differ. Finally, Section 3.5 develops the angular cross-correlation formalism that connects gamma-ray observations to gravitational tracers of dark matter structure at cosmological scales.

3.1 Frequentist and Bayesian Inference

Extracting physical information from noisy gamma-ray data requires a well-defined statistical language. The analyses presented in this thesis draw on two complementary paradigms—frequentist and Bayesian inference—each providing distinct tools for parameter estimation, uncertainty quantification, and hypothesis testing. This section introduces both frameworks from the shared foundation of the likelihood function, develops the key concepts used in the research papers, and discusses when each paradigm is most appropriate (according to personal bias). Readers familiar with statistical inference may proceed directly to the application previews, which connect the general formalism to its specific use in the research chapters of this thesis.

3.1.1 The Inference Problem

The starting point for any quantitative analysis in physics is the problem of inference: given a set of observed data $\mathbf{d}$, a theoretical model $\mathcal{M}$ governed by parameters θ , what can we learn about θ from $\mathbf{d}$? In gamma-ray astrophysics, $\mathbf{d}$ might be a photon-count map recorded by the Fermi LAT, while θ encodes the physical quantities of interest: source fluxes, spectral indices, or the dark matter annihilation cross-section $\langle\sigma v\rangle$. The model $\mathcal{M}$ connects these parameters to the expected data through a forward process that may include astrophysical emission templates, instrumental response functions, and Poisson sampling of photon counts.

The connection between parameters and data is formalized by the likelihood function $\mathcal{L}(\theta) = p(\mathbf{d}|\theta)$, which quantifies how well a given parameter choice explains the observed data [124]. This function is the shared foundation of both major inference paradigms. In the frequentist approach, the parameters θ are treated as fixed but unknown quantities, and inference proceeds by identifying the parameter values that maximize $\mathcal{L}$. In the Bayesian approach, θ are themselves treated as random variables with a prior distribution $p(\theta)$, and the goal is to compute the posterior $p(\theta|\mathbf{d})$ via Bayes’ theorem [125, 126].

These two philosophies are not in competition; they are complementary tools, each with distinct strengths suited to different aspects of the inference problem. In the analyses presented in this thesis, both are used—sometimes on the same dataset—to extract physical information and quantify uncertainty. The following two subsections develop each approach, starting from a common notation and building toward the specific formulations employed in the research papers.

3.1.2 Frequentist Inference

In the frequentist framework, the model parameters θ are fixed but unknown constants of nature. The data $\mathbf{d}$ are treated as one realization of a random process, and all statistical statements are made in terms of the long-run frequency properties of that process. The central object is the likelihood function, $\mathcal{L}(\theta) = p(\mathbf{d}|\theta)$, which quantifies the probability of the observed data as a function of the model parameters [126].

Maximum Likelihood Estimation. The most common frequentist estimator is the maximum likelihood estimator (MLE), which identifies the parameter values that maximize the likelihood:

$$\hat{\theta}_{\text{MLE}} = \underset{\theta}{\operatorname{argmax}} \mathcal{L}(\theta). \quad (3.1.1)$$

Under mild regularity conditions, the MLE is asymptotically unbiased and efficient. As a concrete example, consider counting N photons from a gamma-ray source. If the counts follow a Poisson distribution with expected rate μ , the likelihood is $\mathcal{L}(\mu) = e^{-\mu}\mu^N/N!$, and the MLE is simply

$\hat{\mu} = N$. While this example is trivial, the Poisson likelihood extends naturally to the pixel-by-pixel analysis of gamma-ray sky maps, where the joint likelihood is the product of Poisson terms over all pixels:

$$\mathcal{L}(\boldsymbol{\theta}) = \prod_{a=1}^{N_{\text{pix}}} \frac{\lambda_a(\boldsymbol{\theta})^{k_a} e^{-\lambda_a(\boldsymbol{\theta})}}{k_a!}, \quad (3.1.2)$$

with k_a the observed counts and $\lambda_a(\boldsymbol{\theta})$ the model prediction in pixel a . This product-of-Poisson construction is the standard likelihood used in Fermi-LAT analyses [120].

Nuisance parameters and the profile likelihood. In realistic analyses, the parameter vector $\boldsymbol{\theta}$ contains both the physical parameters of interest $\boldsymbol{\psi}$ and a potentially large set of nuisance parameters $\boldsymbol{\eta}$ (background normalizations, instrumental systematics, source model parameters). The profile likelihood isolates the dependence on $\boldsymbol{\psi}$ by maximizing over the nuisance parameters at each fixed value of $\boldsymbol{\psi}$:

$$\mathcal{L}_{\text{prof}}(\boldsymbol{\psi}) = \max_{\boldsymbol{\eta}} \mathcal{L}(\boldsymbol{\psi}, \boldsymbol{\eta}). \quad (3.1.3)$$

This construction preserves the frequentist interpretation of the likelihood without requiring any prior distribution on $\boldsymbol{\eta}$. An alternative approach is to integrate over the nuisance parameters within the likelihood framework itself, producing an integrated (or marginal) likelihood:

$$\mathcal{L}_{\text{int}}(\boldsymbol{\psi}) = \int \mathcal{L}(\boldsymbol{\psi}, \boldsymbol{\eta}) g(\boldsymbol{\eta}) d\boldsymbol{\eta}, \quad (3.1.4)$$

where $g(\boldsymbol{\eta})$ is a weight function derived from independent auxiliary measurements or physical constraints on the nuisance parameters. Although this resembles Bayesian marginalization in form, the integrated likelihood is interpreted differently: it is a tool for removing $\boldsymbol{\eta}$ from the likelihood rather than a posterior probability statement about $\boldsymbol{\psi}$.

Hypothesis testing and the Test Statistic. To compare competing hypotheses—for instance, whether a gamma-ray source exists at a given position—one generally constructs a likelihood-ratio test statistic (TS):

$$\text{TS} = -2 \ln \frac{\mathcal{L}_0}{\mathcal{L}_1}, \quad (3.1.5)$$

where $\mathcal{L}_0$ is the likelihood under the null hypothesis and $\mathcal{L}_1$ the likelihood under the alternative (or, more commonly, the globally maximized likelihood) [120]. By Wilks' theorem, under certain regularity conditions the TS is asymptotically distributed as a χ^2 random variable with degrees of freedom equal to the difference in free parameters between the two hypotheses [127]. This result allows direct conversion of TS values to statistical significance without explicit Monte Carlo simulations.

The likelihood-ratio test relies on nested hypotheses and on the regularity conditions underlying Wilks' theorem. When these assumptions do not hold—in particular, when the hypotheses are non-nested, or when one wishes to assess the absolute compatibility of the data with a single model—other frequentist tools are available. The Kolmogorov–Smirnov test [128] addresses the latter case by taking as test statistic the supremum of the absolute difference between the empirical cumulative distribution function of the data and that predicted by the model. Its null distribution is independent of the underlying distribution, which makes it a common diagnostic for goodness-of-fit when no specific alternative is being tested. For comparing models that are not nested, information criteria offer a complementary approach. The Akaike Information Criterion [129] is defined as

$$\text{AIC} = -2 \ln \mathcal{L}_{\text{max}} + 2k, \quad (3.1.6)$$

where $\mathcal{L}_{\max}$ is the maximum value of the likelihood and k the number of free parameters. The AIC provides an estimator of the relative Kullback–Leibler divergence between each candidate model and the data-generating process, with the $2k$ term penalising additional parameters and thus discouraging overfitting. Model selection then proceeds by minimising the AIC across the candidates; differences in AIC, rather than absolute values, carry the statistical meaning, and no p -value is produced.

Application preview. The frequentist framework developed above is used extensively in this thesis:

- In Chapter 4, the MSP luminosity function is constrained via MLE applied to an integrated likelihood, and model comparison is performed through $-2\Delta\ln\mathcal{L}$ values.
- In Chapter 5, a profile likelihood test statistic is used with Wilks’ theorem to set 95% CL upper bounds on the dark matter annihilation cross-section; the AIC is additionally used to compare competing models for the unassociated source population.
- In Chapter 7, a two-sample KS test compares the cumulative pixel-wise TS distributions of the real *Fermi*-LAT sky against synthetic realizations drawn from the source-count distribution, providing the statistical criterion that defines sub-threshold source candidates.
- In Chapter 8, a $\Delta\chi^2$ test statistic compares astrophysical-only and astrophysical-plus-DM hypotheses for the cross-correlation angular power spectrum.

3.1.3 Bayesian Inference

The Bayesian framework treats the model parameters θ not as fixed unknowns but as random variables associated with a probability distribution. Before observing any data, our state of knowledge about θ is encoded in a prior distribution $p(\theta)$. After observing data $\mathbf{d}$, this knowledge is updated through Bayes’ theorem:

$$p(\theta|\mathbf{d}) = \frac{p(\mathbf{d}|\theta)p(\theta)}{p(\mathbf{d})}, \quad (3.1.7)$$

where $p(\mathbf{d}|\theta) = \mathcal{L}(\theta)$ is the same likelihood function used in the frequentist approach, and $p(\mathbf{d}) = \int p(\mathbf{d}|\theta')p(\theta')d\theta'$ is the evidence, a normalization constant that integrates the likelihood weighted by the prior over the full parameter space [125]. Within the assumed model, the posterior $p(\theta|\mathbf{d})$ is the complete answer to the inference problem: it encodes not only the best-fit values but also the full uncertainty structure, including correlations between parameters.

Prior, posterior, and evidence. The choice of prior $p(\theta)$ is the defining feature of the Bayesian approach, and the source of much of its flexibility. Informative priors can incorporate physical constraints—for example, requiring fluxes to be non-negative or restricting spectral indices to physically motivated ranges—and can improve inference when data are sparse. Uninformative or weakly informative priors are used when prior knowledge is limited, though the choice is never fully objective [126]. The evidence $p(\mathbf{d})$ is typically intractable to compute, as it requires integration over the full parameter space, but it plays an important role in Bayesian model comparison, where the ratio of evidences between competing models (the Bayes factor) provides a natural criterion for model selection [130].

Credible intervals versus confidence intervals. In the Bayesian framework, uncertainty is quantified through credible intervals: a 95% credible interval is a region of parameter space containing 95% of the posterior probability. This contrasts with the frequentist confidence interval, which guarantees that 95% of intervals constructed from repeated experiments will contain the true parameter value. The distinction is subtle in practice—for well-constrained problems, the two often agree numerically—but the conceptual difference matters: a credible interval makes a direct probability statement about the parameter given the data, while a confidence interval makes a statement about the procedure.

Marginalization over nuisance parameters. As in the frequentist case, realistic Bayesian analyses involve nuisance parameters. The Bayesian approach handles them by marginalizing the joint posterior:

$$p(\boldsymbol{\psi} | \mathbf{d}) = \int p(\boldsymbol{\psi}, \boldsymbol{\eta} | \mathbf{d}) d\boldsymbol{\eta}, \quad (3.1.8)$$

which integrates out the nuisance parameters $\boldsymbol{\eta}$ to yield the marginal posterior for the parameters of interest $\boldsymbol{\psi}$. This is conceptually analogous to the frequentist integrated likelihood (see Section 3.1.2), but the probabilistic interpretation differs: the Bayesian marginal posterior is a fully coherent probability distribution over $\boldsymbol{\psi}$, conditioned on both the data and the prior.

When the likelihood is intractable—as is often the case for complex forward models involving high-dimensional simulations—neural networks offer a route to Bayesian inference through simulation-based methods, in which a density estimator learns the posterior directly from simulated data–parameter pairs. These approaches are developed in Section 3.2.

Application preview. In Chapter 6, the Bayesian framework is implemented via simulation-based inference to reconstruct the source-count distribution dN/dS from gamma-ray sky maps, with posterior uncertainty estimates validated against an independent frequentist cross-check (Section 3.1.4).

3.1.4 When to Use Which

For most inference tasks encountered in practice—parameter estimation, upper-limit setting, model comparison—both the frequentist and Bayesian frameworks can reach the same scientific conclusion. The choice between them is usually pragmatic rather than fundamental, driven by the scientific question at hand, computational considerations, and the conventions of the relevant research community. Community practice already carries considerable weight: high-energy particle physics and the Fermi-LAT collaboration have traditionally favored frequentist methods—source detection via the TS, upper limits via the profile likelihood and Wilks’ theorem, catalog pipelines built on maximum-likelihood fitting [120]—while cosmology and large-scale-structure analyses have largely adopted Bayesian Markov Chain Monte Carlo (MCMC) methods [126]. Neither convention reflects a judgment on the foundations of probability; both reflect what works well for the typical dimensionality and structure of the problems each field encounters most often.

Beyond community defaults, the central question is whether the analysis requires a full characterization of uncertainty or whether a point estimate with a simple error bar suffices. When the posterior structure matters—correlations between parameters, degeneracies, multi-modality, or the marginalization of many nuisance parameters—the Bayesian framework provides this information as its default output (Section 3.1.3). In high-dimensional parameter spaces, however, MCMC itself becomes expensive: samplers suffer from poor mixing, correlated samples, and

trapping in local modes. Frequentist point estimation can remain tractable in the same regime through gradient-based optimization or iterative algorithms such as EM (Section 3.3.3), which converge efficiently without exploring the full parameter space—but at the cost of forgoing the posterior structure that motivates the Bayesian approach in the first place. When a one-dimensional confidence interval or upper limit is the target, as in standard source detection or cross-section bounds, the frequentist route is typically more direct.

A second discriminating factor is the role of external information. Bayesian inference provides a coherent mechanism for incorporating physical constraints—positivity of fluxes, calibration ranges from auxiliary measurements, results from complementary experiments—through the prior. Frequentist methods are preferred when the goal is to present a result with minimal sensitivity to external assumptions, particularly in the data-sparse regime where the prior can considerably shape the posterior. This prior dependence extends to model comparison: the Bayes factor offers a principled comparison between arbitrary model pairs but is sensitive to the prior volume, especially when priors are vague or the models differ substantially in complexity [130]; the frequentist likelihood-ratio test (Section 3.1.2) is prior-free but restricted to nested hypotheses; information criteria such as the AIC apply more broadly but produce only relative rankings without significance levels.

In the limit of abundant, constraining data, these distinctions fade: the likelihood dominates any reasonable prior, and Bayesian credible intervals coincide numerically with frequentist confidence intervals. The choice becomes consequential precisely when data are sparse, priors are informative, or the parameter space is high-dimensional. In the analyses presented in this thesis, both paradigms are used, and in some cases they are applied to the same problem as independent cross-checks. In Chapter 6, a simulation-based inference approach provides Bayesian posterior estimates on the reconstructed dN/dS , which are validated against a frequentist analysis of the same data where confidence intervals are extracted from the empirical distribution of residuals across the validation set. The frequentist cross-check also provides a formal estimate of the model’s bias, which is found to be negligible across the flux range of interest. The compatibility of the two independent error estimates provides confidence that the uncertainty quantification is robust.

3.2 The Simulation-Based Inference Paradigm

The inference framework presented in Section 3.1 assumes that the likelihood function can be evaluated explicitly. This assumption holds for many standard analyses, but it breaks down when the data-generating process involves a complex simulator whose internal stochastic structure prevents the likelihood from being written in closed form. Simulation-based inference (SBI) provides a principled alternative: rather than computing the likelihood analytically, one trains machine learning models on simulated data to approximate the posterior, the likelihood, or their ratio. This section introduces the SBI paradigm, the neural estimation strategies that underpin it, and the validation techniques required to ensure the reliability of the results.

3.2.1 The Intractable Likelihood Problem

The inference methods presented in Section 3.1 share a common requirement: the ability to evaluate the likelihood function $p(\mathbf{x}|\boldsymbol{\theta})$ for a given set of parameters. In many problems of interest, however, the data-generating process is described by a complex computer simulation rather than by an analytically tractable probability distribution. Such models are known as *implicit models*, in contrast to *prescribed models* for which the likelihood can be written in closed form [124, 131].

The resulting inference problem has historically been called *likelihood-free inference*, although the term is somewhat misleading: these methods typically estimate the intractable likelihood rather than dispensing with it entirely. The more descriptive term *simulation-based inference* (SBI) is now preferred [124], and we adopt it throughout this thesis.

The source of the difficulty is straightforward to state. A typical simulator takes as input a vector of physical parameters θ , samples a series of internal stochastic states or latent variables z_i , and produces synthetic data $\mathbf{x}$ as output. The exact likelihood of this output requires marginalizing over the joint density of data and latent variables:

$$p(\mathbf{x}|\theta) = \int p(\mathbf{x}, \mathbf{z}|\theta) d\mathbf{z}, \quad (3.2.1)$$

where the integral runs over every possible execution trace of the simulator [124]. For realistic simulations involving billions of latent variables, such as particle transport through a detector, the stochastic evolution of astrophysical populations, or the formation of large-scale structure, this high-dimensional integral is computationally intractable.

The asymmetry between the forward and inverse problems is a defining feature of the SBI setting. Running the simulator to produce synthetic observations from a given θ is computationally cheap: one simply executes the program. Inverting this process to obtain $p(\mathbf{x}|\theta)$ analytically, however, would require enumerating all possible internal configurations that could have led to the same observable output. This forward–inverse asymmetry motivates an alternative strategy: rather than evaluating the likelihood directly, one can generate large numbers of simulated datasets and use them to learn the relationship between parameters and observables.

A concrete astrophysical example arises in the measurement of the gamma-ray source-count distribution dN/dS below the Fermi-LAT detection threshold. As discussed in detail in Chapter 6, the observed photon-count map results from the superposition of thousands of unresolved point sources. Writing a closed-form likelihood for this full sky map would require integrating over every possible spatial configuration, flux, and number of the unresolved sources. Generating a synthetic map from a given set of dN/dS parameters, by contrast, amounts to a sequence of well-defined sampling steps: draw source fluxes from the dN/dS , scatter them on the sky, and apply additional well-understood sources of noise. The forward simulation is straightforward; the inverse problem is not. SBI addresses exactly this kind of problem, and the following subsections describe how the inversion is accomplished.

3.2.2 Simulation-Based Inference

Given the impossibility of evaluating the likelihood directly, SBI replaces explicit likelihood computation with learned surrogates trained on simulated data. The idea has a long history: Diggle and Gratton [131] proposed using kernel density estimation to approximate intractable likelihoods from simulator output as early as 1984, while Approximate Bayesian Computation (ABC) developed in parallel as an entirely sampling-based alternative (see [124] for a review). Both approaches, however, suffer from the curse of dimensionality: they require hand-crafted summary statistics to compress the data into low-dimensional representations, inevitably discarding information about the parameters of interest. ABC additionally lacks amortization, meaning the full inference procedure must be repeated for each new observation. Modern neural SBI methods overcome these limitations by using deep neural networks as flexible density estimators that operate directly on high-dimensional data [124]. Three main neural strategies have emerged, distinguished by which statistical quantity the neural network learns to approximate.

Neural Posterior Estimation (NPE) trains a conditional density estimator—typically a normalizing flow [132], though a Laplace approximation can serve a similar role as in Chapter 6—to directly approximate the posterior distribution $p(\boldsymbol{\theta}|\mathbf{x})$ [133]. By conditioning the density estimator on the observed data $\mathbf{x}$, the network learns to output the posterior density for any input observation. Because NPE directly provides the posterior, it is the most straightforward approach for Bayesian inference. However, because the prior enters the training, changing the prior requires retraining. The mathematical details of normalizing flow architectures are deferred to Chapters 6 and 9.

Neural Likelihood Estimation (NLE) instead approximates the likelihood function $p(\mathbf{x}|\boldsymbol{\theta})$, conditioning the density estimator on $\boldsymbol{\theta}$ rather than on $\mathbf{x}$ [124]. Once the likelihood surrogate is trained, Bayesian inference proceeds through standard MCMC sampling with any choice of prior. This separation of the likelihood from the prior provides additional flexibility, at the cost of an extra sampling step.

Neural Ratio Estimation (NRE) takes a different approach altogether: rather than learning a probability density, a classifier is trained via supervised learning to discriminate between synthetic datasets generated at different parameter points. The classifier output can be converted into the likelihood-to-evidence ratio $r(\mathbf{x}, \boldsymbol{\theta}) = p(\mathbf{x}|\boldsymbol{\theta})/p(\mathbf{x})$, which is proportional to the posterior up to a normalizing constant [124]. A key advantage of NRE is its ability to target arbitrary low-dimensional marginal posteriors directly, without first estimating the full joint posterior [124, 134]. This makes the approach particularly attractive when one- and two-dimensional marginals suffice, as is common in cosmological and astrophysical parameter estimation. Evaluating the marginal posterior density at any point is fast once the classifier is trained; however, drawing samples from the learned posterior still requires MCMC, making full posterior sampling more expensive than for NPE.

All three strategies provide *amortized inference*: the computationally expensive simulation and training steps are performed once upfront, after which posterior evaluation for new observed data is nearly instantaneous [124]. The neural network also implicitly marginalizes over the vast latent space of the simulator, so that the trained surrogate preserves information that hand-crafted summary statistics would discard.

Application preview. In this thesis, NPE is the SBI strategy most directly relevant to our work. In Chapter 6, a convolutional neural network trained on synthetic gamma-ray sky maps estimates the source-count distribution dN/dS and its uncertainty for any input map, without evaluating an explicit likelihood. The approach operates in the spirit of NPE: the network directly maps an observed sky map to posterior summaries (the mean and variance for each flux bin). Once trained, the inference is amortized: the network can be applied to the real Fermi-LAT sky map to immediately obtain the source-count distribution.

3.2.3 Validation of Simulation-Based Inference

A neural density estimator trained on finite simulations can, in principle, produce posteriors that are poorly calibrated: the reported credible intervals may be systematically too narrow or too wide. Validating SBI results is therefore an integral part of any simulation-based analysis, and the simulator itself provides the means for doing so.

The standard validation framework is *simulation-based calibration* (SBC), introduced by Talts et al. [135]. The procedure is conceptually simple. One samples parameters $\boldsymbol{\theta}^*$ from the prior, generates synthetic data $\mathbf{x}^* \sim p(\mathbf{x}|\boldsymbol{\theta}^*)$ using the simulator, and then applies the trained estimator to obtain the approximate posterior $\hat{p}(\boldsymbol{\theta}|\mathbf{x}^*)$. Because the true parameters are known by

construction, one can check whether the posterior places the correct amount of probability mass around θ^* . Repeating this test across hundreds or thousands of independent simulations yields a statistical picture of the estimator’s calibration: if the $(1 - \alpha)$ credible intervals contain the true parameters in exactly a fraction $(1 - \alpha)$ of trials, the estimator is well-calibrated. Systematic deviations indicate over- or underdispersion of the posterior [124, 135].

SBC is particularly well-suited to amortized estimators, since evaluating the posterior for each new synthetic observation is nearly free once the network has been trained. For non-amortized methods, the computational cost of fitting a new posterior for each of the hundreds of required test datasets would be prohibitive.

A complementary approach is *coverage testing*: for a given confidence level, one directly measures the fraction of test cases in which the true parameters fall within the corresponding credible region. The empirical coverage can be compared to the nominal level across a range of credibility thresholds, and the discrepancy is formally quantified through the empirical calibration error [124]. Departures from the diagonal in a coverage-vs-credibility plot flag potential miscalibration.

In practice, many SBI analyses supplement these general calibration checks with problem-specific diagnostics. In Chapter 6, the internally estimated Bayesian uncertainties are cross-checked against frequentist error bars derived from the distribution of residuals over the validation set (see also Section 3.1.4). The close agreement between the two independent error estimates provides confidence in the reliability of the uncertainty quantification.

These three validation strategies—calibration, coverage, and problem-specific diagnostics—are applied in each of the analyses presented in the subsequent chapters of this thesis.

3.3 Machine Learning in Astrophysics

Section 3.2 introduced simulation-based inference as a framework for problems with intractable likelihoods; this section presents the concrete machine-learning architectures and techniques that implement it, and that underpin data-driven inference in astrophysics and cosmology more broadly. In practice, the data that these methods must process arrive at one of two levels. *Field-level data*—images, sky maps of photon counts, light curves—preserve the full spatial or temporal structure of the signal and call for architectures that can exploit that structure directly. *Summary statistics*—energy spectra, variability indicators, flux ratios—compress the raw signal into a low-dimensional feature set that is easier to model but inevitably discards some information in the process. The two approaches are complementary: field-level methods retain more information at the cost of architectural complexity, while summary statistics trade information content for a simpler, lower-dimensional input space. Both appear in this thesis: Chapter 6 works at the field level, feeding gamma-ray sky maps into a convolutional network to reconstruct the source-count distribution, while Chapter 5 operates on spectral and variability features to search for a dark matter subhalo population among unassociated Fermi-LAT sources.

This section introduces the machine-learning concepts and architectures that underpin both analyses: the canonical learning tasks (Section 3.3.1); neural networks, their training, and uncertainty estimation (Section 3.3.2); convolutional architectures and their spherical extensions for sky-map data (Section 3.3.3); and classical non-parametric density estimation for population-level inference on feature vectors (Section 3.3.4).

3.3.1 Learning Tasks

Machine learning provides tools for extracting information from data when explicit physical models are either unavailable or computationally intractable [125, 126]. The problems addressed in this thesis fall into three canonical learning tasks.

The first is *classification*: assigning a discrete label to an input based on its observed features. In the gamma-ray context, this amounts to determining the astrophysical nature of a detected source. Roughly one-third of the sources in Fermi-LAT catalogs lack firm multi-wavelength associations [136], and supervised classification algorithms—trained on populations of identified AGNs and pulsars—are used to predict the class membership of these unassociated objects. The input features are quantities derived from the gamma-ray signal itself: spectral curvature significance, variability index, hardness ratios, and energy flux [136]. In Chapter 5, this paradigm is extended beyond standard source classes through a generative mixture model that accommodates an entirely new population—dark matter subhalos—among the unassociated sources.

The second task is *regression*: predicting a continuous quantity from observational data. In Chapter 6, a convolutional neural network estimates the source-count distribution dN/dS directly from gamma-ray sky maps. By adopting a heteroscedastic loss function (described in Section 3.3.2), the network simultaneously performs regression on both the dN/dS values and their associated uncertainties, predicting the variance of each output alongside its mean [137].

The third task is *density estimation*: approximating the probability distribution from which observed source features are drawn. While classification asks “what class is this source?”, density estimation asks the complementary question “what does the overall population look like?” This shift from individual classification to population-level inference underpins the generative mixture model developed in Chapter 5, where Kernel Density Estimation—used to model the feature distributions of associated sources—is combined with mixture model fitting to infer the energy spectrum composition of the unassociated gamma-ray sources. Section 3.3.4 returns to this task with classical non-parametric methods, complementing the neural surrogate approaches introduced in Section 3.2.

3.3.2 Neural Networks and Deep Learning

A neural network is a parametric function $f_{\mathbf{w}} : \mathbb{R}^n \rightarrow \mathbb{R}^m$ composed of a sequence of affine transformations, each followed by a non-linear activation function. In a network with L layers, the output of layer ℓ is $\mathbf{h}^{(\ell)} = \sigma(\mathbf{W}^{(\ell)}\mathbf{h}^{(\ell-1)} + \mathbf{b}^{(\ell)})$, where $\mathbf{W}^{(\ell)}$ and $\mathbf{b}^{(\ell)}$ are trainable weight matrices and bias vectors, and σ is a non-linear function such as the rectified linear unit $\text{ReLU}(x) = \max(0, x)$. Other activations have been used historically—sigmoid and hyperbolic tangent were standard in earlier architectures—while GELU has become widespread in modern transformer-based models. Given a sufficiently large number of parameters, neural networks can approximate any continuous function to arbitrary accuracy [125, 138].

Training consists of adjusting the weights $\mathbf{w}$ to minimize a cost function $\mathcal{J}(\mathbf{w})$ using gradient-based optimisation. Backpropagation provides the necessary gradients by applying the chain rule recursively through the network layers, yielding the partial derivative of $\mathcal{J}$ with respect to every weight in a single backward pass. In practice, gradients are estimated from mini-batches rather than the full dataset, giving rise to stochastic gradient descent (SGD) and its modern adaptive variants such as Adam and AdamW [139], which maintain per-parameter learning rates and momentum estimates. The empirical success of deep learning over the past decade

rests on algorithmic advances together with network depth, large datasets, and GPU-accelerated compute [140].

The choice of cost function determines what the network learns, and all of the losses used in this thesis are best understood as instances of maximum-likelihood estimation under an appropriate probabilistic model [125, 141]. For a standard regression problem, the mean squared error $\mathcal{J} = \sum_i (y_i^{\text{true}} - y_i^{\text{pred}})^2$ penalises deviations between predictions and targets uniformly and coincides, up to constants, with the negative log-likelihood of the targets under a homoscedastic Gaussian noise model. For classification, the cross-entropy loss measures the divergence between predicted class probabilities and one-hot labels and is equivalently the negative log-likelihood of the observed labels under a categorical model. Density estimation is most commonly cast as minimising the forward Kullback–Leibler divergence between the data distribution and a parametric model; because the data-side entropy is independent of the model parameters, this minimisation reduces to maximising the log-likelihood of the data—or, for latent-variable models in which the marginal likelihood is intractable, the evidence lower bound (ELBO). When the noise level varies across the output space, the Gaussian variance can itself be parameterised as a network output, yielding the heteroscedastic Gaussian negative log-likelihood [137]; the formalism is presented in Section 6.5.3.4, where Chapter 6 puts it to use.

Beyond a point estimate, predictions carry two distinct sources of error: *aleatoric* uncertainty, intrinsic to the data (irreducible noise that more training data cannot remove), and *epistemic* uncertainty, reflecting the model’s ignorance about which parameter values best describe the data (reducible in the limit of infinite data and capacity). Bayesian neural networks aim to recover both jointly by treating the network weights as random variables rather than fixed parameters, in the same spirit as the simulation-based inference framework of Section 3.2. Chapter 6 presents one practical realisation of this idea, in which MC dropout and a heteroscedastic Gaussian likelihood are combined to produce per-output mean and variance estimates; the technical formalism is detailed there (Section 6.5.3.4). Gal and Ghahramani [142] showed that a neural network with dropout applied before every weight layer is mathematically equivalent to approximate variational inference in a deep Gaussian process; at test time, T stochastic forward passes with dropout active yield a sample mean as the predictive estimate and a sample variance as the epistemic uncertainty.

Overparameterised models can memorise the training set; regularisation keeps them honest. In practice, several complementary techniques are typically used in combination: weight decay (an L_2 penalty on the weights), dropout—which randomly zeros a fraction of activations during training—batch and layer normalisation [143], data augmentation, and early stopping. Dropout doubles as an uncertainty quantifier when left active at test time, as described above, and its self-tuning variant *concrete dropout* [144] replaces the discrete Bernoulli mask with a continuous relaxation that makes the drop probability differentiable, so that the optimal rate for each layer is learned jointly with the network weights during training. Chapter 6 adopts concrete dropout as its regularisation and uncertainty calibration strategy, to which we defer the details of the implementation.

3.3.3 Convolutional Neural Networks

For image-domain analyses, the input is a pixel grid—or, in astrophysics, a HEALPix sky map—and convolutional neural networks (CNNs) exploit this spatial structure directly. CNNs have been applied to astrophysical problems such as galaxy morphology classification [145], strong gravitational lens identification [146], and electromagnetic transient detection. In this thesis, Chapter 6 adopts a CNN to reconstruct the source-count distribution from gamma-ray sky maps.

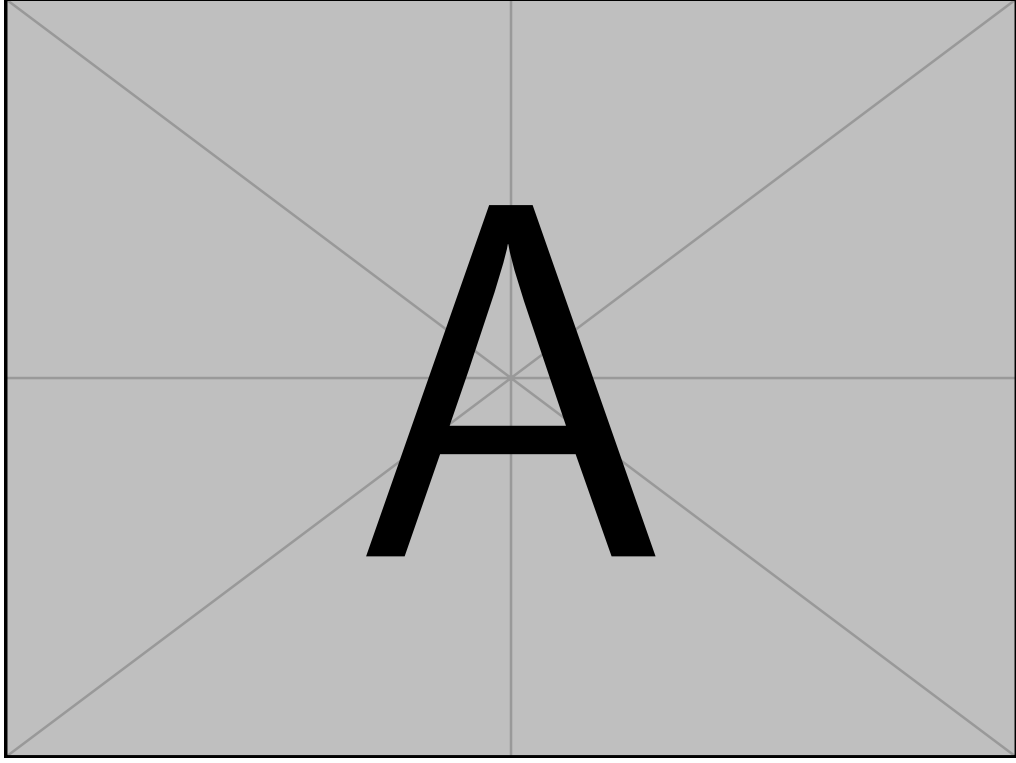


Fig. 3.1. [AA: TODO: need to make this figure, or find a good one] Generic convolutional neural network architecture: a kernel slides over the input image producing feature maps; successive convolution and pooling layers reduce spatial extent while increasing channel depth, before a fully connected head maps the learned features to the output.

A convolutional layer applies a set of learnable filters (kernels) to local patches of the input, producing feature maps that respond to spatial patterns at specific scales. For a two-dimensional input, the discrete convolution of a filter g with an input f is

$$(f * g)[i, j] = \sum_m \sum_n f[m, n] g[i - m, j - n]. \quad (3.3.1)$$

Because the same filter is applied everywhere across the input, the number of parameters per layer is independent of the input size, and the response to a given pattern is the same regardless of its location—a property known as translation equivariance.

Stacking convolutional layers builds progressively larger receptive fields: shallow layers detect local features such as edges and peaks, while deeper layers combine these into more abstract representations. Pooling operations (max or average over local windows) downsample the feature maps between successive convolutional stages, providing approximate translation invariance at the cost of spatial resolution [138, 140]. This hierarchy of local-to-global features allows CNNs to capture information distributed across spatial scales.

Modern CNNs build on a few architectural ideas: VGG-style stacks of small convolutions [147] and the residual connections introduced by ResNet [148], which enabled training of networks hundreds of layers deep. Contemporary computer-vision architectures often combine convolutional building blocks with attention mechanisms borrowed from transformers, as in vision transformers [149] and hybrid models.

A specific challenge arises when the data live on the sphere rather than on a flat grid. Sky maps live on S^2 , not $\mathbb{R}^2$: standard CNNs assume Euclidean translation symmetry, whereas the sphere has rotational symmetry instead, and any flat parametrisation introduces distortion that breaks weight sharing. Three broad strategies have been developed to extend convolutional architectures to spherical data.

The first strategy performs convolutions in the spherical harmonic domain. Because the spherical harmonics are the natural eigenbasis of the Laplace–Beltrami operator on S^2 , a spectral convolution can be defined in exact analogy with the classical convolution theorem on the real line. DeepSphere [150] implements this idea using a graph-based approximation, representing the sphere as a graph whose nodes are HEALPix pixels and whose edges encode pixel adjacencies; spectral graph convolutions then replace the standard spatial filters. This approach preserves the global topology of the sphere and handles irregular pixelisations.

The second strategy applies convolutions to local projections of the sphere. Rather than operating on the curved manifold directly, these methods project spherical patches onto tangent planes or polyhedral faces, where standard 2D convolutions apply without modification. Icosahedral projections [151] decompose the sphere into 20 equilateral triangular faces that cover it nearly uniformly, limiting the distortion inherent in any flat projection.

The third strategy, and the one most directly relevant to this thesis, operates on the HEALPix pixelisation itself. NNHealpix [152] exploits the fixed neighbourhood structure of HEALPix pixels by unrolling each pixel and its eight nearest neighbours into a nine-element vector and applying standard 1D convolutions with stride nine (Fig. 3.2). This preserves the full topology of the sphere at the cost of higher computational overhead and sensitivity to the orientation of pixel neighbourhoods. The map2patch algorithm developed in Chapter 6 subdivides the sphere into the 12 equal-area base pixels of the HEALPix scheme and processes them jointly as a stack of flat tiles, exploiting the exact equal-area property of the base faces so that standard 2D convolutions apply without resampling distortion. The spherical topology is preserved across patch boundaries through a padding scheme that borrows pixels from the adjacent base faces, so that no spatial information is lost where neighbouring tiles meet. The construction preserves the exact area and photon content of every pixel; details of the implementation are deferred to Chapter 6. For statistically isotropic, small-scale signals such as the point-source population studied in Chapter 6, map2patch and NNHealpix produce compatible results, while map2patch achieves more than an order of magnitude improvement in execution speed. More recently, HEAL-SWIN [153] adapted the Swin Vision Transformer to the HEALPix sphere: shifted-window attention operates within local HEALPix patches, and the window boundaries are shifted between successive transformer blocks so that the model captures both local and global spherical structure while retaining the computational efficiency of window-based attention.

Chapter 6 combines the map2patch tiling with concrete dropout and a heteroscedastic Gaussian negative log-likelihood to produce both the reconstructed source-count distribution and calibrated uncertainty estimates; the full architecture and loss formulation are presented in Section 6.5.3.4.

3.3.4 Density Estimation

Having introduced the convolutional architectures used for image-domain inference in Chapter 6, we now turn to the tools underlying the summary-statistic approach of Chapter 5. The simulation-based inference framework of Section 3.2 relies on neural density estimators—normalising flows, score-based diffusion models, and flow matching [132, 154, 155]—that are well suited to high-dimensional or complex distributions [124]. For low-dimensional, well-defined feature spaces,

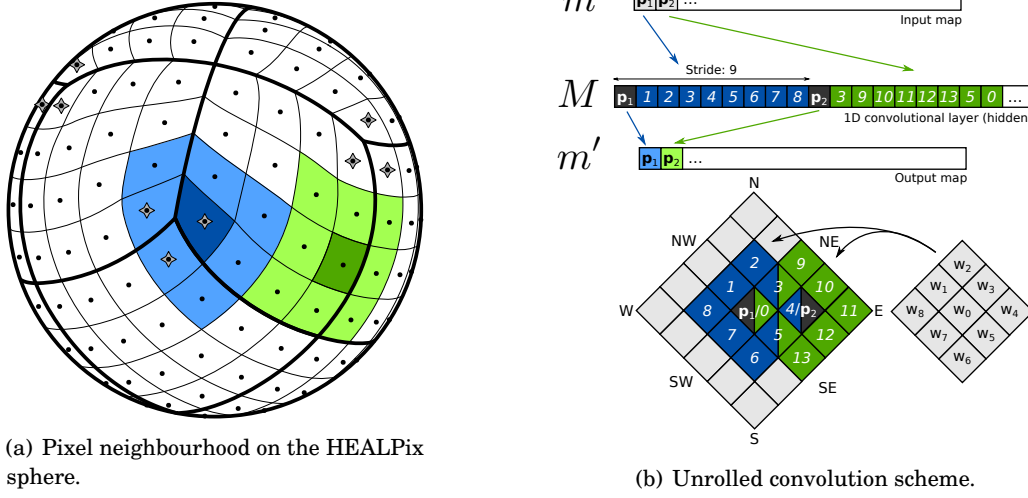


Fig. 3.2. Convolution on the HEALPix sphere. *Left:* each pixel and its nearest neighbours define a local window that can be mapped to a flat patch for convolution. *Right:* in the NNHealpix scheme, each pixel and its eight neighbours are unrolled into a nine-element vector, and a standard one-dimensional convolution with stride nine is applied. Credit: Figs. 2 and 3 of [152].

classical non-parametric methods such as Kernel Density Estimation and mixture models with Expectation-Maximisation are sufficient and often preferable. Chapter 5 sits squarely in this regime: the input features are a handful of spectral and variability quantities per source, and the rest of this subsection introduces the classical tools used there.

Given a set of N observed data points $\{\mathbf{x}_1, \dots, \mathbf{x}_N\}$ drawn from an unknown distribution, the goal is to estimate the probability density function $p(\mathbf{x})$ that generated them. Unlike classification, which learns a decision boundary in feature space, density estimation reconstructs the full probability distribution and can therefore reveal populations that were absent from the training set. A non-parametric approach to this problem is provided by Kernel Density Estimation (KDE); we refer the interested reader to Section 2.5.1 of [125] for a comprehensive treatment. Given N samples, the KDE approximation to the density at a point $\mathbf{x}$ is

$$\hat{p}(\mathbf{x}) = \frac{1}{N} \sum_{i=1}^N K_h(\mathbf{x} - \mathbf{x}_i), \quad (3.3.2)$$

where K_h is a kernel function with bandwidth h . For a Gaussian kernel, $K_h(\mathbf{u}) \propto \exp(-\|\mathbf{u}\|^2/2h^2)$, and the bandwidth controls the trade-off between bias and variance: too small a bandwidth overfits the sample, while too large a bandwidth oversmooths the underlying structure. The optimal bandwidth is determined through cross-validation, selecting the value that maximises the model likelihood on held-out data. Separate KDE models are computed independently for each source class (Galactic and extragalactic), yielding the class-conditional densities $p_{\text{assoc}}(\mathbf{x}|k)$ that form the building blocks of the mixture model described below.

When the observed data arise from multiple overlapping populations, the total density can be written as a mixture model,

$$p(\mathbf{x}) = \sum_{k=1}^K \pi_k p(\mathbf{x}|k), \quad (3.3.3)$$

where π_k are the mixing weights (class prevalences) satisfying $\sum_k \pi_k = 1$, and $p(\mathbf{x}|k)$ is the class-conditional density. The mixing weights and any free parameters of the component densities are typically determined by the Expectation-Maximisation (EM) algorithm [156]: an iterative procedure that alternates between computing the posterior class responsibilities $\gamma_{ik} = p(k|\mathbf{x}_i)$ for each data point (the E-step), and updating the model parameters to maximise the expected log-likelihood under those responsibilities (the M-step); a detailed derivation, including the general formulation of EM and its application to Gaussian mixtures, is given in Chapter 9 of [125]. For the mixing weights, the M-step admits a closed-form solution: $\pi_k^{\text{new}} = N^{-1} \sum_i \gamma_{ik}$, i.e., each weight is updated as the average posterior responsibility over the dataset.

In Chapter 5, this framework is extended to search for dark matter subhalos among the unassociated Fermi-LAT sources. The mixture model combines astrophysical class-conditional densities with a covariate shift correction (Section 3.4) and a possible dark matter component, with class prevalences determined via the EM algorithm following the approach of Saerens et al. [157].

3.4 The Domain Shift Challenge

The machine learning classification methods introduced in Section 3.3 are built on the implicit assumption that the training data faithfully represent the conditions under which the model will be deployed. When this assumption breaks down—as it does for several astrophysical classification tasks—the standard approaches described in the previous sections require modification. We now turn to the problem of *dataset shift*, a systematic mismatch between training and target distributions that, if left unaddressed, can bias inference results. We first formalize the problem (Section 3.4.1) and then describe its principal types and the strategies used to correct them (Section 3.4.2).

3.4.1 The Problem Statement

Supervised classification algorithms rely on the assumption that the training data and the target data share the same joint probability distribution of input features $\mathbf{x}$ and output classes k :

$$p_{\text{train}}(\mathbf{x}, k) = p_{\text{target}}(\mathbf{x}, k). \quad (3.4.1)$$

When this condition holds, a classifier trained on labeled examples can be applied directly to unlabeled data with the expectation that its performance will generalize. In practice, however, this assumption is frequently violated. A mismatch between the training and target distributions—known as *dataset shift* [158]—invalidates the direct application of the trained model and can produce systematically biased estimates of class prevalences and, consequently, incorrect statistical constraints on physical parameters.

This problem is directly relevant for astrophysical source classification. In the Fermi-LAT catalogs, sources are divided into *associated* and *unassociated* categories based on whether a counterpart at other wavelengths has been identified. Machine learning classifiers are typically trained on associated sources—whose class labels (e.g., pulsar or active galactic nucleus) are known from their multi-wavelength counterparts—and then applied to unassociated sources to infer their nature. However, several observational biases break the equal-distribution assumption. Bright sources at high Galactic latitudes are easier to associate with known counterparts, so the associated sample is not a representative subsample of the full source population [159]. As a consequence, the spectral parameter distributions—flux, spectral index α , and spectral curvature

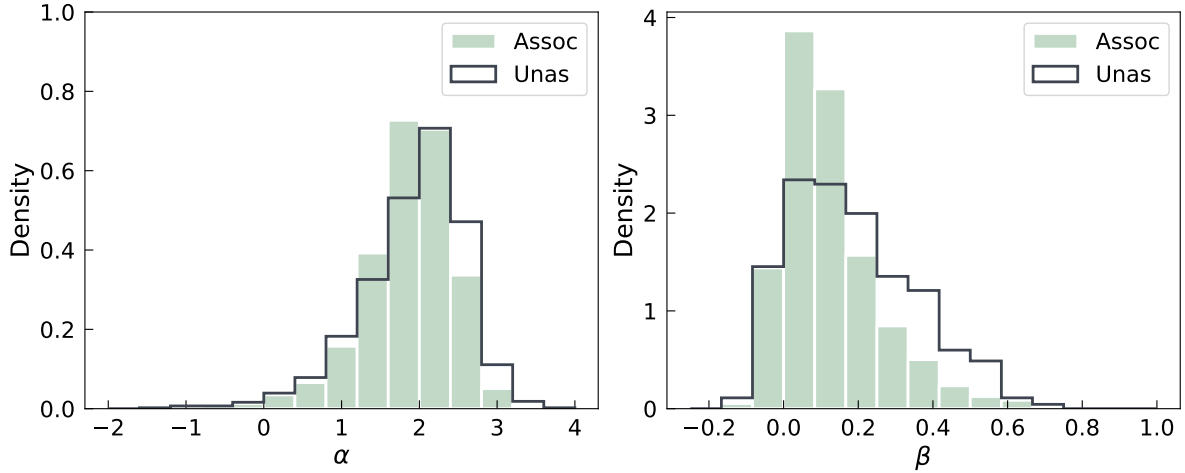


Fig. 3.3. Distributions of the spectral index α and spectral curvature β for associated (green, filled) and unassociated (gray outline) Fermi-LAT sources from the 4FGL-DR4 catalog. The visible mismatch between the two populations illustrates the dataset shift between the associated and unassociated samples discussed in Section 3.4.1. The full analysis is presented in Chapter 5.

β —differ measurably between associated and unassociated sources, as illustrated in Figure 3.3 (see Chapter 5 for the full analysis). Whether this discrepancy arises from observational selection effects, from genuinely different class compositions, or from a combination of both, cannot be determined a priori; we formalize the relevant types of dataset shift in Section 3.4.2. Any analysis that ignores this mismatch risks producing biased class prevalence estimates and, in the context of dark matter searches, unreliable upper bounds on the annihilation cross-section.

3.4.2 Types of Dataset Shift

The joint distribution of features and class labels can be decomposed in two equivalent ways:

$$p(\mathbf{x}, k) = p(k|\mathbf{x})p(\mathbf{x}) = p(\mathbf{x}|k)p(k). \quad (3.4.2)$$

These two factorizations give rise to two distinct types of dataset shift, depending on which factor changes between the training and target domains [158]. Each type has distinct statistical consequences and correction strategies.

Covariate shift. Covariate shift is most naturally understood through the first factorization, $p(\mathbf{x}, k) = p(k|\mathbf{x})p(\mathbf{x})$. Under this type of shift, the posterior class probabilities given the features remain unchanged between the training and target distributions:

$$p_{\text{train}}(k|\mathbf{x}) = p_{\text{target}}(k|\mathbf{x}), \quad (3.4.3)$$

while the marginal distribution of the features differs: $p_{\text{train}}(\mathbf{x}) \neq p_{\text{target}}(\mathbf{x})$ [158]. The class-conditional relationship between features and labels is preserved, but the relative abundance of feature values changes. Covariate shift typically arises from selection effects or observational biases that influence which data points enter the training sample. In the Fermi-LAT context, the association procedure preferentially identifies sources in low-background regions far from the Galactic plane, where multi-wavelength counterparts are easier to match. Proximity to the Galactic plane is, in fact, the dominant source of association bias: the high density of overlapping

sources and diffuse emission near the plane causes multi-wavelength identification methods to fail more frequently [159]. As a result, the associated sample is biased toward high-latitude, brighter sources relative to the full population. The ratio $C(\mathbf{x}) = p_{\text{target}}(\mathbf{x})/p_{\text{train}}(\mathbf{x})$ between the target and training feature distributions characterizes the covariate shift [158]. When this ratio is known or can be modeled, the training-set predictions can be corrected by importance weighting.

In practice, $C(\mathbf{x})$ is rarely known analytically, but it can be approximated parametrically. If the ratio is monotonic in each feature dimension, a product of sigmoid functions provides a physically constrained model that can still accommodate a range of shift shapes:

$$\tilde{C}(\mathbf{x}; \boldsymbol{\theta}_{\text{cov}}) = \prod_{d=1}^D f(x_d; b_d, c_d), \quad f(x_d; b_d, c_d) = \frac{1}{1 + e^{-b_d(x_d - c_d)}}, \quad (3.4.4)$$

where b_d controls the steepness and c_d the midpoint of the modulation along feature dimension d (see Chapter 5 for the derivation and application). These parameters are optimized jointly with the rest of the model by maximizing the likelihood of the target data.

Prior shift. The prior shift is most clearly seen in the second factorization, $p(\mathbf{x}, k) = p(\mathbf{x}|k)p(k)$. Here, the class-conditional feature distributions are invariant:

$$p_{\text{train}}(\mathbf{x}|k) = p_{\text{target}}(\mathbf{x}|k), \quad (3.4.5)$$

but the class prevalences differ: $p_{\text{train}}(k) \neq p_{\text{target}}(k)$ [158]. This type of shift is common when the population composition changes between the training and deployment environments. Among Fermi-LAT sources, for instance, the fraction of Galactic sources is considerably larger in the unassociated sample than in the associated one, because a higher proportion of unassociated sources lie near the Galactic plane [159]. Standard classifiers trained with balanced or representative class fractions from the associated data will therefore assign incorrect class probabilities when applied to the unassociated population. The quantification learning literature [160] addresses this problem by fitting the mixture weights π_k of a generative model directly to the target data, rather than relying on the class fractions of the training set [157, 161].

Combined shift. In general, covariate and prior shifts act simultaneously, and separating their effects is difficult. Changes in the marginal class prevalences $p(k)$ produce corresponding changes in the marginal feature distribution $p(\mathbf{x})$, and vice versa, so that the same observed mismatch between training and target distributions can be explained by either mechanism or by both acting together [158]. Resolving this degeneracy requires additional physical constraints or a model flexible enough to fit both effects at once. A purely covariate-shift correction, for example, might absorb prevalence changes into the reweighting function, while a purely prior-shift model would attribute selection biases to changes in class fractions. Neither gives the correct answer when both effects are present.

Application preview. Addressing this challenge requires a model that treats both effects simultaneously. In Chapter 5, we construct the first such statistical model for Fermi-LAT sources. The unassociated source distribution is modeled as a mixture of Galactic and extragalactic astrophysical components—whose class-conditional densities $p_{\text{assoc}}(\mathbf{x}|k)$ are estimated from the associated sources via kernel density estimation—plus a possible dark matter subhalo component derived from Monte Carlo simulations. The prior shift is handled by fitting the class prevalences π_k to the unassociated data, while the covariate shift is corrected by the sigmoid modulation functions $\tilde{C}(\mathbf{x}; \boldsymbol{\theta}_{\text{cov}})$ described above. All parameters are optimized simultaneously using the

Expectation-Maximization algorithm (Section 3.3.3). This approach draws on the classical dataset shift taxonomy of Moreno-Torres et al. [158] and the quantification learning methods developed by González et al. [160] and Moreo et al. [161], combining them into a unified framework that can be used both for the probabilistic classification of unassociated sources—its primary astrophysical application—and for setting statistically rigorous upper bounds on the dark matter annihilation cross-section. The full analysis is presented in Chapter 5.

3.5 Cross-Correlations as a Complementary Probe

The methods discussed so far in this chapter—frequentist and Bayesian inference, simulation-based approaches, machine learning, and domain adaptation—all operate on data from a single instrument, extracting information about source populations from the gamma-ray sky alone. We now consider a qualitatively different approach: correlating the gamma-ray emission with an independent gravitational tracer of the dark matter distribution, such as a galaxy catalog. ~~The angular cross power spectrum between these two observables provides sensitivity to the collective dark matter signal from unresolved cosmological structure, offering a complementary probe that goes beyond what any single dataset can achieve.~~

Cross-correlations are powerful ~~precisely~~ because they exploit the fact that dark matter, ~~regardless of its particle nature,~~ is the dominant source of gravitational clustering. If dark matter annihilates or decays into gamma rays, the resulting emission must trace the same large-scale structure [AA: check this section ref](cf. Section 1.1.3) ~~that is independently mapped by galaxy surveys. By measuring the spatial correlation between the two fields, one can isolate the collective dark matter contribution from the aggregate of unresolved cosmological structure—a statistical signal that no individual source produces, and one that would be undetectable in either dataset alone.~~

We introduce the cross-correlation formalism in two steps. Section 3.5.1 defines the angular power spectrum, the Limber approximation that projects three-dimensional clustering into two-dimensional observables, and the window functions that encode the redshift distribution of each signal. Section 3.5.2 presents ~~the statistical tools~~ the signal-to-noise ratio and the $\Delta\chi^2$ test statistic—used to forecast the sensitivity of a given experiment to dark matter. The full application of this formalism, including the halo model computation and the detailed sensitivity forecasts for the Cherenkov Telescope Array Observatory, is deferred to Chapter 8.

3.5.1 The Angular Power Spectrum

The observable sky maps produced by gamma-ray telescopes and galaxy surveys are two-dimensional projections of the three-dimensional ~~distribution of matter and radiation~~ in the Universe. To study the spatial correlations between these projected fields, we expand them in ~~the natural basis of the sphere: the spherical harmonics~~ $Y_{\ell m}(\hat{n})$. The angular power spectrum (APS) C_ℓ ~~then~~ quantifies the amplitude of these correlations as a function of the angular scale, parameterized by the multipole ℓ .

We begin by considering a generic intensity field $I_g(\hat{n})$ along the direction $\hat{n}$, which can be written as an integral over the radial comoving distance $\chi(z)$ [162, 163]:

$$I_g(\hat{n}) = \int d\chi g(\chi, \hat{n}) W(\chi), \quad (3.5.1)$$

where $g(\chi, \hat{n})$ is the density field of the source and $W(\chi)$ is the window function encoding the redshift-dependent emissivity or source distribution. The intensity fluctuations $\delta I_g(\hat{n}) \equiv I_g(\hat{n}) -$

$\langle I_g \rangle$ can then be expanded in spherical harmonics as

$$\delta I_g(\hat{n}) = \sum_{\ell m} a_{\ell m} Y_{\ell m}(\hat{n}), \quad (3.5.2)$$

where the coefficients $a_{\ell m}$ encode the angular structure of the fluctuation field [162].

For two intensity fields labelled i and j (for instance, the gamma-ray flux and the galaxy number counts), the angular power spectrum is defined as the ensemble average of the product of their expansion coefficients [162, 163]:

$$C_\ell^{ij} = \frac{1}{2\ell+1} \left\langle \sum_{m=-\ell}^{\ell} a_{\ell m}^{(i)} a_{\ell m}^{(j)*} \right\rangle. \quad (3.5.3)$$

When $i = j$, the expression gives the auto-correlation APS, which measures the intrinsic clustering of a single field. When $i \neq j$, we obtain the cross-correlation APS, which measures the spatial overlap between two distinct observables. By construction, $C_\ell^{ij} = C_\ell^{ji}$.

To connect the measured two-dimensional APS to the underlying three-dimensional power spectrum $P_{ij}(k, \chi)$, which contains the physics of the source distribution, we apply the Limber approximation [164]. This approximation, valid for multipoles $\ell \gg 1$, replaces the oscillatory integrals over spherical Bessel functions with a single line-of-sight integral by identifying the wavenumber $k = \ell/\chi$. The result is a compact expression for the angular power spectrum [162, 163]:

$$C_\ell^{ij} = \int \frac{d\chi}{\chi^2} W_i(\chi) W_j(\chi) P_{ij}\left(k = \frac{\ell}{\chi}, \chi\right). \quad (3.5.4)$$

The amplitude of the cross-correlation signal depends on the overlap between the window functions $W_i(\chi)$ and $W_j(\chi)$ and on the three-dimensional cross-power spectrum at the relevant scales. A significant cross-correlation therefore requires that both observables trace the same large-scale structure at similar redshifts [162]. This idea of correlating a gamma-ray signal with a gravitational tracer of the dark matter distribution was pioneered by Camera et al. [163], who first showed that the cross-correlation of gamma-ray emission with weak-lensing cosmic shear provides a viable channel for indirect dark matter searches.

The window functions are key ingredients of this formalism, as they encode how each observable is distributed along the line of sight. Their specific form depends on the physical origin of the signal. For a galaxy catalog acting as a gravitational tracer of the large-scale structure, the window function is simply the normalized redshift distribution of the surveyed galaxies, $W_g(z) = dN_g/dz$ [162]. The angular power spectrum from annihilating dark matter was first formulated by Ando and Komatsu [165], and the formalism was later extended to multi-wavelength and multi-tracer cross-correlations by Fornengo and Regis [162].

For dark matter annihilation, the gamma-ray window function takes the form [162, 165]

$$W_{\gamma, \text{ann}}(E, z) = \frac{(\Omega_{\text{DM}} \rho_c)^2}{4\pi} \frac{\langle \sigma v \rangle}{2m_\chi^2} (1+z)^3 \Delta^2(z) \frac{dN_a}{dE} [E(1+z)] e^{-\tau(E, z)}, \quad (3.5.5)$$

where $\langle \sigma v \rangle$ is the velocity-averaged annihilation cross section, m_χ is the dark matter particle mass, $\Delta^2(z) = \langle \rho^2 \rangle / \bar{\rho}^2$ is the clumping factor accounting for the enhancement due to substructure within halos, dN_a/dE is the photon spectrum per annihilation event, and $\tau(E, z)$ is the optical depth for absorption on the extragalactic background light. For decaying dark matter, the window function scales linearly with the density rather than quadratically, and the clumping factor does not appear [162]. The annihilation window function is strongly peaked at low redshifts, where

the local dark matter density is highest and the clumping factor most effective. By contrast, the *unresolved* astrophysical source populations that contribute to the diffuse gamma-ray background have markedly different redshift distributions: virtually all blazars at $z \lesssim 0.1$ have already been individually detected and cataloged, so the redshift distribution of the remaining unresolved population drops sharply toward $z \sim 0$, peaking instead at moderate redshifts. This separation in redshift is what makes the cross-correlation with low-redshift galaxy catalogs particularly powerful for isolating a dark matter signal: the dark matter window function peaks precisely where the astrophysical foreground is most suppressed [162].

The three-dimensional power spectrum $P_{ij}(k, \chi)$ in the Limber integral describes the spatial clustering of the correlated fields. It is computed within the halo model framework, which partitions all matter in the Universe into discrete, collapsed dark matter halos [166]. Under this assumption, the power spectrum decomposes into two contributions [162, 166]:

$$P_{ij}(k, z) = P_{ij}^{\text{1h}}(k, z) + P_{ij}^{\text{2h}}(k, z). \quad (3.5.6)$$

The one-halo term P^{1h} accounts for correlations between two points within the same individual halo; it probes the internal mass distribution of halos and dominates on small physical scales (large k). The two-halo term P^{2h} captures correlations between points in two distinct halos; it traces the underlying linear matter power spectrum modulated by the halo bias factors and dominates on large scales (small k). Both terms are expressed as integrals over the halo mass function dn/dM and the Fourier-transformed density profiles of the relevant tracers [162, 166]. The detailed computation of these terms for the specific gamma-ray and galaxy populations considered in this thesis is deferred to Chapter 8, where the full cross-correlation analysis is presented.

3.5.2 Statistical Sensitivity

Given the cross-correlation angular power spectrum $C_\ell^{\gamma g}$ defined in the previous section, we now turn to the statistical tools used to assess how well a given experiment can detect or constrain a dark matter signal. Two quantities are central to this assessment: the signal-to-noise ratio (SNR) and the $\Delta\chi^2$ test statistic.

The detectability of the cross-correlation signal from a given source population (whether astrophysical or dark matter) is quantified by the signal-to-noise ratio [162]:

$$\text{SNR} = \sqrt{\sum_{\ell, i} \left(\frac{C_{\ell, i}^{\gamma g}}{\Delta C_{\ell, i}^{\gamma g}} \right)^2}, \quad (3.5.7)$$

where the sum runs over the multipoles ℓ included in the analysis and the energy bins i . We note that the statistical formalism presented here is tailored to cross-correlations with galaxy catalogs, which is the gravitational tracer relevant for this thesis; cross-correlations with other tracers, such as weak-lensing cosmic shear or HI intensity maps, involve analogous expressions but with modified noise terms and window functions that reflect the specifics of each probe and instrument. The variance of the cross-correlation spectrum, assuming Gaussian-distributed fluctuations, is given by [162]

$$(\Delta C_\ell^{\gamma g})^2 = \frac{1}{(2\ell + 1)f_{\text{sky}}} \left[(C_\ell^{\gamma g})^2 + \left(C_\ell^{\gamma\gamma} + \frac{C_N}{B_\ell^2} \right) (C_\ell^{gg} + C_{N_g}) \right], \quad (3.5.8)$$

where f_{sky} is the observed sky fraction, $C_\ell^{\gamma\gamma}$ and C_ℓ^{gg} are the gamma-ray and galaxy auto-correlation spectra, C_N is the photon noise of the gamma-ray detector, C_{N_g} is the shot noise of

the galaxy catalog, and B_ℓ is the beam window function encoding the finite angular resolution (point-spread function) of the instrument. No beam correction enters the galaxy term, as the angular resolution of galaxy catalogs (typically arcseconds) is negligible at the scales relevant for this analysis (Chapter 8).

The structure of the variance formula reveals a fundamental advantage of cross-correlation over auto-correlation for dark matter searches. The corresponding auto-correlation variance is [162]

$$(\Delta C_\ell^{\gamma\gamma})^2 = \frac{2}{(2\ell+1)f_{\text{sky}}} \left(C_\ell^{\gamma\gamma} + \frac{C_N}{B_\ell^2} \right)^2, \quad (3.5.9)$$

where the photon noise C_N now enters quadratically and adds directly to the signal. In an auto-correlation measurement, $C_\ell^{\gamma\gamma} + C_N$ must be measured and the noise accurately estimated and subtracted to recover the true clustering spectrum. In cross-correlation, by contrast, the noise sources in the gamma-ray map and in the galaxy catalog are physically independent: their cross-noise vanishes on average. The photon noise C_N and galaxy shot noise C_{N_g} therefore contribute only to the statistical uncertainty (the error bars) of the cross-correlation measurement, not to the signal itself. This allows faint correlated signals—such as a subdominant dark matter contribution—to be extracted even in noisy datasets where the auto-correlation would be completely dominated by instrumental noise [162, 163]; a quantitative comparison of auto- and cross-correlation SNR for CTAO observations is given in Table 8.2.

To assess the sensitivity to a dark matter signal specifically, we perform a nested-model comparison in the spirit of Wilks’ theorem [162]. The null hypothesis includes only astrophysical sources, while the alternative hypothesis adds a dark matter component parameterized by a single additional degree of freedom—the annihilation cross section $\langle\sigma v\rangle$ or, in the decay case, the lifetime τ_χ . The test statistic is

$$\Delta\chi^2 = \sum_{\ell,i} \left(\frac{C_{\ell,i}^{\gamma_{\text{astro}}+\text{DM } g}}{\Delta C_{\ell,i}^{\gamma_{\text{astro}}+\text{DM } g}} \right)^2 - \sum_{\ell,i} \left(\frac{C_{\ell,i}^{\gamma_{\text{astro}} g}}{\Delta C_{\ell,i}^{\gamma_{\text{astro}} g}} \right)^2, \quad (3.5.10)$$

where $C_\ell^{\gamma_{\text{astro}} g}$ is the predicted cross-correlation signal under the null hypothesis (astrophysical sources only) and $C_\ell^{\gamma_{\text{astro}}+\text{DM } g}$ includes an additional dark matter component. The test statistic measures how much the inclusion of a dark matter signal improves the fit relative to the astrophysical-only model. In Chapter 8, this formalism is applied to forecast the sensitivity of the Cherenkov Telescope Array Observatory (CTAO) to dark matter annihilation and decay, by cross-correlating the predicted gamma-ray emission with the 2MASS galaxy catalog.

3.6 Summary

This chapter has established the statistical and computational framework referenced throughout the remainder of this thesis. The presentation has been deliberately conceptual: each method was introduced with its definitions, key equations, and standard examples, while the full technical implementations are deferred to the chapters where they are first applied.

Section 3.1 developed the frequentist and Bayesian inference paradigms from the shared foundation of the likelihood function. On the frequentist side, we introduced maximum likelihood estimation, the profile and integrated likelihoods for handling nuisance parameters, and the likelihood-ratio test statistic with Wilks’ theorem—tools that underpin the analyses in Chapters 4, 5, and 8. On the Bayesian side, we presented Bayes’ theorem, posterior marginalization, and

the heteroscedastic Gaussian loss function that enables neural networks to estimate their own predictive uncertainty, as applied in Chapter 6.

Section 3.2 introduced simulation-based inference as a solution to the intractable likelihood problem. When the forward model involves complex simulations with many latent variables, the three neural estimation strategies—NPE, NLE, and NRE—replace explicit likelihood evaluation with learned surrogates trained on synthetic data. We also discussed the validation tools, simulation-based calibration and coverage testing, that are necessary to ensure the reliability of these learned posteriors.

Section 3.3 covered the machine learning methods used for regression, classification, and density estimation on gamma-ray data. Convolutional neural networks, with their weight-sharing and translation-equivariant properties, are well suited to the spatial structure of photon-count maps. Kernel density estimation and the Expectation-Maximization algorithm for mixture models provide the population-level inference framework that forms the basis of the subhalo search in Chapter 5.

Section 3.4 formalized the dataset shift problem—the mismatch between training and target distributions that arises when classifiers trained on associated Fermi-LAT sources are applied to unassociated ones. We distinguished between covariate shift, where the feature distributions change while the class-conditional relationships are preserved, and prior shift, where the class prevalences differ. In general, both act simultaneously, and correcting for one without accounting for the other can produce biased results.

Finally, Section 3.5 introduced the angular cross-correlation formalism. By correlating the gamma-ray sky with an independent gravitational tracer such as a galaxy catalog, one gains access to the collective dark matter signal from unresolved cosmological structure—a statistical signature that no individual source produces and that benefits from the physical independence of the noise sources in the two datasets.

Together, these methods form a coherent toolkit that is applied in various combinations throughout the thesis. The following chapters present the physics results: Chapter 4 introduces the Galactic Center Excess and applies the frequentist framework of Section 3.1 to constrain the millisecond pulsar luminosity function, providing the first direct application of the statistical vocabulary developed here.

THE GALACTIC CENTER GAMMA-RAY EXCESS

[AA: Review this introduction when I finalize the content of the chapters...] The preceding chapters established the theoretical basis for indirect dark matter detection through gamma rays (Chapter 1), described the instrumental and astrophysical landscape of the Fermi Large Area Telescope (Chapter 2), and introduced the statistical and machine-learning methods used to disentangle overlapping emission components (Chapter 3). We now turn to the most persistent anomaly in GeV gamma-ray observations: the Galactic Center Excess (GCE).

Since 2009, analyses of Fermi-LAT data have consistently revealed a bright, statistically significant excess of gamma-ray emission from the region surrounding the Galactic Center [62, 100, 101, 167]. This signal peaks at energies between roughly 1 and 3 GeV, extends out to at least 10 degrees from Sgr A*, and persists across all existing models of the astrophysical diffuse emission [167]. Its spectrum, angular distribution, and overall intensity are each consistent with the signal predicted from the annihilation of $\sim 40\text{--}70$ GeV dark matter particles distributed according to a contracted Navarro–Frenk–White profile [5, 62]. The best-fit annihilation cross section, $\langle\sigma v\rangle \sim (1\text{--}2) \times 10^{-26} \text{ cm}^3/\text{s}$, is compatible with the canonical thermal relic value [62].

The leading astrophysical alternative attributes the excess to a large, unresolved population of millisecond pulsars (MSPs) in the Inner Galaxy [168]. MSPs produce gamma-ray emission with a spectral shape similar to the GCE, and early statistical analyses appeared to support a point-source origin [169, 170]. However, the robustness of these results has since been questioned [171, 172], and the MSP hypothesis faces independent observational challenges: few pulsars have been detected in the Inner Galaxy, and their expected number based on known luminosity functions substantially exceeds the handful of candidates identified in current catalogs [173–175].

After more than fifteen years, the origin of the GCE remains an open question. This chapter traces the observational history of the excess and its consistency with dark matter annihilation (Section 4.1), examines the millisecond pulsar hypothesis and its challenges (Section 4.2), and describes the systematics-driven impasse that has stalled progress (Section 4.3). We then motivate an independent, orthogonal approach: constraining the MSP luminosity function through observations of the Milky Way’s globular cluster system (Section 4.4). Section 4.5 then presents this analysis in full, reproducing the published study of MSP gamma-ray luminosity functions in globular clusters and its implications for the GCE.

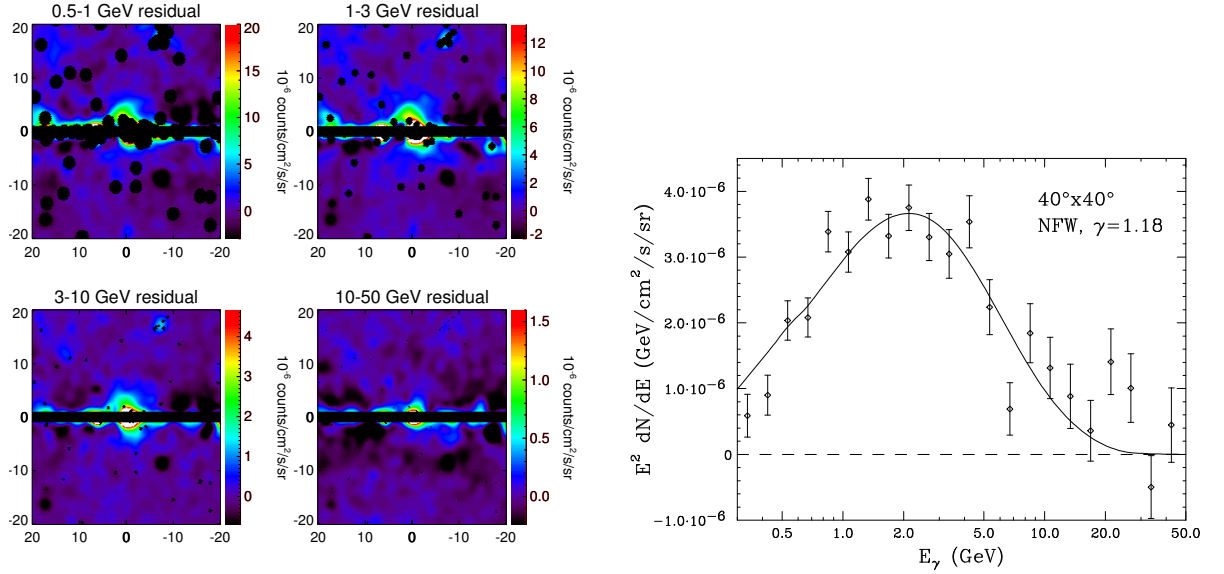


Fig. 4.1. *Left:* Residual gamma-ray intensity maps (in Galactic coordinates) after subtracting the best-fit Galactic diffuse model, Fermi bubbles, and isotropic templates. The dark-matter-like excess is clearly visible around the Galactic Center at energies between ~ 0.5 and 5 GeV. *Right:* Spectrum of the excess for a generalized NFW template with inner slope $\gamma = 1.18$, compared with the prediction for a 43 GeV dark matter particle annihilating to $b\bar{b}$ (solid line). Figures from Daylan et al. [62].

4.1 Discovery and Characterization of the GCE

This section traces the observational history of the Galactic Center Excess, from its first identification in early Fermi-LAT data through the systematic studies that refined its spectral and morphological properties. By the mid-2010s, these studies had converged on a set of observational properties that would frame the interpretive debate discussed in subsequent sections. We close by examining the consistency of these properties with dark matter annihilation.

4.1.1 First Identification

The possibility that dark matter annihilation could produce a detectable gamma-ray signal from the Galactic Center had been discussed extensively in the theoretical literature well before the launch of the Fermi Gamma-Ray Space Telescope. The high dark matter density predicted by N-body simulations in the inner kiloparsec of the Milky Way, combined with the ρ^2 dependence of the annihilation rate, had made this region one of the most promising targets for indirect detection (see Section 1.4).

The first observational evidence for an anomalous gamma-ray component from this region was reported by Goodenough and Hooper in 2009 [100], using roughly one year of Fermi-LAT data. By modeling the angular distribution and energy spectrum of gamma rays collected from the direction of the Galactic Center, they identified a distinctive spectral feature — a bump peaking between approximately 1 and 5 GeV — that could not be absorbed by known diffuse emission templates or cataloged point sources. The angular distribution of this excess emission was centered on the dynamical center of the Milky Way and extended across the inner few degrees. Goodenough and Hooper found that both the spectrum and the spatial distribution were well

described by the annihilation of a relatively light dark matter particle with mass $m_\chi \approx 25\text{--}30$ GeV, annihilating primarily into $b\bar{b}$ quarks. The required halo profile was a Navarro–Frenk–White distribution with a mildly steepened inner slope, $\rho(r) \propto r^{-\gamma}$ with $\gamma = 1.1$, consistent with the adiabatic contraction expected from baryonic dynamics in the inner Galaxy (see Section 1.4.3). The corresponding annihilation cross section, $\langle\sigma v\rangle \approx 9 \times 10^{-26}$ cm³/s, was within a factor of a few of the value predicted for a thermally produced relic. A subsequent, refined analysis by Hooper and Goodenough [101], incorporating additional data and improved background modeling, confirmed these findings and sharpened the parameter estimates.

While these early results were suggestive, the authors were careful to note that an astrophysical origin for the excess — such as emission from a concentrated population of unresolved sources with a similar spectral shape — could not be excluded. Distinguishing between these possibilities would require both improved angular resolution and a systematic understanding of the diffuse background uncertainties, a challenge taken up by several groups in the following years.

4.1.2 Improved Characterization

The initial detection by Goodenough and Hooper prompted several independent groups to revisit the Galactic Center with more sophisticated analyses. Three studies, in particular, established the core observational properties of the excess and defined the systematic landscape that would dominate the debate for the following decade.

Daylan et al. [62] exploited a then-new Fermi event parameter, CTBCORE, which quantifies the reliability of each photon’s directional reconstruction. By selecting only the top two quartiles of front-converting Ultraclean events ranked by this parameter, they suppressed the tails of the point spread function substantially, particularly at low energies where the PSF is broadest. The resulting high-resolution maps (Figure 4.1, left) mitigated the leakage of bright astrophysical emission from the Galactic Plane into the regions of interest, and allowed a cleaner separation of the various gamma-ray components. Within these maps, Daylan et al. confirmed the excess at high statistical significance and measured its angular distribution to be approximately spherically symmetric, centered within $\sim 0.05^\circ$ of Sgr A*, and extending to at least $\sim 10^\circ$ from the Galactic Center. The improved angular resolution also shifted the best-fit dark matter mass upward: the data favored a 36–51 GeV particle annihilating into $b\bar{b}$ with a cross section of $\langle\sigma v\rangle = (1\text{--}3) \times 10^{-26}$ cm³/s, normalized to a local dark matter density of 0.4 GeV/cm³ (Figure 4.1, right). The spatial profile was consistent with a contracted generalized NFW distribution.

Taking a complementary approach, Abazajian et al. [176] focused on constructing empirical models of the diffuse gamma-ray background toward the Galactic Center. Recognizing that the standard Fermi diffuse model might miss localized features, they included a spatial template tracing molecular gas as mapped by 20 cm radio emission. This template accounted for nonthermal bremsstrahlung produced by cosmic-ray electrons interacting with dense molecular clouds in the region. While the inclusion of this additional component confirmed that the extended excess remained highly significant, it also demonstrated a strong dependence of the inferred low-energy spectrum on the choice of background model. In particular, the molecular gas template absorbed some emission that had previously been attributed to the central point source Sgr A*, substantially altering its apparent spectral shape. For the excess itself, Abazajian et al. found the $b\bar{b}$ and $\tau^+\tau^-$ annihilation channels to provide statistically equivalent fits, with the $b\bar{b}$ channel yielding a best-fit mass of $m_\chi = 39.4^{+3.7}_{-2.9}$ (stat.) ± 7.9 (sys.) GeV.

The most comprehensive assessment of background model uncertainties was carried out by Calore, Cholis, and Weniger [167], who evaluated the excess against 60 distinct Galactic diffuse

emission models built using the GALPROP cosmic-ray propagation code. These models spanned a deliberately wide range of physical parameters — cosmic-ray source distributions, gas maps, diffusion coefficients, re-acceleration, convection, interstellar radiation field normalizations, and magnetic field configurations — to bracket the theoretical uncertainty envelope.

Two results from this study proved particularly important. First, the spectral shape of the excess was remarkably stable: regardless of the adopted diffuse model, it consistently displayed a peak at 1–3 GeV, a steep rise at sub-GeV energies, and a falling power-law tail with index ~ -2.7 above the peak, with the overall flux varying by less than a factor of two to three above 1 GeV. Second, when the large correlated uncertainties from empirical residuals were folded into spectral fits, the data were equally well described by a $49^{+6.4}_{-5.4}$ GeV dark matter particle annihilating into $b\bar{b}$, and by a generic broken power-law with break energy $E_{\text{break}} = 2.1 \pm 0.2$ GeV. Because such a broken power-law can readily parameterize astrophysical emission — from unresolved millisecond pulsars, for instance — this analysis made clear that spectral information alone could not distinguish between dark matter and astrophysical origins.

The Fermi-LAT Collaboration’s own analysis of the inner Galaxy [177] essentially confirmed these findings, although the appearance of similar residuals elsewhere along the Galactic plane raised questions about the completeness of existing diffuse models [167].

4.1.3 Established Properties of the Excess

By 2015, the combined results of these and other analyses had established a set of properties for the excess that are broadly agreed upon, even by groups that differ on the question of its origin.

The spectrum of the GCE displays a pronounced peak at energies of 1–3 GeV. Below the peak, the emission rises steeply, with a spectral index harder than ~ 2 down to the lowest energies accessible to the analysis (~ 300 MeV), though the precise shape of this sub-GeV component depends on the adopted diffuse model [167]. Above the peak, the spectrum follows a power-law tail with index ~ -2.7 [62, 167]. The behavior at energies above ~ 10 GeV remains the subject of some debate: early analyses suggested that the spectrum might drop sharply to zero at these energies, but the systematic study by Calore, Cholis, and Weniger [167] showed that this conclusion was not robust, and that a finite, model-dependent flux persists into the tens-of-GeV range. The behavior of the spectrum above 10 GeV would later become relevant to distinguishing between astrophysical and dark matter interpretations, as discussed in Section 4.1.4.

The spatial morphology of the excess is approximately spherically symmetric. Daylan et al. [62] showed that the data strongly disfavor any elongation along or perpendicular to the Galactic Plane by more than $\sim 20\%$, and that the centroid of the emission lies within $\sim 0.05^\circ$ of the position of Sgr A*. The excess has been consistently identified in concentric rings out to at least $\sim 10^\circ$ (corresponding to roughly 1.5 kpc) from the Galactic Center, beyond which systematic uncertainties in the diffuse model become dominant [62, 167]. Some background models support an extension to $\sim 15^\circ$ [167]. The radial fall-off of the emission is consistent with a generalized NFW profile with an inner slope in the range $\gamma \approx 1.1\text{--}1.3$ [62, 167, 176, 178], with different analyses finding specific best-fit values of $\gamma = 1.12 \pm 0.05$ [176], 1.18 [62], and 1.26–1.28 over larger regions of interest [62]. The steepening beyond the canonical $\gamma = 1.0$ NFW slope is qualitatively consistent with the adiabatic contraction of the dark matter profile due to the baryonic potential in the inner Galaxy (see Section 1.4.3).

The robustness of these properties is one of the most striking features of the excess. The $\sim 1\text{--}3$ GeV spectral peak persists across all 60 diffuse models tested by Calore, Cholis, and Weniger [167], with the flux above 1 GeV varying by less than a factor of a few. No known combination of astrophysical diffuse emission processes — including cosmic-ray interactions with

gas and radiation fields, emission from the Fermi Bubbles, or known point source populations — reproduces the observed spherically symmetric morphology [62]. The spherical symmetry, in particular, does not trace any standard astrophysical component (gas, dust, starlight, or star formation), all of which are concentrated along the Galactic Plane. While the signal cannot by itself identify its origin, the existence of a genuine, spatially extended, spectrally peaked excess centered on the Galactic Center was firmly established as a consensus result by 2015.

4.1.4 Consistency with Dark Matter Annihilation

The observational properties established above align closely with the predictions of WIMP dark matter annihilation, a coincidence that has sustained interest in this interpretation for over a decade. The strength of the dark matter case rests on three independent lines of agreement—spectral, morphological, and normalization—each of which is tightly constrained by a small number of parameters.

The expected gamma-ray flux from dark matter annihilation in the Galactic halo takes the factorized form

$$\frac{d\Phi_\gamma}{d\Omega dE} = \frac{\langle\sigma v\rangle}{8\pi m_\chi^2} \frac{dN_\gamma}{dE} \times \int_{\text{l.o.s.}} \rho^2(r) dl, \quad (4.1.1)$$

where the particle physics content—the annihilation cross section $\langle\sigma v\rangle$, the WIMP mass m_χ , and the photon yield per annihilation dN_γ/dE —is cleanly separated from the astrophysical J-factor, the line-of-sight integral of the squared dark matter density (see Section 1.4 for a detailed treatment). Because annihilation occurs essentially at rest¹, the observed spectrum is fixed by the mass and annihilation channel, while the angular distribution on the sky is determined entirely by the density profile. With only three free parameters— m_χ , the annihilation channel, and the inner slope γ of the halo profile—the dark matter model must simultaneously reproduce both the energy spectrum and the spatial morphology of the excess [62]. Additional parameters of the halo profile—the scale radius r_s and the local dark matter density $\rho_\odot$ —are conventionally fixed using dynamical constraints external to the gamma-ray data [62, 176], but their uncertainties propagate into the inferred cross section, as discussed below.

This minimal model succeeds on all three fronts. The GCE spectrum, morphology, and spatial extent are each independently consistent with annihilation of a ~ 30 – 50 GeV WIMP into $b\bar{b}$ final states, distributed according to a contracted NFW profile with inner slope $\gamma \approx 1.1$ – 1.3 (Section 4.1.2 for the spectral fits; Section 4.1.3 for the morphological characterization). No tuning of separate parameters is required to accommodate different observables: the same mass and channel that fit the spectrum also predict the correct spatial distribution, and the same halo profile that matches the morphology yields the observed flux level.

A notable feature of the dark matter fit is its normalization. The annihilation cross section required to produce the observed flux falls in the range $\langle\sigma v\rangle \sim (1\text{--}3) \times 10^{-26} \text{ cm}^3/\text{s}$ [62, 167], which is consistent with the canonical thermal relic value of $\langle\sigma v\rangle_{\text{cosmo}} \approx 2.2 \times 10^{-26} \text{ cm}^3/\text{s}$ [5]. In other words, the very cross section needed to explain the GCE brightness today is the same cross section that would cause a WIMP to freeze out in the early universe with an abundance equal to the observed cosmological dark matter density. No Sommerfeld enhancement, substructure boost factors, or non-thermal production mechanisms are required [62]. The fact that three independent observables—spectrum, morphology, and flux normalization—converge on a simple, internally consistent dark matter model with parameters close to the thermal relic prediction has made the GCE one of the most compelling indirect detection signals in the literature. This agreement

¹WIMPs are a cold dark matter candidate and are therefore non-relativistic.

should, however, be interpreted with care. The inferred cross section is degenerate with the assumed local dark matter density, since the annihilation flux scales as $\rho_\odot^2$; varying $\rho_\odot$ within its observationally allowed range (0.3–0.4 GeV/cm³) shifts $\langle\sigma v\rangle$ by a factor of ~ 1.8 [62, 176]. Additionally, Fermi-LAT observations of dwarf spheroidal galaxies place upper limits on the annihilation cross section that are in tension with this interpretation for some channel and mass combinations [102].

4.2 The Millisecond Pulsar Hypothesis

The observational properties of the GCE are consistent with dark matter annihilation (Section 4.1.4), but they also allow for an astrophysical explanation. The leading alternative attributes the excess to a large population of unresolved millisecond pulsars (MSPs) concentrated in the Galactic bulge. Unlike the dark matter interpretation, which requires a new particle, this hypothesis invokes a known class of astrophysical source; the question is whether a sufficiently large and concentrated population exists in the Galactic Center region. This section reviews the MSP hypothesis, its supporting evidence, and the independent constraints that challenge it.

4.2.1 The Hypothesis and Its Initial Evidence

First proposed by Abazajian in 2011 [168], the MSP hypothesis builds on the observation that MSPs produce gamma-ray spectra naturally similar to the GCE: a stacked analysis of 61 Fermi-LAT-detected MSPs yields a spectrum well described by a power law with spectral index $\alpha \approx 1.57$ and an exponential cutoff at $E_{\text{cut}} \approx 3.78$ GeV, parameters that fall within the 99% confidence contours of the GCE spectral fit [167].

The spatial distribution of these hypothetical MSPs presents the first challenge. The known gamma-ray MSP population follows a disk-like distribution, which would produce a signal elongated along the Galactic plane with an axis ratio of $\sim 1:6$ —incompatible with the observed spherical morphology of the GCE [62]. A standard disk MSP population can account for at most 5–10% of the excess flux [167]. To reproduce the concentrated, roughly spherical GCE morphology, the MSP number density would need to increase steeply toward the Galactic Center as approximately $r^{-2.5}$, requiring a distinct “bulge” population rather than the familiar disk population [167]. Such a bulge component is not unreasonable—the high stellar densities in the inner Galaxy could produce MSPs through the same dynamical channels operating in globular clusters—but it cannot easily account for the signal’s extension to $\sim 10\text{--}15^\circ$ from the Galactic Center [167].

Two independent statistical analyses published in 2016 provided the strongest evidence favoring the point-source interpretation. Lee et al. developed the Non-Poissonian Template Fitting (NPTF) method, which generalizes standard template fitting to account for the photon-count statistics expected from unresolved point sources [169]. While a smooth dark matter halo produces Poissonian photon counts in each pixel (the variance equals the mean), a population of faint point sources introduces additional variance through localized photon clusters. By modeling the source-count distribution dN/dS as a broken power law, Lee et al. found that point sources distributed according to an NFW-squared spatial template fully absorbed the GCE, with a Bayes factor of $\sim 10^6$ favoring the point-source model over smooth dark matter emission. Their inferred source-count function rose sharply just below Fermi’s detection threshold, with the break corresponding to a luminosity of $L_\gamma \sim 2\text{--}3 \times 10^{34}$ erg/s. Extrapolating this distribution, they estimated that ~ 400 near-threshold sources could account for the entire excess within 10° of the Galactic Center [169].

In a complementary approach, Bartels, Krishnamurthy & Weniger applied a wavelet decomposition to the Fermi-LAT maps, which is sensitive to small-scale spatial fluctuations produced by unresolved point sources. Their analysis independently detected excess power at small angular scales in the inner Galaxy, consistent with a population of sub-threshold sources contributing to the GCE [170].

A separate line of evidence came from morphological studies. Macias et al. argued that the GCE spatial distribution is better described by the stellar mass of the Milky Way's boxy/X-shaped bulge than by a spherically symmetric NFW profile [179, 180]. Using near-infrared maps to trace the three-dimensional structure of the central stellar bar, they found that a bulge template absorbed the excess emission with a statistically better fit than the NFW template. Because MSPs are expected to trace the old stellar population that forms the bulge, this morphological preference was interpreted as further evidence for an astrophysical origin. Similar conclusions were reached by Bartels et al. [181], who found that the GCE traces the stellar mass distribution in the inner Galaxy, suggesting a close link between the excess and the Galactic bulge population.

Taken together, the NPTF photon statistics, wavelet power, and stellar bulge morphology built a case that appeared to decisively favor the MSP interpretation by the late 2010s. As we will see in Section 4.3, however, subsequent work revealed that each of these lines of evidence is more fragile than initially appreciated.

4.2.2 Independent Challenges

Despite the statistical evidence from NPTF and wavelet analyses, several independent lines of reasoning have challenged the plausibility of the MSP interpretation. These arguments do not rely on the photon statistics of the GCE itself, but instead constrain the expected properties of any MSP population large enough to produce the excess.

The first constraint comes from scaling the observed relationship between low-mass X-ray binaries (LMXBs) and MSPs. Because MSPs evolve directly from LMXBs, stellar populations of similar age should contain a predictable ratio between the two classes. Cholis, Hooper & Linden exploited this connection by measuring the gamma-ray output of 16 Fermi-detected globular clusters hosting a total of 5 bright ($L > 10^{36}$ erg/s) LMXBs, finding that their combined luminosity amounts to only 4.8% of the GCE flux within the inner 5° [182]. If MSPs were responsible for the entire excess, the inner Galaxy should contain ~ 100 bright LMXBs; INTEGRAL observations have detected only 7 candidates in this region. Even without correcting for selection biases that inflate the globular cluster gamma-ray yield, this comparison limits the MSP contribution to $\sim 4\text{--}11\%$ of the GCE. After accounting for these biases, the authors estimated that only $\sim 1\text{--}5\%$ of the excess can reasonably be attributed to unresolved MSPs [182].

A second challenge arises from the dynamics of MSP formation. Neutron stars receive natal velocity kicks of order $\sim 10^2$ km/s during their formation, and while the kicks received by recycled pulsars are typically smaller than those of young isolated pulsars, they remain large enough to significantly broaden the spatial distribution of the resulting MSP population over gigayear timescales [183]. Brandt & Kocsis showed that this kick energy should disperse a locally formed MSP population well beyond the concentrated profile inferred for the GCE, particularly in the innermost $\sim 1^\circ$ (~ 150 pc) where the observed emission requires a steep volumetric density scaling as $\rho \propto r^{-2.4}$ to $r^{-2.6}$ [183]. A kicked population would produce a significantly flatter, more extended profile, inconsistent with the observed concentration of the signal.

The third and perhaps most direct argument relies on the luminosity function of known MSP populations. If the pulsars responsible for the GCE share the luminosity function of MSPs observed in either the Galactic plane or in globular clusters, then the bright tail of this distribution

should extend above Fermi’s detection threshold, producing individually resolved sources. Hooper & Linden first quantified this constraint using the globular cluster MSP luminosity function, finding that Fermi should have detected a significant number of individual pulsars from the GCE population [173]. A complementary analysis by Holst & Hooper, adopting the luminosity function of Galactic plane MSPs, reached a similar conclusion: Fermi should have already detected dozens of MSPs from the GCE population, yet only a handful of candidates have been identified with directions and distances compatible with an inner Galaxy origin [174]. This discrepancy implies that the GCE can originate from MSPs only if those pulsars are systematically less luminous than any known MSP population—a requirement for which no astrophysical mechanism has been identified. This “missing bright pulsars” problem represents one of the strongest empirical challenges to the MSP hypothesis. In Section 4.5, we present a dedicated analysis that updates and extends this line of argument using 15.8 years of Fermi-LAT data and a comprehensive sample of 157 globular clusters, placing one of the most stringent constraints to date on pulsar interpretations of the GCE.

4.3 The Systematics Stalemate

The evidence reviewed in the preceding section, both for and against the MSP hypothesis, was obtained through template-based analyses that share a common vulnerability: their conclusions depend on modeling choices for which no unique, model-independent prescription exists. The choice of interstellar emission model, the treatment of the Fermi Bubbles at low latitudes, the masking of known point sources, and the parameterization of the source-count function at the faint end each contribute irreducible systematic uncertainties that propagate into the final answer. Over the past decade, a series of exchanges has demonstrated that these uncertainties are large enough to shift the conclusion between “predominantly point sources” and “predominantly smooth emission” depending on the analysis setup. This section traces how the debate reached an impasse and how recent methodological advances are beginning to chart a path forward.

4.3.1 The NPTF Crisis

The evidence for point sources presented in Section 4.2.1 rested largely on the Non-Poissonian Template Fitting (NPTF) framework, which exploits the difference in photon statistics between smooth (Poissonian) emission and populations of unresolved point sources. In 2019, a pair of analyses by Leane and Slatyer called the reliability of these results into question.

The first challenge came from a conceptually simple diagnostic: injecting a known, artificially generated dark matter signal into the real Fermi-LAT data and testing whether the NPTF pipeline could recover it [171]. The injected signal was drawn from a Poisson distribution tracing a contracted NFW profile, exactly as expected from WIMP annihilation. When the standard NPTF analysis was applied to this modified dataset, the injected dark matter flux was completely absorbed by the NFW point-source template, with the reconstructed dark matter component remaining consistent with zero. This misattribution persisted even when the injected signal was scaled to a flux roughly five times larger than the GCE itself. Only at injection levels approaching an order of magnitude above the observed excess did the fit begin to recover a dark matter component, and even then it returned only a fraction of the true injected value.

To understand the origin of this failure, Leane and Slatyer constructed a proof-of-principle simulation containing both a smooth dark matter signal and a population of point sources tracing the Fermi Bubbles. When they deliberately excluded the Fermi Bubbles point-source template from the fit, leaving this population unmodeled, the NFW point-source template absorbed the

mismodeled emission. The simultaneously present dark matter signal was driven to zero, despite being the dominant component in the simulation. This demonstrated that any unmodeled source population with a spatial morphology not fully degenerate with existing templates could fatally bias the NPTF, causing it to absorb and mask a genuine dark matter signal.

A deeper investigation followed in 2020, when Leane and Slatyer discovered a pronounced north-south asymmetry in the GCE [172, 184]. Within a 10° region of interest, the smooth excess in the northern hemisphere was found to be substantially brighter than in the south, with the data preferring this asymmetric model at enormous statistical significance (Bayes factors up to $\sim 10^{27}$ in favor of asymmetry). As shown analytically in their companion paper [172], an overly rigid symmetric template that cannot capture this spatial variance in the data will generate excess pixel-to-pixel scatter that the NPTF interprets as point-source emission. When the authors subdivided the GCE template into independent northern and southern components and allowed both to float freely, the flux attributed to the point-source templates dropped to values consistent with zero. The Bayes factor in favor of point sources, which had stood at $\sim 10^{15}$ under the assumption of a symmetric GCE, collapsed by fourteen orders of magnitude to less than 10, rendering the preference for point sources statistically negligible.

These results did not go unchallenged. Buschmann et al. [185] argued that the signal injection failures reported by Leane and Slatyer were driven primarily by the use of the older p6v11 Galactic diffuse emission model, which suffers from severe over-subtraction in the inner Galaxy. In this model, the diffuse template absorbs more flux than it should, driving the smooth GCE normalization to unphysical negative values and creating a mathematical “hole” that swallows any injected dark matter signal before it can be recovered. To address this, Buschmann et al. introduced several improvements: a state-of-the-art hydrodynamic diffuse model (Model O) that provided a substantially better fit to the data; a spherical-harmonic marginalization technique that corrects for large-scale mismodeling in a data-driven manner without erasing small-scale structure; and a truncated source-count function that conservatively excludes point sources below the one-photon detection threshold, thereby breaking the mathematical degeneracy between ultra-faint sources and smooth emission [186]. With these modifications, the signal injection test worked correctly (injected dark matter signals were properly recovered), yet the data still yielded a Bayes factor of $\sim 10^{3.4}$ in favor of point sources. Similar conclusions were reached independently by Calore et al. [187] using the 1-point Probability Density Function (1pPDF) technique. The 1pPDF is an analytical framework mathematically equivalent in its goals to the NPTF, leveraging pixel-count statistics to reconstruct the source-count distribution. By combining 1pPDF with adaptive template fitting, they demonstrated that point sources constitute a robust sub-threshold component of the GCE, provided the diffuse emission is accurately modeled.

The exchange exposed a fundamental tension: the NPTF results depend sensitively on analysis choices that remain debatable. The preference for point sources can be made to appear or vanish depending on the diffuse model, the treatment of spatial asymmetries, and the parameterization of the source-count function at the faint end. No analysis to date has achieved a resolution that all groups accept, and the community consensus has shifted accordingly: the NPTF, as currently implemented, cannot robustly distinguish between smooth and point-source emission in the inner Galaxy.

4.3.2 The Morphological Ambiguity

Beyond the statistical question of smooth versus point-source emission, the spatial morphology of the GCE itself has become a second axis of contention. As discussed in Section 4.1.3, the excess was initially characterized as approximately spherically symmetric, centered on Sgr A*, and

consistent with a generalized NFW profile with inner slope $\gamma \approx 1.1\text{--}1.3$. This morphology has no obvious astrophysical counterpart among known Galactic structures and was therefore regarded as a strong indicator of dark matter annihilation. However, a series of analyses beginning with Macias et al. [179, 180] and Bartels et al. [181] challenged this picture by arguing that the GCE traces the boxy, X-shaped stellar bulge of the Milky Way rather than a smooth, spherically symmetric halo. Using near-infrared stellar mass maps as spatial templates, these studies found that the stellar bulge template absorbed the GCE flux with a statistically better fit than a standard NFW profile, supporting an astrophysical origin tied to the old stellar population of the inner Galaxy.

The significance of this claim, however, depends sensitively on the analysis setup. Whether the GCE prefers an NFW or bulge morphology can shift depending on the interstellar emission model adopted, the masking procedure applied to the Galactic plane and known point sources, and how the Fermi Bubbles emission is treated at low latitudes [5, 180]. Because the true diffuse background in the inner Galaxy is not known to the precision required, the template fitting procedure is inherently subject to systematic uncertainties that propagate directly into the morphological inference. Different groups, using equally defensible analysis choices, have reached opposite conclusions: some find a strong preference for the bulge template, while others recover a spherically symmetric signal fully consistent with dark matter. No interstellar emission model achieves statistical fluctuation-level agreement with the data in the inner Galaxy; persistent large-scale residuals remain regardless of the morphological template adopted for the GCE. As a result, all morphological conclusions drawn from current analyses should be regarded as provisional.

Beyond the morphological debate, which remains unresolved at the spatial level, an independent spectral observation further complicates the astrophysical interpretation. Cholis et al. [188] examined the high-energy behavior of the GCE spectrum above ~ 10 GeV and found emission extending well beyond the energies at which millisecond pulsars are expected to contribute. The gamma-ray spectra of known MSPs exhibit exponential cutoffs at a few GeV, a characteristic feature of curvature radiation in pulsar magnetospheres. The presence of a power-law tail at higher energies is difficult to reconcile with a purely MSP origin and suggests that either a second emission component contributes at high energies or that the GCE is not entirely astrophysical. This spectral argument operates independently of the morphological debate and was initially cited as evidence against a complete MSP explanation.

However, this high-energy argument has recently been directly challenged. Manconi et al. [189] extended the 1pPDF technique to the > 10 GeV regime, combining it with adaptive template fitting to evaluate both the morphology and the photon-count statistics of the high-energy tail. They found strong evidence for the GCE at these elevated energies, noting that the emission strongly prefers a bulge morphology over a dark matter template. Crucially, their 1pPDF analysis reconstructed a significant population of faint, sub-threshold point sources at these high energies. This provides robust evidence that the high-energy tail of the GCE can be explained by a population of point sources—likely MSPs with harder spectra or a distinct sub-population—rather than requiring a dark matter contribution or an exotic secondary component.

Taken together, the morphological and spectral ambiguities reinforce the conclusion that the current generation of template-based analyses cannot definitively identify the source of the GCE. The answer depends on modeling choices for which there is no clear, model-independent criterion to adjudicate.

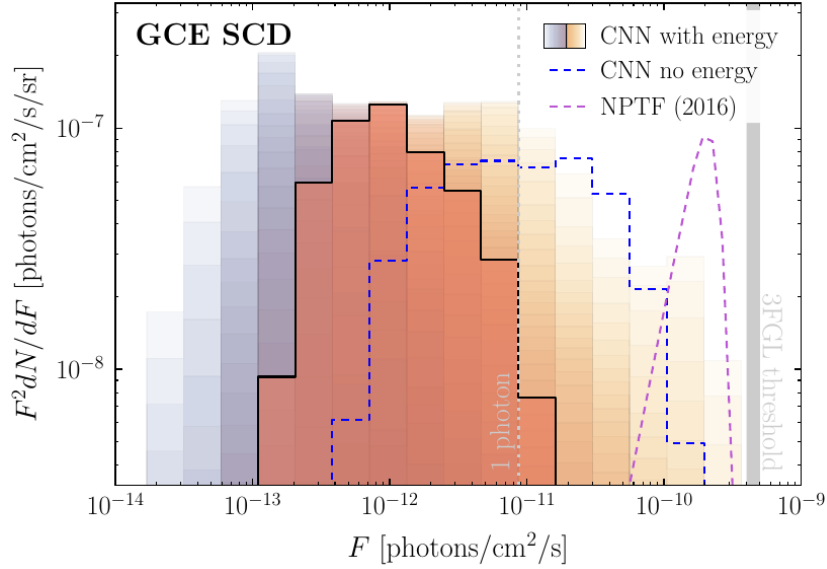


Fig. 4.2. Source-count distribution (SCD) of the GCE inferred via three approaches. The NPTF (dashed purple) placed the emission just below the Fermi 3FGL detection threshold; an energy-independent CNN (dashed blue) preferred a somewhat fainter population. Including energy information (solid black, with quantile bands from purple to orange) drives the SCD below the one-photon line. At this level, at 95% confidence, only 3% of the emission can be excluded as Poissonian. From [190], Figure 1 (right panel).

4.3.3 Recent Revival of the Dark Matter Hypothesis

While the debates over the NPTF and the GCE morphology exposed deep vulnerabilities in established methods, a recent analysis has reframed the question entirely by incorporating energy information into the point-source inference for the first time. List et al. [190] developed a CNN-based simulation-based inference framework that simultaneously fits the spatial and spectral properties of the inner Galaxy across ten logarithmically spaced energy bins from 2 to 20 GeV. Unlike the original NPTF, which discarded all spectral information and neglected pixel-to-pixel correlations from the finite point-spread function, this approach exploits the fact that the GCE has a distinct energy spectrum from the dominant diffuse backgrounds, providing an additional lever arm to separate signal from astrophysical contamination (see Chapter 3 for background on simulation-based inference).

The inclusion of energy information substantially alters the inferred source-count distribution (SCD). Earlier analyses, both the 2016 NPTF [169] and energy-independent CNN approaches [191], had placed the hypothetical point-source population near or just below the Fermi detection threshold, typically predicting populations of a few hundred sources (the 2016 NPTF preferred approximately 200). The energy-dependent CNN drives the SCD to far fainter fluxes: the inferred distribution resides almost entirely below the one-photon threshold, meaning that a typical source would not even produce a single detected photon. To account for the total observed GCE flux with such faint sources, the median prediction requires approximately 200,000 sources, with a 90% confidence lower bound of roughly 35,000 [190].

This result has a fundamental consequence. A population of point sources so faint that individual members produce fewer than one detected photon is mathematically indistinguishable

from smooth, Poissonian emission, which is exactly the spatial signature predicted by dark matter annihilation. The distinction between the point-source and dark matter hypotheses is a nested one: an infinite number of infinitely dim sources produces exactly Poisson statistics. To quantify how much of the GCE can actually be attributed to point sources, List et al. trained a dedicated neural network to distinguish Poissonian from point-source emission given the inferred SCD. For the median SCD prediction, at 95% confidence only 3% of the GCE emission can be excluded as having a Poissonian origin [190]. The remaining 97% is fully consistent with the smooth emission expected from dark matter. This represents a marked shift from the earlier energy-independent CNN analysis, which had been able to exclude 34% of the emission as Poissonian at the same confidence level.

The implications are twofold. First, the key historical evidence that the GCE originates from point sources, including the NPTF's strong Bayes factor and the inferred SCD peaking near the detection threshold, is substantially weakened when spectral information is properly incorporated. Second, if the GCE is nonetheless produced by astrophysical point sources, those sources must be extraordinarily faint and numerous, forming a population with no direct precedent among known Galactic source classes. As we will discuss in Section 4.4, this ultra-faint regime is precisely where independent constraints from globular cluster observations become most informative: the MSP luminosity function measured in globular clusters predicts sources that are far too bright and too few to match the distribution required by the energy-dependent analysis.

4.4 Breaking the Stalemate: Independent Constraints from Globular Clusters

The analyses discussed in the preceding sections share a common vulnerability: they all operate directly within the complex gamma-ray environment of the Galactic Center, where the results depend sensitively on the adopted interstellar emission model, masking procedure, and treatment of the Fermi Bubbles. Whether one uses template fitting, non-Poissonian statistics, or neural network-based inference, the systematic uncertainties associated with the Galactic diffuse emission propagate into every conclusion about the nature of the GCE.

To make further progress, independent constraints from environments that do not suffer from these systematics are needed.

Globular clusters are well suited for this purpose. The high stellar densities in these systems drive frequent dynamical interactions, producing large populations of low-mass X-ray binaries and, through recycling, millisecond pulsars [175]. The number of MSPs in a given cluster correlates with observable macroscopic properties — the stellar encounter rate Γ_e and the visible-band luminosity L_V (a proxy for stellar mass) — allowing the total MSP content to be estimated from quantities that are measured independently of the gamma-ray data [175]. Because globular clusters are discrete, spatially isolated systems far from the confused Galactic plane, their gamma-ray emission can be measured cleanly, without the severe interstellar emission model uncertainties that affect analyses of the GCE.

The age of the stellar populations in globular clusters is especially relevant to the GCE debate. The MSP hypothesis requires the pulsars responsible for the excess to be systematically fainter than those observed in the Galactic Plane, since otherwise Fermi-LAT should have already resolved many of them individually [173, 174]. One natural explanation is that the Galactic Bulge formed its stars early, so its pulsars are very old and have lost much of their rotational kinetic energy, becoming intrinsically dimmer [175]. Globular clusters test this hypothesis directly: their stellar populations are comparably old to the Bulge, and if extreme age naturally produces faint MSPs, that faintness should be observable in these systems as well.

Building on an earlier study by Hooper and Linden [173] that used 7 years of data, in Amerio, Hooper, and Linden [175] we measured the gamma-ray luminosity function of MSPs across the Milky Way’s globular cluster system using 15.8 years of Fermi-LAT data. We detected statistically significant gamma-ray emission from 56 of the 157 known globular clusters, 8 of which had not appeared in previous source catalogs. By fitting the observed gamma-ray luminosities of 87 clusters against predictions based on their stellar encounter rates and visible luminosities, we constrained the underlying log-normal luminosity function. The analysis consistently favors a mean pulsar luminosity of $\langle L_\gamma \rangle \sim (1-8) \times 10^{33}$ erg/s (integrated between 0.1 and 100 GeV), with a log-normal width $\sigma_L \sim 1.4-2.8$. These values are consistent with the luminosity function measured independently for MSPs in the Galactic Plane [174], demonstrating that old stellar populations do not produce systematically fainter pulsars.

The implications for the GCE are direct. If the excess is generated by MSPs with the luminosity function measured in globular clusters, Fermi-LAT’s source catalogs should contain $N_{\text{MSP}} \sim 17-37$ individually resolved pulsars from the Inner Galaxy population. Only three MSP candidates have been identified in this region [174], and systematic radio follow-up campaigns targeting unassociated Fermi sources have not so far uncovered a hidden population [175]. The GCE can therefore originate from MSPs only if they are significantly less luminous, on average, than those found in any other observed environment.

This independent constraint connects directly to the recent findings of List et al. [190]. As discussed in Section 4.3.3, their energy-dependent CNN analysis pushes the source-count distribution of any GCE point sources to extremely faint levels, requiring on the order of 2×10^5 sources — each producing far less than one detected photon — to account for the excess. Such an ultra-faint population would naturally explain the absence of individually resolved pulsars. However, the globular cluster results remove the astrophysical motivation for its existence: if old age does not produce the required ultra-faint MSP populations, no known mechanism can generate $\mathcal{O}(10^5)$ sources of the necessary faintness in the Inner Galaxy. The tension identified from globular clusters and the tension from the energy-dependent SCD analysis thus converge from independent lines of evidence. The remainder of this chapter presents our analysis in full.

[AA: all the rest comes directly from the paper]

HEALPix map order	11
Weeks	9 – 826
Emin	100 MeV
Emax	100 GeV
Instrument Response Functions (IRFs)	P8R3_SOURCE_V3
EVCLASS	128 (Source)
EVTTYPE	3 (Front + Back)
ZMAX	90°
Rocking angle cut	52°
Galactic diffuse emission template	gll_iem_v07
Isotropic background template	iso_P8R3_SOURCE_V3_v1
Source catalog	gll_psc_v35

Table 4.1. Fermi Science Tools settings used in this analysis.

4.5 Millisecond Pulsars in Globular Clusters and Implications for the Galactic Center Gamma-Ray Excess

In this study, we revisit a previous work [173] attempting to constrain the gamma-ray luminosity function of MSPs in globular clusters. To this end, we use the publicly available Fermi data to measure or constrain the gamma-ray emission from each of the Milky Way’s 157 globular clusters (for related work, see Refs. [123, 192–199]). We detect statistically significant gamma-ray emission from 56 of these globular clusters, 8 of which are not contained in previous gamma-ray source catalogs. We then use this data, in combination with the visible luminosities and previously calculated stellar encounter rates of the globular clusters, to constrain the gamma-ray luminosity function of the MSPs in these systems. We find that this MSP population has a luminosity function that is consistent with that previously reported for MSPs in the Galactic Plane, featuring a mean luminosity of $\langle L_\gamma \rangle \sim (1 - 8) \times 10^{33}$ erg/s. If the GCE is generated by pulsars with the same gamma-ray luminosity function and other characteristics as those found in globular clusters, Fermi’s source catalogs would have contained $N_{\text{MSP}} \sim 17 - 37$ pulsars associated with this population. From this, we conclude that the GCE could be generated by pulsars only if they are significantly lower in luminosity or otherwise more difficult to identify than those in globular clusters (or in the Galactic Plane).

4.5.1 Fermi Data Analysis

In this section, we describe our procedure for measuring the spectrum and intensity of the gamma-ray emission from each of the Milky Way’s 157 globular clusters [200]. We employ the python package, Fermipy [201], as provided by the Fermi collaboration, to process the data collected by the Fermi-LAT. We work with HEALPix maps of order 11, corresponding to approximately 0.03° bins, and utilize 15.8 years of data.² We consider photons in the range of 100 MeV to 100 GeV, divided into 5 logarithmic bins per decade. We adopt the standard event selection for gamma-ray point sources, using the SOURCE event class and FRONT + BACK event types. We also employ a zenith angle cut of 90° and a rocking angle cut of 52° in order to reduce the contribution from the Earth’s limb. We require data quality=1 and only consider data that was taken while in survey mode. We do not use data that was collected while passing through the South Atlantic Anomaly. These and other aspects of our analysis are summarized in Table 4.1.

For each globular cluster, we study a $14^\circ \times 14^\circ$ region-of-interest (ROI) centered around the nominal position of the source. We then fit the energy spectrum of the globular cluster, adopting

²MET range: 239557414 - 733276805

a parameterization in the form of a power law with an exponential cut-off:

$$\phi(E) = N_0 E^{-\alpha} e^{-E/E_c}. \quad (4.5.1)$$

In fitting these sources, we follow the standard procedure used in constructing Fermi-LAT source catalogs. We include in our fit any sources in the 4FGL-DR4 catalog that are located within in ROI [123], excluding any globular cluster, pulsar, or unassociated gamma-ray source that is located within 0.2° of the fiducial position of the globular cluster in question. We then add a new point source in the position of the globular cluster, as specified in Ref. [200], and reoptimize the model of the known sources within the ROI. Following standard procedure, we run the source localization pipeline, readjust the centroid of the globular cluster, and look for new point sources with a power-law spectrum. We then move to the characterization of the globular cluster, performing a 3D likelihood scan in the (α, E_c, N_0) parameter space. During this procedure, we model the Galactic diffuse emission and isotropic background contributions according to the templates provided by the Fermi collaboration (see Table 4.1). We allow their normalization to float, as well as the energy spectrum normalization of any other resolved source within 3 degrees. This allows us to identify, for each value of the integrated flux, the combination of these parameters that yields the greatest likelihood.

In Table 4.4, which we report in Appendix 4.A, we list each of the 56 globular clusters for which our analysis identified a statistically significant test statistic, $TS > 25$, along with the best-fit values for α , E_c , and the total flux (integrated between 0.1 and 100 GeV). Note that our previous study, which utilized 7 years of Fermi data, detected only 25 globular clusters at a level of $TS > 25$ [173]. We have compared our results to those reported in the 4FGL catalog [123] and find them to be in overall good agreement. The only significant differences are found in the cases of NGC 5139, Palomar 6, Terzan 5, and NGC 6218, for which our analysis yields integrated fluxes which are larger than those given in the 4FGL catalog by $2 - 3\sigma$.

This appears to be due to the spectral parameterization we have adopted, as these four sources are fit in the 4FGL to a log-parabola or power-law parameterization, rather than that of a power-law with an exponential cutoff.

In addition to the 56 sources with $TS > 25$ listed in Table 4.4, we found an additional 7 sources which yielded $TS > 25$, but that had their highest TS values in bins below 0.4 GeV. Due to the large point spread function and related uncertainties that are encountered at these low energies, we do not consider these sources (which are listed in Table 4.5) to be unambiguous detections, and only report upper limits on their flux.

Finally, in Table 4.6, we report upper limits on the gamma-ray fluxes from each of the remaining globular clusters considered in our analysis, each of which yielded $TS < 25$.

4.5.2 Millisecond Pulsars in Globular Clusters

Due to the high stellar densities found within globular clusters, dynamical interactions between stars take place in these systems much more frequently than they do among stars in the Galactic Plane. As a result, large numbers of low-mass X-ray binaries (LMXBs) and millisecond pulsars (MSPs) are each found within globular clusters. At present, a total of 340 MSPs have been identified within 45 different Milky Way globular clusters (see <https://www3.mpifr-bonn.mpg.de/staff/pfreire/GCpsr.html>, and Refs. [202, 203]). The majority of these pulsars have only been observed at radio wavelengths, as they are too distant to be detected by Fermi. In fact, Fermi's Third Pulsar Catalog [203] contains only four MSPs that are located within a globular cluster: PSR J1717+4308A in NGC 6341, PSR J1823-3021A in NGC 6624, J1824-2452A in NGC 6626, and PSR J1835-3259B in NGC 6652.

The number of MSPs within a given globular cluster is correlated with that cluster's overall visible-band luminosity, L_V (a proxy for stellar mass), and stellar encounter rate, Γ_e . The stellar encounter rate within a globular cluster is given by [204, 205]

$$\Gamma_e = \frac{4\pi}{\sigma_c} \int \rho^2(r) r^2 dr, \quad (4.5.2)$$

where $\rho(r)$ is the stellar density profile of the cluster and σ_c is the velocity dispersion near its core. Throughout this analysis, we make use of the stellar encounter rates tabulated in Ref. [206] (and as we list in Table 4.7).

In this study, we will consider the following five hypotheses for estimating the number of MSPs that will, on average, be found within a given globular cluster:

1. A case in which the mean number of MSPs in a globular cluster is proportional to its stellar encounter rate, $\langle N_{\text{MSP}} \rangle = R_{\text{MSP}} \Gamma_e$. This approximate scaling might be expected if MSPs in globular clusters are formed largely through stellar interactions.
2. A case in which the relationship between the stellar encounter rate and the expected number of pulsars within a globular cluster is not linear, but instead scales as $\langle N_{\text{MSP}} \rangle = R_{\text{MSP}} \Gamma_e^{0.7}$. Empirically speaking, this hypothesis is preferred over that of a strictly linearly proportional relationship [197, 207].
3. A case in which the mean number of MSPs in a globular cluster is proportional to its visible luminosity, $\langle N_{\text{MSP}} \rangle = R_{\text{MSP}} L_V$.
4. A case in which the mean number of MSPs in a globular cluster is related to a combination of its stellar encounter rate and visible luminosity, $\langle N_{\text{MSP}} \rangle = R_{\text{MSP}} \Gamma_e + R'_{\text{MSP}} L_V$.
5. A case in which the mean number of MSPs in a globular cluster is related to a combination of a non-linear scaling to its stellar encounter rate and its visible luminosity, $\langle N_{\text{MSP}} \rangle = R_{\text{MSP}} \Gamma_e^{0.7} + R'_{\text{MSP}} L_V$.

For further discussion regarding the dynamics of globular clusters and empirical relationships between their observed parameters, see Refs. [197, 207–209].

Throughout the remainder of this study, we will focus on the 87 globular clusters that have either $L_V > 10^5 L_\odot$ or $\Gamma_e > 0.01$ (in units such that $\Gamma_e = 1$ for the case of NGC 104). In Figure 4.3, we plot the number of MSPs that are known to exist in each of these 87 globular clusters, as a function of their stellar encounter rate, visible luminosity, or combinations of these parameters. We also show bands which represent the predicted relationships between these quantities, for selected values of $R_{\text{MSP}}^{\text{Radio}}$.³ The inner (outer) bands are calculated such that they cover 68.27% (95.45%) of a Poisson distribution for a mean value of $\langle N_{\text{MSP}}^{\text{Radio}} \rangle$. Even by eye, it is clear that the correlation is less tight for the case of $\langle N_{\text{MSP}} \rangle = R_{\text{MSP}} \Gamma_e$ than for $\langle N_{\text{MSP}} \rangle = R_{\text{MSP}} \Gamma_e^{0.7}$, or for $\langle N_{\text{MSP}} \rangle = L_V$. The tightest correlations are found for combinations of Γ_e and L_V (as shown in the final two frames). One should bear in mind that the current catalog of the MSPs in globular clusters is incomplete, limiting the utility of any quantitative comparisons. That being said, the results presented in Figure 4.3 provide us with motivation to consider the five hypotheses described above for the anticipated number of MSPs in globular clusters.

³While we take R_{MSP} and R'_{MSP} to be related to the predicted number of gamma-ray MSPs in a given globular cluster, the same quantities with the superscript “Radio” correspond to the predicted number of radio MSPs. Due to the different beaming angles of pulsars in the gamma-ray and radio bands, these quantities are not necessarily the same.

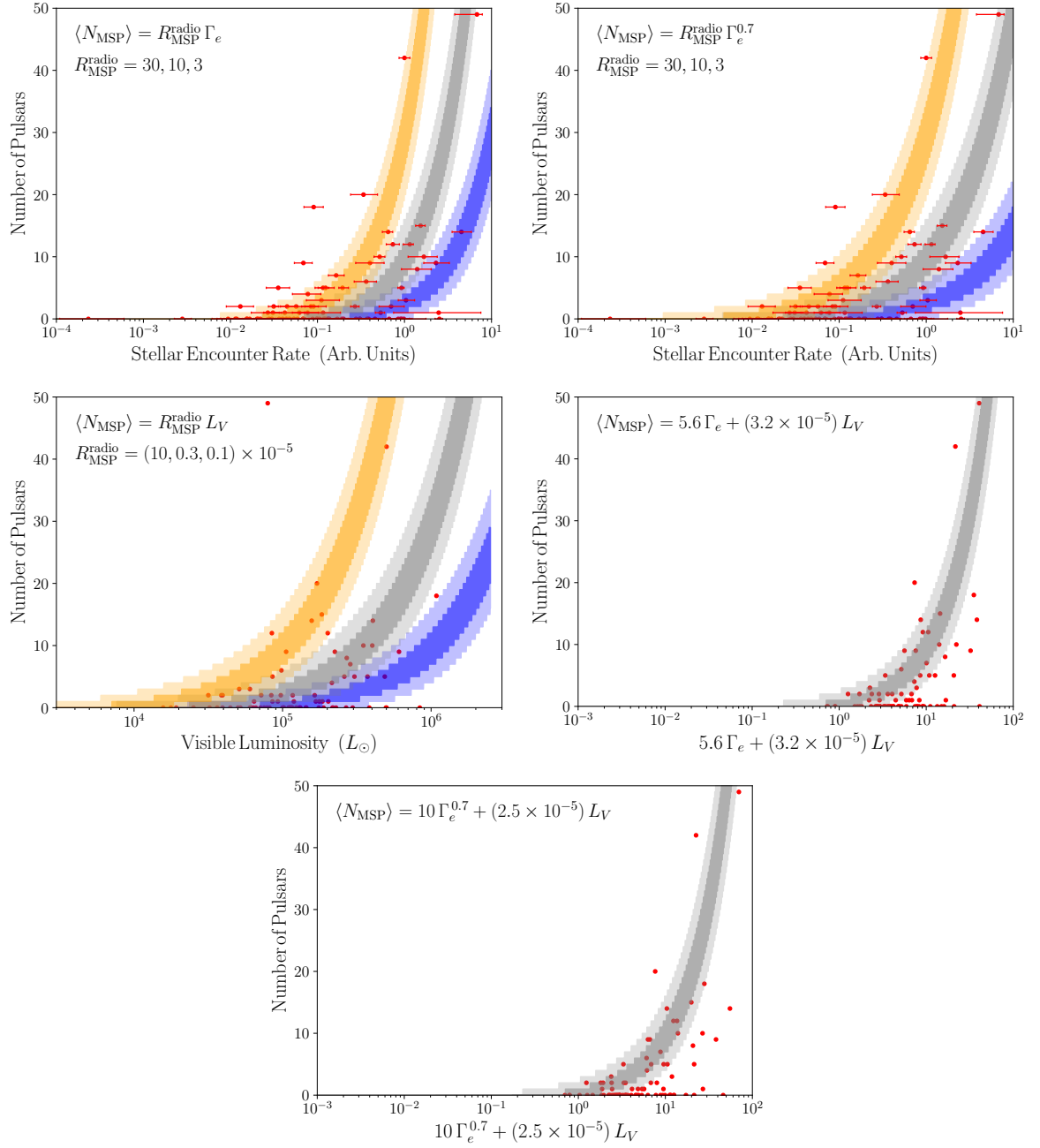


Fig. 4.3. The number of known millisecond pulsars in each of the 87 globular clusters considered in this study (those with $L_V > 10^5 L_\odot$ or $\Gamma_e > 0.01$) as a function of combinations of the stellar encounter rate, Γ_e , and visible luminosity, L_V (in units of solar luminosities). Also shown are colored bands representing the predicted number of MSPs (68.27% and 95.45% containment) in each case, for selected values of $R_{\text{MSP}}^{\text{Radio}}$. The stellar encounter rate is normalized such that $\Gamma_e = 1$ for NGC 104.

4.5.3 The Millisecond Pulsar Gamma-Ray Luminosity Function

For the gamma-ray luminosity function of MSPs, we adopt a log-normal parameterization:

$$\frac{dP}{dL_\gamma}(L_\gamma) = \frac{1}{L_\gamma \sigma_L \sqrt{2\pi}} \exp\left(-\frac{(\ln L_\gamma - \ln L_0)^2}{2\sigma_L^2}\right), \quad (4.5.3)$$

where L_0 and σ_L characterize the center and width of this distribution, respectively. We will often refer to this luminosity function in terms of the mean luminosity, which is given by $\langle L_\gamma \rangle = L_0 e^{\sigma_L^2/2}$.

The luminosity function of a globular cluster that contains a single MSP is simply equal to the expression given in Eq. 4.5.3,

$$\frac{dP}{dL_{\text{tot}}}(L_{\text{tot}}, N_{\text{MSP}} = 1) = \frac{dP}{dL_1}(L_1 = L_{\text{tot}}). \quad (4.5.4)$$

For a cluster with two MSPs, $N_{\text{MSP}} = 2$, it is instead given by

$$\frac{dP}{dL_{\text{tot}}}(L_{\text{tot}}, N_{\text{MSP}} = 2) = \int_0^{L_{\text{tot}}} \frac{dP}{dL_1}(L_1) \frac{dP}{dL_2}(L_2 = L_{\text{tot}} - L_1) dL_1, \quad (4.5.5)$$

and for $N_{\text{MSP}} = 3$,

$$\frac{dP}{dL_{\text{tot}}}(L_{\text{tot}}, N_{\text{MSP}} = 3) = \int_0^{L_{\text{tot}}-L_1} \int_0^{L_{\text{tot}}} \frac{dP}{dL_1}(L_1) \frac{dP}{dL_2}(L_2) \frac{dP}{dL_3}(L_3 = L_{\text{tot}} - L_1 - L_2) dL_1 dL_2, \quad (4.5.6)$$

and so on.

For a globular cluster containing a given mean number of expected MSPs, $\langle N_{\text{MSP}} \rangle$, its gamma-ray luminosity function will be given by

$$\frac{dP}{dL_{\text{GC}}}(L_{\text{GC}}, \langle N_{\text{MSP}} \rangle) = \sum_{N_{\text{MSP}}=0}^{\infty} P(N_{\text{MSP}}) \frac{dP}{dL_{\text{tot}}}(L_{\text{tot}}, N_{\text{MSP}}), \quad (4.5.7)$$

where

$$P(N_{\text{MSP}}, \langle N_{\text{MSP}} \rangle) = \frac{\langle N_{\text{MSP}} \rangle^{N_{\text{MSP}}} e^{-\langle N_{\text{MSP}} \rangle}}{N_{\text{MSP}}!} \quad (4.5.8)$$

is the Poisson probability that the globular cluster will contain N_{MSP} pulsars.

In order to constrain the gamma-ray luminosity function of MSPs in globular clusters, we closely follow the analysis from Ref. [173]. We repeat these steps here for clarity:

1. For a single globular cluster, i , we fit the energy spectrum to the Fermi-LAT data, as detailed in Section 4.5.1. This gives us the observational likelihood profile for the flux, ϕ_i , which we can write as $\mathcal{L}_{\text{obs}}(\phi_i)$. This function quantifies the measurement uncertainty of the flux.
2. We define the theoretical model for the gamma-ray luminosity function as in Eq. 4.5.7. This is a probability density function for the luminosity, $L_{\text{GC},i}$, which depends on a set of model parameters which we call θ . We write this dependence explicitly as $p(L_{\text{GC},i}|\theta)$.

3. Given the relation between the flux and luminosity, $\phi_i = L_{\text{GC},i}/4\pi d_i^2$, where d_i for each globular cluster is given in Table 4.7, it is possible to rewrite the luminosity function as function of the flux: $p_i(\phi_i|\theta)$. While the parameters θ would not depend on the specific globular cluster, the distance d_i does. For this reason, given a gamma ray luminosity function, $p_i(\phi_i|\theta)$ will be different for each globular cluster.
4. We combine the likelihood derived for a specific observation with the probability to observe a certain flux, and marginalize (integrate) over the flux in order to obtain the likelihood as a function of the parameter θ : $\mathcal{L}_{\text{obs}}^i(\theta) = \int_0^\infty \mathcal{L}_{\text{obs}}(\phi_i) p_i(\phi_i|\theta) d\phi_i$
5. We notice how each observation is independent from each other. For this reason, the joint likelihood on the entire set of globular clusters adopted in the analysis will be given by $\mathcal{L}_{\text{obs}}(\theta) = \prod_i \mathcal{L}_{\text{obs}}^i(\theta)$.
6. Given the likelihood of the entire dataset $\mathcal{L}_{\text{obs}}(\theta)$, it is now possible to derive a maximum likelihood estimation of the parameters θ .

In Figure 4.4, we show several examples of the results we obtain following this procedure. In each frame, we plot the measured gamma-ray luminosities of the 87 globular clusters considered in this portion of our study (those with $L_V > 10^5 L_\odot$ or $\Gamma_e > 0.01$), compared to their stellar encounter rates, visible luminosities, or selected combinations thereof. Then, as dark (light) grey bands, we plot the 68.27% (95.45%) range of the total gamma-ray luminosity from a given cluster that is predicted by the model in question. In each frame, we show the results as found adopting the best-fit values for $\langle L_\gamma \rangle$, σ_L , R_{MSP} , and/or R'_{MSP} (see Table 4.2).

In Figure 4.5, we plot the constraints on the luminosity function parameters, $\langle L_\gamma \rangle$ and σ_L , after marginalizing over the values of R_{MSP} and/or R'_{MSP} . The best-fit point in each frame is shown by a dot, while the contours reflect the regions that are preferred at the levels of 1, 2 and 3σ confidence. We show the results for each of the five hypotheses described in Section 4.5.2 for the relationship between the value of $\langle N_{\text{MSP}} \rangle$ and those of L_V and/or Γ_e , finding each case to yield qualitatively similar results. Namely, this procedure consistently favors scenarios with $\langle L_\gamma \rangle \sim (1-8) \times 10^{33}$ erg/s and $\sigma_L \sim 1.4-2.7$. In Table 4.2, we show the values of the parameters favored by our analysis, including those for R_{MSP} and R'_{MSP} . These parameters are directly related to the total number of gamma-ray MSPs contained within this sample of 87 globular clusters: $N_{\text{MSP}}^{\text{tot}} = 42.4 R_{\text{MSP}} + 1.55 \times 10^7 R'_{\text{MSP}}$ (for cases with $\langle N_{\text{MSP}}^{\text{tot}} \rangle \propto \Gamma_e$), or $N_{\text{MSP}} = 41.7 R_{\text{MSP}}^{0.7} + 155 \times 10^7 R'_{\text{MSP}}$ (for cases with $\langle N_{\text{MSP}} \rangle \propto \Gamma_e^{0.7}$). For the parameters favored by our analysis, these 87 globular clusters contain a total of $\sim 400-1500$ gamma-ray emitting MSPs.

We note that the different hypotheses we have considered for the relationship between the value of $\langle N_{\text{MSP}} \rangle$ and those of L_V and/or Γ_e can yield significantly different values for the overall likelihood. In Table 4.3, we show the difference between these resulting likelihoods (for the best-fit parameters in each case). Note that these hypotheses have different numbers of degrees-of-freedom, and this should be taken in to account when comparing their relative likelihoods.

So far, we have assumed that the mean number of MSPs in a given globular cluster is determined entirely by its luminosity and/or stellar encounter rate, without any departures from this prediction (other than the scatter inherent in each cluster's draw from the Poisson distribution for N_{MSP}). In reality, our models for determining $\langle N_{\text{MSP}} \rangle$ cannot possibly contain all of the relevant information and, therefore, should not be perfectly predictive. This leads us to

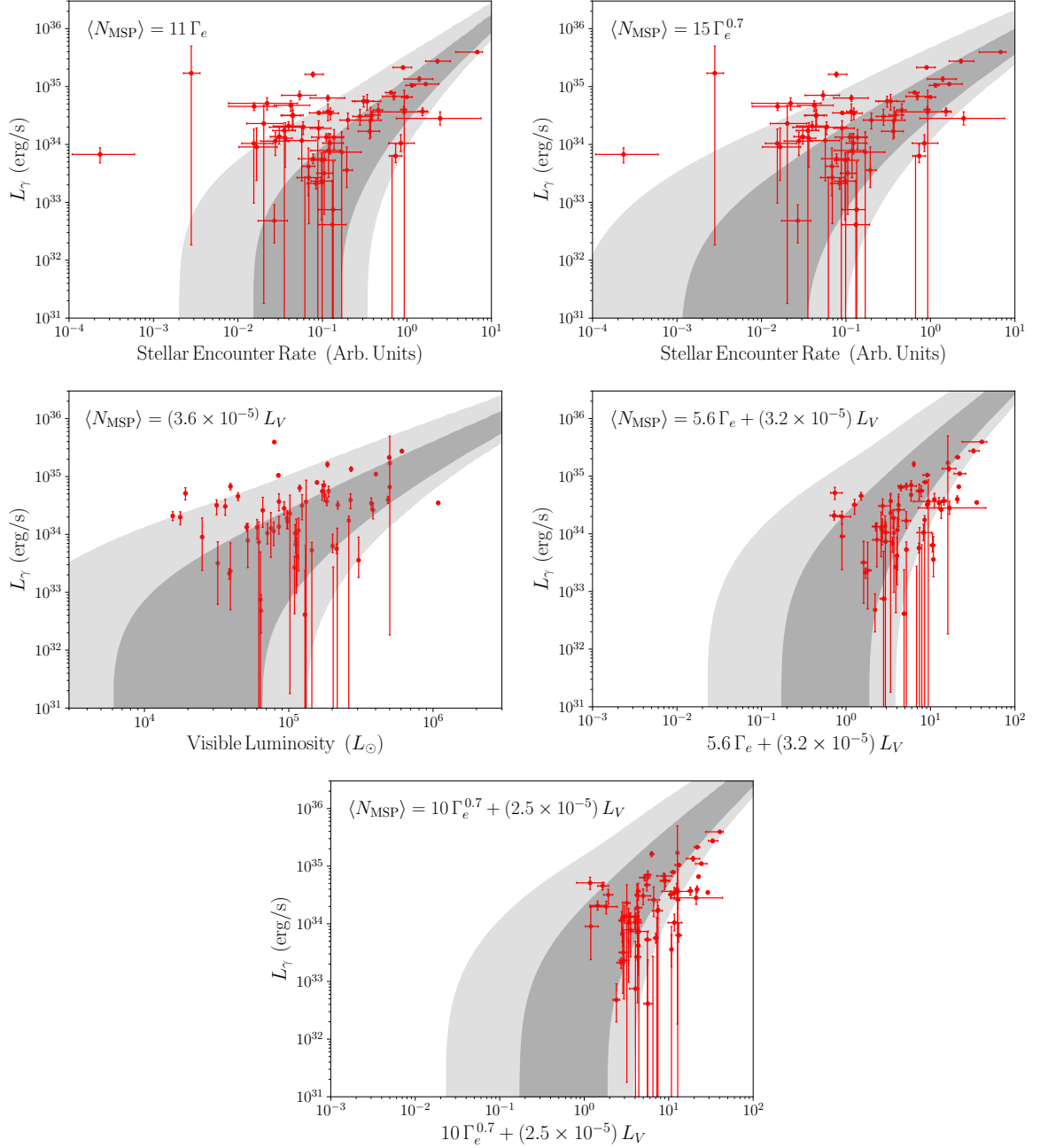


Fig. 4.4. The measured gamma-ray luminosities of the 87 globular clusters considered in this study (those with $L_V > 10^5 L_\odot$ or $\Gamma_e > 0.01$), compared to combinations of their stellar encounter rates and/or visible luminosities. As dark (light) grey bands, we show the 68.27% (95.45%) confidence interval predicted by the model for the total gamma-ray luminosity. In each frame, we show the results that were obtained using the best-fit values for $\langle L_\gamma \rangle$, σ_L , R_{MSP} , and/or R'_{MSP} (see Table 4.2).

expect that there should be some degree of variation in the mean number of MSPs in given cluster, beyond that which can be predicted by the values of L_V and Γ_e alone.

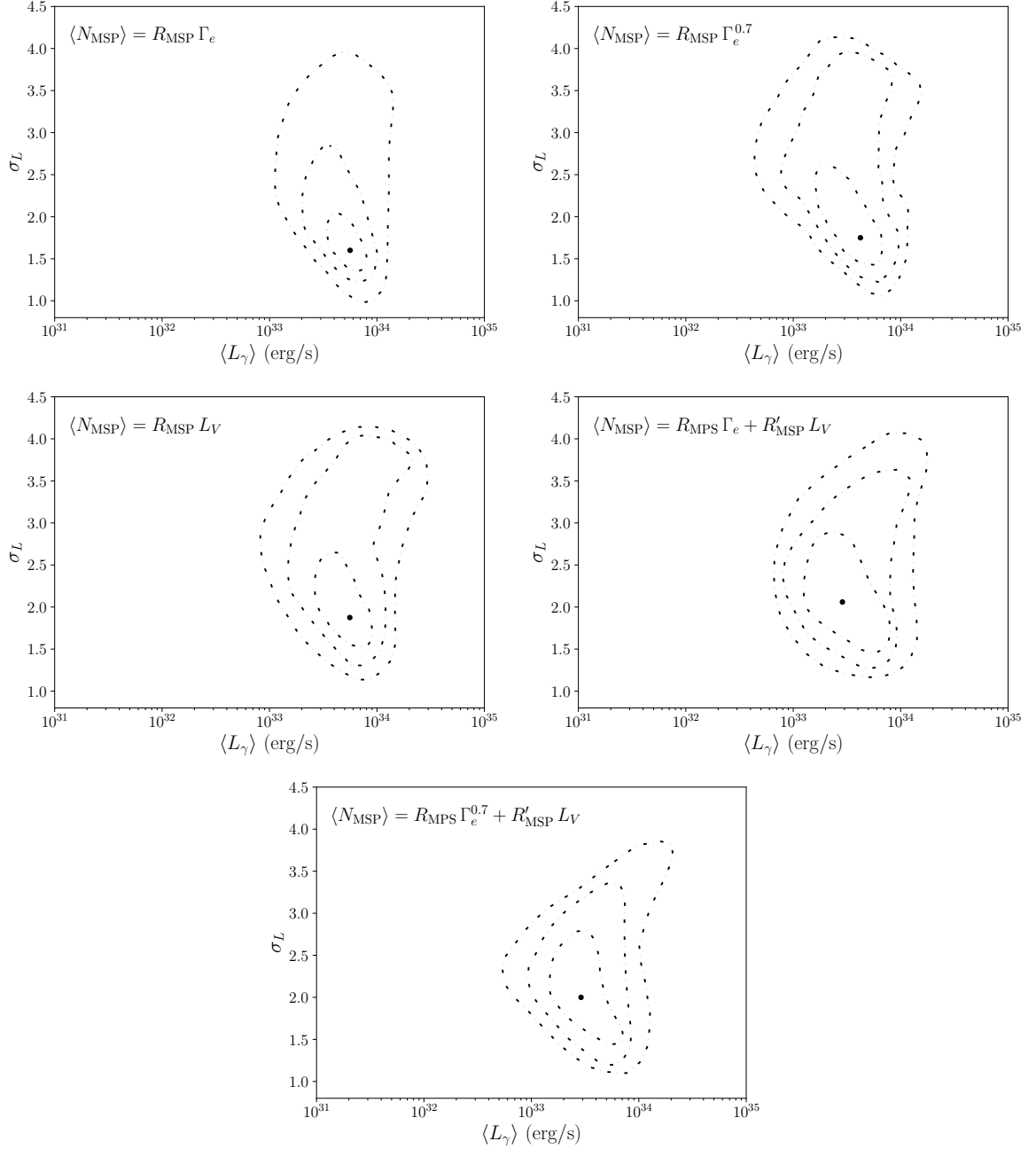


Fig. 4.5. Constraints on the MSP gamma-ray luminosity function parameters, after marginalizing over R_{MSP} and/or R'_{MSP} . The best-fit point in each frame is shown as a dot, while the contours reflect the regions that are preferred at the levels of 1, 2 and 3σ confidence.

To assess the impact of such cluster-to-cluster variation, we take the mean number of expected MSPs to have an intrinsic degree of scatter, which we model according to a log-normal distribution with a width, σ_N . This more complex model represents a potentially more conservative approach, and allows us to accounting for potential unmodeled systematic uncertainties. By rerunning our analysis, we find that in the case of our best fit model, $\langle N_{\text{MSP}} \rangle = R_{\text{MSP}} \Gamma_e^{0.7} + R'_{\text{MSP}} L_V$, the

$\langle N_{\text{MSP}} \rangle$	Parameter	Best Fit	1σ Range	2σ Range	3σ Range
$R_{\text{MSP}} \Gamma_e$	$\langle L_\gamma \rangle$	5.6×10^{33}	$(3.4 - 8.1) \times 10^{33}$	$(2.0 - 10) \times 10^{33}$	$(1.1 - 13) \times 10^{33}$
	σ_L	1.6	1.4 – 2.0	1.2 – 2.8	1.0 – 3.9
	R_{MSP}	11	9.5 – 15	7.6 – 32	6.4 – 110
$R_{\text{MSP}} \Gamma_e^{0.7}$	$\langle L_\gamma \rangle$	4.2×10^{33}	$(1.8 - 6.6) \times 10^{33}$	$(0.8 - 9.2) \times 10^{33}$	$(0.4 - 15) \times 10^{33}$
	σ_L	1.8	1.4 – 2.6	1.1 – 3.9	1.0 – 4.1
	R_{MSP}	15	11 – 36	8.5 – 130	7.1 – 220
$R_{\text{MSP}} L_V$	$\langle L_\gamma \rangle$	5.6×10^{33}	$(2.7 - 8.7) \times 10^{33}$	$(1.4 - 21) \times 10^{33}$	$(0.8 - 29) \times 10^{33}$
	σ_L	1.9	1.5 – 2.7	1.3 – 4.0	1.1 – 4.1
	R_{MSP}	3.6×10^{-5}	$(2.7 - 6.5) \times 10^{-5}$	$(1.8 - 22) \times 10^{-5}$	$(1.4 - 35) \times 10^{-5}$
$R_{\text{MSP}} \Gamma_e + R'_{\text{MSP}} L_V$	$\langle L_\gamma \rangle$	2.9×10^{33}	$(1.3 - 5.5) \times 10^{33}$	$(0.9 - 8.4) \times 10^{33}$	$(0.7 - 14) \times 10^{33}$
	σ_L	2.1	1.7 – 2.6	1.3 – 3.5	1.2 – 4.0
	R_{MSP}	5.6	3.2 – 14	1.8 – 21	1.2 – 39
	R'_{MSP}	3.2×10^{-5}	$(1.8 - 8.0) \times 10^{-5}$	$(1.4 - 11) \times 10^{-5}$	$(1.1 - 18) \times 10^{-5}$
$R_{\text{MSP}} \Gamma_e^{0.7} + R'_{\text{MSP}} L_V$	$\langle L_\gamma \rangle$	2.9×10^{33}	$(1.4 - 7.2) \times 10^{33}$	$(0.9 - 8.7) \times 10^{33}$	$(0.5 - 21) \times 10^{33}$
	σ_L	2.0	1.4 – 2.8	1.2 – 3.4	1.1 – 3.9
	R_{MSP}	10	6.4 – 23	3.9 – 66	2.9 – 110
	R'_{MSP}	2.5×10^{-5}	$(1.6 - 5.7) \times 10^{-5}$	$(1.1 - 8.0) \times 10^{-5}$	$(0.8 - 11) \times 10^{-5}$

Table 4.2. The parameter values favored by our analysis. Values for the average MSP gamma-ray luminosity, $\langle L_\gamma \rangle$, are given in erg/s.

$\langle N_{\text{MSP}} \rangle$	$-2\Delta \ln \mathcal{L}$
$R_{\text{MSP}} \Gamma_e$	146.0
$R_{\text{MSP}} \Gamma_e^{0.7}$	45.4
$R_{\text{MSP}} L_V$	37.7
$R_{\text{MSP}} \Gamma_e + R'_{\text{MSP}} L_V$	13.5
$R_{\text{MSP}} \Gamma_e^{0.7} + R'_{\text{MSP}} L_V$	0.0

Table 4.3. The change in twice the log-likelihood of our analysis, relative to the best-fit case shown in the bottom row of this table.

overall fit is improved by a maximum of $-2\Delta \ln \mathcal{L} \approx 8$ for $\sigma_N \approx 0.5$. In this new best fit, the values of $\langle L_\gamma \rangle$ and σ_L are shifted modestly from the case without this intrinsic scatter, from $\langle L_\gamma \rangle \approx 2.9 \times 10^{33}$ erg/s and $\sigma_L \approx 2.0$ to $\langle L_\gamma \rangle \approx 1.8 \times 10^{33}$ erg/s and $\sigma_L \approx 1.4$.

While an improvement of $-2\Delta \ln \mathcal{L} \approx 8$ corresponds to a statistical preference between 2σ and 3σ , this more complex model also introduces new modeling uncertainties. Given this, we cannot exclude that this improvement is due to statistical fluctuations or unknown systematic uncertainties. We therefore refrain from adopting the model with cluster-to-cluster variations as our fiducial baseline, and instead opt for the simpler fitting formula without variations, which

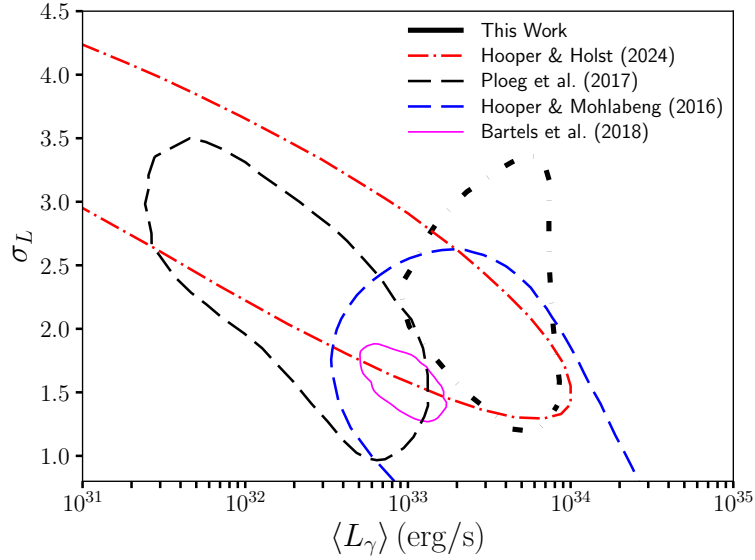


Fig. 4.6. A comparison of the 2σ constraints on the MSP gamma-ray luminosity found in this study (for the best-fitting case of $\langle N_{\text{MSP}} \rangle = R_{\text{MSP}} \Gamma_e^{0.7} + R'_{\text{MSP}} L_V$) to those published in previous studies of MSPs in the Galactic Plane. The luminosity function preferred by our analysis is in good agreement with those derived by Holst and Hooper [174], and by Hooper and Mohlabeng [173], but favor somewhat larger values for the average MSP luminosity than Ploeg *et al.* [212], or Bartels *et al.* [213].

has a stronger basis in the literature.

Before moving on, we will use these results to comment on the previous studies that have interpreted the gamma-ray emission observed from NGC 5139 [210] and NGC 104 [211] to be possible signals of annihilating dark matter. Although this interpretation cannot be easily ruled out, our results in Figure 4.4 demonstrate why no such exotic emission is required. In fact, these clusters have values of Γ_e and L_V that are entirely compatible with their observed gamma-ray luminosities. No dark matter annihilation or other exotic contributions are required.

4.5.4 Comparisons with Previous Studies

A number of previous studies have attempted to measure or constrain the gamma-ray luminosity function of MSPs [173, 174, 198, 212–215]). In Figure 4.6, we compare the results of this study (for the best-fitting case of $\langle N_{\text{MSP}} \rangle = R_{\text{MSP}} \Gamma_e^{0.7} + R'_{\text{MSP}} L_V$) with those published in previous studies of the MSP population within the Galactic Plane. The luminosity function preferred by our analysis is in good agreement with those recently derived by Holst and Hooper [174], and previously by Hooper and Mohlabeng [173]. Our results, however, favor somewhat larger values of the average MSP luminosity, $\langle L_\gamma \rangle$, than the analyses of Ploeg *et al.* [212] or Bartels *et al.* [213].

We note that the analysis of Bartels *et al.* [213] made use of pulsar distance measurements, which are often subject to significant and difficult to quantify systematic uncertainties. For most pulsars, distance estimates are derived from measurements of their dispersion measure (ie., the frequency-dependent time delay of the radio pulses, which can be used to determine the column density of free electrons along the pulsar’s line-of-sight). When the dispersion measure of a pulsar is combined with a model of the free electron distribution, one can obtain an estimate of its

distance. Such estimates are known to vary significantly, however, depending on the electron distribution model that is adopted [216, 217], and sometimes disagree with more reliable parallax determinations by factors of up to ~ 3 [203].

It is possible that the MSPs population found within globular clusters could feature a gamma-ray luminosity function that is different from those pulsars in the Galactic Plane. This could be motivated by the fact that globular clusters are known to contain an old stellar population, and their pulsars could plausibly have already lost much of rotational kinetic energy, compared to that of more recently formed MSPs. This difference, however, would lead us to expect MSPs in globular clusters to be systematically *fainter* than those in the Galactic Plane, which is opposite to that found in the studies of Ploeg *et al.* [212] and Bartels *et al.* [213].

We can more directly compare our results to those of the 2016 analysis by Hooper and Linden [173], which also studied MSPs in globular clusters, using in their case 7 years of Fermi data (compared to the 15.8 years of data used in this analysis). The 2016 study favored a MSP gamma-ray luminosity function with $\langle L_\gamma \rangle \sim (0.7 - 5) \times 10^{34}$ erg/s and $\sigma_L \sim 1 - 3$. While the results presented here are statistically consistent with those of that study (at the approximately $\sim 2\sigma$ level), our analysis tends to prefer somewhat smaller values of $\langle L_\gamma \rangle$ and larger values of R_{MSP} , by a factor of $\sim 2 - 8$. For a more detailed comparison, see Appendix 4.A.1.

4.5.5 Implications for the Galactic Center Gamma-Ray Excess

As stated in the introduction, very few pulsars have been detected in the vicinity of the Inner Galaxy. If the GCE is generated by MSPs, this fact requires the MSPs responsible for this signal to be faint enough to have gone undetected. This, in turn, would require the MSPs in the Inner Galaxy to be systematically lower in luminosity than those found in other environments [174] (see also, Refs. [173, 198, 212–215, 218]).

The possibility that the Inner Galaxy could contain a large population of low-luminosity pulsars has been motivated by the large proportion of old stars that are found in the Galactic Bulge relative to those in the Galactic Plane. The stellar populations found in globular clusters, however, are comparably old to those in the Bulge. By measuring the gamma-ray luminosity function of MSPs in globular clusters, we can hope to shed some light on the question of whether the Inner Galaxy might contain a very large abundance of very low-luminosity MSPs.

In Figure 4.7, we plot the number of MSPs that should have been detected by Fermi and contained within their source catalogs [203] if they were responsible for producing the GCE, as a function of the luminosity function parameters [174]. If the GCE is generated by MSPs with the same luminosity function and other characteristics as those found in globular clusters, Fermi's source catalogs would have contained $N_{\text{MSP}} \sim 17 - 37$ individual members of this population. Given that only three millisecond pulsars have been detected with a direction and distance that does not preclude it from being part of an Inner Galaxy population, the luminosity function derived here is incompatible with pulsar interpretations of the GCE, unless a large number of unassociated Fermi sources are, in fact, unidentified MSPs. Given systematic efforts that have been conducted to search for radio pulsations from unassociated Fermi sources, this possibility seems unlikely. Alternatively, the observed gamma-ray excess could be generated by MSPs if those pulsars are significantly less luminous, on average, than those located in globular clusters or in the Galactic Plane.

As mentioned in Section 4.5.3, our models for the mean number of millisecond pulsars in a given globular cluster are subject to intrinsic uncertainties. To account for this, we also considered a more flexible model that includes an intrinsic cluster-to-cluster variation in the expected number of MSPs. The purpose of this approach is to derive a more conservative lower

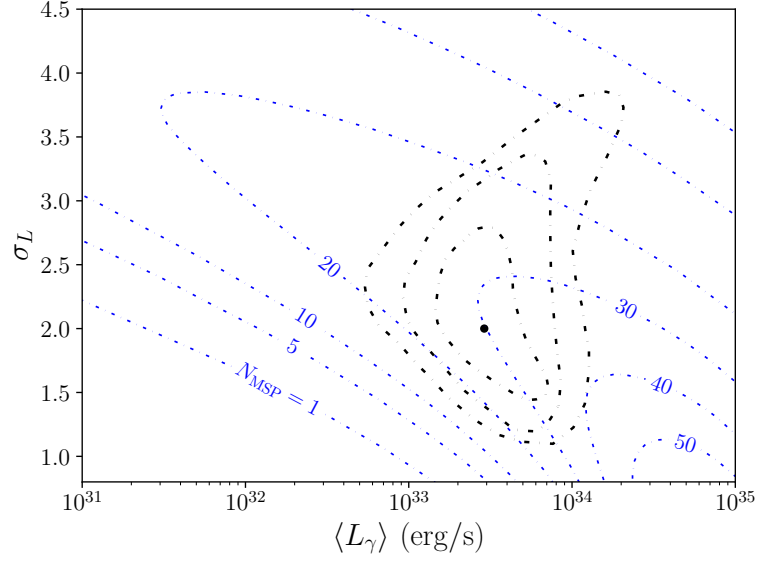


Fig. 4.7. The constraints on the MSP gamma-ray luminosity found in this study (shown for the case of $\langle N_{\text{MSP}} \rangle = R_{\text{MSP}} \Gamma_e^{0.7} + R'_{\text{MSP}} L_V$), shown along with contours representing the number of Inner Galaxy MSPs that should appear in the Fermi source catalogs, N_{MSP} , assuming that the Galactic Center Gamma-Ray Excess (GCE) is generated by MSPs [174]. As there are only three Inner Galaxy pulsars candidates in the 3PC, MSPs can generate the excess gamma-ray emission only if that source population consists of members that are significantly less luminous than those present in globular clusters.

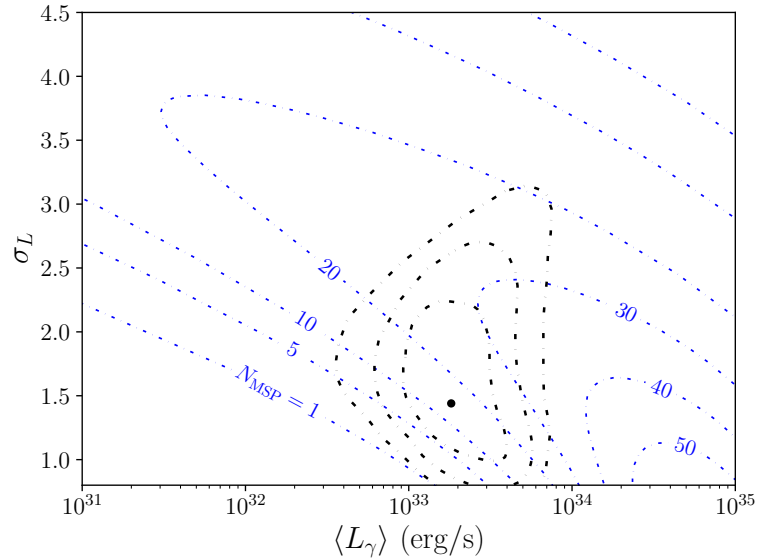


Fig. 4.8. As in Figure 4.7, but for the best fit model including cluster-to-cluster variations. The results are compatible with the estimates in Figure 4.7 within the 1σ band, although the model with cluster-to-cluster variations favors lower values for $\langle L_\gamma \rangle$ and σ_L .

bound on the number of pulsars that should have been detected in the Inner Galaxy if they are responsible for the Galactic Center Excess. In Figure 4.8, we show the results of this analysis. By allowing for a wider range of pulsar populations, particularly those that are fainter on average, this model lowers the number of expected detections to the range of $N_{\text{MSP}} \approx 3 - 30$. Although this bound is less constraining by construction, the results nonetheless remain in tension at a level of at least 2σ with the three identified pulsar candidates.

4.5.6 Summary and Conclusions

In this study, we have analyzed the gamma-ray emission from the Milky Way’s globular clusters, using 15.8 years of publicly available data collected by the Fermi Gamma-Ray Space Telescope. We report the robust detection of 56 globular clusters with a statistically significant test statistic of $\text{TS} > 25$, 8 of which are not contained in previous gamma-ray source catalogs.

Making use of previously established correlations between the mean number of millisecond pulsars in a globular cluster and the stellar encounter rate or visible luminosity of that cluster, we perform a determination of the gamma-ray luminosity function of the millisecond pulsars in the Milky Way’s globular cluster system. To this end, we have considered 87 globular clusters with high stellar encounter rates and/or visible luminosities, and combined this information with our measurements of their gamma-ray luminosities. We then fitted the data to a log-normal luminosity function, finding it to favor a mean pulsar luminosity of $\langle L_\gamma \rangle \sim (1 - 8) \times 10^{33} \text{ erg/s}$ (integrated between 0.1 and 100 GeV), and a width of $\sigma_L \sim 1.4 - 2.7$. Our analysis also indicates that the Milky Way’s globular cluster system contains a total of $\sim 400 - 1500$ gamma-ray emitting millisecond pulsars. The luminosity function favored by this analysis is consistent with that exhibited by millisecond pulsars found in the Galactic Plane [174].

The leading astrophysical interpretation of the Galactic Center Gamma-Ray Excess has long been that this signal could be generated by large population of centrally-located millisecond pulsars. Very few pulsars have been detected in the Inner Galaxy, however, forcing us to place strong constraints on the gamma-ray luminosity function of any pulsar population that could be responsible for the excess emission. In particular, such a pulsar population would have to be systematically less bright than those found in the Galactic Plane [174].

It has been suggested that the Inner Galaxy pulsar population could be older and therefore less luminous, on average, than pulsars found in the Galactic Plane. The stars in globular clusters are also very old, however. Thus if the Inner Galaxy pulsar population is very faint, one would expect the same to be true among the pulsars found in globular clusters. The results of this study indicate that this is not the case. This result poses a challenge to pulsar interpretations of the Galactic Center Gamma-Ray Excess and, by extension, potentially bolsters dark matter interpretations of this long-standing signal.

4.A Additional Tables

In this section, we provide several tables which contain information regarding each of the globular clusters considered in our study, as well as the results of our gamma-ray analysis for each of these sources.

Table 4.4

Globular Cluster	Alternate Name	Flux (erg/cm ² /s)	α	E_c (GeV)	TS
NGC 104	47 Tuc	$2.773^{+0.041}_{-0.051} \times 10^{-11}$	1.37	2.85	10801.3
Terzan 5	Terzan 11	$9.245^{+0.215}_{-0.181} \times 10^{-11}$	1.75	4.24	5955.9
NGC 6266	M62	$1.984^{+0.091}_{-0.097} \times 10^{-11}$	1.44	2.79	1504.2
NGC 6388		$1.823^{+0.091}_{-0.090} \times 10^{-11}$	1.41	2.31	1399.8
NGC 6626	M28	$2.170^{+0.111}_{-0.118} \times 10^{-11}$	1.52	2.35	1346.0
NGC 5139	Omega Centauri	$1.100^{+0.056}_{-0.054} \times 10^{-11}$	1.21	2.11	1168.9
NGC 6624		$1.405^{+0.086}_{-0.090} \times 10^{-11}$	1.60	4.75	902.9
GLIMPSE01		$4.305^{+0.256}_{-0.238} \times 10^{-11}$	1.55	4.22	686.5
NGC 6441		$1.710^{+0.112}_{-0.120} \times 10^{-11}$	1.85	3.91	675.9
NGC 6440		$1.586^{+0.141}_{-0.142} \times 10^{-11}$	1.90	6.41	399.2
NGC 6752		$3.954^{+0.444}_{-0.419} \times 10^{-12}$	1.07	1.46	341.5
NGC 6316		$1.248^{+0.115}_{-0.119} \times 10^{-11}$	1.86	5.12	336.2
GLIMPSE02		$2.568^{+0.233}_{-0.238} \times 10^{-11}$	1.78	2.81	203.7
NGC 6652		$5.664^{+0.694}_{-0.676} \times 10^{-12}$	1.87	4.78	191.8
2MS-GC01	2MASS-GC01	$2.666^{+0.226}_{-0.238} \times 10^{-11}$	1.66	1.86	191.8
NGC 2808		$3.625^{+0.489}_{-0.484} \times 10^{-12}$	1.69	4.06	182.1
NGC 6541		$4.747^{+0.666}_{-0.670} \times 10^{-12}$	1.85	4.80	154.8
NGC 6093	M80	$4.402^{+0.857}_{-0.807} \times 10^{-12}$	1.67	6.78	146.0
NGC 6569		$4.984^{+0.821}_{-0.779} \times 10^{-12}$	1.62	6.64	133.8
NGC 7078		$3.608^{+0.488}_{-0.517} \times 10^{-12}$	2.04	2.19	127.4
NGC 6717	Palomar 9	$3.454^{+0.619}_{-0.560} \times 10^{-12}$	1.29	3.26	121.2
NGC 6656	M22	$4.557^{+0.882}_{-0.784} \times 10^{-12}$	0.82	1.00	120.5
NGC 1904	M79	$3.176^{+0.373}_{-0.362} \times 10^{-12}$	2.81	56.20	100.9
Palomar 6		$1.137^{+0.173}_{-0.162} \times 10^{-11}$	2.20	≥ 100	94.9
Terzan 1	HP 2	$6.774^{+0.862}_{-0.871} \times 10^{-12}$	0.00	1.61	91.0
NGC 6139		$4.583^{+0.957}_{-0.911} \times 10^{-12}$	1.64	3.65	84.5
NGC 6254	M10	$2.295^{+0.407}_{-0.332} \times 10^{-12}$	0.00	0.92	79.7
Terzan 2	HP 3	$7.670^{+1.844}_{-1.672} \times 10^{-12}$	1.97	17.66	76.4
NGC 1851		$2.144^{+0.312}_{-0.314} \times 10^{-12}$	2.45	≥ 100	76.2
NGC 6544		$1.280^{+0.184}_{-0.193} \times 10^{-11}$	1.46	0.65	76.2
NGC 6341	M92	$1.334^{+0.340}_{-0.307} \times 10^{-12}$	1.38	1.95	73.7
NGC 6402	M14	$3.379^{+0.775}_{-0.706} \times 10^{-12}$	1.52	2.54	68.0

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Table 4.4 – continued from previous page

Globular Cluster	Alternate Name	Flux (erg/cm ² /s)	α	E_c (GeV)	TS
Terzan 9		$5.727^{+1.324}_{-0.947} \times 10^{-12}$	0.00	0.58	67.5
NGC 6397		$3.363^{+0.671}_{-0.653} \times 10^{-12}$	2.08	5.21	67.3
NGC 5904	M5	$1.552^{+0.432}_{-0.374} \times 10^{-12}$	1.15	2.30	66.8
Terzan 6	HP 5	$5.145^{+1.434}_{-1.163} \times 10^{-12}$	0.38	1.21	59.4
NGC 6273	M19	$2.853^{+0.837}_{-0.808} \times 10^{-12}$	1.88	21.03	53.1
Whiting 1		$1.156^{+0.298}_{-0.277} \times 10^{-12}$	1.37	15.60	52.4
Liller 1		$9.263^{+2.229}_{-2.224} \times 10^{-12}$	1.63	5.60	51.1
NGC 6528		$4.068^{+1.252}_{-1.128} \times 10^{-12}$	1.08	1.20	47.2
UKS 1		$1.068^{+0.199}_{-0.193} \times 10^{-11}$	0.23	0.25	45.6
NGC 362		$7.162^{+3.977}_{-1.624} \times 10^{-13}$	0.35	1.27	44.6
NGC 6637	M 69	$2.061^{+0.678}_{-0.675} \times 10^{-12}$	1.45	5.22	43.8
NGC 6723		$2.529^{+0.623}_{-0.612} \times 10^{-12}$	1.75	5.50	43.6
NGC 6218	M12	$3.024^{+0.657}_{-0.730} \times 10^{-12}$	2.39	≥ 100	42.9
NGC 6539		$6.541^{+1.290}_{-1.355} \times 10^{-12}$	2.66	≥ 100	39.0
NGC 5286		$2.418^{+0.660}_{-0.657} \times 10^{-12}$	2.23	18.15	35.8
NGC 6304		$2.557^{+1.101}_{-0.853} \times 10^{-12}$	0.92	1.69	34.8
NGC 6342		$3.671^{+0.874}_{-1.082} \times 10^{-12}$	2.32	≥ 100	33.1
NGC 6712		$2.381^{+1.123}_{-0.561} \times 10^{-12}$	0.00	0.67	31.4
NGC 6380	Ton 1	$2.617^{+0.886}_{-0.773} \times 10^{-12}$	0.75	1.77	29.9
NGC 6517		$4.119^{+1.208}_{-1.183} \times 10^{-12}$	1.76	2.06	29.2
IC 1257		$1.529^{+0.657}_{-0.468} \times 10^{-12}$	0.58	2.33	29.0
NGC 6522		$2.389^{+0.968}_{-0.588} \times 10^{-12}$	0.00	0.75	28.1
NGC 6540	Djorg 3	$2.841^{+1.163}_{-0.937} \times 10^{-12}$	1.16	1.95	28.1
NGC 6352		$2.532^{+0.596}_{-0.635} \times 10^{-12}$	0.00	0.25	27.9

Table 4.4. The gamma-ray flux (integrated between 0.1 – 100 GeV) and the best-fit spectral parameters for each of the 56 globular clusters that were detected in our analysis with TS > 25.

Globular Cluster	Alternate Name	Flux (erg/cm ² /s)	TS
NGC 6838	M 71	$< 5.65 \times 10^{-12}$	62.84
Lynga 7	BH184	$< 7.18 \times 10^{-12}$	54.87
AM 1	E 1	$< 1.47 \times 10^{-12}$	36.99
AM 4		$< 2.88 \times 10^{-12}$	34.07
Palomar 1		$< 2.00 \times 10^{-12}$	31.94
NGC 6809	M 55	$< 3.42 \times 10^{-12}$	27.05
Palomar 10		$< 6.40 \times 10^{-12}$	27.04

Table 4.5. The 2σ upper limits on the gamma-ray flux (integrated between 0.1 – 100 GeV) for globular clusters that were detected in our analysis with TS > 25, but with spectra that peak at energies below 0.4 GeV.

Table 4.6

Globular Cluster	Alternate Name	Flux (erg/cm ² /s)	TS
IC 4499		$< 1.83 \times 10^{-12}$	15.53
E 3		$< 2.31 \times 10^{-12}$	23.25
NGC 4372		$< 2.35 \times 10^{-12}$	20.94
NGC 6101		$< 1.88 \times 10^{-12}$	21.99
NGC 4833		$< 3.88 \times 10^{-12}$	0.00
NGC 6362		$< 2.52 \times 10^{-12}$	0.00
NGC 1261		$< 1.97 \times 10^{-12}$	0.00
NGC 6584		$< 1.04 \times 10^{-12}$	4.92
Rup 106		$< 3.80 \times 10^{-12}$	0.90
NGC 5927		$< 1.57 \times 10^{-12}$	3.78
NGC 5946		$< 4.57 \times 10^{-12}$	0.00
BH 176		$< 2.92 \times 10^{-12}$	7.08
FSR 1735		$< 1.35 \times 10^{-11}$	0.00
ESO-SC06	ESO280-SC06	$< 5.01 \times 10^{-12}$	0.00
NGC 3201		$< 6.69 \times 10^{-13}$	1.51
NGC 6496		$< 4.16 \times 10^{-12}$	0.00
Ton 2	Pismis 26	$< 3.39 \times 10^{-12}$	18.01
NGC 5986		$< 2.77 \times 10^{-12}$	0.00
Pyxis		$< 3.64 \times 10^{-12}$	1.86

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Globular Cluster	Alternate Name	Flux (erg/cm ² /s)	TS
NGC 6256		$< 7.57 \times 10^{-12}$	0.00
NGC 2298		$< 2.72 \times 10^{-12}$	0.14
Terzan 3		$< 5.80 \times 10^{-12}$	0.06
Terzan 7		$< 3.59 \times 10^{-12}$	0.00
NGC 6453		$< 3.23 \times 10^{-12}$	10.92
Terzan 8		$< 4.20 \times 10^{-12}$	2.46
NGC 5824		$< 7.52 \times 10^{-13}$	2.91
Djorg 1		$< 1.45 \times 10^{-11}$	20.69
NGC 6681	M 70	$< 1.75 \times 10^{-12}$	17.92
NGC 6558		$< 1.61 \times 10^{-12}$	2.42
Terzan 4	HP 4	$< 1.30 \times 10^{-11}$	0.00
NGC 6715	M 54	$< 2.17 \times 10^{-12}$	20.52
Arp 2		$< 2.81 \times 10^{-12}$	15.95
HP 1	BH 229	$< 5.26 \times 10^{-12}$	22.36
BH 261	AL 3	$< 2.09 \times 10^{-12}$	6.06
1636-283	ESO452-SC11	$< 2.73 \times 10^{-12}$	23.07
Djorg 2	ESO456-SC38	$< 2.86 \times 10^{-12}$	3.22
NGC 4590	M 68	$< 2.93 \times 10^{-12}$	0.00
NGC 288		$< 2.36 \times 10^{-12}$	0.03
NGC 6293		$< 7.33 \times 10^{-12}$	20.31
NGC 5694		$< 3.57 \times 10^{-12}$	5.87
NGC 6121	M 4	$< 1.90 \times 10^{-12}$	6.55
NGC 6355		$< 9.45 \times 10^{-12}$	0.00
Terzan 10		$< 9.44 \times 10^{-12}$	0.00
NGC 6144		$< 4.89 \times 10^{-12}$	0.00
NGC 6553		$< 1.66 \times 10^{-12}$	1.34
NGC 6638		$< 2.04 \times 10^{-12}$	9.93
NGC 6284		$< 3.67 \times 10^{-12}$	0.00
NGC 6401		$< 4.23 \times 10^{-12}$	15.46
NGC 6325		$< 7.45 \times 10^{-12}$	1.42
NGC 6642		$< 7.30 \times 10^{-12}$	0.66
NGC 7099	M 30	$< 3.44 \times 10^{-12}$	7.64

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Table 4.6 – continued from previous page

Globular Cluster	Alternate Name	Flux (erg/cm ² /s)	TS
Terzan 12		$< 1.34 \times 10^{-11}$	0.00
NGC 6287		$< 6.53 \times 10^{-12}$	5.51
NGC 6235		$< 5.46 \times 10^{-12}$	0.00
NGC 6864	M 75	$< 3.06 \times 10^{-12}$	20.48
Palomar 12		$< 1.04 \times 10^{-12}$	11.81
Eridanus		$< 6.93 \times 10^{-13}$	6.00
NGC 5897		$< 3.05 \times 10^{-12}$	0.00
2MS-GC02	2MASS-GC02	$< 2.97 \times 10^{-12}$	0.59
Palomar 8		$< 7.56 \times 10^{-12}$	1.39
NGC 6333	M 9	$< 3.95 \times 10^{-12}$	0.00
NGC 6356		$< 4.14 \times 10^{-12}$	0.00
NGC 7492		$< 7.41 \times 10^{-13}$	8.29
NGC 6171	M 107	$< 4.21 \times 10^{-12}$	0.00
NGC 6981	M 72	$< 2.81 \times 10^{-12}$	0.00
Palomar 11		$< 4.69 \times 10^{-12}$	0.00
IC 1276	Palomar 7	$< 9.51 \times 10^{-12}$	0.00
NGC 5634		$< 2.62 \times 10^{-12}$	0.00
NGC 6366		$< 1.91 \times 10^{-12}$	0.77
NGC 7089	M 2	$< 2.52 \times 10^{-12}$	0.75
Palomar 15		$< 1.60 \times 10^{-12}$	13.98
NGC 6535		$< 4.04 \times 10^{-12}$	18.22
Palomar 5		$< 8.43 \times 10^{-13}$	1.28
Palomar 3		$< 2.82 \times 10^{-12}$	2.17
NGC 6760		$< 8.63 \times 10^{-12}$	0.00
NGC 6749		$< 1.01 \times 10^{-11}$	0.00
NGC 6426		$< 5.67 \times 10^{-12}$	4.99
NGC 6934		$< 2.73 \times 10^{-12}$	0.00
Ko 1		$< 2.73 \times 10^{-12}$	0.06
Palomar 13		$< 3.99 \times 10^{-12}$	0.26
Palomar 14	AvdB	$< 2.68 \times 10^{-12}$	0.01
NGC 7006		$< 1.31 \times 10^{-12}$	7.41
NGC 5053		$< 2.04 \times 10^{-12}$	17.89

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Table 4.6 – continued from previous page

Globular Cluster	Alternate Name	Flux (erg/cm ² /s)	TS
NGC 5024	M 53	$< 3.63 \times 10^{-12}$	0.58
NGC 4147		$< 6.24 \times 10^{-13}$	3.16
Ko 2		$< 2.96 \times 10^{-12}$	1.22
NGC 5272	M 3	$< 2.70 \times 10^{-12}$	10.28
NGC 5466		$< 2.33 \times 10^{-12}$	6.05
Palomar 4		$< 2.58 \times 10^{-12}$	0.00
NGC 6779	M 56	$< 1.75 \times 10^{-12}$	9.53
Palomar 2		$< 4.40 \times 10^{-12}$	0.00
NGC 6205	M 13	$< 5.64 \times 10^{-13}$	12.24
NGC 2419		$< 2.76 \times 10^{-12}$	0.00
NGC 6229		$< 4.51 \times 10^{-13}$	4.16

Table 4.6. The 2σ upper limits on the gamma-ray flux (integrated between 0.1 – 100 GeV) for those globular clusters that were not significantly detected in our analysis (TS < 25).

Table 4.7

Globular Cluster	Alt. Name	Distance (kpc)	Luminosity ($L_{\odot}$)	Γ_e
NGC 104	47 Tuc	4.46	5.01×10^5	$1.00^{+0.15}_{-0.13}$
NGC 362		8.61	2.01×10^5	$0.74^{+0.14}_{-0.12}$
NGC 1261		16.29	1.13×10^5	$0.015^{+0.011}_{-0.004}$
Palomar 2		27.11	1.32×10^5	$0.93^{+0.84}_{-0.56}$
NGC 1851		12.07	1.84×10^5	$1.53^{+0.20}_{-0.19}$
NGC 1904	M79	12.94	1.19×10^5	$0.12^{+0.019}_{-0.014}$
NGC 2419		82.49	5.01×10^5	$0.0028^{+0.0008}_{-0.0005}$
NGC 2808		9.59	4.88×10^5	$0.92^{+0.067}_{-0.083}$
NGC 4147		19.30	2.51×10^4	$0.017^{+0.013}_{-0.006}$
NGC 4372		5.81	1.12×10^5	$0.0002^{+0.0004}_{-0.0001}$
NGC 5024	M53	17.85	2.61×10^5	$0.035^{+0.012}_{-0.010}$
NGC 5139	Omega Centauri	5.17	1.09×10^6	$0.090^{+0.027}_{-0.020}$
NGC 5272	M3	10.18	3.05×10^5	$0.19^{+0.033}_{-0.018}$
NGC 5286		11.67	2.68×10^5	$0.46^{+0.058}_{-0.061}$
NGC 5634		25.18	1.02×10^5	$0.020^{+0.014}_{-0.008}$

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Table 4.7 – continued from previous page

Globular Cluster	Alt. Name	Distance (kpc)	Luminosity ($L_{\odot}$)	Γ_e
NGC 5694		35.01	1.16×10^5	$0.19^{+0.052}_{-0.034}$
NGC 5824		32.17	2.96×10^5	$0.98^{+0.17}_{-0.16}$
NGC 5904	M5	7.47	2.86×10^5	$0.16^{+0.039}_{-0.030}$
NGC 5927		7.67	1.14×10^5	$0.068^{+0.013}_{-0.010}$
NGC 5946		10.55	6.37×10^4	$0.13^{+0.034}_{-0.045}$
NGC 5986		10.43	2.03×10^5	$0.062^{+0.016}_{-0.010}$
NGC 6093	M80	10.01	1.67×10^5	$0.53^{+0.059}_{-0.069}$
NGC 6121	M4	2.22	6.43×10^4	$0.027^{+0.012}_{-0.010}$
NGC 6139		10.12	1.89×10^5	$0.31^{+0.094}_{-0.089}$
NGC 6205	M13	7.14	2.25×10^5	$0.069^{+0.018}_{-0.015}$
NGC 6229		30.46	1.43×10^5	$0.048^{+0.031}_{-0.009}$
NGC 6218	M12	4.83	7.18×10^4	$0.013^{+0.0054}_{-0.0040}$
NGC 6254	M10	4.39	8.39×10^4	$0.031^{+0.0043}_{-0.0041}$
NGC 6256		10.28	6.19×10^4	$0.17^{+0.119}_{-0.060}$
NGC 6266	M62	6.83	4.02×10^5	$1.67^{+0.71}_{-0.57}$
NGC 6273	M19	8.80	3.84×10^5	$0.20^{+0.067}_{-0.039}$
NGC 6284		15.29	1.31×10^5	$0.67^{+0.12}_{-0.11}$
NGC 6287		9.38	7.52×10^4	$0.036^{+0.0077}_{-0.0077}$
NGC 6293		9.48	1.11×10^5	$0.85^{+0.37}_{-0.24}$
NGC 6304		5.88	7.11×10^4	$0.12^{+0.054}_{-0.022}$
NGC 6316		10.45	1.85×10^5	$0.077^{+0.025}_{-0.015}$
NGC 6341	M92	8.27	1.64×10^5	$0.27^{+0.030}_{-0.029}$
NGC 6325		7.83	5.20×10^4	$0.12^{+0.045}_{-0.046}$
NGC 6333	M9	7.91	1.29×10^5	$0.13^{+0.059}_{-0.042}$
NGC 6342		8.53	3.16×10^4	$0.045^{+0.014}_{-0.013}$
NGC 6356		15.08	2.17×10^5	$0.088^{+0.020}_{-0.014}$
NGC 6355		9.22	1.45×10^5	$0.10^{+0.041}_{-0.026}$
Terzan 2	HP 3	7.49	1.92×10^4	$0.022^{+0.029}_{-0.014}$
NGC 6380	Ton 1	10.88	8.55×10^4	$0.12^{+0.068}_{-0.045}$
NGC 6388		9.92	4.97×10^5	$0.90^{+0.24}_{-0.21}$
NGC 6402	M14	9.25	3.73×10^5	$0.12^{+0.032}_{-0.030}$
NGC 6401		10.56	1.24×10^5	$0.044^{+0.011}_{-0.011}$

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Table 4.7 – continued from previous page

Globular Cluster	Alt. Name	Distance (kpc)	Luminosity ($L_{\odot}$)	Γ_e
NGC 6397		2.30	3.87×10^4	$0.084^{+0.018}_{-0.018}$
Palomar 6		5.79	4.45×10^4	$0.016^{+0.013}_{-0.008}$
Terzan 5	Terzan 11	5.98	7.94×10^4	$6.80^{+1.04}_{-3.02}$
NGC 6440		8.45	2.70×10^5	$1.40^{+0.63}_{-0.48}$
NGC 6441		11.60	6.08×10^5	$2.30^{+0.97}_{-0.64}$
Terzan 6	HP 5	6.78	9.29×10^4	$2.47^{+5.07}_{-1.72}$
NGC 6453		11.57	6.61×10^4	$0.37^{+0.13}_{-0.09}$
NGC 6517		10.63	1.71×10^5	$0.34^{+0.15}_{-0.10}$
NGC 6522		7.70	9.82×10^4	$0.36^{+0.11}_{-0.10}$
NGC 6528		7.93	3.63×10^4	$0.28^{+0.11}_{-0.05}$
NGC 6539		7.79	1.77×10^5	$0.042^{+0.029}_{-0.015}$
NGC 6544		2.96	5.11×10^4	$0.11^{+0.068}_{-0.037}$
NGC 6541		7.54	2.19×10^5	$0.39^{+0.095}_{-0.063}$
NGC 6553		5.96	1.10×10^5	$0.069^{+0.027}_{-0.019}$
NGC 6558		7.37	3.22×10^4	$0.11^{+0.026}_{-0.019}$
NGC 6569		10.90	1.75×10^5	$0.054^{+0.030}_{-0.021}$
NGC 6584		13.49	1.02×10^5	$0.012^{+0.0054}_{-0.0034}$
NGC 6624		7.91	8.47×10^4	$1.15^{+0.11}_{-0.18}$
NGC 6626	M28	5.52	1.57×10^5	$0.65^{+0.084}_{-0.091}$
NGC 6638		9.41	6.03×10^4	$0.14^{+0.039}_{-0.027}$
NGC 6637	M69	8.80	9.73×10^4	$0.090^{+0.036}_{-0.018}$
NGC 6642		8.13	3.94×10^4	$0.10^{+0.031}_{-0.025}$
NGC 6652		10.00	3.94×10^4	$0.70^{+0.29}_{-0.19}$
NGC 6656	M22	3.23	2.15×10^5	$0.078^{+0.032}_{-0.026}$
NGC 6681	M 70	9.01	6.03×10^4	$1.04^{+0.27}_{-0.19}$
NGC 6712		6.93	8.55×10^4	$0.031^{+0.0056}_{-0.0066}$
NGC 6715	M54	26.49	8.39×10^5	$2.52^{+0.23}_{-0.27}$
NGC 6717	Palomar 9	7.11	1.57×10^4	$0.040^{+0.022}_{-0.014}$
NGC 6723		8.65	1.16×10^5	$0.011^{+0.0080}_{-0.0044}$
NGC 6752		3.99	1.06×10^5	$0.40^{+0.18}_{-0.13}$
NGC 6760		7.36	1.17×10^5	$0.057^{+0.027}_{-0.019}$
NGC 6779	M 56	9.44	7.87×10^4	$0.028^{+0.012}_{-0.009}$

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Table 4.7 – continued from previous page

Globular Cluster	Alt. Name	Distance (kpc)	Luminosity ($L_\odot$)	Γ_e
Palomar 10		5.93	1.77×10^4	$0.059^{+0.043}_{-0.036}$
Palomar 11		13.40	5.01×10^4	$0.021^{+0.011}_{-0.007}$
NGC 6864	M 75	20.84	2.29×10^5	$0.31^{+0.095}_{-0.082}$
NGC 6934		15.63	8.17×10^4	$0.030^{+0.012}_{-0.008}$
NGC 7006		41.21	1.00×10^5	$0.0094^{+0.0049}_{-0.0033}$
NGC 7078	M 15	10.38	4.06×10^5	$4.51^{+1.36}_{-0.99}$
NGC 7089	M 2	11.56	3.50×10^5	$0.52^{+0.078}_{-0.071}$
NGC 7099	M 30	8.12	8.17×10^4	$0.32^{+0.12}_{-0.08}$

Table 4.7. The distance, visible luminosity, and stellar encounter rate [206] for each of the 87 globular clusters with either $L_V > 10^5 L_\odot$ or $\Gamma_e > 0.01$. Throughout this paper, we use values of the stellar encounter rate that are normalized such that $\Gamma_e = 1$ for NGC 104.

4.A.1 Comparison With Previous Work

In this section, we provide a direct comparison, illustrated in Figure 4.9, between our results (Figure 4.7) and those from the prior 2016 work by Hooper and Linden [173]. Their analysis was based on 7 years of Fermi data, in contrast to the 15.8 years of data utilized in our work. The 2016 study favored an MSP gamma-ray luminosity function with parameters of $\langle L_\gamma \rangle \sim (0.7 - 6) \times 10^{34} \text{ erg/s}$ and $\sigma_L \sim 1 - 3$.

While our results are statistically consistent with theirs at the $\sim 2\sigma$ level, our analysis prefers smaller values for $\langle L_\gamma \rangle$ and larger values for R_{MSP} by a factor of $\sim 2 - 8$. Furthermore, although the best-fit values for both distributions lie near the $N_{\text{MSP}} = 30$ curve, our findings suggest a narrower range of $N_{\text{MSP}} \sim 17 - 37$, compared to the previous estimate of $N_{\text{MSP}} \sim 12 - 50$. Our updated analysis refines the prior work by both encompassing the findings of Ref. [173] as far as N_{MSP} is concerned, and providing tighter constraints on the number of MSPs required to explain the Galactic Center Excess.

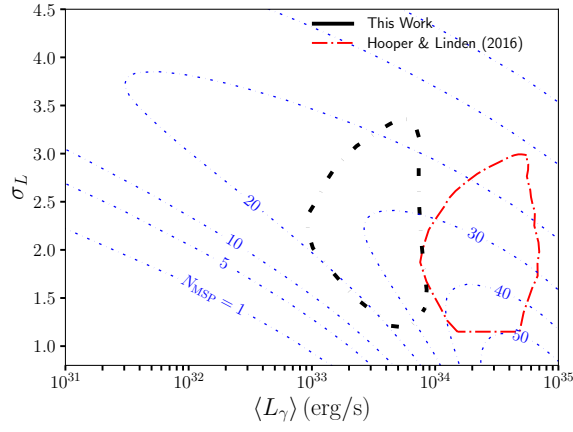


Fig. 4.9. As in Figure 4.7, for the 2σ contours, but comparing the results from the prior 2016 work from Hooper and Linden [173] with our most recent findings.

SEARCHING FOR DARK MATTER SUBSTRUCTURES

The preceding chapter examined the Galactic Center Excess, the brightest expected signature of dark matter annihilation in gamma rays. That analysis confronted a difficulty common to all Galactic Center studies: disentangling a potential dark matter signal from the complex astrophysical emission of the inner Milky Way, where overlapping source populations and uncertain diffuse backgrounds prevent definitive conclusions. We now turn to a complementary search strategy, one that targets dark matter in a very different environment.

Rather than analyzing a diffuse excess embedded in the bright Galactic foreground, this chapter asks whether any of the point sources detected by the Fermi Large Area Telescope could be dark matter subhalos. The Λ CDM cosmological model predicts that the Milky Way’s dark matter halo hosts a vast population of gravitationally bound substructures [95, 96]. The great majority of these subhalos have masses well below the threshold for retaining baryonic matter [219] and consequently produce no electromagnetic radiation through conventional astrophysical processes. If, however, the dark matter particle annihilates into Standard Model final states, the densest of these subhalos could generate detectable gamma-ray emission. They would then appear in Fermi-LAT catalogs as unassociated point sources lacking any multi-wavelength counterpart.

This chapter proceeds as follows. We first review the theoretical predictions for the Milky Way’s subhalo population within Λ CDM (Section 5.1) and their expected gamma-ray properties (Section 5.2). We then introduce the unassociated source problem in the fourth Fermi-LAT source catalog, survey previous attempts to identify dark matter subhalo candidates through classification-based methods, and explain why the dataset shift between associated and unassociated source populations necessitates the quantification learning framework developed in Section 3.4 (Section 5.3). The remainder of the chapter presents the full analysis, following [220].

5.1 Dark Matter Substructure in Λ CDM

Before searching for dark matter subhalos among gamma-ray sources, we must first understand what Λ CDM predicts for the subhalo population of a Milky Way-like galaxy: how many subhalos are expected, at what masses, and how their abundance is shaped by baryonic processes. This

section reviews these predictions, establishing the theoretical foundation for the detectability analysis that follows in Section 5.2.

5.1.1 Hierarchical Structure Formation

A defining prediction of Λ CDM cosmology is that cosmic structure assembles in a bottom-up fashion. The primordial power spectrum, processed through the radiation-dominated epoch, retains more power on small spatial scales, so that smaller overdensities reach the nonlinear regime and gravitationally collapse before larger ones [221]. The resulting bound objects—dark matter halos—then grow through the continuous accretion of diffuse matter and the merging of pre-existing smaller halos, a process known as hierarchical structure formation [222].

When a smaller halo is accreted by a larger host, it does not dissolve entirely into a smooth background. Because early-forming halos are highly concentrated, their dense inner cores can withstand the tidal forces of the host and survive as distinct, gravitationally bound substructures [19]. The outer layers of the infalling halo are preferentially stripped away and absorbed into the ambient host, but the tightly bound core persists as a subhalo. As a result, Λ CDM predicts that Milky Way-mass halos are populated by a large number of subhalos spanning a wide range of masses.

The quantitative characterization of this subhalo population has been one of the major achievements of cosmological N-body simulations. Shortly before the turn of the century, numerical resolution reached the level necessary to robustly demonstrate the survival of subhalos within their host halos [19]. Since then, dedicated high-resolution simulations have mapped the substructure of Milky Way-like galaxies in detail. Two benchmark projects are the Via Lactea II (VL-II) simulation [96], which resolves subhalos down to approximately $10^4 M_\odot$, and the Aquarius project [95], which achieves comparable resolution and has been widely used to calibrate subhalo abundance predictions.

These simulations reveal a subhalo mass function that rises steeply toward lower masses, implying that the total number of subhalos in a Milky Way-like galaxy is vast [5, 19]. The theoretical mass function extends far below the resolution floor of current simulations; for a canonical 100 GeV WIMP, the smallest bound structures could have masses as low as $\sim 10^{-8} M_\odot$, with the precise value spanning $\sim 10^{-11}$ – $10^{-3} M_\odot$ depending on the WIMP model [5, 223]. The quantitative predictions for the subhalo population, including the J-factor distributions used in this chapter, are derived from the VL-II simulation extended through repopulation techniques [224] and are discussed in detail in Section 5.4.6.

5.1.2 Luminous Satellites vs. Dark Subhalos

Not all subhalos are alike from an observational standpoint. Above a mass threshold of approximately $10^8 M_\odot$, a subhalo possesses sufficient gravitational potential to accrete and retain baryonic matter, enabling star formation and producing a visible dwarf galaxy [219]. Roughly 50 such satellite galaxies are currently known within the virial radius of the Milky Way [19], ranging from the classical dwarf spheroidals—Draco, Sculptor, Fornax—to the ultra-faint systems discovered over the past two decades by digital sky surveys such as the Sloan Digital Sky Survey and the Dark Energy Survey [19]. These dwarf spheroidal galaxies are among the most dark-matter-dominated objects known, with mass-to-light ratios exceeding ~ 1000 in some cases, and they have been extensively used as targets for indirect dark matter searches in gamma rays (cf. Section 1.4.5).

Below $\sim 10^8 M_\odot$, however, galaxy formation becomes progressively inefficient. The ultraviolet background produced by cosmic reionization suppresses gas accretion in halos with virial masses

below $\sim 10^9 M_\odot$, while atomic cooling ceases to operate below $\sim 10^8 M_\odot$ [19]. The vast majority of dark matter subhalos therefore contain no stars, no gas, and produce no conventional electromagnetic signature. These are the truly “dark” subhalos: structures whose existence is predicted with high confidence by Λ CDM but which cannot be detected through traditional astronomical observations. If dark matter annihilates, however, these objects could shine in gamma rays, and it is precisely this population that we target in the analysis presented in this chapter.

The apparent discrepancy between the predicted abundance of subhalos and the observed number of satellite galaxies was historically known as the “missing satellite problem” [19]. This tension has been largely resolved on two fronts. Observationally, improved surveys have steadily increased the census of ultra-faint dwarf galaxies, and statistical corrections for sky coverage and detection efficiency suggest that the total satellite population is larger than the pre-2004 sample indicated [19]. Theoretically, hydrodynamical simulations including baryonic physics—UV feedback, supernova-driven outflows, reionization—naturally suppress galaxy formation in low-mass halos, bringing predictions into broad agreement with observations once detection efficiencies are accounted for [5].

Baryonic processes also modify the survival of the subhalos themselves. Tidal stripping and gravitational shocks from the Galactic disk preferentially deplete the inner Galaxy of substructure [5, 225, 226], and the survival rate of low-mass subhalos below $\sim 10^6 M_\odot$ remains uncertain [227]. This uncertainty propagates directly into predictions for the number of gamma-ray-detectable subhalos and is discussed further in Section 5.4.6.

5.2 Dark Matter Subhalos as Gamma-Ray Targets

Having established that Λ CDM predicts a vast population of dark subhalos orbiting the Milky Way, we now turn to the question of whether any of these objects could be observed in gamma rays. The gamma-ray flux from an annihilating subhalo depends on both the particle-physics parameters— m_{DM} , $\langle\sigma v\rangle$, and the annihilation channel—and on the astrophysical properties of the subhalo, encoded in its J-factor. The general formalism for the annihilation flux and J-factor was developed in Section 1.4.4; here we highlight those features of the expected signal that are specific to the subhalo population and that underpin the search strategy of this chapter.

The most important such feature is *spectral uniformity*. For a fixed dark matter mass and annihilation channel, all subhalos share the same intrinsic energy spectrum; only the overall normalization differs, set by the J-factor of each individual object. Astrophysical source classes, by contrast, exhibit broad, heterogeneous spectral distributions driven by diverse physical conditions: extragalactic sources display a wide range of nearly power-law spectra, while Galactic sources such as pulsars present highly curved spectra spanning a similarly broad parameter range [228]. A population of dark matter subhalos would therefore occupy a localized region in the spectral parameter space, with the observed scatter arising mainly from statistical measurement uncertainties rather than intrinsic physical differences. This population-level coherence is what enables the statistical approach developed later in this chapter.

The expected sky distribution of dark matter subhalos provides a complementary handle. Because subhalos trace the roughly spherical dark matter halo of the Milky Way, their projected distribution at high Galactic latitudes ($|b| > 10^\circ$) is approximately isotropic [229, 230]. This contrasts with Galactic source populations, which concentrate along the Galactic plane, but resembles the distribution of extragalactic sources. The spectral properties introduced above therefore become the primary discriminant: the combination of isotropic spatial distribution and spectral homogeneity provides the statistical foundation for the population-level search.

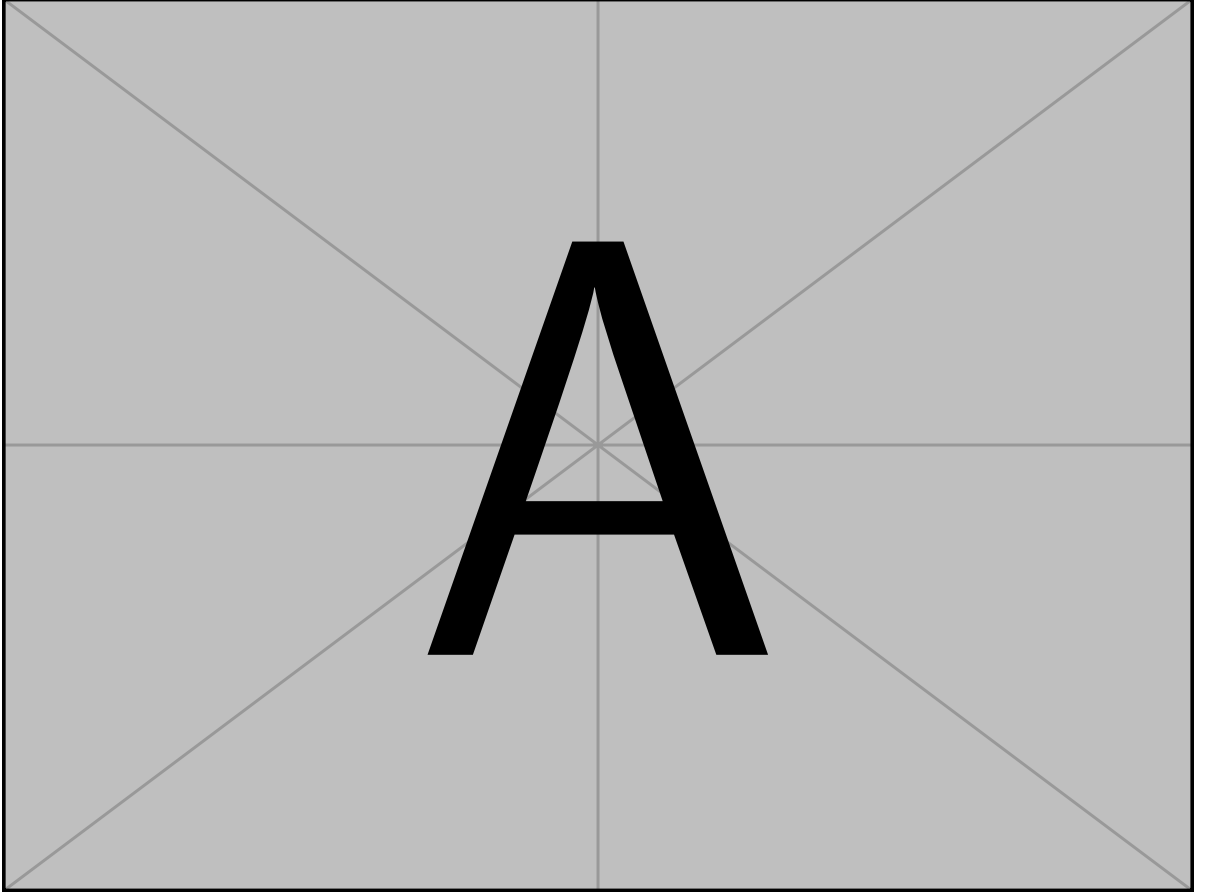


Fig. 5.1. Dark matter substructure in a Milky Way-like halo from an N-body cosmological simulation, illustrating the vast population of subhalos predicted by Λ CDM. Subhalos above $\sim 10^8 M_\odot$ can host dwarf satellite galaxies, while the far more numerous low-mass subhalos remain dark. From [96].

The expected number of Fermi-LAT-detectable subhalos spans several orders of magnitude, depending on the assumed annihilation cross section, the J-factor model, and the survival rate of low-mass subhalos in the presence of baryonic tidal effects [229–231]. Published predictions range from zero to several tens of detectable objects. This wide spread is not a weakness but a motivation: it defines the parameter space that a data-driven search must explore. The quantitative J-factor distributions and the detection pipeline are presented in the paper body (Section 5.4.6 and Section 5.4.7).

The analysis proceeds in two stages. First, we ask whether the observed population of unassociated Fermi-LAT sources is better described by a model that includes a dark matter subhalo component or by one containing only known astrophysical source classes. If no statistically significant subhalo contribution is found, the profile likelihood is used to set 95% confidence upper bounds on $\langle\sigma v\rangle$ as a function of m_{DM} . Both possible outcomes are scientifically informative: a positive detection would provide direct evidence for annihilating dark matter in the Λ CDM substructure, while a null result constrains $\langle\sigma v\rangle$ and provides an independent cross-check of the limits obtained from dwarf spheroidal galaxies [103]. Dark subhalos are, in this respect, exceptionally clean targets: unlike the Galactic Center (Chapter 4), a subhalo below $\sim 10^8 M_\odot$

contains no stars or gas, and any detected gamma-ray emission would originate entirely from dark matter annihilation.

5.3 The Unassociated Source Problem

Having established the theoretical grounds for dark matter subhalo searches in the previous sections, we now turn to the observational reality. The Fermi-LAT catalogs contain a large population of gamma-ray sources with no identified counterpart at other wavelengths, and these unassociated sources have long been recognized as the natural testing ground for DM subhalo searches.

5.3.1 Unassociated Sources in Fermi-LAT Catalogs

The latest incremental release, the 4FGL-DR4 catalog [123, 228], contains approximately 7,200 sources detected above a significance threshold of 4σ . Of these, **2,428 sources**—roughly one third—remain unassociated, meaning that no plausible counterpart has been identified at other wavelengths [123]. Among the unassociated population, **1,282 sources** lie at Galactic latitudes $|b| > 10^\circ$, where contamination from diffuse Galactic emission is reduced and the search for exotic source populations becomes more tractable. [AA: check these number again in the paper]

Association completeness depends on both source brightness and Galactic latitude: brighter sources are far easier to link to multi-wavelength counterparts, while the association fraction drops from approximately 85% at high latitudes to around 40% near the Galactic plane, where dense source fields and X-ray absorption impede counterpart identification [121]. The majority of unassociated sources are therefore expected to be ordinary astrophysical objects whose association has failed for instrumental or geometric reasons. Their spectral properties offer direct clues to their nature: at high latitudes, the average photon index ($\Gamma \approx 2.24$) matches that of blazars of unknown type, while closer to the Galactic plane the spectra soften ($\Gamma \approx 2.42$), consistent with pulsars [121]. Further discussion of the Fermi-LAT association procedure and its completeness can be found in Section 2.3.1. [AA: update link in the future when ch2 is finalized]

To date, no unassociated Fermi-LAT source has been confirmed as a dark matter subhalo [121, 228, 229, 232]. [AA: recently, there have been some claims of DM subhalo detection, though the community remains skeptical. Papers to check and cite: arXiv:2507.16932 and arXiv:2409.19493. Once added, revisit the “no confirmed” statement in the line above — may need softening to “no unambiguously confirmed” or a qualifying clause about contested candidate claims.] As discussed in Section 5.2, even under optimistic assumptions the most detectable subhalos would produce gamma-ray spectra similar to those of pulsars or soft-spectrum blazars, making individual identification on spectral grounds alone extremely difficult. Nevertheless, the large unassociated population has motivated a sustained effort to search for DM subhalo candidates within it.

5.3.2 Previous DM Subhalo Searches and the Failure of Classify-and-Count

[AA: this is a strong statement, review it and maybe tone it down or better scope it] All prior searches for dark matter subhalos among Fermi-LAT unassociated sources share a common strategy, referred to as *classify-and-count*: a set of DM subhalo candidates is selected from the unassociated population based on their gamma-ray properties, and upper limits on $\langle\sigma v\rangle$ are derived under the assumption that the number of true subhalos cannot exceed the candidate count [229–246]. Early analyses relied on hand-crafted spectral and spatial cuts applied to successive catalog releases [232–235, 239, 240, 242], with some studies emphasizing spatial

extension as a potential discriminant [232, 239, 244]. More recently, machine learning classifiers have automated and extended the candidate selection [230, 241, 245, 246]. In all cases, none of the identified candidates could be unambiguously distinguished from standard astrophysical sources.

The classify-and-count approach, whether implemented with hand-crafted cuts or ML classifiers, suffers from three interrelated problems that compromise the statistical rigor of the resulting constraints. First, no dark matter subhalo has ever been identified, so the “DM” class used in supervised classification has no empirical anchor and the resulting classification probabilities cannot be calibrated against reality. [AA: review this statement] Second, the posterior probability $p(k|\mathbf{x})$ produced by any discriminative classifier absorbs the class prevalences of its training set (Section 3.4.2). When classifiers are trained on balanced datasets—with artificially equal fractions of AGNs, pulsars, and simulated DM sources—the resulting predictions reflect these imposed priors rather than the true composition of the gamma-ray sky. The same prior shift problem arises when a model trained on associated sources, where extragalactic objects dominate, is applied to the unassociated population, which has a substantially higher Galactic fraction. Correcting this bias requires a realistic testing set that mirrors the target population, but in the classify-and-count framework such a set cannot be constructed: the true class prevalences among unassociated sources are unknown, and for the DM class no confirmed example exists at all. Third, the number of DM candidates depends on an ad hoc probability threshold $p_{\text{DM}} > p_{\text{threshold}}$ whose value carries no statistical justification: different thresholds yield candidate counts that can differ by an order of magnitude and, consequently, different physical bounds on $\langle\sigma v\rangle$. The resulting upper limits lack a well-defined confidence level—they represent an assumption ($N_{\text{DM}} \leq N_{\text{candidates}}$) rather than a statistical inference.

5.3.3 The Quantification Learning Alternative

The failure of classify-and-count is conceptual, not technical: the approach attempts to answer a population-level question—how many DM subhalos contribute to the unassociated catalog?—using a tool designed for individual-level decisions. If we had access to an already well established population of DM subhalos, we could train a machine learning model to identify new ones, but that is currently not the case. A different framework is needed, one that models the composition of the unassociated population as a whole and fits the DM contribution directly to the data.

The formal machinery for this problem was introduced in Section 3.4: *quantification learning*. Standard classifiers estimate the posterior probability $p(k|\mathbf{x})$ —the probability that a source belongs to class k given its observed features—and are therefore designed to label individual objects. However, as shown in Section 3.4.2, the posterior absorbs the class prevalences of the training set: through Bayes’ theorem, $p(k|\mathbf{x}) \propto p(\mathbf{x}|k)p(k)$, so any change in the underlying class fractions between the associated and unassociated populations invalidates the classifier’s predictions. Quantification learning inverts the question. Instead of asking “to which class does this source belong? ”, it asks “what is the distribution of observed features within each class?”—that is, it estimates the class-conditional densities $p(\mathbf{x}|k)$. These densities describe the intrinsic physical profile of each source population: the spectral shape, curvature, and flux distribution that characterize, say, blazars or pulsars as a class. Because the physical properties of a source class do not change when its relative abundance changes, $p(\mathbf{x}|k)$ is invariant under prior shift ($p_{\text{assoc}}(\mathbf{x}|k) = p_{\text{unas}}(\mathbf{x}|k)$; see Equation 3.4.5). This invariance is what makes the class-conditional densities learnable from the associated catalog and directly portable to the unassociated population.

The shift from $p(k|\mathbf{x})$ to $p(\mathbf{x}|k)$ carries a deeper conceptual consequence: the unit of analysis moves from the individual source to the population. A classifier assigns each source a label and is only as good as the labels in its training set—labels that, for a class never observed (dark matter subhalos), simply do not exist. A generative model, by contrast, characterizes each class through its feature distribution and then models the target data as a mixture of these distributions, $\tilde{p}_{\text{unas}}(\mathbf{x}) = \sum_k \pi_k p(\mathbf{x}|k)$. In this formulation, the class prevalences π_k are free parameters determined by fitting the model to the data rather than imposed during training [160, 161]. Maximizing the product likelihood over all unassociated sources simultaneously constrains every mixing fraction, naturally accommodates a new class (dark matter subhalos) whose prevalence was zero in the training set, and supports statistically rigorous confidence intervals through the profile likelihood [220].

In practice, prior shift and covariate shift between the associated and unassociated populations act simultaneously (Section 3.4.2). The covariate shift arises from observational biases—it is easier to associate bright, high-latitude sources—and is therefore an instrumental effect that depends on the measured features of a source rather than on its intrinsic physical class. Accordingly, the sigmoid modulation functions $\tilde{C}(\mathbf{x}; \boldsymbol{\theta}_{\text{cov}})$ that correct for this bias are shared across all astrophysical classes, and their parameters are fitted jointly with the class prevalences. The results of the analysis show that prior shift is the dominant effect driving the differences between the two populations: the relative fraction of Galactic sources rises from 6% in the associated sample to 29% among the unassociated sources (table 5.1), consistent with the higher density of unassociated sources near the Galactic plane. A quantitative comparison presented in appendix 5.C.1 confirms that the prior shift alone is preferred at 7σ significance over a model with no dataset shift, while the covariate shift alone reaches 4.5σ ; the covariate shift correction is thus statistically significant but acts as a secondary refinement. The full construction of this mixture model, including the kernel density estimation of astrophysical templates, the Monte Carlo simulation of the DM component, and the optimization via the Expectation-Maximization algorithm, is presented in the paper body that follows.

[AA: Follows the paper:]

5.4 Search for dark matter subhalos among unassociated Fermi-LAT sources in presence of dataset shift

In this paper, we construct for the first time a statistical model for the distribution of unassociated sources at latitudes $|b| > 10^\circ$.¹ It consists of a mixture model, i.e. a linear combination of probability distribution functions (PDFs) of classes of astrophysical sources plus a contribution of DM subhalos. The PDFs of classes of astrophysical sources are determined from the associated sources, while the distribution of DM subhalos is determined from Monte Carlo simulations assuming a model of J-factors from N-body DM simulations and particular parameters for DM (mass, annihilation cross section, and channel). The prior shift problem is addressed by fitting the relative contribution of the different components to the data (the distribution of unassociated

¹We exclude sources with $|b| < 10^\circ$, since the main background in this analysis is the astrophysical sources among the unassociated ones. Almost half of the unassociated sources (1146 out of 2428) are within this range of latitudes and most of them are expected to be the astrophysical Galactic sources. The decrease in the area, on the other hand, is about 17%, i.e., assuming an isotropic distribution of DM subhalos around the Earth, the increase in signal-to-noise ratio due to masking of the Galactic plane can be estimated as $(1 - 0.17)\sqrt{2428/1146} \approx 1.14$. This increase is likely larger due to higher flux detection threshold (smaller detectability volume for DM subhalos) toward the Galactic plane, i.e., the decrease in signal is less than 17% as a result of masking the Galactic plane.

sources), while the covariate shift problem is addressed by adding a modulation to the PDFs of the classes of associated sources (the parameters of the modulation are also fit to the data simultaneously with the class prevalence).

In more formal terms, we construct PDFs of astrophysical classes, denoted as $p(\mathbf{x}|k)$, which represent the probability of a source with features $\mathbf{x}$ given a source class k . The PDFs for unassociated sources are determined by modulating the PDFs of the associated ones by functions of input features $\mathbf{x}$ (these functions are the same for all classes k). Previous supervised learning studies have focused on estimating the conditional probability $p(k|\mathbf{x})$ of the class of source k given the input features [136, 159, 230, 241, 245–254], as typically done by probabilistic classification models. Estimating $p(\mathbf{x}|k)$ instead brings several crucial advantages. First of all (see section 5.4.1), it allows us to model the difference between distributions of associated and unassociated sources as a combination of both covariate and prior shifts. The latter allows us to model the different prevalence of classes of astrophysical sources for training (associated sources) and target (unassociated sources) datasets as well as to include a new class of sources among the unassociated ones, which in this analysis is modeled as DM subhalos. Moreover, upon specification of the class prior $p(k)$, the joint distribution $p(k, \mathbf{x}) = p(\mathbf{x}|k)p(k)$ can be sampled to generate mock data, which can be used in Monte Carlo analyses.² Finally, the joint PDF $p(\mathbf{x}, k)$ can be marginalized over the class k to estimate $p(\mathbf{x})$, i.e., the marginal distribution of the source features. The quantity $p(\mathbf{x})$ represents a probability density for a source with parameters $\mathbf{x}$. The product of $p(\mathbf{x}_i)$ over unassociated sources is the model likelihood, which we maximize to obtain the optimum model parameters both for the prior and covariate shifts. In particular, we use such likelihood to estimate the best fit and upper bounds on the DM annihilation cross section $\langle\sigma v\rangle$. Although generative models have been constructed previously in analyses of *Fermi*-LAT source distributions, e.g. [246], they were not used for construction of maximum likelihood models, which we do in the current work.

The paper is organized as follows. In section 5.4.2, we describe the selection of the data. We discuss the covariate and prior shifts in section 5.4.3. We present our statistical model in section 5.4.4. The model of J-factors used in this paper is reviewed in section 5.4.6, while the simulation of gamma-ray sources corresponding to DM subhalos is presented in section 5.4.7. We optimize the model in section 5.4.9, while the corresponding limits on DM annihilation determined from the observed gamma-ray sources are presented in section 5.4.10. Section 5.4.11 contains discussion and conclusions. We provide details about the statistical model and perform several ~~model performance and~~ consistency tests in appendices.

5.4.1 Statistical analysis

5.4.2 Data selection

In this work we use the 4FGL-DR4 catalog [123, 228]. We group the known types of associated sources into two classes:³

- Galactic ([dominated by millisecond pulsars at high latitudes](#) [68, 104, 121, 198, 203, 255]): pulsars, binaries, star-forming regions, supernova remnants, pulsar wind nebulae, globular clusters, and novae;

²That’s why probabilistic models estimating $p(\mathbf{x}|k)$ are called “generative models”, as opposed to “discriminative” models focusing on $p(k|\mathbf{x})$.

³Cf. table 5 in ref. [228] for details about the types of sources. Galactic sources: psr, hmb, sfr, snr, pwn, gc, gal, bin, msp, lmb, spp, glc, nov. Extragalactic sources: bll, sbg, rdg, css, bcu, ssrq, fsrq, sey, nlsy1, agn.

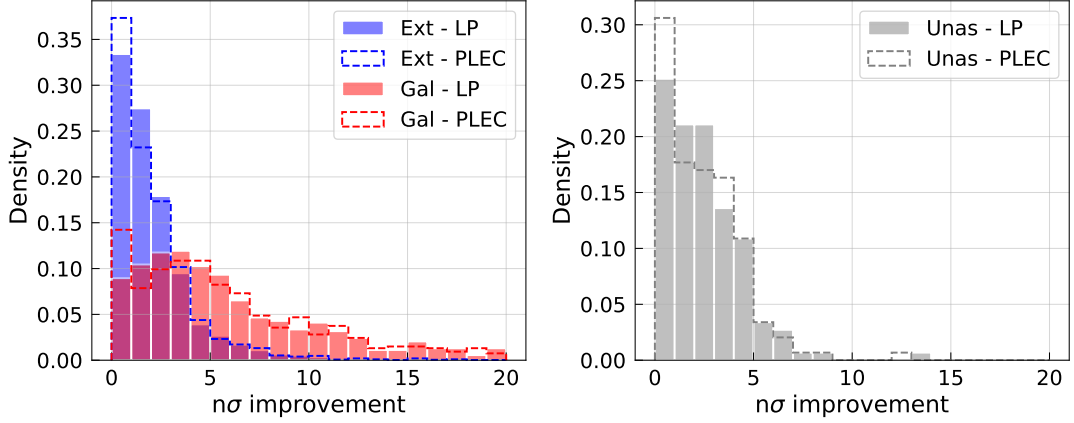


Fig. 5.2. Distribution of the significance of fit improvement, as reported in the 4FGL-DR4 catalog, when replacing a power-law energy spectrum with either a log parabola (LP) or a power law with an exponential cutoff (PLEC). We display the histograms for associated sources (left panel) and unassociated sources (right panel).

- Extragalactic (mostly AGNs [85, 104, 111, 114, 121, 256, 257]): blazars, quasars, starburst galaxies, normal galaxies, radio galaxies and radio sources, Seyfert galaxies, and non-blazar active galaxies.

This choice is made to group sources based on their fundamentally different physical origins and radiation environments. By separating the magnetospheric emission characteristic of Galactic pulsars from the jet-driven processes of extragalactic AGNs, we ensure that the classification reflects the intrinsic physical diversity of the γ -ray sky, which manifests in distinct spectral signatures.

In order to characterise the sources, we focus on their energy spectra modeled with the log-parabola (LP) function

$$\ln \frac{F(E)}{K} = -\alpha \ln \frac{E}{E_0} - \beta \left(\ln \frac{E}{E_0} \right)^2, \quad (5.4.1)$$

where the parameters correspond to the following columns in the 4FGL-DR4 catalog: K – LP_Flux_Density, α – LP_Index, β – LP_beta, and E_0 – Pivot_Energy.

The LP model provides a well-motivated spectral choice; it offers a significant improvement over a simple power law, as evidenced in the LP_SigCurv column in the 4FGL-DR4 catalog, and performs at least as well as a power law with exponential cut off both for the unassociated and for the associated Galactic and extragalactic sources (figure 5.2). ~~At the same time, it avoids the complexity and statistical weaknesses of using high-dimensional binned fluxes.~~

Provided that only three parameters are independent and that E_0 in the catalog is different for different sources, we rescale E_0 to a fixed value of 1 GeV (for analysis of $M_{\text{DM}} > 20$ GeV) or 100 MeV (for $M_{\text{DM}} < 20$ GeV). We use the following three parameters as input features:

1. $\log_{10} \phi$ – the logarithm of the flux integrated between 100 MeV and 1 TeV (obtained by summing fluxes in energy bins provided in the catalog);
2. α – LP_Index rescaled to $E_0 = 1$ GeV or $E_0 = 100$ MeV depending on the choice of DM mass, i.e., above or below 20 GeV respectively;

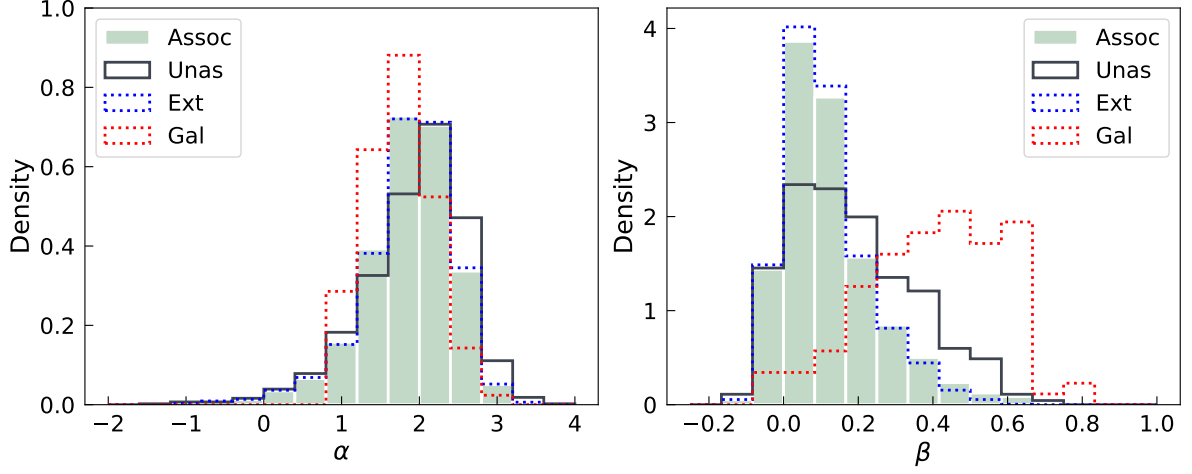


Fig. 5.3. Normalised distributions of α and β parameters for associated and unassociated sources for $E_0 = 1$ GeV, as well as the distributions for extragalactic and Galactic sources with $|b| > 10^\circ$.

3. β – LP_beta.

We collect these variables for each source i inside a vector $\mathbf{x}_i \equiv \{\log_{10}(\phi_i), \alpha_i, \beta_i\}$. In our analysis, we consider only sources with Galactic latitudes $|b| > 10^\circ$.

5.4.3 Covariate and prior shifts

The basic assumption behind a supervised classification task is that the joint distribution of the input features $\mathbf{x}$ and output classes (type of sources) k are the same for the training and target datasets:

$$p_{\text{train}}(\mathbf{x}, k) = p_{\text{target}}(\mathbf{x}, k), \quad (5.4.2)$$

while, in general, a dataset shift represents a situation where the training and target distributions are different $p_{\text{train}}(\mathbf{x}, k) \neq p_{\text{target}}(\mathbf{x}, k)$. In our case, the training dataset is the associated sources, while the target dataset is the unassociated sources. Consequently, we will use associated (assoc) and unassociated (unas) subscripts for the training and target datasets, respectively.

The joint distribution of features $\mathbf{x}$ and class labels k can be expressed as a product of conditional and marginal probabilities in two equivalent ways:

$$p(\mathbf{x}, k) = p(k|\mathbf{x})p(\mathbf{x}) = p(\mathbf{x}|k)p(k). \quad (5.4.3)$$

This decomposition gives rise to two distinct types of dataset shift [158]:

1. **Covariate shift:** the conditional class probabilities remain unchanged, $p_{\text{assoc}}(k|\mathbf{x}) = p_{\text{unas}}(k|\mathbf{x})$, but the feature distributions differ, $p_{\text{assoc}}(\mathbf{x}) \neq p_{\text{unas}}(\mathbf{x})$.
2. **Prior shift:** the feature distributions conditioned on the class are invariant, $p_{\text{assoc}}(\mathbf{x}|k) = p_{\text{unas}}(\mathbf{x}|k)$, but the class prevalence changes, $p_{\text{assoc}}(k) \neq p_{\text{unas}}(k)$.

Note that for the covariate shift, differences in the marginal distribution of the features $p(\mathbf{x})$ generally lead to differences in the marginal class probabilities $p(k)$. Analogously, for prior shift, changes in the marginal class distributions $p(k)$ in general result in changes in the marginal

feature distributions $p(\mathbf{x})$. We show the distributions of the α and β parameters in figure 5.3. Notice that the distributions are different for associated and unassociated sources. In principle, this difference can be due to either a covariate shift or a prior shift. For example, as can be seen from the histograms of the associated extragalactic and Galactic sources, a change in the class prevalence in unassociated sources would certainly lead to a change in the β parameter distribution, which could be explained as a prior shift. On the other hand, if there are differences in the distribution of unassociated sources, which affect all unassociated sources, e.g., larger uncertainties in the reconstructed energy spectrum, then this effect can be addressed by a covariate shift.

5.4.4 Statistical model

In general, our goal is to construct a probabilistic model for the distribution of unassociated sources $\tilde{p}_{\text{unas}}(\mathbf{x})$ under the assumption that it is a mixture of classes of associated sources plus a possible new class of sources.

In the covariate shift assumption (in absence of a prior shift) this task is actually trivial as

$$p_{\text{unas}}(\mathbf{x}) = \sum_k p_{\text{unas}}(k|\mathbf{x})p_{\text{unas}}(\mathbf{x}) = \sum_k p_{\text{assoc}}(k|\mathbf{x})p_{\text{assoc}}(\mathbf{x})C(\mathbf{x}), \quad (5.4.4)$$

where in the second equality we use the covariate shift assumption $p_{\text{assoc}}(k|\mathbf{x}) = p_{\text{unas}}(k|\mathbf{x})$ and define $C(\mathbf{x}) = p_{\text{unas}}(\mathbf{x})/p_{\text{assoc}}(\mathbf{x})$. The function $C(\mathbf{x})$ describes the covariate shift, i.e., the difference in the distribution of training and target datasets.

Moreover, we can add a new class of sources with a PDF $p_{\text{new}}(\mathbf{x})$ and define a model for the unassociated sources as:

$$\tilde{p}_{\text{unas}}(\mathbf{x}) = \sum_k p_{\text{assoc}}(k|\mathbf{x})p_{\text{assoc}}(\mathbf{x})\tilde{C}(\mathbf{x}) + \pi_{\text{new}}p_{\text{new}}(\mathbf{x}), \quad (5.4.5)$$

where $\tilde{C}(\mathbf{x}) = \max(p_{\text{unas}}(\mathbf{x}) - \pi_{\text{new}}p_{\text{new}}(\mathbf{x}), 0)/p_{\text{assoc}}(\mathbf{x})$ can be viewed as a generalized covariate shift that describes the difference in the distribution of training dataset and a part of the target dataset that corresponds to the same classes as in the training dataset. If $p_{\text{unas}}(\mathbf{x}) \geq \pi_{\text{new}}p_{\text{new}}(\mathbf{x})$ for $\forall \mathbf{x}$, then the model in eq. (5.4.4) is equal to the model in eq. (5.4.5), i.e., the two models are indistinguishable until the new class predicts more sources in some region of space than there are unassociated sources. Although mathematically we cannot exclude a situation when all unassociated sources in some region of space belong to a new class, this is not very likely from the physical point of view. In order for the model to be more constraining we put additional physical conditions on the generalized covariate shift function $\tilde{C}(\mathbf{x})$. In particular, we model $\tilde{C}(\mathbf{x})$ as a product of 1-dimentional functions

$$\tilde{C}(\mathbf{x}) = \prod_{d=1}^3 f_d(x_d), \quad (5.4.6)$$

where each $f_d(x_d)$ is monotonous. The product is over the dimensions d of the space of input variables. In particular, we assume here that the covariate shift in each of the input variables is independent of the other input variables. We implement $f_d(x_d)$ using sigmoid functions

$$f(x_d; b_d, c_d) = \frac{1}{1 + e^{-b_d(x_d - c_d)}}, \quad (5.4.7)$$

where each function has two parameters (b_d, c_d) determined from maximizing the likelihood of the model. Our region of interest (ROI) in feature space will be restricted to a window in

every dimension where the ratio $p_{\text{unas}}/p_{\text{assoc}}$ of the empirical distributions are monotonous. We approximate the empirical distributions of associated and unassociated gamma-ray sources using kernel density estimation (KDE) with a Gaussian kernel [125]. In order to find the optimal bandwidth for the kernels, we perform 5-fold cross-validation repeated 20 times. For each of the repetitions, we randomize the dataset, split it into 5 subsets, and repeat the training 5 times: each time, a new subset is used for testing, and the remaining four subsets are used for training. The performance is evaluated by calculating the likelihood of the model on the testing dataset. The optimal bandwidth parameter is obtained from the maximum of the performance averaged over the 100 evaluations (20 repetitions with 5 training/testing splits in each repetition). The optimal bandwidth parameters for associated and unassociated sources are computed independently. We show the corresponding KDE models in the $E_0 = 1$ GeV case in the left panels in figure 5.4. The right panels of figure 5.4 show the ratios of the KDE models, where the bands represent the statistical uncertainty.

The ROI is determined as the intervals in parameter space where the ratio of the KDE models is monotonous within statistical uncertainties. In the $E_0 = 1$ GeV case, we restrict our analysis to $\alpha \in [0.8, 1.7]$. Analogously, in the $E_0 = 100$ MeV case, $\alpha \in [0.2, 0.8]$. The range of the β parameter is the same in both cases, $[0.1, 0.55]$. Finally, we restrict the $\log_{10} \phi$ parameter to the interval $[-10.3, -8.2]$. In addition, we compute the 3D KDE models and identify the optimal bandwidth parameters independently for Galactic and extragalactic classes of sources, which yields the $p_{\text{assoc}}(\mathbf{x}|k)$ used in the analysis.

In general, the weight (prevalence) of classes of sources does not need to be the same for associated and unassociated sources, i.e., there can be a prior probability shift in addition to the covariate shift. In order to allow a change in the prevalence of the classes of sources, we express the terms in the sum in eq. (5.4.5) in the form suitable for the prior shift:

$$p_{\text{assoc}}(k|\mathbf{x})p_{\text{assoc}}(\mathbf{x}) \equiv p_{\text{assoc}}(\mathbf{x}, k) = \pi_k p_{\text{assoc}}(\mathbf{x}|k), \quad (5.4.8)$$

where for associated sources $\pi_k = \pi_k^{\text{assoc}}$ is the prevalence of class k , while for unassociated sources, π_k are in general unknown and will be determined from the fit of the model to the data. Overall, our model of the unassociated source takes the form:

$$\tilde{p}_{\text{unas}}(\mathbf{x}) = \left(\sum_{k=1}^K \pi_k p_{\text{assoc}}(\mathbf{x}|k) \right) \tilde{C}(\mathbf{x}; \boldsymbol{\theta}_{\text{cov}}) + \pi_{\text{DM}}(\boldsymbol{\theta}_{\text{DM}}) p_{\text{DM}}(\mathbf{x}; \boldsymbol{\theta}_{\text{DM}}), \quad (5.4.9)$$

where $K = 2$ is the number of classes of associated sources (Galactic and extragalactic, cf. section 5.4.2). The DM distribution $p_{\text{DM}}(\mathbf{x}; \boldsymbol{\theta}_{\text{DM}})$ is determined below in section 5.4.7.

The optimizable parameters in this model are: π_k , describing the prevalence of the classes of associated sources; $\boldsymbol{\theta}_{\text{cov}} \equiv \{b_d, c_d\}$ for $d = 1, 2, 3$, parametrizing the covariate shift in eq. (5.4.7), and $\langle \sigma v \rangle$, the annihilation cross section, which we have included in the definition of $\boldsymbol{\theta}_{\text{DM}} = \{\langle \sigma v \rangle, M_{\text{DM}}, \text{ch.}\}$. The other parameters, namely the DM mass M_{DM} as well as the annihilation channel ‘ch.’, are kept fixed during the optimization procedure.

Note that for $\tilde{C}(\mathbf{x}; \boldsymbol{\theta}_{\text{cov}}) = 1$, the model reduces to a prior shift model, while for $\pi_k = \pi_k^{\text{assoc}}$ and $\pi_{\text{DM}} = 0$ the model reduces to a covariate shift model. Thus, in general, the model in eq. (5.4.9) is a combination of covariate and prior shift models. We point out that the model in eq. (5.4.9), identified as the likelihood of a given unassociated source, allows us to:

1. determine the unknown parameters of the model by maximizing the likelihood;

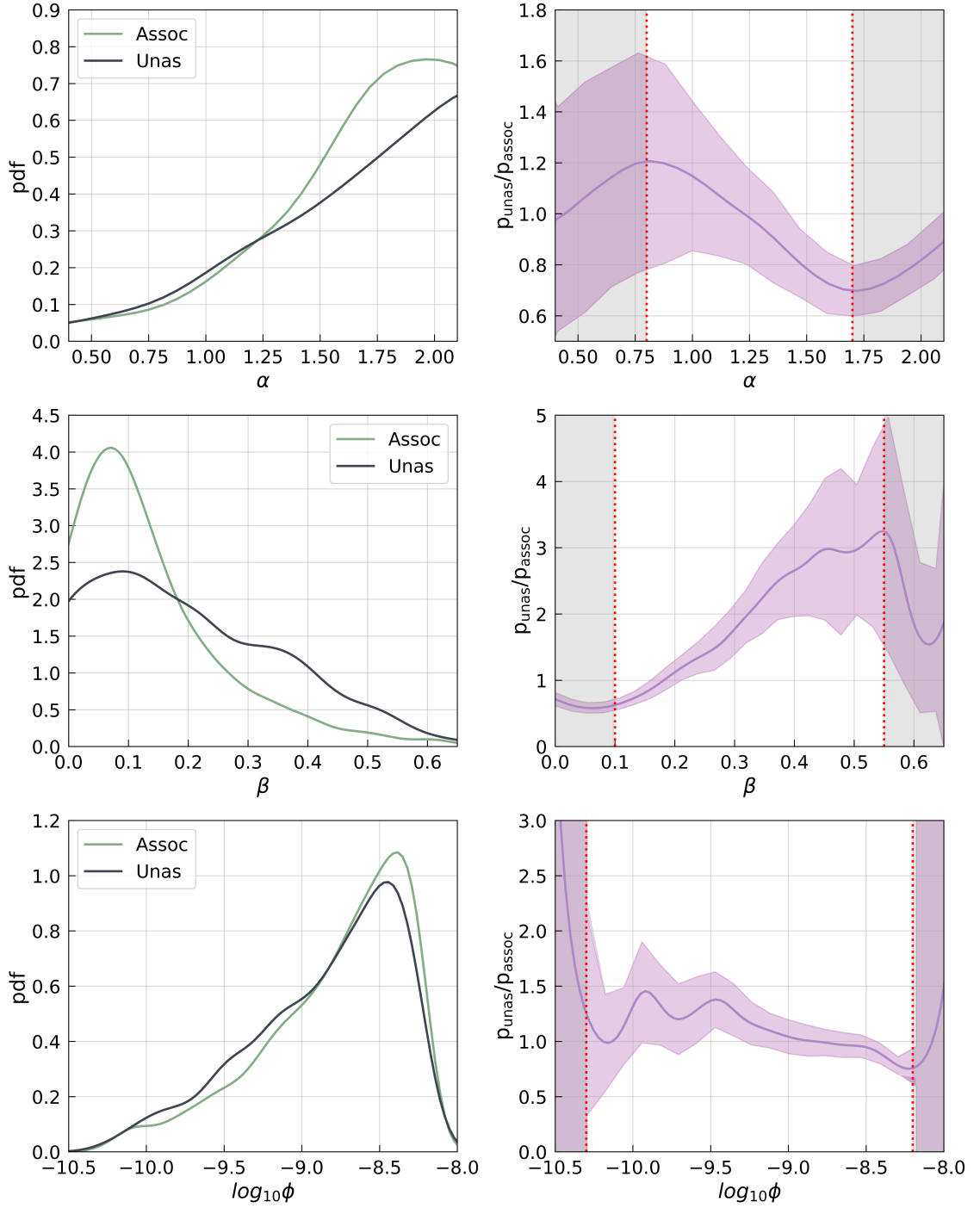


Fig. 5.4. *Left column:* full distributions of associated and unassociated sources modeled with KDE for the case of $E_0 = 1$ GeV before truncation and normalization to the ROI. *Right column:* the ratio of the distributions $p_{\text{unassoc}}/p_{\text{assoc}}$. The red vertical lines show the range of parameters that we use in the analysis (ROI). We estimate the error (purple bands) as the Poisson error in the histograms and compute the relative errors for the associated and unassociated distributions, which we sum in quadrature.

2. detect or put an upper bound at a given level of confidence for a possible new component, such as DM;
3. generate new data (for Monte Carlo tests);
4. determine the conditional class probabilities of individual sources using the probability rule $p(k|\mathbf{x}) = p(\mathbf{x}, k) / \tilde{p}_{\text{unas}}(\mathbf{x})$. In the case of astrophysical classes $p(\mathbf{x}, k) = \pi_k p_{\text{assoc}}(\mathbf{x}|k) \tilde{C}(\mathbf{x}; \boldsymbol{\theta}_{\text{cov}})$, while for the DM class: $p(\mathbf{x}, \text{DM}) = \pi_{\text{DM}} p_{\text{DM}}(\mathbf{x}; \boldsymbol{\theta}_{\text{DM}})$.

We stress that, as far as we are aware of, this is the first time a generative probabilistic model for the unassociated sources has been proposed. Previous studies have focused on modeling directly the conditional probabilities $p(k|\mathbf{x})$, which does not give access to the data likelihood. In this respect, our approach is more informative.

5.4.5 Dark matter subhalos model

5.4.6 Subhalo J-factors from N-body simulations

A key component of our pipeline is the so-called $\tilde{J}$ -factor distributionTM of the Galactic subhalo population. This J-factor, also known simply as $\tilde{\text{astrophysical factor}}^{\text{TM}}$, acts as a normalization factor for the expected annihilation signal and can be computed as the volume integral of the subhalo DM density squared. In this work, we follow ref. [224] and adopt the J-factor values provided for two different prescriptions of the subhalo internal structure⁴: one derived from the mass enclosed within the subhalo tidal radius, $J_{\text{tidal}}^{M_{\text{sub}}}$, and another based on the maximum circular velocity of each subhalo, $J_{\text{tidal}}^{V_{\text{max}}}$. These two prescriptions are advisable here to estimate current J-factor uncertainties. Indeed, measuring subhalo masses is not a trivial task, especially at scales close to the numerical resolution limit of the simulation, where only a few simulation particles are available. Because of this, the computation of masses relies on adopting a specific form of the subhalo DM density profile, which is then fitted to the available simulation data and integrated up to the assumed tidal radius. Unfortunately, even the precise value of the latter can vary significantly among different definitions and is very sensitive to changes in the tidal forces acting on the subhalo as it orbits around the host halo center. In contrast to these mass-related issues, the peak of the circular velocity profile within each subhalo, i.e. V_{max} , has been shown to be less affected by numerical resolution and tidal forces (see, e.g., refs. [224, 258]). Its determination also does not depend on adopting a specific functional form of the DM density profile to model the subhalo structural properties. As a result, J-factor calculations based on the latter approach are considered more reliable. Nevertheless, we use both J-factor sets in this study mainly to enable direct comparison with previous results, where J-factor based on subhalo masses are more common. Also, as said, the exercise will be useful to understand the current level of uncertainties on subhalo modeling.

The calculation of astrophysical J-factors in ref. [224] is based on the Via Lactea II (VL-II) N-body cosmological simulation [259], a DM-only simulation of a Milky-Way-sized halo resolving subhalos down to approximately $10^4 M_{\odot}$. To explore subhalos below the VL-II resolution limit, the simulation was repopulated with millions of smaller subhalos, extending the analysis down by nearly five orders of magnitude to about $10^{-1} M_{\odot}$ [244]. Additionally, by placing the Earth at random positions consistent with its Galactocentric distance, thousands of realizations of a Milky Way-like galaxy were created to obtain statistically robust results. This led to accurate

⁴All data are available at https://projects.ift.uam-csic.es/damasco/?page_id=831.

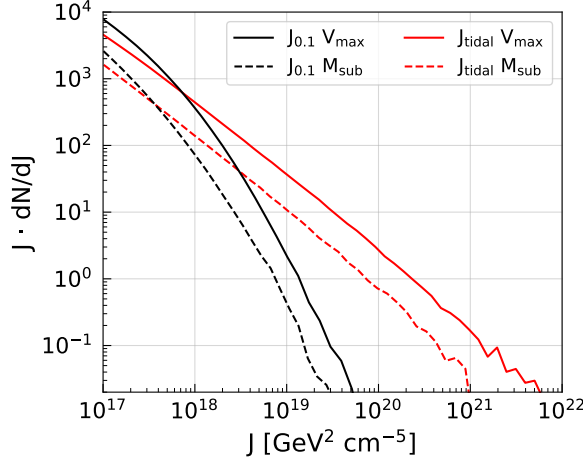


Fig. 5.5. J-factor number density times J for the different J-factor cases considered in this work. See section 5.4.6 for further details.

J-factor distributions (based on both mass and circular velocity) for the entire subhalo population of a galaxy like our own in a Λ CDM universe. A limitation of ref. [224] is the absence of baryonic effects in the parent simulation. Baryons are expected to significantly influence the DM subhalo population, particularly in the Galaxy’s innermost regions (e.g., [260–262]), however none of these hydrodynamical simulations resolve subhalos lighter than $\sim 10^6 M_\odot$. Thus, the survival of such low-mass subhalos to tidal stripping, especially with baryonic influences, remains uncertain [225–227].

In addition to $J_{\text{tidal}}^{M_{\text{sub}}}$ and $J_{\text{tidal}}^{V_{\text{max}}}$ from ref. [224], we also utilize $J_{0.1}^{M_{\text{sub}}}$ and $J_{0.1}^{V_{\text{max}}}$, derived from the same simulation work. These values correspond to J-factors integrated not up to the tidal radius of each subhalo but rather to the corresponding radius subtending 0.1° in the sky. This choice aligns with the typical angular resolution of the *Fermi*-LAT in the GeV range. By using $J_{0.1}^{M_{\text{sub}}}$ and $J_{0.1}^{V_{\text{max}}}$, we focus on point-like subhalo emissions, as would appear in *Fermi*-LAT point-source catalogs. Notably, ref. [244] concludes that the brightest Galactic subhalos in the *Fermi*-LAT gamma-ray sky, if detected, would be extended, with angular sizes around 0.2° – 0.3° . Consequently, for such subhalos, our $J_{0.1}^{M_{\text{sub}}}$ and $J_{0.1}^{V_{\text{max}}}$ values are expected to be significantly lower than their corresponding $J_{\text{tidal}}^{M_{\text{sub}}}$ and $J_{\text{tidal}}^{V_{\text{max}}}$ values. We adopt $J_{0.1}^{V_{\text{max}}}$ as our nominal case in this paper, while we use the other J-factor values to compare with previous literature, e.g., in Calore et al. (2019) [231] a $J_{0.1}^{M_{\text{sub}}}$ model was used.

Figure 5.5 illustrates the four sets of J-factor distributions discussed in this section, along with their corresponding number density, which will be the relevant quantity for our pipeline.

5.4.7 Gamma-ray emission from dark matter subhalos

Given a DM setup θ_{DM} (i.e., a DM mass M_{DM} , annihilation channel, and annihilation cross section $\langle\sigma v\rangle$), and an expected distribution of J-factors determined from DM N-body simulations (described in section 5.4.6), we estimate the expected distribution of DM annihilating subhalos that can be detected by *Fermi* LAT for a certain observation time and instrument response functions (IRFs). The procedure is as follows:

1. We start with a distribution $p(J) = dN/dJ$ of J-factors expected from N-body DM simulations (cf. figure 5.5 in section 5.4.6). We sample the values of J from $p(J)$.

2. Given a J-factor and DM parameters $\theta_{\text{DM}} = \{\langle\sigma v\rangle, M_{\text{DM}, \text{ch.}}\}$, we determine the expected gamma-ray spectrum as

$$\phi_{\text{DM}}(E; \theta_{\text{DM}}, J) = \frac{J}{4\pi} \frac{\langle\sigma v\rangle}{2M_{\text{DM}}^2} \left. \frac{dN_\gamma}{dE} \right|_{\text{ch.}}, \quad (5.4.10)$$

where dN_γ/dE is the theoretical spectrum of gamma rays produced in an annihilation of a pair of DM particles from ref. [84].

3. We place DM subhalos at random positions in the sky with J-factors sampled in step 1 and with the spectra defined in eq. (5.4.10).
4. For the diffuse gamma-ray background, we use the official *Fermi*-LAT diffuse model template employed in the construction of the 4FGL-DR4 source catalog (gll_iem_v7) and the corresponding isotropic background.
5. We fit the gamma-ray data around each of the simulated DM subhalos with $|b| > 10^\circ$, where we use the log-parabola model in eq. (5.4.1) with fitting parameters $\{\log_{10} \phi, \alpha, \beta\}$. The fit is performed within a circular region centered on the target DM source. The radius of this region is energy-dependent, corresponding to the 95% containment angle of the PSF, and is constrained to the range $[0.5^\circ, 5^\circ]$. This range is chosen to ensure sufficient statistics in the highest energy bins while simultaneously avoiding excessive background contamination in the lowest energy bins. We use the same Gaussian priors for the β parameter as in the 4FGL-DR4 catalog with mean 0.1 and sigma 0.3 [123]. For details see appendix 5.A.
6. We keep DM subhalos detected with significance $\text{TS} > 25$, where $\text{TS} = -2\ln \mathcal{L}/\mathcal{L}_*$ is the likelihood-ratio test statistic between the models with and without the subhalo. Such a threshold corresponds to the significance for PS detection in the *Fermi*-LAT catalogs. The fraction of subhalos passing the significance cut defines the detection efficiency $\varepsilon_{\text{eff}}(\phi_{\text{DM}})$, which is a function of the DM flux.
7. In order to get a good estimate of the DM distribution, $p_{\text{DM}}(\log_{10} \phi_{\text{DM}}, \alpha, \beta | \theta_{\text{DM}})$, we simulate realizations of the gamma-ray sky until we detect at least 20,000 DM subhalos across all simulations or the number of simulations reaches 1,000. The latter can happen for parameters, where DM subhalos have very small detection probability.⁵

In order to reduce the computation time, we perform simulations in flux bins. At first, we determine the distribution of α and β parameters, given the flux ϕ_{DM} and DM parameters θ_{DM} , $p_{\text{DM}}(\alpha, \beta | \phi_{\text{DM}}, \theta_{\text{DM}})$, by modeling the distribution of recovered α and β parameters in this flux bin (see appendix 5.A for details). Then we multiply this distribution by the distribution of fluxes, which depends on the distribution of J-factors $n(\phi_{\text{DM}}) = \frac{dN}{d\phi_{\text{DM}}}(\phi_{\text{DM}}; \theta_{\text{DM}}) = \frac{dN}{dJ} \frac{dJ}{d\phi_{\text{DM}}}$ and by detection efficiency as a function of flux $\varepsilon_{\text{eff}}(\phi_{\text{DM}})$ to determine the probability density times the DM subhalo weight in eq. (5.4.9):

$$\pi_{\text{DM}}(\theta_{\text{DM}}) p_{\text{DM}}(\mathbf{x}; \theta_{\text{DM}}) = \frac{1}{N_{\text{unas}}^{\text{ROI}}} p_{\text{DM}}(\alpha, \beta | \phi_{\text{DM}}; \theta_{\text{DM}}) n(\phi_{\text{DM}}) \varepsilon_{\text{eff}}(\phi_{\text{DM}}), \quad (5.4.11)$$

where $N_{\text{unas}}^{\text{ROI}}$ is the number of unassociated sources in the ROI.

⁵The limit of a thousand simulations is needed to avoid infinite iterations for low-efficiency parameter combinations. For details on simulation and detection of DM subhalos, see appendix 5.A.

We note that given the DM parameters, the contribution of DM subhalos to unassociated sources is fixed. In particular, the number of subhalos in our ROI is

$$N_{\text{sub}}^{\text{ROI}}(\boldsymbol{\theta}_{\text{DM}}) = \int_{\text{ROI}} d\alpha d\beta d\phi_{\text{DM}} p_{\text{DM}}(\alpha, \beta | \phi_{\text{DM}}; \boldsymbol{\theta}_{\text{DM}}) n(\phi_{\text{DM}}) \varepsilon_{\text{eff}}(\phi_{\text{DM}}), \quad (5.4.12)$$

while the DM weight π_{DM} in eq. (5.4.9) is

$$\pi_{\text{DM}}(\boldsymbol{\theta}_{\text{DM}}) = \frac{N_{\text{sub}}^{\text{ROI}}(\boldsymbol{\theta}_{\text{DM}})}{N_{\text{unas}}^{\text{ROI}}}. \quad (5.4.13)$$

In figure 5.6 we show the distributions of DM subhalos in α and β parameters (we integrate the distributions in the flux dimension for presentation purposes) for different DM masses at some characteristic $\langle\sigma v\rangle$ values. We note that the DM distributions are not fully contained inside the ROI that we selected in this analysis, which is indicated as a green box in the plots.

In particular, a significant fraction of the DM subhalo distribution is truncated for DM masses of 10 GeV, 30 GeV, and 1 TeV. ~~For these masses our analysis can be further improved by adopting a covariate shift model suitable for a larger region in feature space.~~ Although choosing a larger ROI that covers a larger fraction of the DM subhalo distribution could increase statistics, our assumption that the covariate shift can be modeled by a monotonous function would not be valid anymore. Introducing a more flexible covariate shift model to account for this would require additional parameters, potentially leading to larger uncertainty in π_{DM} due to correlations with these new parameters (i.e., a larger parameter covariance matrix) and, consequently, weaker limits. Moreover, the gain from larger statistics would be limited by the significant background of astrophysical sources outside of our current ROI, particularly unassociated millisecond pulsars with $\alpha > 0.7$ and $\beta \sim 0.4$ (upper right panel of figure 5.6). Conversely, regions with small α values (bottom panel) contain very few astrophysical sources, necessitating extrapolation that carries large systematic uncertainties (see the wide uncertainty band for small alpha values in the upper left panel in figure 5.4). Finally, our limits on DM annihilation are conservative: positive residuals after fitting the data with astrophysical components can be absorbed by the DM component, resulting in a positive best-fit π_{DM} and weaker upper limits than in the $\pi_{\text{DM}} = 0$ case.

In figure 5.7 we show the contour levels for the half maximum of the DM subhalo distribution for different choices of $\langle\sigma v\rangle$ and J-factor models for $M_{\text{DM}} = 100$ GeV. The distribution shifts to higher values of β for larger $\langle\sigma v\rangle$ values. Analogously, choosing J_{tidal} shifts the distribution to higher β values compared to $J_{0.1}$. This is to be expected, since for faint sources (small $\langle\sigma v\rangle$ or small J-factors) the fit has large statistical uncertainties and the best-fit value of the β parameter is more affected by the priors on β than in case of bright sources.

5.4.8 Mixture model of gamma-ray sources and limits on DM annihilation

5.4.9 Model optimization

Our general aim is to either detect a population of DM subhalos among the *Fermi*-LAT unassociated sources or to put an upper limit on $\langle\sigma v\rangle$ for different DM masses. For a fixed DM mass and annihilation channel, we determine the best-fit cross section $\langle\sigma v\rangle^*$ (optimized simultaneously with the other parameters) by maximizing the full likelihood built from the model in eq. (5.4.9), defined as:

$$\mathcal{L}(\langle\sigma v\rangle, \pi_k, \boldsymbol{\theta}_{\text{cov}}) = \prod_{i=1}^{N_{\text{unas}}^{\text{ROI}}} \tilde{p}_{\text{unas}}(\mathbf{x}_i | \langle\sigma v\rangle, \pi_k, \boldsymbol{\theta}_{\text{cov}}; M_{\text{DM}}, \text{ch.}). \quad (5.4.14)$$

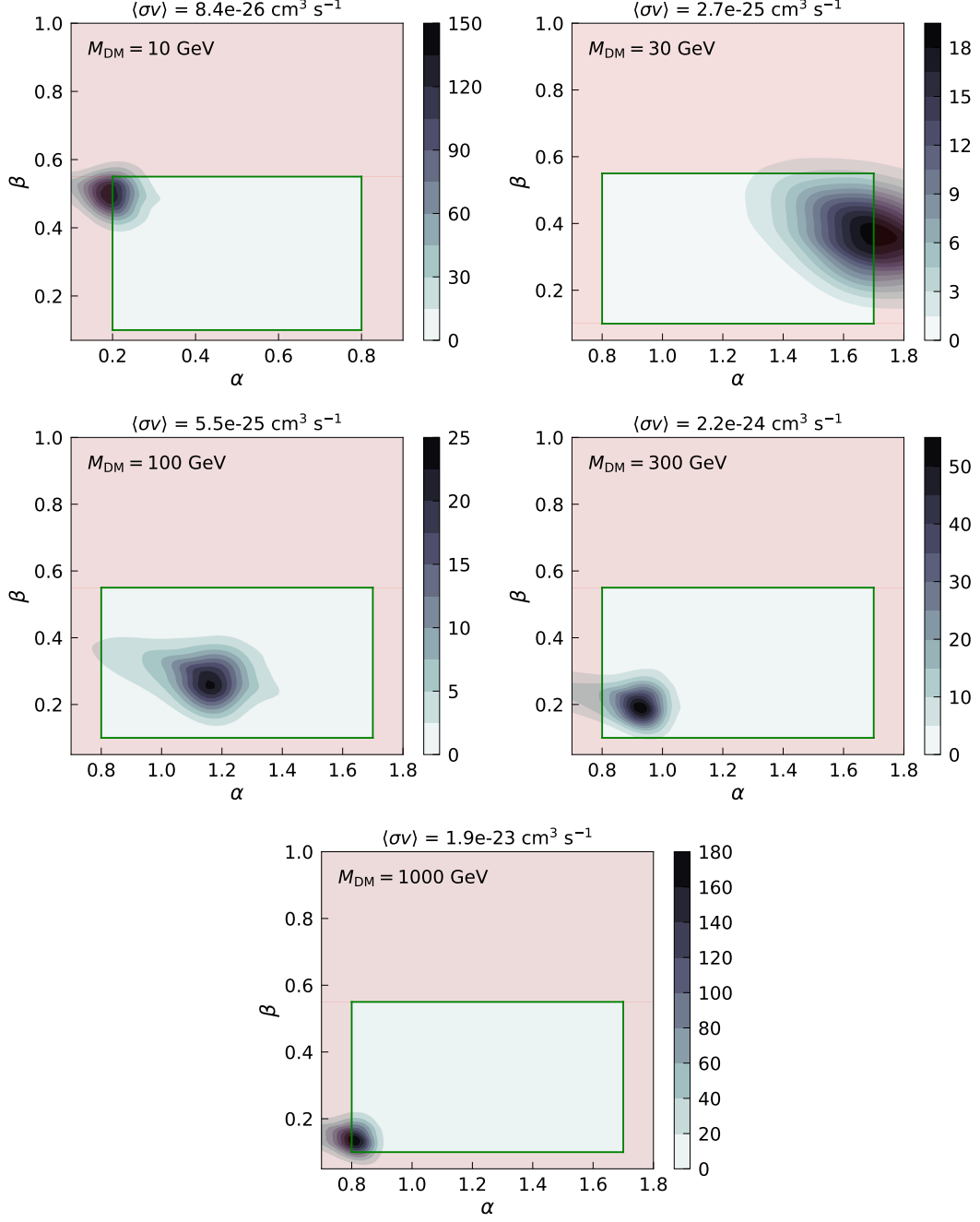


Fig. 5.6. DM distribution at different DM masses, $b\bar{b}$ annihilation channel, integrated in flux for visualization purposes. The $\langle\sigma v\rangle$ values are chosen close to our bound values. The green boxes indicate the parameter ranges (ROI) used for this analysis and define the truncation region for the distribution. The color bars show the value of the DM subhalo PDF (eq. 5.4.11), normalized to unit integral within the ROI.

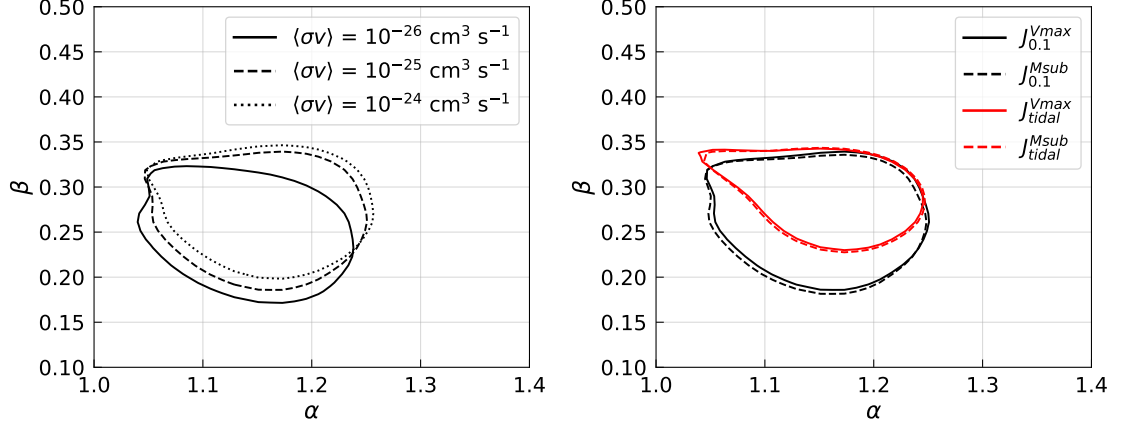


Fig. 5.7. Contour plots at half maximum of the DM subhalo distribution for different values of $\langle\sigma v\rangle$ and J-factor models at $M_{\text{DM}} = 100$ GeV. *Left panel:* contour lines for different choices of $\langle\sigma v\rangle$ for $J_{0.1}^{Vmax}$. *Right panel:* contour lines for different J-factor models for $\langle\sigma v\rangle = 10^{-25} \text{ cm}^3 \text{ s}^{-1}$.

We maximize eq. (5.4.14) with the Expectation-Maximization (EM) algorithm, which is efficient for the cases of probabilistic mixture models [157], see appendix 5.B for more details. From the ML point of view our approach is a semi-supervised learning algorithm.

Indeed, in supervised learning the model is fixed by the training data (including the PDFs for the different classes), while in unsupervised learning, the model is determined solely based on target data. In our case, we use the associated sources (training data) to determine PDFs of the two classes of associated sources (which is similar to supervised learning), but we further modify the PDFs to account for the possible dataset shift between training and target data and add a new class of sources (which is similar to unsupervised learning).

We optimize the likelihood for a set of DM mass values: $M_{\text{DM}} = 10, 30, 100, 300, 1000$ GeV, and determine, for each M_{DM} , the upper bound on $\langle\sigma v\rangle$ at 95% confidence level (CL) by profiling the test-statistic

$$\text{TS}(\langle\sigma v\rangle) = -2 \ln \frac{\mathcal{L}(\langle\sigma v\rangle)}{\mathcal{L}_{\text{max}}}, \quad (5.4.15)$$

where $\mathcal{L}(\langle\sigma v\rangle)$ is the likelihood optimized over the rest of parameters π_k and θ_{cov} , and $\mathcal{L}_{\text{max}} = \mathcal{L}(\langle\sigma v\rangle^*, \pi_k^*, \theta_{\text{cov}}^*)$ is the value of the likelihood for the best-fit parameters. Following the Wilks' theorem, we assume that TS follows the χ^2 distribution. We compute the upper bound on $\langle\sigma v\rangle$ as the value corresponding to the 95% coverage probability (one-sided), which for one degree of freedom corresponds to $\text{TS} = 3.84$.

We maximize the likelihood of the model in eq. (5.4.14) to determine the best-fit parameters. At first, we fix the DM contribution to be zero and determine the best-fit model with the known astrophysical sources only. For reference, we show the model for the associated sources (no fitting) in the left panel of figure 5.8 and the best-fit model for unassociated sources in the middle and right panels. In these plots, we display the distributions evaluated in the analysis ROI for α and β parameters integrating over the flux coordinate for visualization purposes. We decompose the mixture model of eq. (5.4.9) and associate the blue color with the extragalactic component and the red color with the Galactic component, both respectively weighted by their prior weight π_k to highlight their relative contribution. The intensity of the color bar corresponds to the value of the component likelihood weighted by the prior weight, $\pi_k p(x|k)$, and a darker, more intense color is

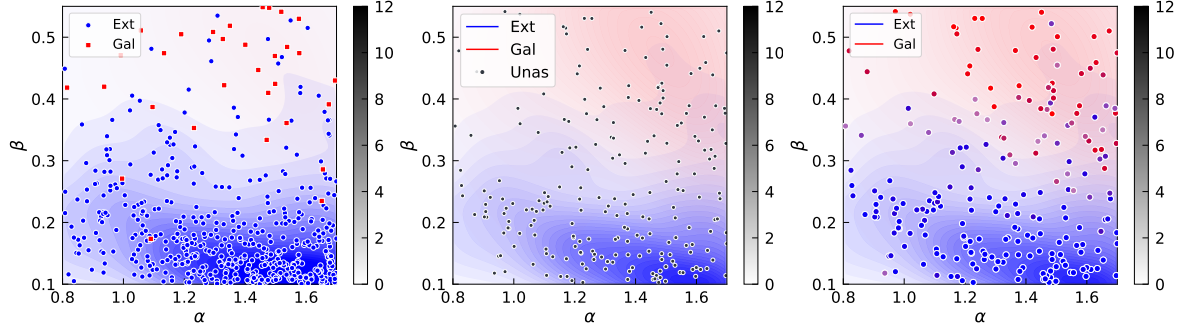


Fig. 5.8. Distributions of sources in the plane of the spectral parameters (α, β) for $E_0 = 1$ GeV. *Left panel:* PDFs of Galactic and extragalactic sources weighted by the corresponding class fractions. The sources are colored according to the catalog class. *Middle panel:* best-fit mixture model for the unassociated sources. The background colors show the individual components of the astrophysical-only model from eq. (5.4.9), where the DM subhalo contribution is set to zero. Specifically, we plot the decomposition of the mixture model into class-conditional PDFs weighted by their best-fit class prevalence $\pi_k p(x|k)$ for the extragalactic (blue) and Galactic (red) populations. For visualization, the distributions are integrated over the flux dimension. The color bar represents the value of the resulting probability density (shown in gray to represent both red and blue colors). *Right panel:* Class probabilities for unassociated 4FGL-DR4 sources (for $E_0 = 1$ GeV), on top of the mixture distribution (same as the middle panel). The color scale is proportional to the class probabilities, according to the formula $\text{color}(x_i) = \text{red} \cdot p(C_k = \text{gal}|x_i) + \text{blue} \cdot p(C_k = \text{ext}|x_i)$, e.g., a purple-colored source is a source for which the posterior class probabilities are similar and the classification is ambiguous.

associated with regions where there is a higher probability of observing a certain source class. We will follow this convention for all upcoming distribution plots.

We note that the extragalactic sources are dominated by AGNs, which usually have close to a power-law spectra (small β values). The Galactic sources often have curved spectra (large β values). The relative contribution of Galactic sources is larger for the unassociated sources compared to the associated ones (brighter red color in the right panel compared to the left panel in figure 5.8). The model on the middle and right panels of figure 5.8 is obtained by optimizing the prior and covariate shift parameters, π_k and θ_{cov} . We display the optimized Galactic and extragalactic components separately in figure 5.9. The corresponding best-fit parameters π_k (the class prevalence) are presented in table 5.1. As qualitatively discussed above, the expected contribution of Galactic sources to unassociated sources at high latitudes (29%) is significantly larger than the fraction of associated Galactic sources (6%) in the ROI. The right panel of figure 5.8 also shows the model's predictions for class probabilities of unassociated sources. We compare the relative importance of prior and covariate shift parts of the model in appendix 5.C.1, where we find that the prior shift plays a more important role than the covariate shift. In particular, for the model without the DM component, we find that the prior shift is preferred at 7σ significance (for one additional degree of freedom) compared to a model without either prior or covariate shifts, while the model with the covariate shift is preferred at 4.5σ significance (with 6 additional degrees of freedom). The model with both prior and covariate shifts is preferred at 6.7σ significance (with 7 degrees of freedom), i.e., most of the improvement is, indeed, due to the prior shift.

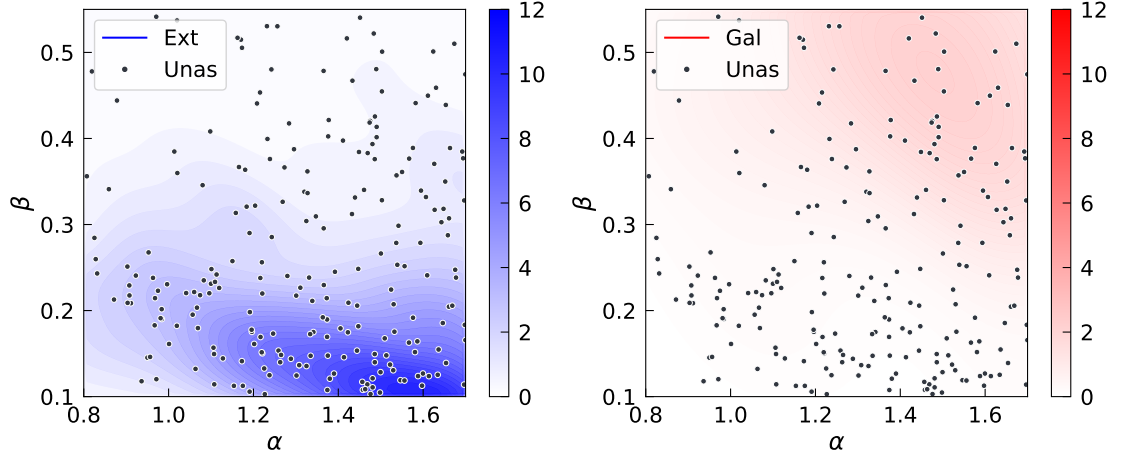


Fig. 5.9. Decomposition of the mixture model for unassociated sources with two classes (Galactic and extragalactic) for the ROI cut associated with $E_0 = 1$ GeV. The model is the same as in the middle and right panels of figure 5.8. *Left panel:* extragalactic component. *Right panel:* Galactic component.

type	N_{ext}	N_{Gal}	% Ext	% Gal	N_{tot}
assoc	466	31	94%	6%	497
unas	160.6	66.4	71%	29%	227

Table 5.1. Class prevalence for Galactic and extragalactic sources in the ROI. For the associated sources (second row) the numbers are derived from the 4FGL-DR4 catalog. For the unassociated sources (third row), the numbers correspond to the best-fit model with astrophysical components only (using $E_0 = 1$ GeV). The percent of galactic (% Gal) and extragalactic (% Ext) sources are given by the best-fit parameters π_k in eq. (5.4.9) for the two astrophysical components. The expected number of Galactic and extragalactic sources among the unassociated ones is derived as $\pi_k \times N_{\text{tot}}$, where $N_{\text{tot}} = 227$ is the total number of unassociated sources in the ROI.

5.4.10 Dark matter limits

The best-fit model including the DM component with DM mass of 100 GeV and $\langle\sigma v\rangle = 3.4 \cdot 10^{-25} \text{cm}^3 \text{s}^{-1}$ is shown in figure 5.10. We simultaneously fit the parameters π_k , θ_{cov} and $\langle\sigma v\rangle$. The corresponding components are presented in figure 5.11. The DM component is shown by the green color.⁶

Provided that all DM subhalos have the same gamma-ray spectra, the corresponding distribution of DM subhalos is much better localized in the α and β plane than the extragalactic and Galactic components. The spread in the DM subhalo distribution comes from statistical uncertainties in the reconstruction of spectral parameters. We note that not only the overall normalization but also the shape of the distribution of DM subhalos in the right panel of figure 5.11 changes for different annihilation cross sections and J-factor models (cf. figure 5.7). The reason is that the distribution depends on the relative contribution of subhalos with different J-factors, e.g., smaller J-factors correspond to proportionally smaller flux and larger uncertainties, which affect

⁶The class probabilities for unassociated sources in the astrophysical only model, as well as DM subhalo class probabilities for the different DM masses, are available online at <https://doi.org/10.5281/zenodo.17463823>.

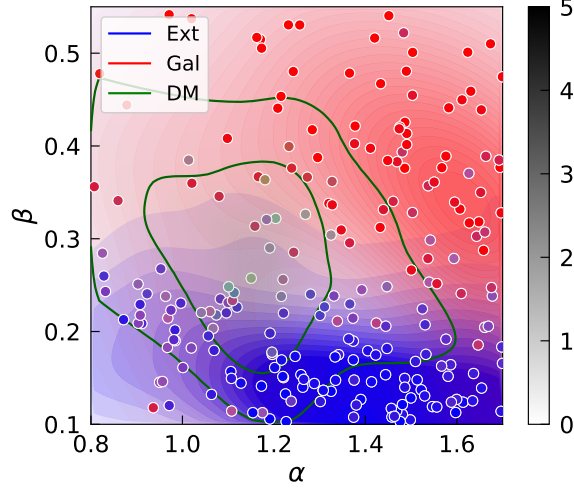


Fig. 5.10. Mixture model for unassociated sources including the DM component with $M_{\text{DM}} = 100$ GeV, following the same conventions as in figure 5.8. The green component represents the best-fit DM distribution for $\langle\sigma v\rangle = 3.4 \cdot 10^{-25} \text{cm}^3 \text{s}^{-1}$. The green lines represent the 68% and 95% containment levels for the DM distribution. The background colors represent the class-conditional distributions weighted by their prior weight: $\pi_k p(x|k)$. Sources are colored according to their posterior class probability following the formula $\text{color}(x_i) = \text{red} \cdot p(C_k = \text{gal}|x_i) + \text{blue} \cdot p(C_k = \text{ext}|x_i) + \text{green} \cdot p(C_k = \text{DM}|x_i)$.

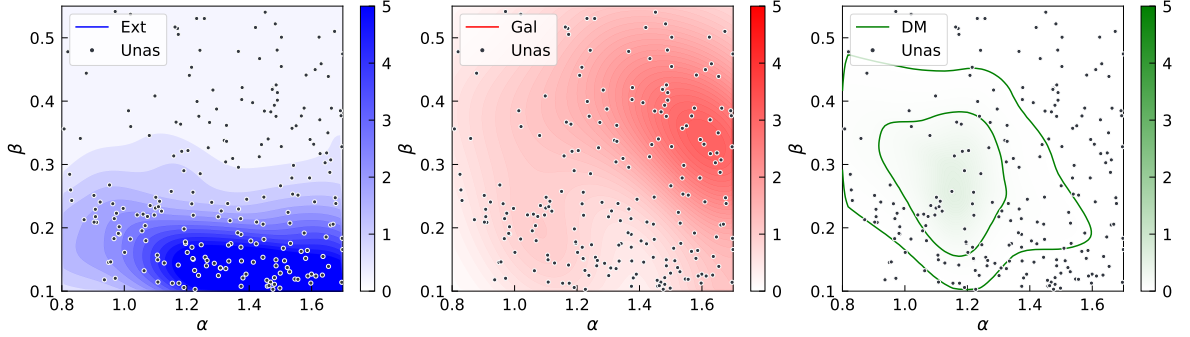


Fig. 5.11. Components of the mixture model in figure 5.10, following the same conventions as in figure 5.8.

the shape of the distribution. Nevertheless, for larger $\langle\sigma v\rangle$ values the expected contribution of DM subhalos also increases.

We present the details of the best-fit model for the DM subhalo component (including upper limits in the annihilation cross section) in table 5.2. Columns 2 to 6 show the best-fit model, e.g., the best-fit DM subhalo prevalence, the expected number of subhalos within our region of interest ($N_{\text{subhalo}}^{\text{ROI}}$) and for the full range of parameters ($N_{\text{sub}}^{\text{total}}$), as well as the best-fit thermally-averaged annihilation cross-section ($\langle\sigma v\rangle$). The value of $\Delta\text{TS}_0 = -2(\log \mathcal{L}_0 - \log \mathcal{L}_{\text{max}})$ quantifies the preference for the best-fit model over a baseline astrophysical-only scenario. For all tested models, the significance is less than 2 sigma. We notice that the largest possible contribution from DM subhalos is for $M_{\text{DM}} = 30$ GeV (although still not significant). **The expected number of DM**

M_{DM} [GeV]	% DM	best-fit model				upper bound		
		$N_{\text{subhalo}}^{\text{ROI}}$	$N_{\text{subhalo}}^{\text{total}}$	$\langle\sigma v\rangle$ [$\text{cm}^3 \text{s}^{-1}$]	ΔTS_0	$N_{\text{subhalo}}^{\text{ROI}}$	$N_{\text{subhalo}}^{\text{total}}$	$\langle\sigma v\rangle$ [$\text{cm}^3 \text{s}^{-1}$]
10	0.0	0.00	0.00	0.0	0.0	2.83	6.56	5.6×10^{-26}
30	6.8	15.37	36.23	2.7×10^{-25}	3.5	27.30	64.20	3.6×10^{-25}
100	2.4	5.45	6.21	3.4×10^{-25}	1.2	19.36	21.86	5.9×10^{-25}
300	2.0	4.59	7.66	1.1×10^{-24}	1.1	13.23	21.46	1.7×10^{-24}
1000	0.0	0.00	0.00	0.0	0.0	3.37	13.75	4.7×10^{-24}

Table 5.2. Expected number of DM subhalos (N_{sub}) for the **best-fit model** and for the **95% confidence upper bound** on the annihilation cross-section ($\langle\sigma v\rangle$). In both cases, we report the subhalo count within our ROI and the total count. The test statistic difference quantifies the significance of the best-fit model over an astrophysical-only model, $\Delta\text{TS}_0 = -2(\log \mathcal{L}_0 - \log \mathcal{L}_{\text{max}})$. The number of subhalos, N_{sub} , is constrained to be non-negative.

subhalos of 15.4 for this DM mass in our ROI is comparable, e.g., to 16 DM subhalo candidates reported in ref. [229]. This comparison should be taken with a grain of salt, as the calculation in ref. [229] was performed for the 3FGL catalog (8 years of data) compared to the 4FGL-DR4 catalog (14 years of data) in our work with the J-factors based on the mass estimate within the tidal radius (cf. $J_{\text{tidal}}^{M_{\text{sub}}}$ in Section 5.4.6) compared to the J-factor integrated within 0.1° from the center base on the maxima velocity used in table 5.2. ~~In this case~~ We also notice that for $M_{\text{DM}} = 30 \text{ GeV}$, the expected contribution from Galactic sources (mostly millisecond pulsars) is the smallest. This is to be expected, since the spectrum of DM annihilation in the $b\bar{b}$ channel is similar to the spectra of pulsars at this DM mass. As a result, there may be a degeneracy between the DM subhalos and Galactic sources for this DM mass. Provided that we do not find significant evidence for a signal from DM, we proceed to set conservative upper bounds on $\langle\sigma v\rangle$ shown in the “upper bound” section of the table.

In order to estimate the overall goodness of fit of the model in eq. (5.4.9) without DM to the data, we compare in appendix 5.C.2 the log likelihood of the fit to the data with the distribution of log likelihood values for the data generated from the best-fit model. We find that the deviation of the log likelihood for the real data is less than 2σ smaller than the mean log likelihood value for the simulated data, i.e., the deviation is consistent with statistical fluctuations within 2σ . We also perform DM injection tests in appendix 5.C.3. We find that our method can recover the DM signal within statistical fluctuations: for all DM masses the $\langle\sigma v\rangle$ value for the injected signal is within the 68% containment of the recovered $\langle\sigma v\rangle$ values around the median recovered $\langle\sigma v\rangle$.

In addition, we determine the precision-recall curves for the classification of individual DM subhalos for 1000 simulations with injected DM signal (in case of $M_{\text{DM}} = 30 \text{ GeV}$) and evaluate the average precision of our model in appendix 5.C.4.

We show the upper bounds on $\langle\sigma v\rangle$ in figure 5.12. ~~where we also compare with the limits obtained previous works. To derive~~ The uncertainty band for the annihilation cross-section bound is derived with ~~we adopt~~ a bootstrap strategy. We identify the best-fit model for each DM mass and use the probability density function for that model (cf. eq. (5.4.9)) to sample 100 new datasets via Hybrid (a.k.a. “Hamilton”) Monte Carlo (HMC) [263] and the NUTS algorithm [264]. For each simulated dataset, we derive the bounds on $\langle\sigma v\rangle$ and determine the 95% containment interval for the resulting distribution of values. We further adjust the upper limit of this interval to conservatively account for potential systematic uncertainties associated with the presence of 4FGL sources, as detailed in Appendix 5.A. This uncertainty range is represented by the

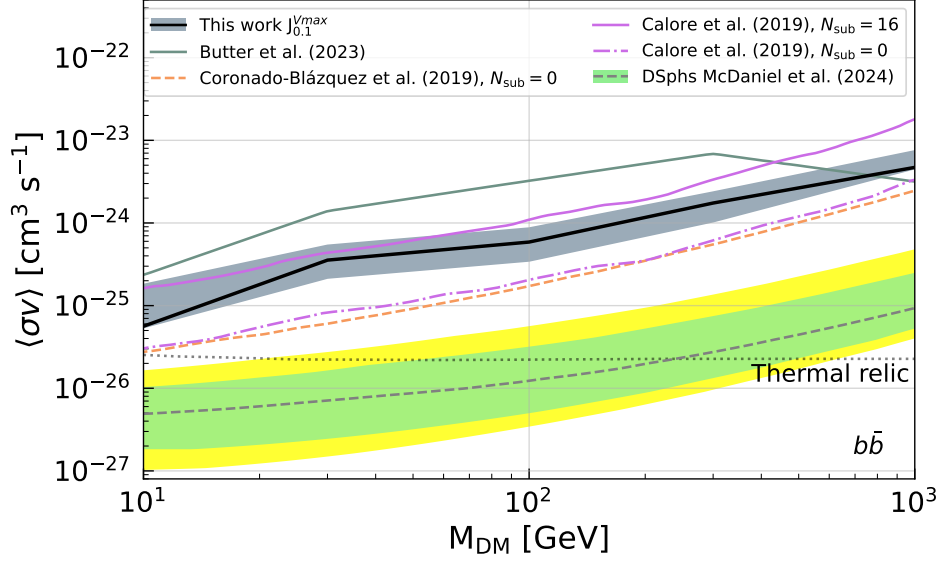


Fig. 5.12. Upper bounds with 95% confidence on the DM annihilation cross section $\langle\sigma v\rangle$ in the $b\bar{b}$ channel. Solid black line: our baseline limits derived with the V_{max} determination of the J-factor integrated within the radius of 0.1° . Gray band: 95% containment of the upper limit due to statistical fluctuations of the distribution of sources. Solid and dashed purple lines: the limits derived by Calore et al. (figure 6 left panel) [231]. Dashed orange line: limits derived by Coronado-Blázquez et al. (figure 5 upper panel) [230]. Solid green line: the limits derived by Butter et al. (figure 11) [245]. Green and yellow bands with dashed gray line: limits derived for dwarf spheroidal Galaxies at 68% and 95% confidence level by McDaniel et al. (figure 6 top left panel) [103]. Dotted black line: the expectation for the WIMP DM thermal relic cross-section from ref. [6].

gray shaded band in figure 5.12. For comparison we show the limit obtained by assuming that there are 16 DM subhalos in the 3FGL catalog by the purple solid line [231]. We notice that our limit for $M_{\text{DM}} = 30$ GeV is close to the limit derived in ref. [231], although the upper limit on the number of DM subhalos in our analysis (27.3) is larger than the number of DM subhalo candidates (16) assumed in [231] following the analysis of ref. [229]. Our limit is slightly better due to longer observation time (14 years vs 8 years), which results in better flux sensitivity. For other DM masses, our limit is better than the limit reported in [231] due to smaller upper limits on the number of DM subhalo candidates. as well as the best possible sensitivity of the 3FGL catalog to DM subhalos (assuming that no unassociated sources correspond to DM subhalos) shown by purple dash-dotted line and the orange dashed line. All of these upper limits on DM subhalo candidates, on the other hand, are larger than 1. As a result, the constraints on the DM annihilation cross sections are worse than the best possible constraints corresponding to zero observed DM subhalos, shown by the purple dash-dotted line [231] and the orange dashed line [230] in figure 5.12. The solid green line corresponds to more recent bounds derived using an ML approach [31]. Our limits are also significantly better for most of DM masses than in ref. [245] due to the very conservative approach in estimating the number of DM subhalos adopted in ref. [245], where the number of possible DM subhalo candidates, e.g., for $M_{\text{DM}} = 30$ GeV is estimated to be 281 in the most conservative scenario.

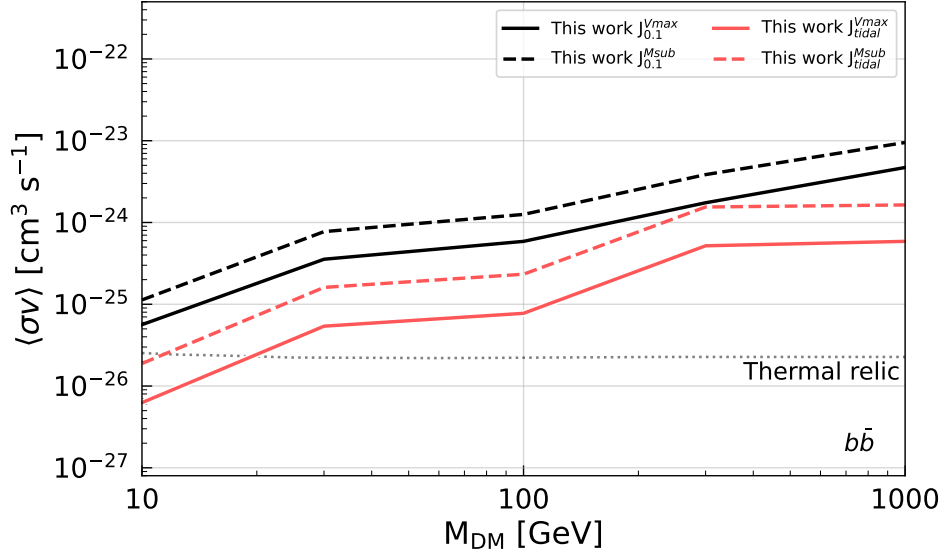


Fig. 5.13. Comparison of the 95% upper bounds on $\langle\sigma v\rangle$ derived in this work for different choices of the J-factor. Solid (dashed) black line: the limit derived with the V_{max} (M_{sub}) determination of the J-factor integrated within the radius of 0.1° . Solid (dashed) red line: the limit derived with the V_{max} (M_{sub}) determination of the J-factor integrated within the tidal radius. The uncertainty bands for the different J-factor models are proportional to the uncertainty band for our baseline model, $J_{0.1}^{V_{\text{max}}}$ (figure 5.12), with the same scaling factors as the ratios of the corresponding upper limits. These uncertainty bands are not plotted for clarity of presentation.

~~In dashed gray, we plot~~ Dashed gray line in figure 5.12 shows a recent bound derived using dwarf spheroidal galaxies [103] with 68% and 95% confidence bands shown by green and yellow areas respectively. We note that the constraints derived in this work are an order of magnitude less stringent than the limits derived using dwarf spheroidal galaxies (even if we consider J-factors integrated to tidal radius for the subhalos). Provided that the values of J-factors $\lesssim 10^{20} \text{GeV}^2 \text{cm}^{-5}$ for the brightest dwarf galaxies [103] are comparable to the brightest subhalos (cf. figure 5.5), the main reason for the less stringent constraints is that the positions and J-factors of the dwarf galaxies are known (although with some uncertainties), while the subhalos can be at any position in the sky.

In figure 5.13, we display a comparison of the upper bounds for different choices of the J-factor. In particular, the limits for the J-factor integrated within 0.1° for the V_{max} and M_{sub} J-factor determination methods are shown by the black solid and dashed lines respectively. The limits corresponding to the J-factor integrated up to the tidal radius for the V_{max} and M_{sub} J-factor determination methods are shown by the red solid and dashed lines, respectively.

The limits presented in figures 5.12 and 5.13 are derived assuming the velocity-independent DM annihilation cross section. In case of velocity-dependent annihilation cross section, e.g., due to Sommerfeld enhancement [61, 265–267] the relative contribution of small subhalos close to the Earth, which have a smaller velocity dispersion than the dwarf galaxies, will be more important. In particular, a significant population of subhalos at distances of a few kpc with typical velocity dispersion of a few km/s is expected [224]. These subhalos can have a factor of a few larger enhancement of annihilation cross section compared to the dwarf galaxies, where the typical velocity dispersion has a peak in the distribution around 10 km/s (e.g., [268]). The actual

enhancement of the limits from subhalos compared to the dwarf galaxies depends on the details of the model (e.g., there should be no saturation of enhancement down to velocities of a few km/s), on the integral of the velocity dispersion for the subhalos, and on the relative contribution of small subhalos in the vicinity of the Earth and large distant subhalos. This detailed analysis is deferred to a future work.

5.4.11 Discussion and conclusions

In this work, we have searched for DM annihilating subhalos among the unassociated *Fermi*-LAT sources. We developed the first fully probabilistic model of gamma-ray sources, incorporating both known astrophysical source classes and a potential new population of gamma-ray sources modeled as DM subhalos annihilating into Standard Model particles.

We characterize gamma-ray sources using their energy spectra fitted by the log-parabola function. We take the parameters $\mathbf{x}$ of the log-parabola fit as the input features. We then construct a probabilistic model $p(\mathbf{x})$ for the unassociated sources as a weighted mixture of three source classes: Galactic and extragalactic astrophysical sources plus the hypothetical DM subhalos. The Galactic and extragalactic components encompass various source types classified by the *Fermi*-LAT collaboration. We analyze data shifts between associated and unassociated sources within a general framework of quantification learning, considering two effects: (1) a covariate shift affecting all source classes in the same way, and (2) a prior probability shift, which allows for changes in class prevalence, including the possible presence of a new source class among the unassociated sources.

The astrophysical components of the mixture model are the conditional PDFs $p(\mathbf{x}|k)$ for the two astrophysical classes k , Galactic and extragalactic. They are built from the corresponding empirical distributions of the associated sources, modified by an optimizable covariate-shift function. The DM subhalo component of the mixture is determined by Monte Carlo simulations based on a set of DM parameters (mass, annihilation channel, and cross section), including the distribution of J-factors coming from N-body simulations. The global likelihood of the model is obtained by multiplying the PDFs $p(\mathbf{x}_i)$ of the unassociated sources i with $\mathbf{x}_i$ inside the range of parameters that we consider.

By optimizing the model likelihood, we extract the best-fit parameters, including $\langle\sigma v\rangle$, for various DM masses in the $b\bar{b}$ annihilation channel. No significant excess corresponding to DM subhalos is observed beyond Galactic and extragalactic sources. Consequently, we derive 95% upper bounds on the DM annihilation cross section. To our knowledge, this is the first time an upper bound on DM annihilation has been obtained by constraining the contribution of DM subhalos to unassociated *Fermi*-LAT sources using a maximum likelihood approach. Our constraints are competitive with previous limits derived from DM subhalo searches using classify-and-count methods. With respect to the latter, our approach, based on quantification learning, offers several advantages :

- (i) It accounts for differences in distributions (data shifts) between associated and unassociated source populations.
- (ii) It determines the potential contribution of a new hypothetical source population, e.g., the DM annihilating subhalos, without imposing ad hoc assumptions on its fraction relative to known sources. This is a key advantage over traditional supervised machine learning classification.

- (iii) The probabilistic model enables one to generate mock data (samples of unassociated sources), which we use to validate our approach and estimate the uncertainties of the upper bounds on the DM annihilation cross section.
- (iv) The model provides both the PDF $p(\mathbf{x}|k)$ of observed source features $\mathbf{x}$ given a source class k (either astrophysical or DM), and the posterior probability $p(k|\mathbf{x})$ of a source belonging to a given class based on observed features.

Thus, our approach allows one not only to determine the conditional probabilities $p(k|\mathbf{x})$, which is the main objective of ML classification models, but also to estimate the overall likelihood of the model given the distribution of unassociated sources. This likelihood serves as a tool to search for, or constrain, a new class of sources absent from the training data, i.e., the associated *Fermi*-LAT sources.

This work bridges the gap between ML approaches for characterizing unassociated sources traditionally focused on classification and the standard astrophysical analysis using maximum likelihood models. As a result, we construct a model capable of both classifying sources and constraining a new source population. Our approach is broadly applicable to searches of anomalies in distributions for datasets exhibiting both covariate and prior shifts, where the target data is unlabeled.

5.A Simulation of gamma-ray signals from dark matter subhalos

In this appendix we provide details about the simulation of gamma-ray signal from DM subhalos. Eq. (5.4.11) describes the probability of observing a DM subhalo given a DM model. This equation includes three terms that need to be estimated through simulations: the number density $n(\phi_{\text{DM}})$ of DM subhalos as a function of expected flux (section 5.4.7), the probability $p_{\text{DM}}(\alpha, \beta|\phi_{\text{DM}}; \theta_{\text{DM}})$ of observing a source with the log-parabola spectrum parameters (α, β) given a DM model, which we obtain by computing and fitting simulations of DM point sources, and the efficiency $\epsilon_{\text{eff}}(\phi_{\text{DM}})$ of the detection of a source, which we obtain from the same set of simulations by identifying the fraction of sources which can be detected by *Fermi*-LAT. We now describe how the DM distribution conditioned on the gamma-ray flux is derived and how we estimate the detection efficiency for DM subhalos.

We compute the DM distribution conditioned on the gamma-ray flux, through realistic photon-counts simulations of DM point sources. We decompose the gamma-ray sky into two components: a diffuse emission component (constituted by the diffuse Galactic emission and an isotropic background) and the signal from the DM point sources. We model the diffuse Galactic emission using the `gll_iem_v7` template provided by the *Fermi*-LAT collaboration as well as the corresponding isotropic background model. We integrate eq. (5.4.10) in 16 logarithmic energy bins between 100 MeV and 1 TeV to obtain the theoretical energy spectrum in units of the photon flux per energy bin. We specify the spectral energy distribution (SED), dN_γ/dE , using the recent estimations from CosmiXs [84].

Furthermore, we notice that the J-factor and annihilation cross-section term factor out of the integral into an overall constant that ultimately defines the total flux normalization:

$$\Phi_i = A \int_{E_{\min,i}}^{E_{\max,i}} \frac{dN_\gamma}{dE} dE. \quad (5.A.1)$$

For this reason, once we fix the SED for a specific DM annihilation channel and mass, the energy spectrum $\phi(E)$ is defined except for an energy-independent normalization term. By fixing

Healpix order	8
Weeks	9 – 795
Emin	100 MeV
Emax	1 TeV
Instrument Response Functions (IRFs)	P8R3_SOURCE_V3
EVCLASS	128 (Source)
EVTTYPE	3 (Front+Back)
ZMAX	90° under 1 GeV, 105° otherwise
Galactic foreground model	gll_iem_v7.fits

Table 5.3. Fermi Science Tools settings used for the 15-year data set analysis.

the overall flux normalization, we obtain the expected gamma-ray flux due to a specific DM source. Experimental uncertainties in the reconstruction of energy and direction of gamma rays induce a spread in the distribution of photons associated with a point source. We simulate sources by convolving with the *Fermi*-LAT point spread function (PSF) and exposure using *Fermi* Science Tools (fermitools) version 2.2.0⁷.

In table 5.3 we summarize the analysis settings, which are similar to the standard event selection cuts used in the construction of gamma-ray source catalogs. Likewise, we convolve the diffuse Galactic foreground and isotropic emission with the PSF and exposure to obtain the diffuse foreground emission. For each simulation, we add the emission from 10,000 DM sources randomly and isotropically distributed in the sky together with the foreground emission. We chose this number of sources to roughly match the number of resolved sources in the 4FGL catalog and to have sufficient statistics for the DM sources, while avoiding excessively demanding simulations. We thus obtain a map for the expected photon counts due to a DM population with a specific mass, annihilation channel, and a common flux.

We employ each simulation (associated with a specific total gamma-ray flux for the DM source) to estimate $p_{\text{DM}}(\alpha, \beta | \phi_{\text{DM}}; \theta_{\text{DM}})$ as follows. First, we perform a Poisson realization of the expected photon counts map. Then we proceed with fitting the energy spectrum of each source with a log parabola function (with a fixed spatial position), yielding the DM distribution conditioned on the flux. Finally, we estimate the detection efficiency $\epsilon_{\text{eff}}(\phi_{\text{DM}})$ by studying the TS of the DM sources for a given map. We define the TS of a source by comparing the overall likelihood with and without the target source. We define the TS as:

$$TS = -2 \ln \left(\frac{\mathcal{L}}{\mathcal{L}_0} \right), \quad (5.A.2)$$

where $\mathcal{L}$ is the likelihood of the best fit model including the source, and $\mathcal{L}_0$ is the likelihood of the model without the source. If the TS of a source is larger than 25 we include that source to a database of resolved sources, for which the SED fit parameters are reliable and can be used for the estimation of $p_{\text{DM}}(\alpha, \beta | \phi_{\text{DM}}; \theta_{\text{DM}})$. Else, we discard the source as a low TS source which would not be measured by the *Fermi* LAT.

The detection efficiency $\epsilon_{\text{eff}}(\phi_{\text{DM}})$ of the *Fermi* LAT to a DM source with a specific flux and SED is then defined as the ratio between the number of detected sources and the overall number of sources included in the simulation:

$$\epsilon_{\text{eff}} = \frac{N_{\text{sources}}(TS > 25)}{N_{\text{sources}}^{\text{tot}}}. \quad (5.A.3)$$

⁷<https://github.com/fermi-lat/Fermitools-conda/wiki>

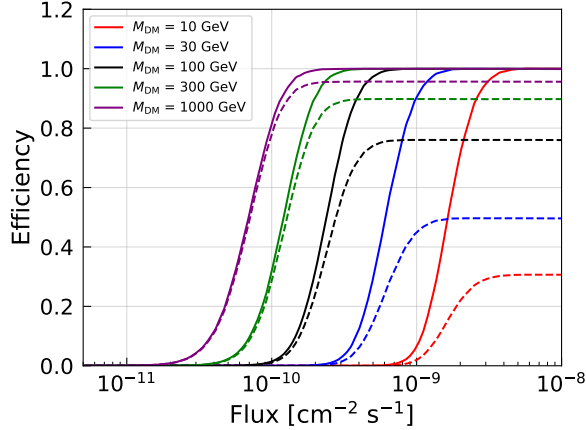


Fig. 5.14. Detection efficiency for the $b\bar{b}$ annihilation channel simulations. Solid lines denote the baseline efficiency used in the analysis, derived from simulations that disregard known 4FGL sources. Dashed lines illustrate a conservatively corrected efficiency, representing a worst-case scenario where a disk around each 4FGL source is masked to estimate the maximum potential degradation caused by their inclusion.

In order to avoid biasing the estimation of the DM distribution due to a low number of detectable sources or due to a very specific realization of the sky, we perform additional simulations of the gamma-ray sky until we detect a total of at least 20,000 DM sources or until 1,000 simulations have been performed. Once we have built a dataset of measurements for the energy spectrum of DM sources, we estimate the probability density function $p_{DM}(\alpha, \beta | \phi_{DM})$ by employing KDE, in analogy with what we did in section 5.4.4 for the astrophysical sources.

Due to the high number of DM sources, and the large number of simulations, we cannot employ the full detection and characterization pipeline which is usually adopted by the *Fermi*-LAT collaboration. We exploit the additional information available to us from simulations to perform a simplified analysis pipeline. The true energy spectrum of each DM source is fixed (and known) in our simulations, and so are the foreground components. The positions of the DM sources are known as well, and the sources are all point-like sources. For these reasons, we make the following assumptions in our analysis:

- We assume that the energy spectrum of each background source (i.e. any source different from the one we are fitting) is known and fixed to the theoretical one (while in general they are jointly optimized together with the target of the fit);
- We fix the diffuse Galactic and the isotropic background components;
- We assume that the localization of the source is known (which in this case is true, but not in the case of source detection on real data);
- We assume that there are no low TS background sources, as we know exactly what we put inside our simulation.
- We assume that contributions from known 4FGL sources are negligible when estimating the DM distribution. While these bright sources can reduce detection efficiency and should ideally be modeled jointly, we estimate the maximum degradation by masking a disk around

each 4FGL source. This mask corresponds to the 68% containment angle of the PSF at the peak energy (E_{peak}) of the DM spectrum for each considered mass. Because the peak of the DM energy spectrum increases with the DM mass, the detection efficiency is affected more significantly at lower DM masses, as displayed in figure 5.14. Specifically, for $M_{DM} \geq 100$ GeV, the narrower PSF results in a masked sky fraction of 20% or less, deteriorating the bounds by approximately 15%. Conversely, for lower masses ($M_{DM} = 10 - 30$ GeV), the broader PSF yields a masked fraction of 50 – 70%, leading to a 35 – 60% degradation of the bounds. Except for the $M_{DM} = 10$ GeV case, this degradation remains within the bootstrap uncertainty band derived for the unmasked analysis. Nonetheless, we conservatively adjust the 95% upper confidence level in figure 5.12 to account for these systematic effects.

By making these assumptions, we fit each DM source individually, regardless of the others.

To perform a maximum likelihood estimation fit of the DM energy spectrum for each source, we define the likelihood as follows:

$$\mathcal{L} = \prod_{e=1}^{N_{bin}} \prod_{j \in \text{disc}_e} \mathcal{P}(k_{e,j}, \lambda_{e,j}(\mathbf{x})). \quad (5.A.4)$$

Specifically, for each energy bin, we identify a disc around the source corresponding to the 95% containment angle of the PSF with a minimal angle of 0.5° and a maximal angle of 5° . This ensures high enough statistics for the high-energy bins, and reduces the background noise for the low-energy bins, while leaving the intermediate-energy bins unaffected.

For each pixel of the disc, we compare the photon counts k_{ij} with the estimated photon counts $\lambda_{ij}(\mathbf{x})$ using the Poisson likelihood. The estimated photon counts are a sum of the Galactic foreground in the disc, the isotropic background, the emission of the other background (fixed) DM sources and the photon counts coming from the target source:

$$\lambda(\mathbf{x}) = \lambda_{gal} + \lambda_{iso} + \lambda_{S \neq S_i} + \lambda_{S_i}(\mathbf{x}). \quad (5.A.5)$$

Specifically, for a source i and pixels j :

$$\lambda_{S_i,e,j} = \left[\int_{E_{min}^e}^{E_{max}^e} \phi_{0,i} \frac{E}{E_0}^{-\alpha_i - \beta_i \log(E/E_0)} dE \right] \cdot \text{PSF}_{ij}^e \cdot \text{exposure}_j^e \cdot \frac{4\pi}{12N_{side}^2}. \quad (5.A.6)$$

We repeat the fitting procedure for each DM source, at a fixed total flux, and we build the DM distribution $p_{DM}(\alpha, \beta | \phi) \equiv p_{DM}^{KDE}(\alpha, \beta | \phi)$ through the KDE.

5.B Details of model optimization with the EM algorithm

Maximum likelihood estimation of the parameters of a mixture model is a challenging task due to the interplay between the mixture weights and the distribution parameters. A standard iterative algorithm for this task is the Expectation Maximization (EM) algorithm [125, 126].

We introduce latent binary vectors $\mathbf{z}_i$, corresponding to each data point $\mathbf{x}_i$, where $\dim(\mathbf{z}_i) = K$, where K is the number of mixture components. These latent variables encode the membership of point $\mathbf{x}_i$ in the k -th component of the mixture, such that all elements of $\mathbf{z}_i$ are zero, except for $z_{ik} = 1$. In other words, $\mathbf{z}_i$ is a categorical random variable represented with the one-hot-encoding scheme. The coefficients π_k of the likelihood in eq. (5.4.9) are interpreted as the parameters

determining the corresponding categorical *prior* distribution of $\mathbf{z}_i$, i.e. $p(z_{ik} = 1) = \pi_k \forall i$, such that

$$p(\mathbf{z}_i) = \prod_k \pi_k^{z_{ik}}. \quad (5.B.1)$$

With this interpretation for $\mathbf{z}_i$, we can identify $p(\mathbf{x}_i | z_{ik} = 1) = p_k(\mathbf{x}_i | \boldsymbol{\theta}_k)$, such that

$$p(\mathbf{x}_i | \mathbf{z}_i) = \prod_k p_k(\mathbf{x}_i | \boldsymbol{\theta}_k)^{z_{ik}}. \quad (5.B.2)$$

We then have all the elements to compute the *posterior* distribution of $\mathbf{z}_i$ as:

$$p(z_{ik} = 1 | \mathbf{x}_i) = \frac{p_k(\mathbf{x}_i | \boldsymbol{\theta}_k) \pi_k}{\sum_j p_j(\mathbf{x}_i | \boldsymbol{\theta}_j) \pi_j} \equiv \gamma(z_{ik}). \quad (5.B.3)$$

In order to find the optimum parameters $\{\boldsymbol{\theta}_k\}$ and weights $\boldsymbol{\pi}$, the idea is to consider the joint likelihood $p(\mathbf{x}_i, \mathbf{z}_i)$ for the whole dataset of points $\mathbf{x}_i$ with associated latent variables $\mathbf{z}_i$:

$$p(\mathbf{X}, \mathbf{Z} | \{\boldsymbol{\theta}_k\}, \boldsymbol{\pi}) = \prod_i \prod_k [\pi_k p_k(\mathbf{x}_i | \boldsymbol{\theta}_k)]^{z_{ik}}. \quad (5.B.4)$$

In spite of this apparent complication of introducing extra variables in the problem, i.e., the latent variables, the likelihood in eq. (5.B.4) is arguably easier to optimize than the original likelihood $p(\mathbf{x} | \{\boldsymbol{\theta}_k\}, \boldsymbol{\pi}) = \prod_i p(\mathbf{x}_i)$, where each term in the product is built as in eq. (5.4.9). One of the reasons for this is that, the log-likelihood of eq. (5.4.9) is a log of a sum, which is computationally challenging to optimize, especially in the large K regime, while the log of the likelihood in eq. 5.B.4 is the sum of logs, i.e., $\log p(\mathbf{X}, \mathbf{Z} | \{\boldsymbol{\theta}_k\}, \boldsymbol{\pi})$ is very similar to the cross-entropy cost function, which is easily optimizable.

An obvious problem is that in eq. (5.B.4) we do not know the z_{ik} variables. The EM algorithm considers instead the expectation value of this likelihood (or more concretely, of the log-likelihood), under the posterior distribution of $\mathbf{Z}$, i.e.

$$\mathbb{E}_{\mathbf{Z}}[\log p(\mathbf{X}, \mathbf{Z} | \{\boldsymbol{\theta}_k\}, \boldsymbol{\pi})] = \sum_{\mathbf{Z}} p(\mathbf{Z} | \mathbf{X}, \{\boldsymbol{\theta}_k\}, \boldsymbol{\pi}) \ln p(\mathbf{X}, \mathbf{Z} | \{\boldsymbol{\theta}_k\}, \boldsymbol{\pi}). \quad (5.B.5)$$

Eq. (5.B.5) is maximized according to the following iterative procedure:

1. Initialize $\{\boldsymbol{\theta}_k\} = \{\boldsymbol{\theta}_k\}_{\text{old}}$, and $\boldsymbol{\pi} = \boldsymbol{\pi}_{\text{old}}$;
2. Evaluate $p(\mathbf{Z} | \mathbf{X}, \{\boldsymbol{\theta}_k\}_{\text{old}}, \boldsymbol{\pi}_{\text{old}})$;
3. Plug the posterior obtained in step 2 into expr.(5.B.5), such that now the only term containing the optimizable parameters $\{\boldsymbol{\theta}_k\}, \boldsymbol{\pi}$ is the log-likelihood. So we get the expression:

$$Q(\{\boldsymbol{\theta}_k\}, \boldsymbol{\pi}) \equiv \sum_{\mathbf{Z}} p(\mathbf{Z} | \mathbf{x}, \{\boldsymbol{\theta}_k\}_{\text{old}}, \boldsymbol{\pi}_{\text{old}}) \ln p(\mathbf{X}, \mathbf{Z} | \{\boldsymbol{\theta}_k\}, \boldsymbol{\pi}).$$

4. Update the parameters as:

$$\{\boldsymbol{\theta}_k\}_{\text{new}}, \boldsymbol{\pi}_{\text{new}} = \underset{\{\boldsymbol{\theta}_k\}, \boldsymbol{\pi}}{\operatorname{argmax}} Q(\{\boldsymbol{\theta}_k\}, \boldsymbol{\pi})$$

5. Repeat above steps until convergence

Note that in step 4 above, the optimization of π is analytical, i.e., using a Lagrange-multiplier optimization with the condition $\sum_j \pi_j = 1$, we obtain:

$$(\pi_k)_{\text{new}} = \frac{1}{N} \sum_i^N \gamma(z_{ik}), \quad (5.B.6)$$

where $\gamma(z_{ik})$ is given by eq. (5.B.3). The optimization of $\{\theta_k\}$ is in general not analytical.

5.C Consistency checks of the model

In this appendix we perform several checks of our model. At first, we investigate the relative importance of prior and covariate shifts by comparing models with only prior or only covariate shift to the full model that contains both prior and covariate shifts. We then proceed to perform consistency checks. We start by generating distributions of sources in parameter space from our best-fit model (including only astrophysical components) and fit the generated data with the same model. The goal is to determine a distribution of $\log \mathcal{L}$ values due to statistical fluctuations (since the data was generated from this model) and to compare to the value of $\log \mathcal{L}$ for the actual data to check if the model without a DM contribution is generally a good fit to the data. As a last consistency check, we perform injection tests, where we add a DM subhalo component to the astro-only model to check if we can recover the injected signal.

5.C.1 Importance of covariate and prior shifts

In order to investigate the importance of the prior and covariate shifts, we first compare the performance of four models in eq. (5.4.9) without the DM component: a model without either prior or covariate shifts (i.e., no free parameters) determined from associated sources, denoted as “as”, models with only prior or covariate shifts, denoted as “p” or “c” respectively, and a model with both prior and covariate shifts, denoted as “p + c”.

The model without either prior or covariate shifts is constructed from the catalog of associated sources following sections 5.4.3 and 5.4.4, where the prior probability of a class is given by the relative class prevalence π_k in the 4FGL-DR4 catalog, while the PDFs of the Galactic and extragalactic components are derived from the corresponding associated sources. The model with either prior or covariate shift follows eq. (5.4.9), where we fix the covariate shift modulation to be equal to 1 (i.e., $\tilde{C}(\mathbf{x}; \theta_{\text{cov}}) = 1$), or fix the prior probabilities to the corresponding values for associated sources.

We compare the performance of the four models evaluated on the unassociated sources in the ROI for $E_0 = 1$ GeV in table 5.4. In column 2, we show the difference in log likelihood between the model specified in the first column and the model based on distributions of associated sources. We report the corresponding additional degrees of freedom in column 3. We find that the prior shift model leads to a larger increase in likelihood for one degree of freedom (we have a constraint that the sum of priors is equal to 1: $\pi_{\text{gal}} + \pi_{\text{ext}} = 1$) compared to a model with covariate shift only. The corresponding significance of exclusion of the null hypothesis (the “as” model) is presented in column 4. It is derived by assuming the χ^2 distribution for $-2\Delta \log \mathcal{L}$ with the corresponding number of degrees of freedom from column 3. In column 5 we show the Akaike information criterion, $\text{AIC} = -2\log \mathcal{L} + 2k$, where k is the number of degrees of freedom. We find that although log likelihood of the full prior + covariate shift model is the largest, both the significance of exclusion of the null hypothesis and the AIC for this model are comparable to the ones for the prior shift only model. Thus, we conclude that the prior shift is the dominant effect compared to

model	$2\Delta\log\mathcal{L}_{as,x}$	$\Delta\text{dof}_{as,x}$	$N\sigma_{as,x}$	$\text{AIC}_{as,x}$	$2\Delta\log\mathcal{L}_{x,p+c}$	$\Delta\text{dof}_{x,p+c}$	$N\sigma_{x,p+c}$
associated	0	0	0.0	-178	64.7	7	6.7
prior	49.5	1	7.0	-225	14.2	6	2.2
covariate	33.4	6	4.5	-199	30.3	1	5.5
prior + covariate	64.7	7	6.7	-227	0.0	0	0.0

Table 5.4. Comparison of models of unassociated sources with Galactic and extragalactic components. “as” denotes the model derived from distributions of associated sources applied to the unassociated sources (without free parameters). “ $p + c$ ” is the full covariate + prior shift model. Subscript “ x ” denotes the model specified in the first column of the table. For example, the value of $2\Delta\log\mathcal{L}_{as,x} = 33.4$ in the fourth row and second column corresponds to the difference between the model derived for associated sources and the model with only covariate shift, i.e., to $x = c$ (covariate) in this case.

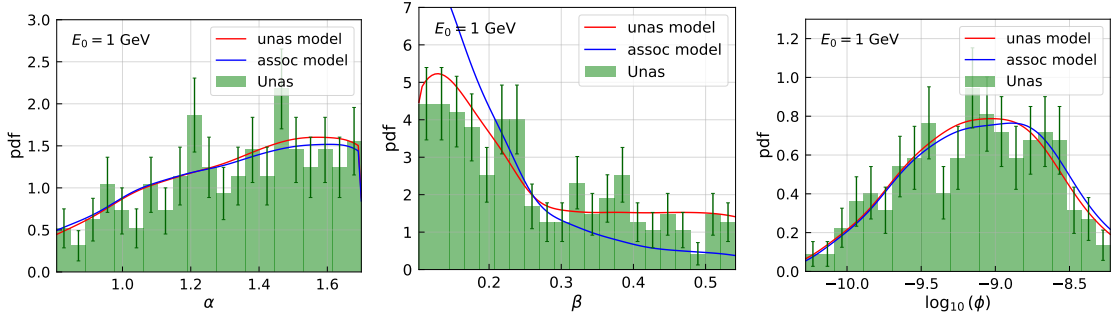


Fig. 5.15. Distributions of the α , β and $\log_{10}(\phi)$ parameters for the unassociated sources in the 4FGL-DR4 catalog. The bars show statistical uncertainty. The best-fit model that includes both covariate and prior shifts is shown by the red line while the model derived for associated sources is shown by the blue line. The 3D model is integrated in two variables to obtain the distribution in the third variable, shown on the plot.

the covariate shift. We further compare the models in column 1 to the full prior + covariate shift model in columns 6, 7, and 8, where we show $2\Delta\log\mathcal{L}$, the additional degrees of freedom, and the significance of exclusion of the null hypothesis, which, in this case, is the model in the first column. In particular, we find that the prior shift model can be excluded only at 2.2 sigma level in favor of the “ $p + c$ ” model.

In figure 5.15, we compare our models with the distributions of parameters α , β , and $\log_{10}(\phi)$ for the unassociated sources. We display the best-fit “ $p + c$ ” model (in red) and compare it with the prior KDE model derived for associated sources (in blue). The “ $p + c$ ” model provides a reasonable fit to the data for all the distribution parameters. On the other hand, the model derived for associated sources fails to capture the β dependence correctly, as expected, since the distribution for unassociated sources differs considerably from the distribution of associated sources for the β parameter (cf. figure 5.3).

5.C.2 Bootstrap MC and overall goodness of fit

We quantitatively estimate the goodness of fit of the “ $p + c$ ” model in figure 5.16. In this figure, we generate samples of sources in the parameter space from the best-fit “ $p + c$ ” model. We then reoptimize the model for each of the samples. The corresponding distribution of the log likelihood values has only statistical fluctuations. The red lines on these plots show the best-fit $\log\mathcal{L}$ value

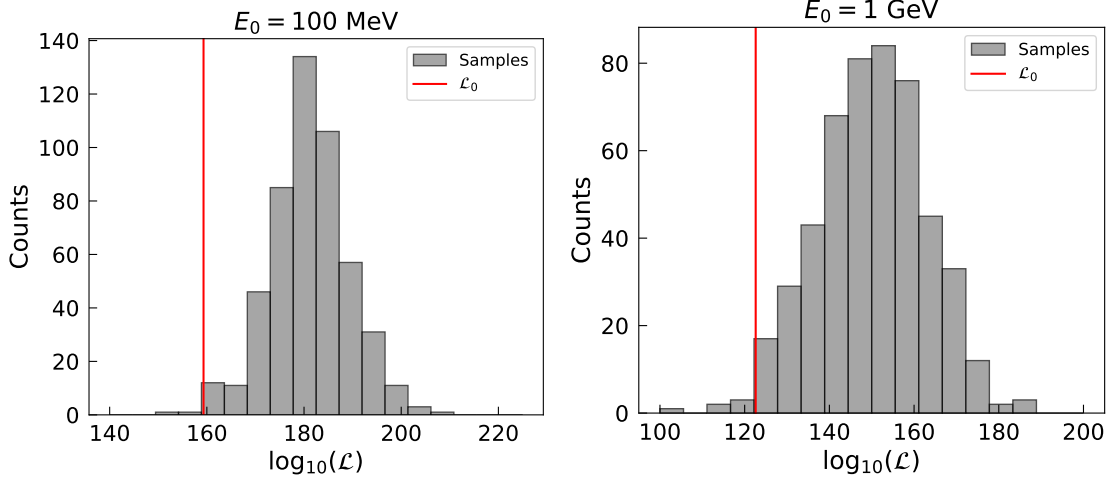


Fig. 5.16. Distribution of log-likelihood for the “ $p + c$ ” model with astro-only components for 1000 datasets simulated from the best-fit “ $p + c$ ” model for $E_0 = 100 \text{ MeV}$ (left panel) and $E_0 = 1 \text{ GeV}$ (right panel). The red line represents the log-likelihood of the “ $p + c$ ” model with astro-only components for the unassociated 4FGL-DR4 sources.

for the real data. These values are about 2 sigma smaller than the mean of the $\log \mathcal{L}$ distribution both for $E_0 = 100 \text{ MeV}$ (left panel) and for $E_0 = 1 \text{ GeV}$ (right panel). On the one hand, this deviation shows that the model is not perfect, i.e., there are possibly some deviations of the model and the data. On the other hand, the deviation has only about 2 sigma statistical significance. This deviation may be connected to the slight excess observed, e.g., in the $M_{\text{DM}} = 30 \text{ GeV}$ DM annihilation (cf. the $M_{\text{DM}} = 30 \text{ GeV}$ row in Table 5.2). **Indeed, if there is an excess at about 2σ significance, which is not accounted by the model, then the model would be excluded at 2σ significance compared to a model that can explain the data without excesses, e.g., the data generated from the model itself (cf. figure 5.16).**

5.C.3 DM subhalo injection test

In order to ensure our model is actually capable of reconstructing a DM signal, we inject a DM signal into simulated data and test the ability of our model to reconstruct it. We choose a significant value of $\langle \sigma v \rangle$ for each DM mass. For $M_{\text{DM}} = 30, 100, 300 \text{ GeV}$, we adopt the best fit $\langle \sigma v \rangle$ value from table 5.2, while for $M_{\text{DM}} = 10, 1000 \text{ GeV}$, since the best-fit value is zero, we inject a signal with a $\langle \sigma v \rangle$ corresponding to the 3σ significance (which should be clearly distinguishable from the null hypothesis).

We show the histogram of the recovered $\langle \sigma v \rangle$ values for the 5 DM masses in figure 5.17. The $\langle \sigma v \rangle$ values for the injected DM signal (red dashed-dotted lines, $\langle \sigma v \rangle_0$) are within the 68% containment interval (green dashed lines) of the histogram relative to the median values for the distributions (solid green lines). Thus, our methodology is capable of reconstructing a simulated signal, when present, in the majority of cases.

Finally, we **showcase illustrate** the model’s predictive capabilities. Using eq. (5.B.3), we derive the posterior probability that each unassociated source belongs to a given class k (e.g., extragalactic, Galactic, or DM). The predicted class label, C_i^{pred} , is then assigned to the class with the highest posterior probability: $C_i^{\text{pred}} = \text{argmax}_k \{\tilde{p}(z_{ik} = 1 | \mathbf{x}_i)\}$. Figure 5.18 displays the posterior probabilities and predicted classes for all unassociated sources. The left panel shows

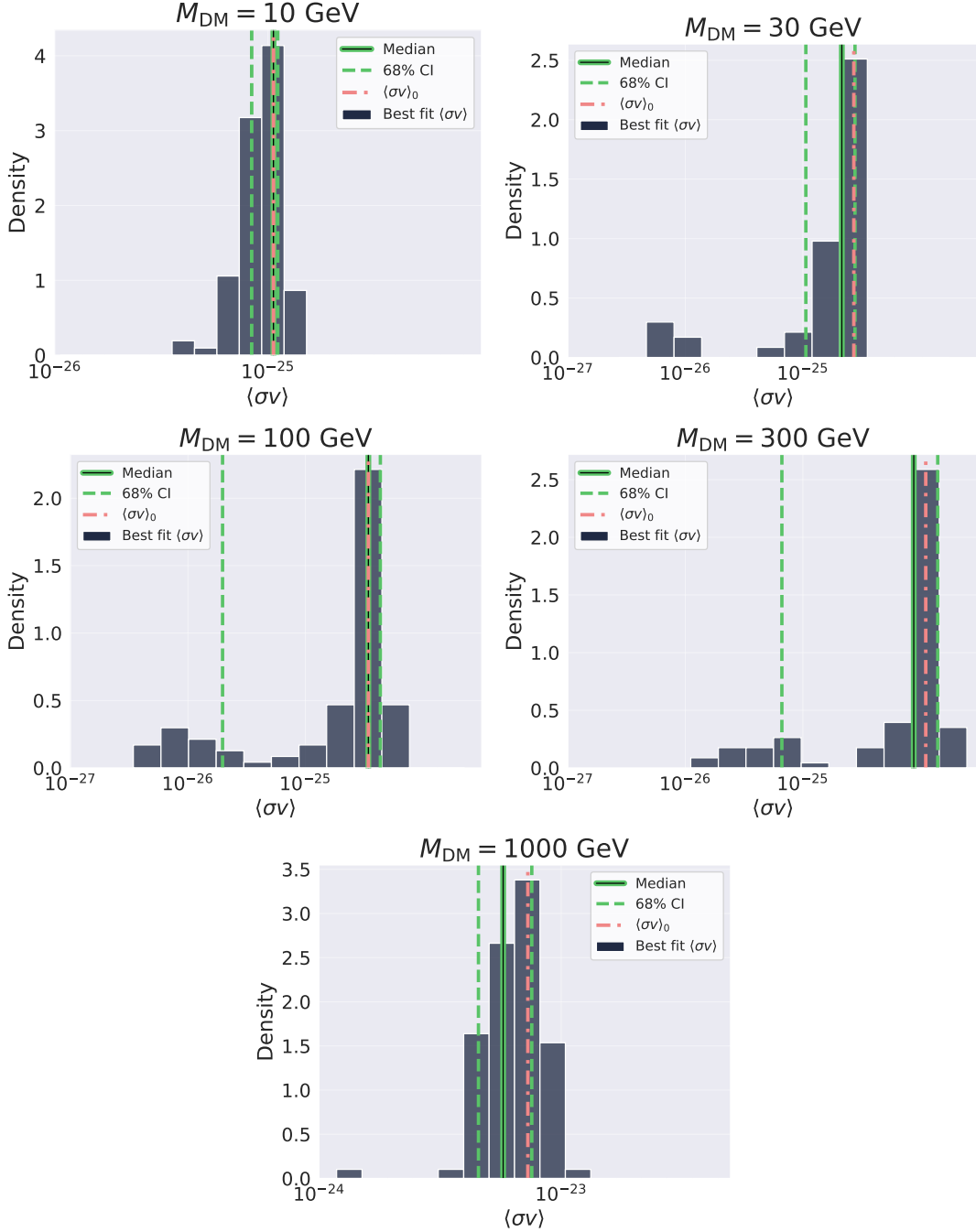


Fig. 5.17. Histograms representing the reconstructed $\langle\sigma v\rangle$ values for simulations with injected DM signals. The red dash-dotted line represents the annihilation cross-section of the injected signal ($\langle\sigma v\rangle_0$). The green solid line represents the median of the distribution, while the green dashed lines show the 68% containment interval. In every case, the true $\langle\sigma v\rangle$ value lies inside the 68% containment region.

results from our best-fit model with only astrophysical classes (for $E_0 = 1 \text{ GeV}$), while the right panel shows the best-fit model including DM (for the $M_{\text{DM}} = 30 \text{ GeV}$ case).

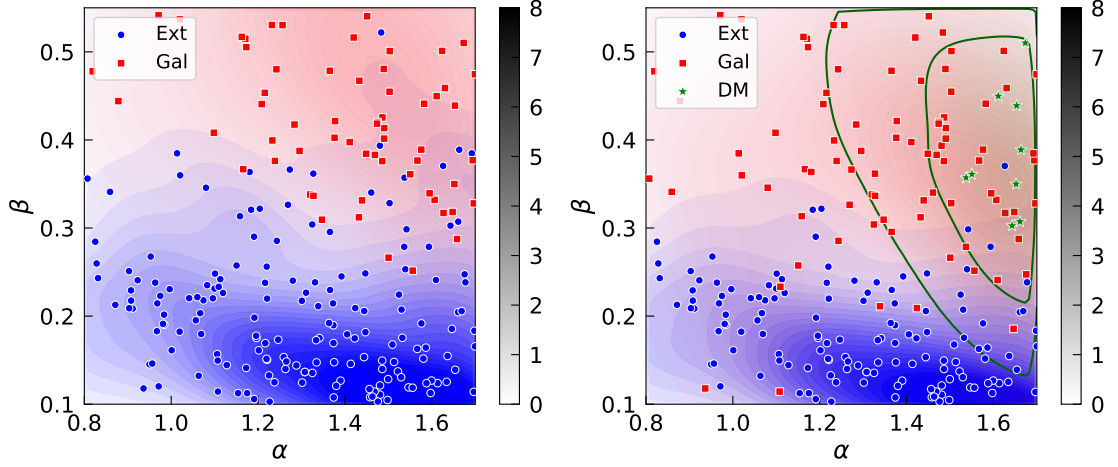


Fig. 5.18. Predicted source classes for unassociated 4FGL-DR4 sources according to the posterior class probabilities for the $E_0 = 1 \text{ GeV}$ case. *Left panel:* best-fit model with astrophysical only classes. *Right panel:* best-fit model for astrophysical classes and DM ($M_{\text{DM}} = 30 \text{ GeV}$, $\langle\sigma v\rangle \sim 2.7 \cdot 10^{-25} \text{ cm}^3 \text{ s}^{-1}$). The green contours represent the 68% and 95% containment levels for the DM distribution.

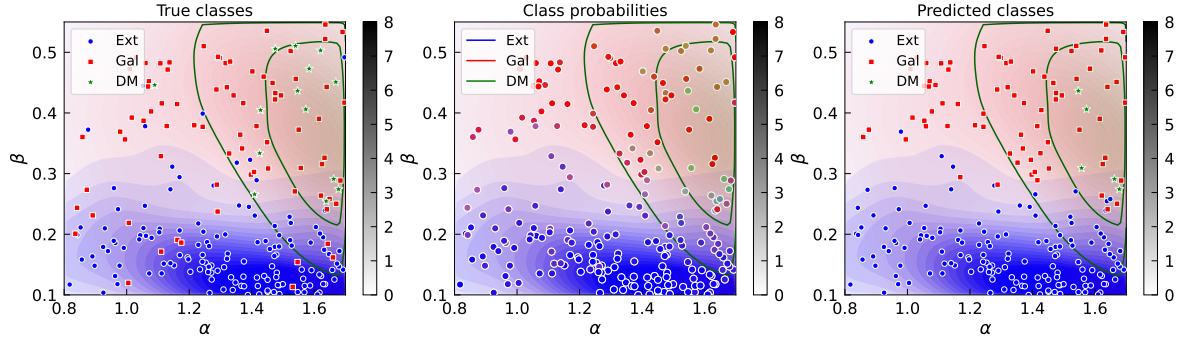


Fig. 5.19. Estimated best-fit distributions and true or predicted source classes for a bootstrap simulation, with $M_{\text{DM}} = 30 \text{ GeV}$ and $\langle\sigma v\rangle \sim 2.7 \times 10^{-25} \text{ cm}^3 \text{ s}^{-1}$. *Left panel:* markers colored according to their true class C_i^{true} in the simulation. *Middle panel:* markers colored according to their posterior class probability following the same convention as figure 5.10. *Right panel:* markers colored according to the predicted class, given by $C_i^{\text{pred}} = \text{argmax}_k \{\bar{p}(z_{ik} = 1 | \mathbf{x}_i)\}$.

To further assess performance, For comparison, we repeat this exercise on a simulated bootstrap sample for the $M_{\text{DM}} = 30 \text{ GeV}$ case in figure 5.19. The figure compares the true class labels (left panel) with the posterior class probabilities (middle panel) and the final predicted class labels (right panel) for a model fit to the bootstrap sample. Access to both true labels and posterior probabilities enables a deeper analysis of the model's performance. For instance, while not all DM sources were correctly classified due to significant parameter overlap with the Galactic source population, their true class often received a high posterior probability, even if it was not the maximum. This highlights shows that to avoid false negatives when identifying a subdominant class, it is crucial important to consider all sources with a reasonably high posterior probability for that class even if this probability is not the highest. ,not only those where it is the highest.

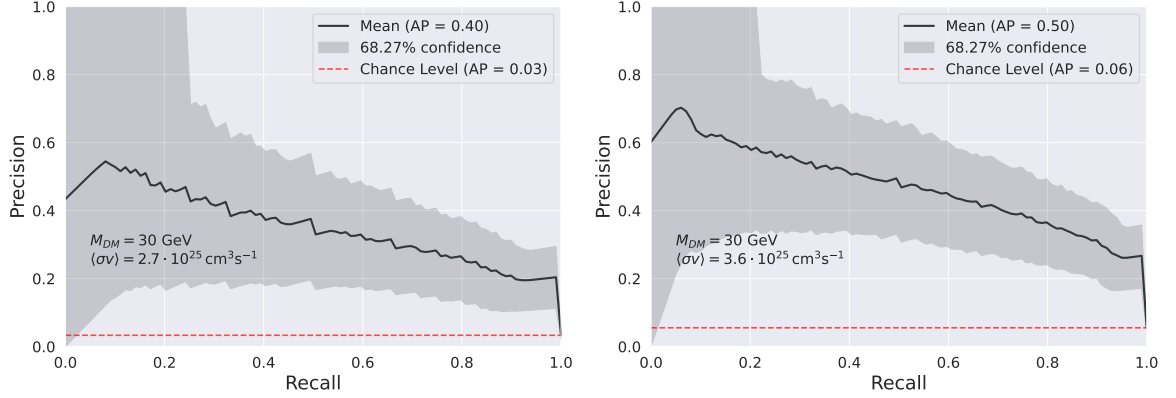


Fig. 5.20. Precision and recall for classification of individual DM subhalos among unassociated *Fermi*-LAT sources. Solid black line (gray band) shows the mean (68.27% statistical containment) for 1000 simulations at $M_{\text{DM}} = 30$ GeV. *Left panel:* performance for the best-fit cross-section in table 5.2 (average precision $AP = 0.40 \pm 0.16$). *Right panel:* performance for a cross-section close to the 95% confidence upper bound in table 5.2 ($AP = 0.50 \pm 0.14$).

5.C.4 Precision-recall curves for classification of individual DM subhalos

To further quantify the model’s ability to distinguish DM subhalos from the astrophysical background, we perform a Precision-Recall (PR) analysis for the case of $M_{\text{DM}} = 30$ GeV. This DM mass yields the highest possible signal due to DM annihilation in subhalos (cf. table 5.2) but it is also challenging due to the extensive spectral overlap with Galactic sources. The PR framework is more relevant here than the Receiver Operating Characteristic curves for highly unbalanced datasets, as it focuses exclusively on the minority class (the signal), providing a rigorous assessment of how many “DM-like” candidates are likely to be actual subhalos without being skewed by the overwhelming abundance of background sources.

We consider two physical scenarios for the annihilation cross-section: the best-fit value obtained in our primary analysis ($\langle\sigma v\rangle = 2.7 \times 10^{-25} \text{ cm}^3/\text{s}$) and a value close to the 95% upper bound ($\langle\sigma v\rangle = 3.6 \times 10^{-25} \text{ cm}^3/\text{s}$). For each scenario, we evaluate the PR performance across a suite of 1000 independent simulations. In figure 5.20, we display the mean PR curve along with the 68.27% asymmetric confidence band derived from the simulation quantiles, representing the expected statistical variance due to background realizations and Poisson noise.

To quantify performance, we compute the Average Precision (AP):

$$AP = \sum_n (R_n - R_{n-1}) P_n \quad (5.C.1)$$

where P_n and R_n are the precision and recall at the n -th classification threshold. We find $AP = 0.40 \pm 0.16$ for the best-fit case and $AP = 0.50 \pm 0.14$ for the upper bound. These scores significantly outperform their respective random chance baselines of 0.03 and 0.06. The fact that the AP remains below unity is a direct reflection of the intrinsic physical confusion between DM subhalos annihilating in the $b\bar{b}$ channel for $M_{\text{DM}} = 30$ GeV and Galactic millisecond pulsars.

EXTRACTING THE GAMMA-RAY SOURCE-COUNT DISTRIBUTION WITH DEEP LEARNING

Part II examined dark matter annihilation signatures in specific, resolved astrophysical structures. Both strategies we explored encountered fundamental limitations. The Galactic Center presents the strongest expected signal, but it remains inextricably entangled with complex diffuse foregrounds and unresolved astrophysical sources. Searching for individual dark matter subhalos among unassociated gamma-ray point sources offers a much cleaner environment, yet is severely limited by the finite sensitivity of the instrument and the large look-elsewhere effect. The inherent ambiguity of the Galactic Center Excess and the rarity of detectable subhalos expose the boundaries of source-by-source identification.

These limitations motivate a shift in strategy. We move from asking whether any specific source is a dark matter subhalo to a broader question: can we recover the statistical properties of the entire faint source population from the collective photon-count distribution of the gamma-ray sky? Below the formal detection threshold lies a vast population of sources that cannot currently be resolved individually. They collectively imprint their abundance and flux distribution onto the measured pixel counts, and characterizing this population requires inference techniques capable of extracting population-level observables — most notably the source-count distribution, dN/dS .

This chapter introduces the framework required to probe the unresolved sky and presents a deep learning approach to measuring the gamma-ray dN/dS . Section 6.1 builds the conceptual argument for transitioning to population studies. Section 6.2 formalizes the source-count distribution as the primary observable (cf. Section 2.2.2 for the astrophysical context) and reviews analytical approaches to measuring it. Section 6.3 details the limitations of these analytical formulations and introduces simulation-based inference (cf. Section 3.2) via convolutional neural networks on spherical maps as a robust alternative. Finally, Section 6.4 frames the original research paper that forms the core of this chapter, where we validate our methodology by recovering the extragalactic dN/dS down to a factor of 50 below the *Fermi*-LAT threshold.

The results of this chapter feed into two downstream analyses. The recovered source-count distribution becomes the empirical prior for the probabilistic cataloging framework developed in Chapter 7. The population-level perspective established here also provides the methodological foundation for the cosmological dark matter searches pursued through cross-correlations in

Chapter 8.

6.1 Limits of Individual-Source Methods

The preceding chapters searched for dark matter in individual, resolved gamma-ray targets. We now step back and assess the broader limitations of that paradigm, before motivating the transition to population-level methods that forms the core of Part III.

6.1.1 The Resolved-Source Paradigm and Its Constraints

The Galactic Center Excess, first reported by Goodenough and Hooper [100] and subsequently confirmed by independent analyses [62, 101, 167], remains the most prominent anomaly in *Fermi*-LAT data. As discussed in Chapter 4 (Section 4.1), disentangling a potential dark matter contribution from astrophysical explanations has proven so far intractable. The difficulty is not a lack of signal strength but an excess of competing models: the morphology of the excess is equally well described by an NFW-squared dark matter profile and by a population of unresolved millisecond pulsars tracing the stellar bulge [169, 170]. The interstellar emission models introduce further degeneracies, as different templates for the Galactic diffuse emission shift the inferred GCE spectrum and morphology [167]. Even the non-Poissonian template fitting technique, initially presented as a way to distinguish point-source from diffuse origins, was shown to be sensitive to mismodeling artifacts [171, 184]. More recently, simulation-based inference approaches incorporating energy information have reopened the question. As discussed in Section 4.3.3, the CNN-based analysis of List et al. [190] pushes the inferred source-count distribution to fluxes so faint that the GCE becomes statistically indistinguishable from the smooth, Poissonian emission expected from dark matter annihilation. The origin of the excess thus remains an open problem, with neither the dark matter nor the millisecond pulsar hypothesis decisively favored by current data. The Galactic Center therefore illustrates a general principle: when the target region contains multiple bright, partially degenerate emission components, attributing any residual to a specific physical origin becomes a problem of systematic uncertainties rather than statistical sensitivity.

Chapter 5 pursued an alternative strategy, searching for dark matter subhalos among the 2428 unassociated point sources in the 4FGL-DR4 catalog [123, 228] at Galactic latitude $|b| > 10^\circ$ (Section 5.3). The environment is far cleaner: high-latitude sources are not contaminated by the Galactic diffuse emission that plagues the GCE. However, our quantification analysis found no statistically significant dark matter subhalo component (Section 5.4.8). This null result reflects two structural constraints. First, only the most massive and nearby subhalos produce fluxes above the *Fermi*-LAT catalog threshold of $TS > 25$; the vast majority of the predicted subhalo population [95] lies well below it. Second, the look-elsewhere penalty is severe: any individual candidate extracted from a catalog of thousands of unassociated sources requires evidence well beyond the 5σ threshold conventionally demanded for discovery.

These two case studies expose a shared limitation. Both the GCE analysis and the subhalo search operate on sources that have already been detected or on emission that is already visible. They are structurally blind to the population of sources below the detection threshold. Yet this sub-threshold population is far from negligible: the very existence of the isotropic diffuse gamma-ray background [111] implies a substantial flux contribution from sources too faint to be individually resolved. The challenge is not to look harder at individual sources, but to develop methods that extract population-level information from the collective photon statistics of the unresolved sky.

6.1.2 The Population-Level Alternative

The limitations of individual source identification motivate a conceptual shift. Rather than asking whether any specific source is a dark matter subhalo, we can ask a different question: what is the statistical distribution of all sources, including those too faint to detect individually?

Below the formal *Fermi*-LAT detection threshold lies a vast population of unresolved sources. Their cumulative emission forms a significant fraction of the unresolved gamma-ray background (UGRB) [104, 111]. While these sources are invisible individually, their collective photon statistics encode recoverable information about the underlying population. This population-level approach complements the strategies explored in Part II: resolved-source methods probe the bright tail of the population, whereas statistical methods probe the faint end. Together, they provide a complete census.

Chapters 6 and 7 focus on characterizing this general unresolved source population — primarily astrophysical objects such as blazars [114, 269], millisecond pulsars [270], and star-forming galaxies [271]. A thorough census of the astrophysical population is a prerequisite for any dedicated dark matter search: only after accounting for the known source classes can a residual exotic component be meaningfully constrained. The search for a collective dark matter signal in the unresolved gamma-ray sky is deferred to Chapter 8, which employs angular cross-correlations with tracers of large-scale structure, a technique complementary to the population statistics developed here.

The primary observable for characterizing the sub-threshold source population is the source-count distribution, denoted dN/dS . We define this distribution, discuss its physical interpretation, and review analytical approaches to measuring it in the following section.

6.2 The Source-Count Distribution

The analysis of the unresolved gamma-ray sky begins with a single observable: the differential source-count distribution, dN/dS . This function describes how many point sources exist at each flux level and provides the most direct statistical characterization of the populations that contribute to the unresolved gamma-ray background. Its reconstruction below the telescope detection threshold — where individual sources can no longer be identified — is the central inference problem addressed in this chapter.

This section introduces the dN/dS and the methodological context necessary to understand the approach developed in Section 6.3. We begin with the formal definition and its connection to the underlying luminosity function (Section 6.2.1), then review the 1-point photon-count distribution, the principal analytical method used to extract dN/dS from sub-threshold data, and discuss how measurements of the total source-count distribution constrain the composition of the unresolved gamma-ray sky (Section 6.2.2).

6.2.1 Definition and Physical Meaning

The differential source-count distribution, dN/dS , gives the number of sources per unit solid angle per unit flux at a given integral photon flux S in a specified energy band. It is the fundamental observable that describes how point sources are distributed as a function of their apparent brightness, and it connects directly to the physics of the emitting populations.

For sources bright enough to be individually cataloged, dN/dS is measured by simply counting objects in the catalog as a function of their flux (see also Section 2.2). The 4FGL-DR3 catalog [228], based on 12 years of *Fermi*-LAT data, contains 6658 resolved sources and

provides a direct measurement of the source-count distribution in the resolved regime [272]. Above the flux threshold where the catalog detection efficiency reaches 100%, this measurement is straightforward. Below this threshold, the detection efficiency degrades, and direct counting must be supplemented by Monte Carlo corrections to reconstruct the true underlying distribution. Further below — in the unresolved regime, where individual sources are too faint to be detected at all — the cumulative emission of these sources forms the unresolved gamma-ray background (UGRB), and the dN/dS must be inferred from the statistical properties of the photon-count map. This reconstruction is the central challenge addressed by this chapter.

The source-count distribution is the observational projection of a more fundamental quantity: the luminosity function. The gamma-ray luminosity function, $\Phi_\gamma(z, L_\gamma)$, gives the comoving number density of sources per unit luminosity L_γ as a function of redshift z . The connection between the two is given by [104] [AA: this equation is correct, I double checked]

$$\frac{dN}{dS} = \int_{z_{\min}}^{z_{\max}} dz \int_{\Gamma_{\min}}^{\Gamma_{\max}} d\Gamma \frac{dN}{dV dL_\gamma d\Gamma}(z, L_\gamma, \Gamma) \frac{dV}{dz d\Omega} \frac{dL_\gamma}{dS}, \quad (6.2.1)$$

where Γ is a parameter that characterizes the energy spectrum shape of the source, $dV/(dz d\Omega)$ is the comoving volume element per unit redshift and solid angle, and dL_γ/dS is the Jacobian relating luminosity to observed flux through the luminosity distance $D_L(z)$. The integrand $dN/(dV dL_\gamma d\Gamma)$ — the comoving density of sources per unit luminosity and spectral index — is the luminosity function itself, which encodes the intrinsic abundance of sources at each cosmic epoch.

This integral makes precise the physical content of dN/dS : it takes the intrinsic distribution of sources in luminosity and redshift, projects it through the flux–distance relation, and integrates over all redshifts and spectral shapes to produce the observed flux distribution. In other words, dN/dS can be interpreted as the integral along the line of sight of the gamma-ray luminosity function. It is what we observe when we look at the sky, while the luminosity function is the astrophysical quantity we ultimately want to constrain.

The source-count distribution therefore bridges two distinct observational regimes. At the bright end, it is anchored by direct measurements from the Fermi-LAT catalog, which shows a flux distribution broadly compatible with an S^{-2} power law (shallower than the Euclidean expectation $S^{-5/2}$) across most of the resolved range [272]. At the faint end, below the detection threshold of approximately $S_{\text{th}} \sim 2 \times 10^{-10} \text{ cm}^{-2} \text{ s}^{-1}$ in the 1–10 GeV band, the dN/dS must be reconstructed statistically from the collective imprint of unresolved sources on the photon-count map. The techniques for performing this reconstruction — and for extending the dN/dS well below the Fermi-LAT threshold — are the subject of the following subsections.

[AA: maybe add a note on that $S^{-5/2}$, where it comes from and why, it can be a footnote as that's a question that pops up often in talks: why do we get S^{-2} and do we expect that?]

6.2.2 Extracting the Source-Count Distribution from Photon Statistics

Recovering the source-count distribution below the detection threshold requires extracting population information from the collective statistical imprint of unresolved sources on the photon-count map. The photon-count distribution — also called the 1-point probability distribution function (1pPDF) — provides a well-established method for doing so.

The idea is straightforward: given a pixelated sky map, one constructs the histogram of the number of pixels containing exactly k photons. Different source populations leave distinct imprints on this histogram [104]. Bright but rare sources, such as blazars just below the detection

limit, produce multiple photons during the observation period; because these photons are spatially correlated, they populate pixels with moderately large photon counts, generating non-Poissonian tails in the distribution. By contrast, faint but numerous sources, such as star-forming galaxies, contribute on average far less than one photon each; their combined signal is statistically indistinguishable from a smooth Poissonian background [273]. The departures from Poisson statistics therefore encode information about the sub-threshold source population and its dN/dS . This connection is made rigorous through a generating-function formalism [104, 273] in which the total photon-count probability factorises into independent contributions from point sources, Galactic foreground, and isotropic background, and the parameters of dN/dS are obtained by fitting the predicted pixel histogram to data via maximum likelihood.

Malyshev and Hogg [273] pioneered this approach for gamma rays, applying it to 11 months of Fermi-LAT data and constraining the contribution of unresolved AGN-like sources to approximately $17 \pm 2\%$ of the total flux at high Galactic latitudes above 1 GeV. Zechlin, Cuoco et al. [274] improved the method by incorporating the spatial variation of the Galactic foreground and exposure at the pixel level, measuring the dN/dS in the 1–10 GeV band down to fluxes approximately one order of magnitude below the detection threshold. Subsequent extensions by the same group [275] and by Lisanti et al. [257] generalised the technique to multiple independent energy bands spanning 1–171 GeV. The same pixel-statistics formalism was also adapted for the Non-Poissonian Template Fit (NPTF) applied to the Galactic Center Excess (Section 4.4), where it served to distinguish a smooth dark matter signal from a population of unresolved point sources such as millisecond pulsars — illustrating the broad applicability of photon-count methods across very different science cases.

These photon-count methods — the 1pPDF and, as we shall see, the CNN-based approach developed in this chapter — reconstruct the *total* source-count distribution: the aggregate dN/dS of all point sources, regardless of their astrophysical class. They measure how many sources exist at each flux level, but they cannot, by themselves, separate the blazar contribution from that of star-forming galaxies, misaligned AGNs, or any other population. This is a fundamental feature of photon-count statistics: they respond to photons, not to source identities. A complementary body of work approaches the problem from the opposite direction, estimating the dN/dS of individual source classes by extrapolating from multi-wavelength correlations. Blazars are modeled through direct luminosity-function fits to the Fermi-LAT catalog and may account for roughly 20–30% of the UGRB intensity below 100 GeV [114]; star-forming galaxies and misaligned AGNs are estimated through empirical scaling relations linking their gamma-ray emission to better-measured infrared and radio luminosities [276–278], with each contributing of order 10–50% depending on the assumed redshift evolution [104]. These estimates suggest that the known astrophysical populations could collectively account for a large fraction of the UGRB intensity, though each carries uncertainties of at least a factor of a few, reflecting the limited samples from which the scaling relations are derived. A more precise measurement of the total dN/dS would therefore help constrain the room remaining for additional contributors such as dark matter annihilation.

6.3 Simulation-Based Inference for dN/dS

The 1pPDF formalism reviewed in the previous section represents the most developed analytical approach to sub-threshold source counts. It has produced the strongest existing measurements of the gamma-ray dN/dS below the *Fermi*-LAT detection limit. However, its underlying assumptions — a parametric model for the source-count distribution, a simplified treatment of spatial

correlations, and a computationally expensive non-amortized likelihood — impose limitations that motivate a complementary approach.

This section discusses those limitations in detail (Section 6.3.1), then introduces the simulation-based inference framework that addresses them (Section 6.3.2), and finally summarizes the handling of the spherical geometry of the input data (Section 6.3.3).

6.3.1 Limitations of the Analytical Framework

The generating function formalism underlying the 1pPDF method rests on several simplifying assumptions, each of which constrains its applicability.

Neglect of spatial correlations. The 1pPDF treats each pixel independently. In practice, however, the point-spread function (PSF) of the *Fermi* LAT smears a point source across multiple pixels, introducing spatial correlations that the generating function approach does not capture. These correlations carry information about the source-count distribution that is discarded in the single-pixel treatment, reducing the effective constraining power and potentially introducing biases.

Parametric dependence. The dN/dS is typically modeled as a multiply broken power law with a fixed number of segments. This assumption is sufficiently flexible for smooth, featureless distributions, but it cannot accommodate spectral features, bumps, or breaks at unforeseen flux values. The reconstruction is therefore conditional on the assumed functional form.

Absence of amortization. Each evaluation of the likelihood requires a fresh computation of the generating function at every pixel, which is computationally expensive. Exploring high-dimensional posterior distributions over the dN/dS parameters and nuisance parameters (such as the Galactic foreground normalization) demands large numbers of these evaluations, making the inference slow and scaling poorly with the number of energy bins or model parameters.

Limited multi-band extension. While multi-energy extensions of the 1pPDF have been performed [257, 275], extending the formalism to a large number of energy bins simultaneously is practically challenging, since the number of model parameters and the computational cost grow rapidly.

The community has recognized some of these issues: Zechlin et al. developed a Compound Poisson Generator (CPG) formalism [274] to address potential biases in the 1pPDF approach, though this alternative still operates within the generating-function framework. A fundamentally different strategy is needed to overcome all four limitations simultaneously.

6.3.2 A Simulation-Based Approach

The limitations identified above share a common root: the 1pPDF relies on an analytic likelihood that incorporates only a subset of the information contained in the photon-count map. An alternative is to bypass the likelihood entirely and employ simulation-based inference (SBI), the general framework introduced in Section 3.2.

As discussed in Chapter ??, the SBI paradigm exploits an asymmetry between the forward and inverse problems (Section 3.2.1): simulating a realistic gamma-ray sky map from a given set of dN/dS parameters is computationally straightforward, whereas evaluating the corresponding likelihood is not. In the forward direction, one draws source fluxes from a parametric dN/dS , places them on the sky following an assumed spatial distribution, convolves with the instrument PSF and exposure, adds the Galactic foreground and isotropic background, and applies Poisson noise to produce a synthetic photon-count map. Each simulation is fast and faithful to the known instrumental and astrophysical effects.

A convolutional neural network (CNN) is then trained on an ensemble of such simulated maps to learn the mapping from a photon-count sky to the underlying dN/dS , using the architecture and training methodology detailed in Section 3.3. The specific configuration adopted for the dN/dS problem discretizes the output into 20 flux bins spanning $[5 \times 10^{-12}, 10^{-7}] \text{ cm}^{-2} \text{ s}^{-1}$, plus the isotropic background level F_{iso} , yielding 21 output parameters. An ensemble of 9×10^5 synthetic maps is generated for training. The discretized representation is critical: unlike the parametric power-law adopted by the 1pPDF, it allows the CNN to reconstruct the dN/dS non-parametrically, without assuming a fixed functional form. The network employs the heteroscedastic Gaussian negative log-likelihood cost function and concrete dropout regularisation—following the Bayesian-neural-network paradigm motivated in Section 3.3.2—to estimate both the dN/dS and its uncertainties; the full formalism is presented in the paper body (Section 6.5.3.4). A frequentist cross-check using the distribution of residuals on held-out validation maps confirms that the Bayesian error bands are well calibrated (Section 6.5.3.6).

This framework addresses the four limitations identified in Section 6.3.1: inter-pixel PSF correlations are captured implicitly by the convolutional architecture, the dN/dS is recovered non-parametrically, inference is amortized (a trained network produces predictions in milliseconds), and the pipeline extends naturally to multiple energy bins by adding input channels.

6.3.3 Inference on Spherical Data

Fermi-LAT data are represented as HEALPix sky maps on the sphere. Applying standard planar CNNs to such data requires projecting the sphere onto a flat image, which inevitably introduces distortions. The map2patches algorithm, described in Section 3.3, addresses this by subdividing the HEALPix sphere into 12 equal-area base patches at order $n = 0$, each of which is independently mapped to a 2D image and processed through standard 3D convolutions. This approach preserves the area and information content of each pixel without spherical distortion, while leveraging the optimized performance of standard convolutional operations.

A fully spherical cross-check using the NNHealpix architecture [152] yields compatible results, confirming that the map2patches approximation is reliable for the isotropic, small-scale signal of interest. The map2patches implementation is approximately an order of magnitude faster, enabling efficient generation of the large training ensembles required by the SBI framework. Full architectural details are presented in the paper body (Section 6.5.3).

6.4 From Methodology to Measurement

The CNN-based framework described in the preceding sections was designed to overcome the limitations of existing analytical methods while retaining their strengths. Recovering the sub-threshold dN/dS non-parametrically and with calibrated uncertainties opens several scientific directions.

First, a robust measurement of the source-count distribution constrains the faint-end population of astrophysical emitters — blazars, star-forming galaxies, and millisecond pulsars — that collectively generate the bulk of the unresolved gamma-ray background. Quantifying their contribution is essential before any residual can be attributed to non-standard sources. Second, the recovered dN/dS provides the empirical prior for the probabilistic source cataloging framework developed in Chapter 7: the flux distribution of sub-threshold sources directly determines the posterior probability that any given sky pixel contains a faint, as-yet-unresolved source. Third, the methodology itself is readily extensible. An energy-dependent dN/dS reconstruction across multiple energy bands would enable spectral decomposition of the unresolved sky into individual

Healpix order	6, 7
Weeks	9 – 745
Emin	1 GeV
Emax	10 GeV
Instrument Response Functions (IRFs)	P8R3_SOURCEVETO_V3
EVCLASS	2048 (Source Veto)
EVTYPE	1 (Front)
ZMAX	90

Table 6.1. Fermi Tools settings used for the 14-year data set analysis.

source classes — and, in principle, a search for spectral signatures inconsistent with known astrophysical populations. This extension, while beyond the scope of the present work, represents a natural next step toward the dark matter cross-correlation searches pursued in Chapter 8.

The analysis presented below should therefore be understood as a proof of principle for the deep-learning approach to population-level gamma-ray inference. We apply the trained CNN to 14 years of *Fermi*-LAT data in the 1–10 GeV energy band and recover the source-count distribution down to fluxes approximately 50 times below the catalog detection threshold, demonstrating that the method produces stable, well-calibrated results across a wide dynamic range. The remainder of this chapter presents the analysis in full.

6.5 Extracting the gamma-ray source-count distribution below the *Fermi*-LAT detection limit with deep learning

In this paper we update the measurement of the dN/dS below the detection threshold to the increased statistics offered by 14 years of *Fermi*-LAT data. However, differently from Refs. [274, 275], we adopt here a method based on machine learning techniques to obtain the dN/dS below the detection threshold. A similar methodology has also been used recently to investigate gamma-ray unresolved sources close to the Galactic center [191, 279, 280]. We train a convolutional neural network (CNN) on synthetic gamma-ray maps, built from a wide variety of source-count distributions and then apply the trained CNN to the 14-year *Fermi*-LAT map for photon energies in the (1,10) GeV band. We show that the CNN is able to reconstruct the dN/dS , thus obtaining an updated result which is fully compatible with the one obtained in [274]. The methodology presented here is also meant to be a proof of principle for the adoption of a CNN to the reconstruction of the source-count distribution of the extragalactic sky, with the future aim of properly investigating additional features of the dN/dS , like energy correlations and the presence of additional components which could be traced to dark matter. An advantage of the CNN method is that it avoids the need to calculate complicated and numerically demanding likelihoods, as in the 1-point PDF method. Furthermore, we will describe an improved version of the treatment of a CNN on a spherical domain which will further optimize the computational aspect.

Data selection and *Fermi*-LAT map generation is discussed in Section 6.5.1, while Section 6.5.2 discusses in detail how synthetic maps for the CNN training and validation are constructed. Section 6.5.3 discusses the neural network architecture that we use and its implementation, including validation and error estimation. Section 6.5.4 discusses the analysis of the *Fermi*-LAT map with the trained CNN and presents the ensuing results for the dN/dS . Section 6.5.5 gives our conclusions.

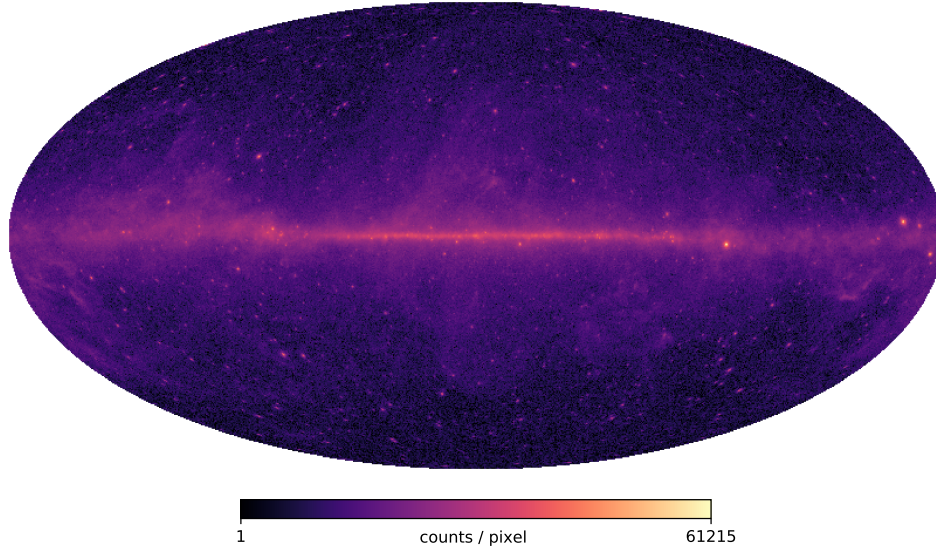


Fig. 6.1. *Fermi*-LAT photon-counts map in units of counts per pixel in the 1-10 GeV energy range for the 14-years dataset, at the Healpix resolution $N_{\text{side}} = 128$ ($n = 7$).

6.5.1 Data selection

We process the *Fermi*-LAT data through the Fermi Tools suite [281] to produce a full-sky map of photon counts for gamma-ray energies in the (1-10) GeV range. We choose this range as a suitable balance between high statistics (which occurs at lower energies) and good angular resolution (which is better the higher the energy) of the detector, as well as being able to confront our results with those of Ref. [274].

We consider the first 14 years of data collected by the *Fermi*-LAT (specifically, from week 9 to week 745 of operation), and Pass 8 event selection [118, 119]. We adopt the P8R3_SOURCEVETO_V3 instrument response functions (IRFs), event class (EVCLASS) 2048 (SOURCEVETO) and event type (EVTYPE) 1 (FRONT). We used standard quality selection criteria, i.e. DATA_QUAL==1 and LAT_CONFIG==1. The atmospheric gamma rays from the Earth limb emission is removed by adopting a cut on the maximum zenith angle ZMAX of 90 degrees. FRONT events refer to gamma-rays detected by the first layers of the tracking module of the *Fermi*-LAT detector, and are optimised in order to have a better angular resolution, which is a crucial requirement for our analysis. The SOURCEVETO class of events achieves good suppression of the charged cosmic-ray (CR) background while still retaining a large event statistics. All those settings are summarised in Table 6.1. Finally, the maps are constructed with equal-area pixels by adopting the Healpix [282] pixelisation scheme of order $n = 6$ ($N_{\text{side}} = 64$ resolution, which corresponds to 0.92° pixel side) although for testing we will also use maps with order $n = 7$ ($N_{\text{side}} = 128$). The photon count map, in counts per pixel, for the 14 year data set is shown in Figure 6.1. This is the map to which we will apply the CNN to extract the source-count distribution of gamma-ray sources.

Using the *Fermi*-LAT tools we further extract the *Fermi*-LAT exposure map and the point-spread-function (PSF) for photon energies in (1, 10) GeV, which will be needed to construct the synthetic maps used to train and validate the CNN. Some care is required, since the PSF is a rapidly varying function of energy. Following previous works [274, 283], we build the (1, 10)

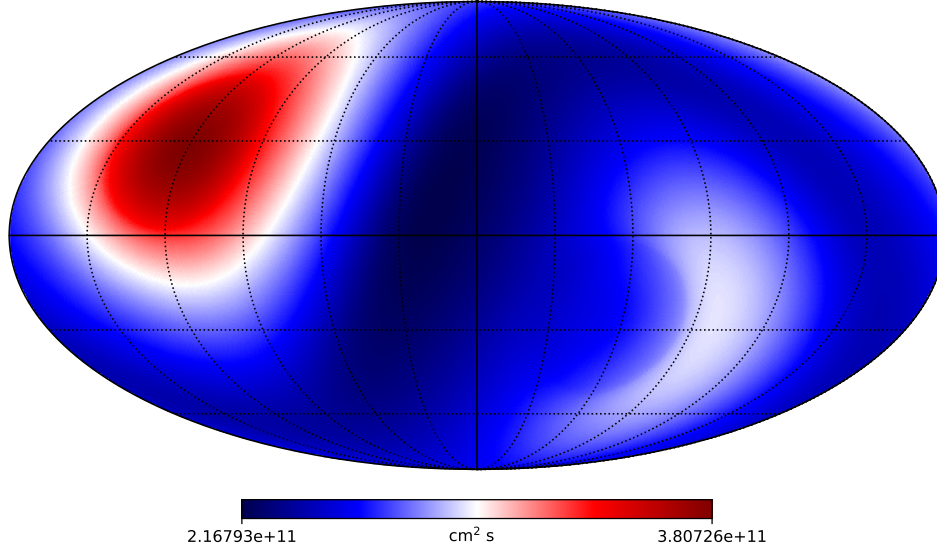


Fig. 6.2. *Fermi*-LAT mean exposure map in the 1-10 GeV energy range for the 14-years data set, in units of $\text{cm}^2 \text{s}$, at the Healpix resolution $N_{\text{side}} = 128$ ($n = 7$).

GeV mean PSF by averaging the energy-dependent PSF in the (1, 10) GeV energy range with a $E^{-2.4}$ weight, corresponding to the approximate energy spectrum of the high Galactic latitude gamma-ray sky. The average exposure map in the (1, 10) GeV energy bin and the averaged PSF profile are shown in Figure 6.2 and in Figure 6.3, respectively. The *Fermi*-LAT exposure slightly changes with energy, although its variation inside our energy bin is small.

A further dataset we will use is the catalog of resolved sources. In particular, we will employ the recently released Data Release 3 of the fourth *Fermi*-LAT catalog of gamma-rays sources (4FGL-DR3) [228], which increments the previous release to 12 years of data taking and contains 6658 resolved sources. From this catalog, we obtain the (1-10) GeV source-count distribution in the resolved regime, shown in Figure 6.4. The detailed procedure with which we obtain the dN/dS from the list of sources is described in Appendix B of [274]. Here and through the rest of the manuscript S will always refer to an integral flux in the range 1-10 GeV. The blue points refer to the regime in which the catalog is fully efficient in detecting sources, while the gray points denote sources for which the *Fermi*-LAT sensitivity to point sources starts degrading. Our aim is to infer the behaviour of the source count in the unresolved regime, i.e., for source fluxes below $S_{\text{th}} \sim 2 \cdot 10^{-10} \text{ cm}^{-2} \text{ s}^{-1}$ by adopting a CNN technique. We stress that in the following the dN/dS of resolved sources will be used only for comparison with the result of the CNN analysis, which are completely independent from it. Thus, the fact that we use 14 years of data while the catalog is based on 12 years does not have an impact on the analysis.

6.5.2 Synthetic map generation

In order to instruct the CNN to extract the dN/dS from the *Fermi*-LAT photon-count map, we need to produce synthetic analogs of the map, constructed from a wide selection of dN/dS and for the same characteristics (detector exposure and PSF, known physical foregrounds) of the data we will be using. These maps will then be partly used to train a CNN and partly used to validate the

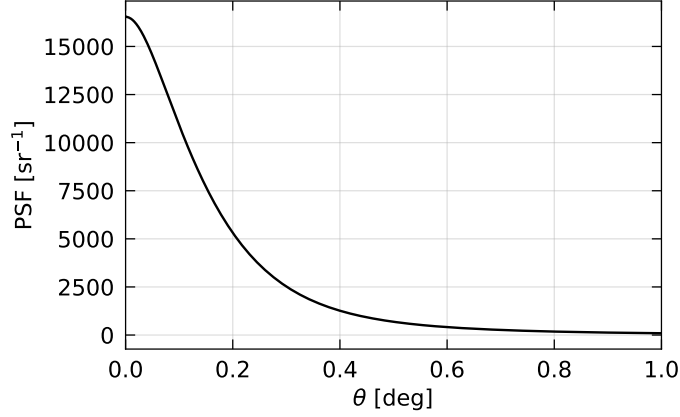


Fig. 6.3. Average *Fermi*-LAT point spread function (PSF) for energies in (1, 10) GeV.

CNN, to determine its performance and to estimate the uncertainty on the reconstructed dN/dS . Finally, we will apply the trained CNN to the *Fermi*-LAT data.

The gamma-ray maps are modeled as the sum of three contributions:

- A population of gamma-ray point sources. Since our analysis will be restricted to the high galactic latitude region (in order to minimize in the analysis the impact of the galactic foreground emission, as discussed in the following sections), these sources will mostly be of extra-galactic origin and can therefore be assumed to be distributed homogeneously across the sky. The high galactic latitude sky also contains a population of millisecond pulsars which can also be approximately assumed to be isotropically distributed [284], and thus are ideally included in our phenomenological dN/dS . In terms of photon flux, the sources are extracted, for each map realization, from the input dN/dS distributions, modeled as in Section 6.5.2.1.
- Diffuse gamma-ray emission from the Milky Way. Galactic emission is the dominant contribution at low galactic latitudes, with a declining intensity at high latitudes. We model the galactic emission as discussed in Section 6.5.2.2, by adopting a template model normalized to the *Fermi*-LAT data. Stability of the results against foreground modeling will be tested through the adoption of different templates.
- An isotropic background, which integrates all source emissions which are too faint to be processed by the CNN and possible truly diffused emission mechanisms, such as gamma-rays from cosmological cascades from ultra high energy cosmic rays or a possible contribution from dark matter annihilation or decay, as well as a residual instrumental CR background contamination. This term is modeled as a constant.

A flux map $\mathcal{M}$ will therefore be obtained as the sum of three contributions:

$$\mathcal{M} = S + A_{\text{gal}} \cdot \mathcal{G} + F_{\text{iso}} \quad (6.5.1)$$

where S is the map due to the underlying distribution of sources (obtained from a dN/dS model as described in the next subsection), $\mathcal{G}$ is a template map for the foreground emission and F_{iso} is a constant that denotes the isotropic component discussed above. The constant A_{gal} is a normalization constant for the galactic foreground template.

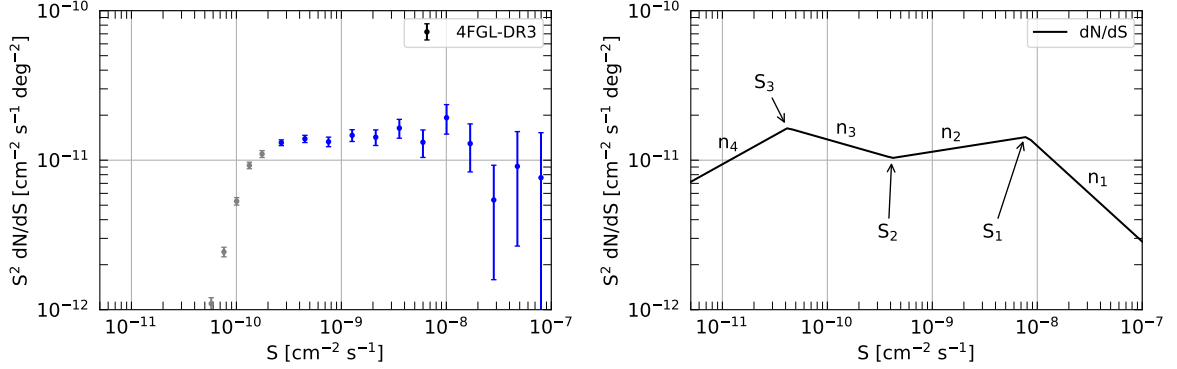


Fig. 6.4. Left: The 1-10 GeV dN/dS for resolved *Fermi*-LAT sources, obtained from the 4FGL-DR3 [228] catalog. Right: An example of dN/dS with 3 breaks S_j ($j = 1, 3$) and four slopes, where n_i ($i = 1, 4$) is the inclination of the slope.

The count map $\mathcal{N}$ is then obtained by multiplying the flux map by the *Fermi*-LAT exposure map $\mathcal{E}$, depicted in Figure 6.2, and by taking into account the steradian-to-pixel conversion factor. Specifically, the counts in pixel a are obtained as:

$$\mathcal{N}_a = \mathcal{M}_a \cdot \mathcal{E}_a \cdot \frac{4\pi}{N_{\text{pix}}} \quad (6.5.2)$$

where the number of pixels in the Healpix scheme is $N_{\text{pix}} = 12 \cdot N_{\text{side}}^2$ with $N_{\text{side}} = 2^n$ with n the pixeling order.

Since the exposure slightly changes with energy, the above procedure could be improved by performing a convolution in energy of the photon flux with the exposure, instead of directly multiplying the two quantities as performed in Eq. (6.5.2). However, the gradient in energy of the exposure in the (1, 10) GeV energy range is small, and the two procedures would provide very similar results. We, thus, use the simpler procedure outlined above. Eq. 6.5.2 also implies that we do not adopt energy dispersion. In the energy range 1-10 GeV the energy dispersion is about 10% [118] and may induce normalization changes of similar size. Thus, as explained in the next section, we will re-derive the galactic foreground normalization with a dedicated fit in order to avoid possible biases in the normalization.

From the count map $\mathcal{N}$, which contains the number of photon counts in each pixel as determined by the model, we finally produce a Poisson realization $\mathcal{N}_{\mathcal{P}}$, by extracting in each pixel a photon count from a Poisson distribution with mean equal to the photon number of the corresponding pixel in $\mathcal{N}$. $\mathcal{N}_{\mathcal{P}}$ constitutes our synthetic map realization, which mimics a count map of an experiment with the *Fermi*-LAT specifications, where the underling model is the one of Eq. 6.5.1.

In the next Sections we discuss in more detail how the components of Eq. 6.5.1 are modeled.

6.5.2.1 Differential source-count distribution

Following [274], the differential source-count distribution dN/dS is modeled as a multi-break power-law (MBPL). A MBPL with N_b breaks located at fluxes S_j ($j = 1, 2, \dots, N_b$) is defined as:

Parameter	Prior	Range
A_S	log-flat	[1, 15]
F_{iso}	log-flat	$[0.5 \cdot 10^{-7}, 7.0 \cdot 10^{-7}]$
S_1	log-flat	$[3 \cdot 10^{-9}, 5.0 \cdot 10^{-8}]$
S_2	log-flat	$[5 \cdot 10^{-11}, 3.0 \cdot 10^{-9}]$
S_3	log-flat	$[5 \cdot 10^{-12}, 5 \cdot 10^{-11}]$
n_1	flat	[2, 4]
n_2	flat	[1.80, 2.15]
n_3	flat	[1.5, 2.5]
n_4	flat	[1.0, 3.0]

Table 6.2. Ranges of the parameters and their sampling priors adopted in the map generation with $N_b = 3$ breaks. The normalization A_S is in units of $10^7 \text{ cm}^2 \text{ s sr}^{-1}$. F_{iso} is in units of $\text{cm}^{-2} \text{ s}^{-1} \text{ sr}^{-1}$. The breaks S_1 , S_2 and S_3 are given in units of $\text{cm}^{-1} \text{ s}^{-1}$. All other quantities are dimensionless.

$$\frac{dN}{dS} = A_S \times \begin{cases} \left(\frac{S}{S_0}\right)^{-n_1}, & S > S_1 \\ \left(\frac{S_1}{S_0}\right)^{-n_1+n_2} \left(\frac{S}{S_0}\right)^{-n_2}, & S_2 < S \leq S_1 \\ \vdots & \\ \left(\frac{S_1}{S_0}\right)^{-n_1+n_2} \left(\frac{S_2}{S_0}\right)^{-n_2+n_3} \dots \left(\frac{S}{S_0}\right)^{-n_{N_b+1}}, & S \leq S_{N_b} \end{cases} \quad (6.5.3)$$

where S_0 is a reference flux value which we choose to be $S_0 = 3 \cdot 10^{-8} \text{ cm}^{-2} \text{ s}^{-1}$, and n_j are the power-law indices. S denotes the source flux in the energy interval (1,10) GeV. The dN/dS distribution is normalized with an overall factor A_S .

For definiteness, in this work we adopt 3 breaks and 4 slopes. The ranges in which the parameters are varied to produce different source-count distributions is reported in Table 6.2. The ranges have been chosen in order to intercept the expected features of the dN/dS and are broad enough to allow for a good variability in the modeling of the source counts, in order to expose the CNN to a wide set of options. In fact, the prior ranges of the three break positions are chosen relative to the *Fermi*-LAT threshold for resolving point sources, $S_{\text{th}} \sim 2 \cdot 10^{-10} \text{ cm}^{-2} \text{ s}^{-1}$, such that S_1 , S_2 , S_3 are respectively larger, across and below S_{th} . The flux interval over which we will try to reconstruct the dN/dS is set as $[5 \cdot 10^{-12}, 1 \cdot 10^{-7}] \text{ cm}^{-2} \text{ s}^{-1}$. The upper limit of this interval is set by the brightest sources in the catalog, while the lower limit is set at about 1.6 orders of magnitude below the *Fermi*-LAT threshold for resolving sources. Sources fainter than the lower limit of this interval are assumed to just contribute to the isotropic component. This lower limit, indeed, is also close to the theoretical sensitivity given by the flux of a point source contributing exactly one photon, which we calculate as $F_{\text{sens}} = 1/\mathcal{E}_{av} = 3.74 \cdot 10^{-12} \text{ cm}^{-2} \text{ s}^{-1}$ where $\mathcal{E}_{av}$ is the average exposure in the region $|b| > 30^\circ$ where we perform the analysis.

The prior ranges for the power-law slopes are anchored to the results from the 4FGL catalog in the resolved regime, and allow for progressively wider variability moving toward the unresolved regime. Since catalog data exhibit a source count compatible with S^{-2} , except for large fluxes, where the slope increases, we adopt priors for n_1 and n_2 around these behaviours. For n_3 and n_4 , we progressively enlarge the prior ranges, in order to allow for more variability especially in the sub-threshold (unresolved) regime we are interested in. An example of a dN/dS generated from this procedure is shown in the right panel of Figure 6.4, which also illustrates the notation.

The maps $\mathcal{S}$ are generated according to the following procedure:

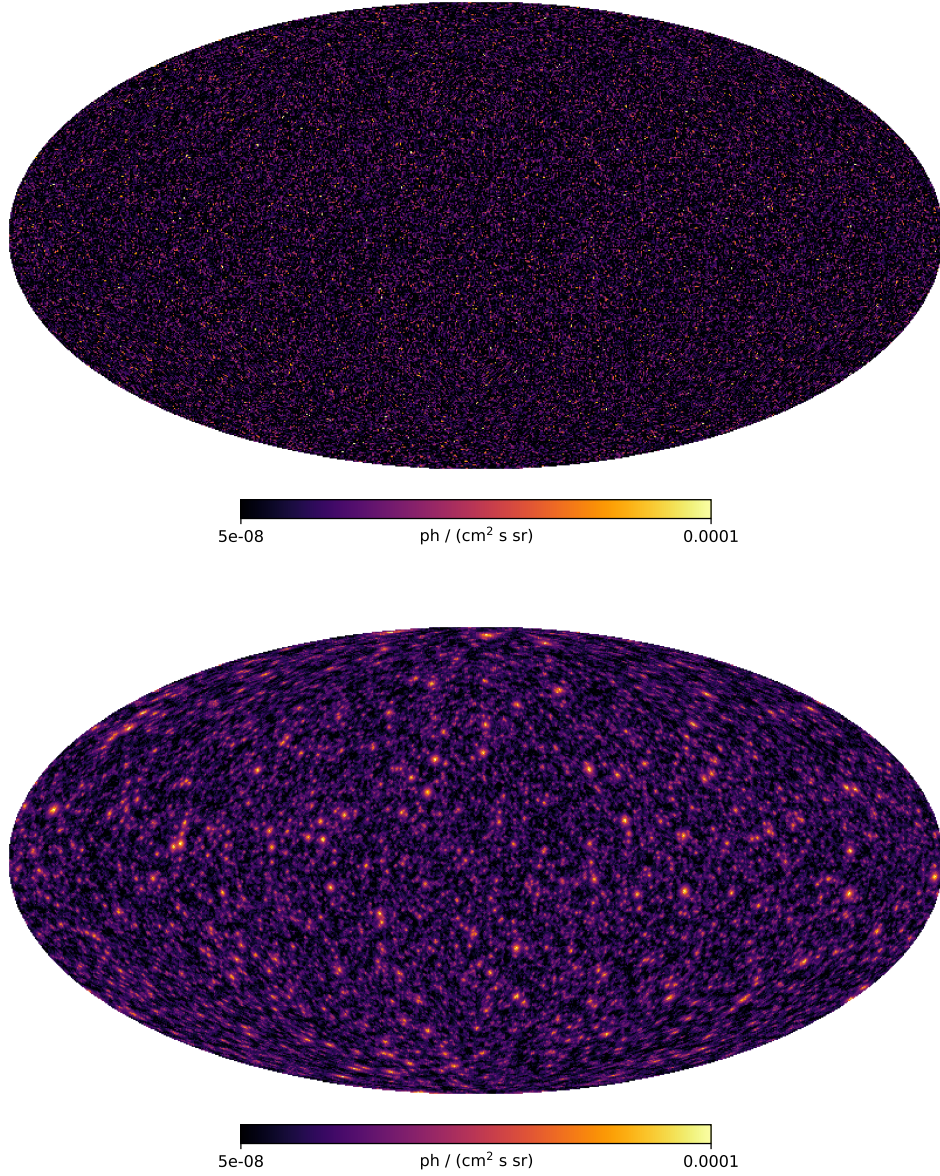


Fig. 6.5. Simulated flux map of point sources extracted from a specific source-count distribution, before (upper panel) and after (lower panel) the PSF is applied, in units of photons/(cm² s sr). The maps are shown at Healpix resolution $N_{\text{side}} = 128$ ($n = 7$).

1. A dN/dS is selected by random sampling of its parameters in the intervals and with the priors defined in Table 6.2;
2. The number N_i of sources in the i th flux bin is calculated as $N_i = 4\pi(S_{i,\text{max}} - S_{i,\text{min}})dN/dS|_i$ where $dN/dS|_i$ is the value at the center of the i th bin. The flux bins are determined by subdividing the interval of interest $[5 \cdot 10^{-12}, 1 \cdot 10^{-7}] \text{ cm}^{-2} \text{ s}^{-1}$ in 10^3 log-intervals per decade;

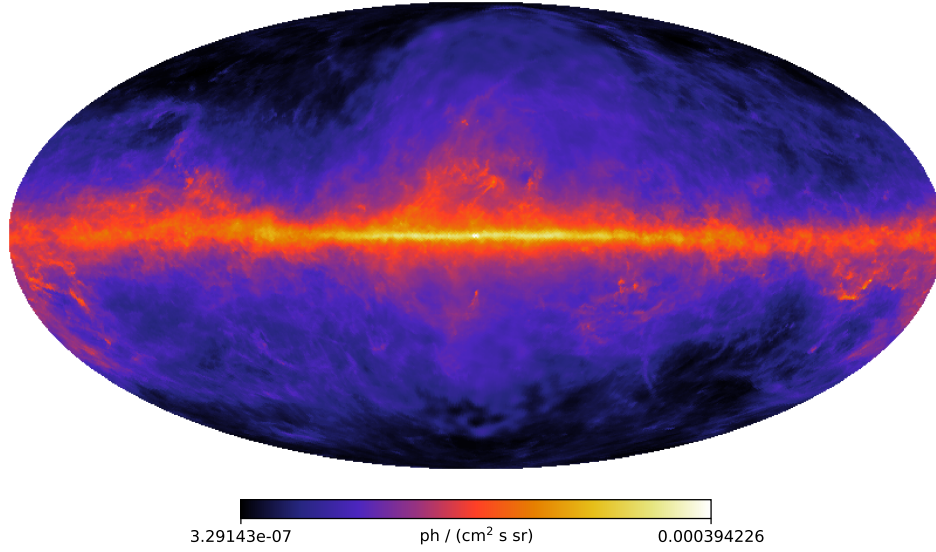


Fig. 6.6. gll_iem_v07 galactic foreground model [121], integrated in the (1, 10) GeV energy bin, in units of photons/(cm² s sr), at the Healpix resolution $N_{\text{side}} = 128$ ($n = 7$).

3. A Poisson extraction on N_i determines the number of sources $N_{P,i}$ in each flux interval;
4. The sources are randomly positioned on the sphere by uniformly sampling $\cos\theta$ in $[-1, 1]$ and φ in $[0, 2\pi]$, θ and φ being the spherical coordinates. The selected angular position is then converted into a pixel position in the Healpix pixeling scheme with the ang2pix routine. An example of a point-source map is shown in the upper panel of Figure 6.5;
5. Finally, convolution with the *Fermi*-LAT PSF shown in Figure 6.3 is applied to each point source. Since sources are assumed to be point-like, we simply replace each point source with an extended circular object with a radial profile equal to the PSF, and whose total flux is equal to the one of the original source. Since the image formation process is linear in the intensity, in case of overlapping sources (either because of their spatial position, or because of the PSF smearing) the resulting flux is additive. An example of flux map with the PSF applied is shown in the lower panel of Figure 6.5.

In order to avoid possible biases in the generated maps, due to the resolution of the angular binning, the maps generated with the procedure described above have been produced at a resolution higher than the one adopted for the analysis. We have therefore built our maps with $N_{\text{side}} = 128$. These maps have then been downgraded to $N_{\text{side}} = 64$ for the analysis.

6.5.2.2 Galactic foreground and isotropic background

The galactic foreground is implemented through the adoption of the *Fermi*-LAT template model gll_iem_v07¹. The map of the integrated flux in the (1, 10) GeV energy range is shown in Figure 6.6. To check consistency and stability of our results, we will also adopt a different version of the galactic template, namely model gll_iem_v05¹. The extragalactic isotropic background instead

¹<https://fermi.gsfc.nasa.gov/ssc/data/access/lat/BackgroundModels.html>

Foreground template	Latitude cut	A_{gal}	F'_{iso}
gll_iem_v07	$ b < 30^\circ$	0.888 ± 0.005	$(4.91 \pm 0.04) \cdot 10^{-7}$
gll_iem_v07	$ b < 40^\circ$	0.874 ± 0.008	$(4.90 \pm 0.06) \cdot 10^{-7}$
gll_iem_v07	$ b < 50^\circ$	0.838 ± 0.013	$(5.05 \pm 0.08) \cdot 10^{-7}$
gll_iem_v05	$ b < 30^\circ$	1.030 ± 0.006	$(2.57 \pm 0.06) \cdot 10^{-7}$

Table 6.3. Values of the A_{gal} and F'_{iso} parameters obtained from a fit to the *Fermi*-LAT map, for different galactic foreground template models and latitude cuts. F'_{iso} is in units of $\text{cm}^{-2} \text{s}^{-1} \text{sr}^{-1}$.

is modeled as a constant flux F'_{iso} , which captures all the unresolved emission which is too faint to be associated to the emission modeled through the dN/dS (i.e., which refers to fluxes $S < 5 \cdot 10^{-12} \text{cm}^{-2} \text{s}^{-1}$, below our region of interest).

As shown in Figure 6.6, the galactic foreground dominates at low latitudes, although it extends to high latitudes with a smaller flux. For these reasons, low latitudes are unsuitable for the study of the extra-galactic background. Our baseline analysis will therefore apply a mask to the maps to cover the dominant part of the galactic emission, for which we adopt a latitude cut $|b| < 30^\circ$. In order to check the stability of the results against the foreground treatment, in Section 6.5.4 we will show the results obtained with two additional latitude cuts: $|b| < 40^\circ$ and $|b| < 50^\circ$.

Given the various approximations we use, instead of using the nominal normalization of the galactic foreground models, we prefer to use a data-driven approach and normalize the galactic templates gll_iem_v07 and gll_iem_v05 directly to the observed *Fermi*-LAT map. This is realized by performing a maximum-likelihood Poisson fit, using as likelihood:

$$\mathcal{L} = \prod_{a=1}^{N_{\text{pix}}} \frac{\lambda_a^{k_a} e^{-\lambda_a}}{k_a!} \quad (6.5.4)$$

where k_a is the number of photons in pixel a of the *Fermi*-LAT count map and λ_a the corresponding quantity from the model count-map, given by $\lambda_a = (A_{\text{gal}} \mathcal{G}_a + F'_{\text{iso}}) \cdot \mathcal{E}_a \cdot \frac{4\pi}{N_{\text{pix}}}$, where A_{gal} and F'_{iso} are the two fit parameters. In performing the fit we mask the sky with the low-latitude cut and with a 2° circular mask around each of the sources in the 4FGL catalog. We added a prime on the F'_{iso} derived from the maximum-likelihood fit above to explicitly distinguish it from the F_{iso} introduced earlier. Confronting with Eq. (6.5.1), the F'_{iso} term denotes the sum of the purely isotropic contribution F_{iso} and the contribution of sources, $\mathcal{S}$ term, below the 4FGL detection threshold.

The results of the fit are shown in Table 6.3, for three latitude cuts for the gll_iem_v07 template and for $|b| < 30^\circ$ for the alternative gll_iem_v05 model. We see that both the normalisation parameter A_{gal} and F'_{iso} are consistent among themselves for a fixed galactic foreground model, implying consistency with respect to the latitude cut. The model gll_iem_v05 instead requires a larger normalization, with an ensuing reduced value for F'_{iso} by a factor of 2. This reflects a certain amount of degeneracy which is present at high galactic latitude between the foreground emission and the isotropic emission, allowed by the uncertainties in the foreground model. The normalization of the v07 model different from 1 is related to the fact that we do not include energy dispersion. The v05 model, instead, has been created originally without including the effect of energy dispersion², and indeed we obtain a normalization compatible with 1.

Throughout the rest of the analysis we will fix the normalization parameter A_{gal} to the value determined from the above fit. The values of F'_{iso} are, instead, used to define the range of the

²<https://fermi.gsfc.nasa.gov/ssc/data/access/lat/BackgroundModels.html>

prior for F_{iso} in Table 6.2 and *a posteriori* to check consistency of the results. One illustration of a flux map obtained as the sum of all components is shown in Figure 6.7. Instead of fixing A_{gal} , in principle, we could also leave it as a free parameter and instruct the network to predict its value. We found, however, that the above procedure constrains A_{gal} very precisely and it is thus preferable. Furthermore, we also found that fixing A_{gal} provides a better stability of the neural network.

Nonetheless, to better assess the impact of fixing the A_{gal} value, we also performed a further test using different values in the range (0.78, 1.0) (i.e., a range centered on our fiducial value 0.888 ± 0.005 , thus encompassing many standard deviations away from its central value.) We found that the dN/dS reconstructed by the CNN is largely unaffected by the employed value of A_{gal} , while the specific A_{gal} choice mostly affects F_{iso} , as expected because of the high galactic latitude degeneracy between the foreground emission and the isotropic emission. Example plots are shown in the Appendix 6.6.4.

6.5.2.3 Count maps

With a flux map $\mathcal{M}$ available, we can now produce a synthetic photon-count map $\mathcal{N}$ as described in Eq. 6.5.2. One example is shown in Figure 6.8. In order to generate maps compatible with a counting experiment like *Fermi*-LAT, we then add Poisson noise to the map performing a Poisson realization of the number of photons in each pixel of the map $\mathcal{N}$, and we dub this final map as $\mathcal{N}_{\mathcal{P}}$ count map. One example is shown in Figure 6.9, which also shows the same map with the latitude cut applied: these are the type of maps that we use to train and validate the neural network. We have generated in total 1 million maps, 90% of which have been used for training and 10% for validation. The maps have been stored as a TFRecord dataset³ for use with Tensorflow [285]. The maps have been generated in Healpix format with $N_{\text{side}} = 128$ resolution (order $n = 7$) and then downsampled to $N_{\text{side}} = 64$ (order $n = 6$) for the training and validation (this resolution corresponds to 49152 pixels).

The training of the CNN and the subsequent analysis of the *Fermi*-LAT data can be performed equivalently with maps in units of photon-count or in units of flux. Nonetheless, we will work with maps in units of flux, which are easily obtained from the $\mathcal{N}_{\mathcal{P}}$ count maps by using the exposure map and by inverting Eq. (6.5.2). The reason to convert the $\mathcal{N}_{\mathcal{P}}$ count maps in units of flux is motivated by the fact that count maps contain a spurious large-scale angular dependence introduced by the exposure map (see Figure 6.2) which might confuse the CNN, while this potential issue is avoided using flux maps. Finally, to further ease the learning process of the CNN we also apply a foreground subtraction procedure, i.e., to the above flux maps we subtract the original galactic template flux map. In this way the CNN can focus on learning the dN/dS , seeing the exposure and galactic foregrounds only as sources of noise, instead of trying to learn their detailed structure. These foreground-subtracted flux maps are the maps we use to train the CNN.

6.5.3 Neural network architecture and training

In this Section we describe the architecture of the CNN we adopt and how data are handled. The input of the CNN are synthetic foreground-subtracted flux maps, and the output vector $\hat{y}_i = (dN/dS|_j, F_{\text{iso}})$ contains the reconstructed $dN/dS|_j$ in 20 flux bins in $[5 \cdot 10^{-12}, 1 \cdot 10^{-7}] \text{ cm}^{-2} \text{ s}^{-1}$, plus the reconstructed value of F_{iso} . The index i , thus, has values $i = 1, \dots, 21$. A possible alternative would be to choose as output the breaks and the slopes of the dN/dS , i.e., the same

³https://www.tensorflow.org/tutorials/load_data/tfrecord

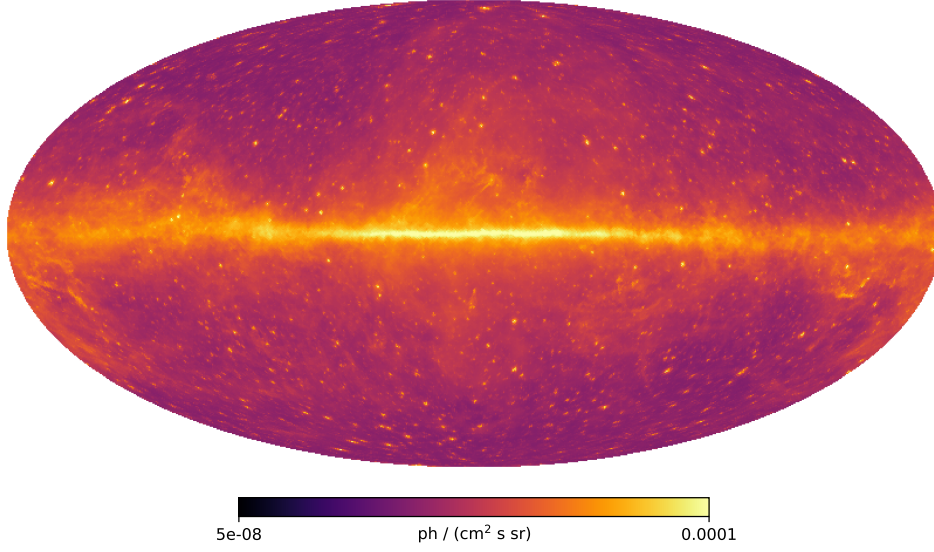


Fig. 6.7. The same as in Figure 6.5, with the galactic foreground model normalized with $A_{\text{gal}} = 0.886$ and an isotropic background component $F_{\text{iso}} = 4 \cdot 10^{-7} \text{ cm}^{-2} \text{ s}^{-1} \text{ sr}^{-1}$. Units are photons/(cm² s sr).

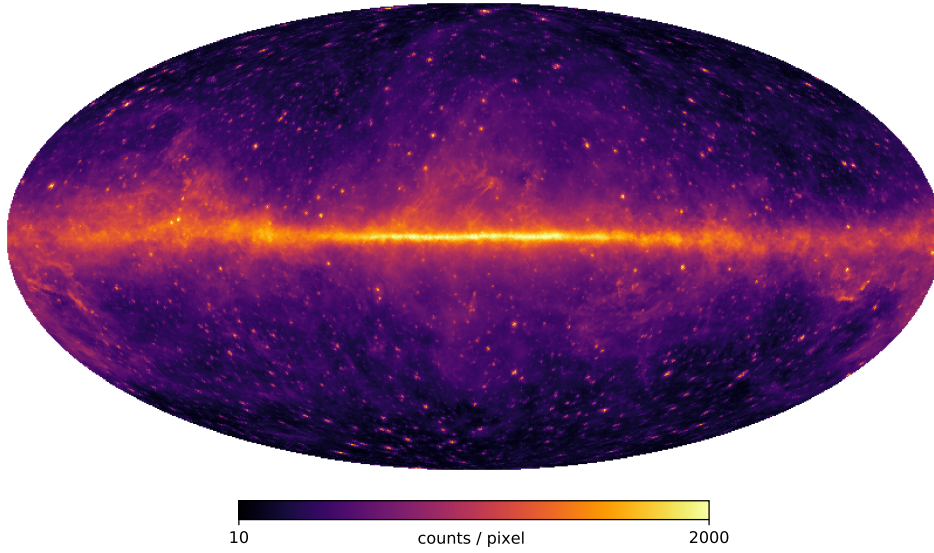


Fig. 6.8. Simulated count map in units of counts per pixel, at the Healpix resolution $N_{\text{side}} = 128$ ($n = 7$).

parameters used to model the input dN/dS . Nonetheless, we found a better stability of the CNN using as output directly the dN/dS function as discretized above, and we will thus use this as output.

As further described below, we will use an approach in which we will let the CNN determine

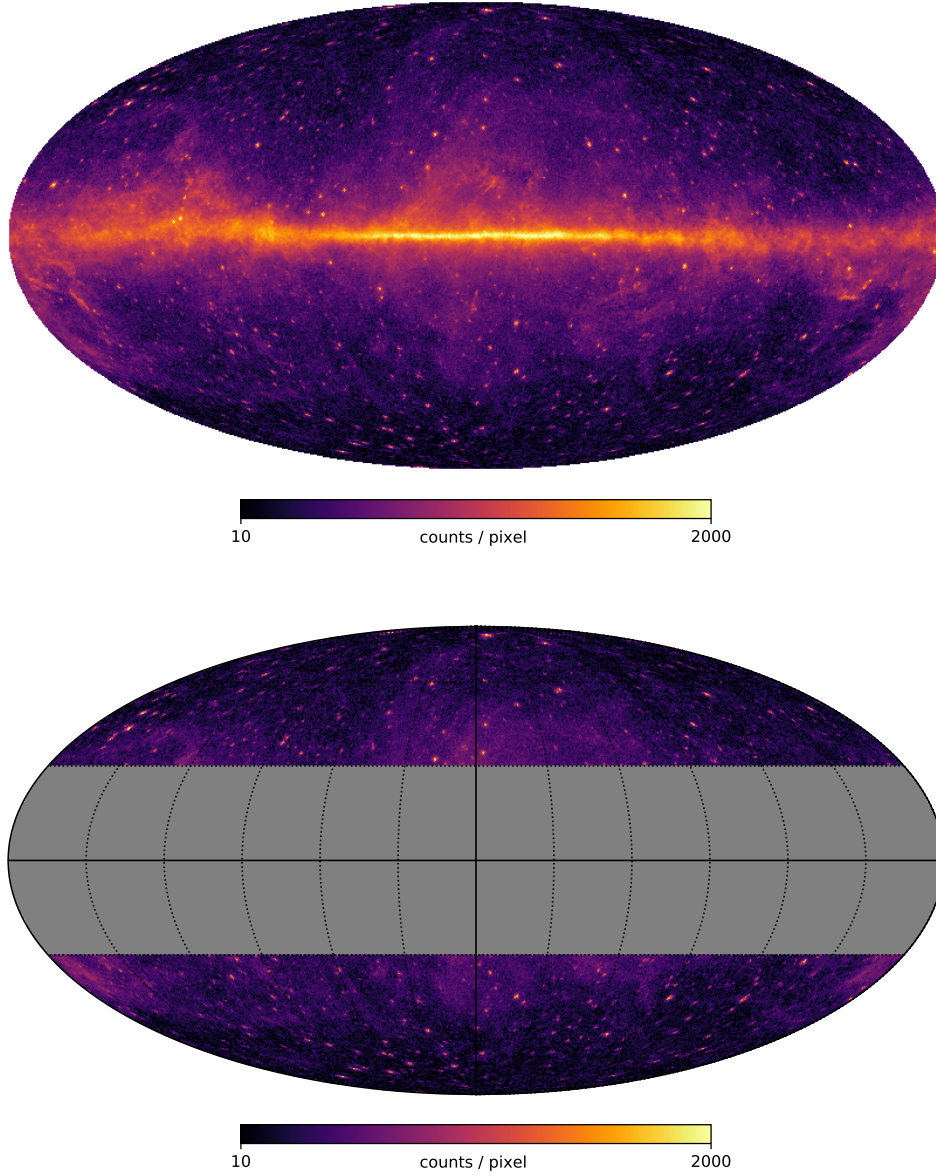


Fig. 6.9. Simulated map in units of counts per pixel after adding Poisson noise from the synthetic model map of Figure 6.8, at the same Healpix resolution $N_{\text{side}} = 128$ ($n = 7$). The lower panel shows the map with a $|b| < 30^\circ$ latitude cut.

also the errors on the output parameters. The output vector will thus be effectively doubled. Beside the $\hat{y}_i$, we thus have also $\hat{\sigma}_i$ the output vector of the errors.

Since we are dealing with sky maps, we can approach the problem from the point of view of image analysis. However, the image here is in spherical projection. We therefore need to adopt a method able to deal with this situation.

6.5.3.1 Spherical neural networks

In the last years, the machine learning community has realised the importance of developing neural network architectures on complex topologies. One of the most interesting non-trivial domains is that of a sphere, since it cannot be mapped on a plane without introducing distortion. This is the case we are confronting here, since we are dealing with sky maps. There are many efforts to implement CNN architectures that can do convolutions on the sphere [150, 152, 286–288], and while there have been many proposals, there is not yet agreement on a standard algorithm.

In dealing with information extraction from spherical images, one has to cope with a set of issues: i) the algorithm needs to properly deal with the topology of the sphere without distorting the image or losing information; ii) the algorithm has to be computationally efficient; iii) it would be desirable to be able to use the same neural network architectures which already work for flat images, which have been proven to be very efficient and effective.

In fact, a full spherical architecture is usually slower than a flat architecture when applied to images of the same size, and it often happens that it is not possible to adopt pre-existing models without re-implementing them *ad-hoc*. On the other hand, the adoption of a flat-image architecture either introduces spherical distortions, due to the mapping of a spherical image on a plane, or may lose some of the large scale information contained in the image.

In this work, we implemented a fast and reliable spherical architecture on the Healpix pixelisation of the sphere, taking inspiration from the idea proposed in [288] for the icosahedron. This method proved to be reliable when the information we need to retrieve from the maps is a small-scale effect, like in our case the distribution of sources, which are point-like (although smeared by the PSF) and isotropically distributed.

The method is based on the fact that, in the Healpix pixelisation scheme, each pixel has equal area and contains the information of the underlying field (in our case, the photon counts) averaged over that pixel. As discussed above, we train our neural network on maps of order $n = 6$ ($N_{\text{side}} = 64$), which contain 49152 pixels. Considering instead the Healpix sphere with base pixelisation of order $n = 0$ ($N_{\text{side}} = 1$), we can subdivide the map into 12 equal-area patches, each of which containing 4096 pixels. Each patch is an independent realisation of the underlying isotropic field and can be mapped into a 2D flat image. We call this parsing algorithm `map2patch`. Notice that this is not a re-sampling of the map: it's just a subdivision of the spherical image into 12 patches, in a way that preserves the area and content of each pixel. This allows us to use standard convolutional network techniques that work efficiently on flat images, while preserving the whole available statistics of the full map.

Convolution is performed on each patch separately. In order to be able to extract information from each patch in parallel, we define 3D convolutions, where the first two dimensions are the spatial dimensions, and the third dimension specifies the patch. In order to force the neural network to learn from each individual sky patch, we employ convolution filters of size $(N, N, 1)$, where N is the dimension of the filter. By doing so, we effectively apply 2D filters to the whole sky. Note that by employing such filter size, the output of each convolution will still have 12 separate images on the third axis, associated to each sky patch, while the number of channels will vary according to the number of filters employed. This technique will thus naturally scale for images with different color channels, such as RGB images, or spherical images for which several channels are available, as is the case where multiple energy bands are considered at once.

We stress that with our method we partially lose large-scale information and information stored at the border of the patches. Thus our algorithm would not be suitable in the case one is

interested in recovering large-scale features. In our case, however, we are interested in exploring information which is statistically isotropic and/or lies at scales much smaller than the size of a patch. In this case our architecture is certainly viable and extremely efficient. Also, since we build our patch on the pixels of the Healpix pixelisation scheme, there is no spherical distortion in our images, unlike other architectures which employ the euclidean projection of the sphere.

This approach is extremely fast, since it leverages on optimised 3D convolutions. Furthermore the sphere-to-patch operation is performed asynchronously by the CPU while the accelerator is training the model, allowing for virtually no bottlenecks in the processing pipeline.

In order to cross-check our method, we also implemented the full-sky truly-spherical convolutions of [152], obtaining very similar results. In Appendix 6.6.5, we report some examples of the results and checks performed with a fully spherical convolution model. The two methods are therefore interchangeable for the type of information we want extract from the *Fermi*-LAT maps: however, our implementation allowed us to gain more than a factor of 10 in time of execution of the whole pipeline of training and validation. This convergence of results with the fully spherical method of [152] gives us confidence on the reliability of our model, and the speed gain sets the basis for a future efficient extension of our analysis to several energy bins in the full *Fermi*-LAT energy range, in order to determine also the energy dependence of the dN/dS of unresolved sources.

6.5.3.2 Data pre-processing

The maps are constructed with the procedure outlined in Section 6.5.2. As explained more in detail there, even though we apply a latitude cut to the maps, in order to further reduce the impact of the galactic foreground (and as it is customary in analyses of the extragalactic gamma-ray background), we subtract the galactic foreground (normalized by the A_{gal} constant) from the input maps. This will also facilitate to some extent the training of the CNN, whose goal is to reconstruct the source-count distribution. Photon-counts maps are converted in units of flux, by using the exposure map. In order to reduce the variability of the inputs seen by the CNN and stabilize its behaviour, we take the logarithm of the flux values in the pixels and then map them in $[-1, 1]$ range. Finally, each map is parsed into 12 patches using the `map2patch` algorithm discussed in the previous Sections.

Given the approximate S^{-2} behavior of the dN/dS , for the output vector, we reconstruct $y_i = S^2 dN/dS|_i \times 10^{11} \text{ cm}^2 \text{ s deg}^2$ ($i = 1, 20$) and $y_{21} = F_{\text{iso}} \cdot 10^7 \text{ cm}^2 \text{ s sr}$, in order to work with (pure) numbers of order unity. We adopt this strategy in order to avoid that the mean square error could favour the optimisation of any flux bin over another, making the deviations more even across the range of S .

6.5.3.3 Network architecture

In recent years, we have seen the rise of many convolutional neural network models for computer vision tasks [289–291]. Modern convolutional neural network architectures can have a very large numbers of internal parameters. This is due to the complex links between each neural unit, which allows for these models to be highly expressive and effectively behave as universal approximators for many kinds of tasks [292]. Unfortunately this comes with a downside: these models are prone to fitting too well the training dataset, which would lead to poorer performance on new data. In the field of Machine Learning, this problem is known as "overfitting" [293]. In order to prevent overfitting, a possible solution is to train the model on a big enough dataset. This will render impossible for the NN to learn exactly the representation of the training set, as

well as showing a richer feature space, which will result in better performance on any additional sample for which we may want to get a prediction, like the real case scenario with the Fermi photon counts map. Another solution is to apply regularisation layers which force the NN to learn a more general representation of the data, by promoting several metrics. Some of the most common regularisation techniques include L1 and L2 regularisation, batch normalisation, and dropout [143, 294–296].

We choose as our convolutional neural network architecture the EfficientNet V2M model [297]. This model has been proven to be faster to train than other models for computer vision tasks, while retaining the same expressivity and predictive power. Furthermore, its low memory footprint helps in training with more data at the same time (larger batch sizes), which in turn reduces the risk of overfitting. In this work, following the original implementation of EfficientNet, we will adopt batch normalisation layers after each convolution.

We additionally employ a fairly new trainable implementation of dropout, called concrete dropout [144], before each convolution layer. Dropout [294] has been proven to be an efficient way to reduce overfitting in a model, by randomly turning off neurons or connections inside a model. Unfortunately, the dropout probability for each dropout layer has to be either hand picked or optimised through grid search, which is practically unfeasible for very large models. If the dropout probability is too small, the regularisation effect will not be very significant, while on the other hand too high a dropout probability will result in slower training and worse overall performance (underfitting). We therefore adopt the implementation of dropout as a concrete Bernoulli distribution [298], which allows the self-learning of the optimal dropout probability, effectively saving a conspicuous amount of time while training and ensuring that the model will be less prone to overfitting.

6.5.3.4 Bayesian error estimation and cost function

In order to correctly estimate the error on a prediction, it is necessary to correctly estimate both the systematic error (sometimes called epistemic error [299]) and aleatory error components of a measurement. One way to estimate such components together is through a Bayesian approach, and in particular by defining a Bayesian neural network capable of estimating the posterior distribution of the desired observables. It has been shown [142] that a NN with arbitrary depth and non-linearities, with dropout applied before every weight layer, is mathematically equivalent to an approximation of a Bayesian model.

In a subsequent analysis [137] it has been shown how to estimate the first and second momenta (mean and variance) of the posterior distribution, under the assumption of a Gaussian likelihood of unknown variance. In particular [137] shows how to define a neural network capable of determining the appropriate variances of such a multivariate Gaussian. In our case, training a neural network with such an approach is advantageous, as the neural network will learn to weight less the component of the measurement at low fluxes, which we expect to be more uncertain, which helps in obtaining stable measurements for the test dataset.

Following [137] we define the heteroscedastic Gaussian negative log-likelihood:

$$NLL = \sum_{i=1}^N \left[\frac{(y_i^{\text{true}} - y_i^{\text{pred}})^2}{2\sigma_i^2} + \frac{1}{2} \log(\sigma_i^2) + \frac{1}{2} \log(2\pi) \right] \quad (6.5.5)$$

where we compare the true dN/dS and F_{iso} (transformed as discussed in Section 6.5.3.2) used to generate the simulated map with the prediction of the neural network, and thus $N = 20 + 1$. For the training of the neural network, we omit the last term of Eq. (6.5.5) which is an irrelevant

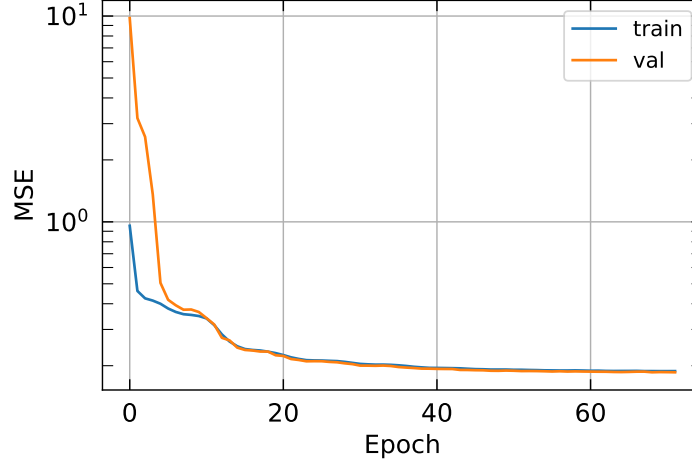


Fig. 6.10. Epoch evolution of the MSE for the training (blue line) and validation (orange line) sets.

constant. Regarding the errors, since the first term in Eq. (6.5.5) is a decreasing function of the σ_i while the second is increasing, this ensures that there will be a minimum which will provide the optimal estimate for the σ_i themselves.

In principle, the above cost function can easily be generalized to account for the covariance among the different bins and F_{iso} . In practice, however, this implies estimating also the off-diagonal terms of a 21×21 symmetric matrix. Such matrix has to be inferred autonomously from data without any supervision, making the problem very challenging to optimise in this form. We thus refrain from attempting estimating the covariance. For CNN applications where the covariance between a set of parameters has been estimated, see e.g. Ref. [191, 279].

6.5.3.5 Training and validation

We train our model using a Tensor Processing Unit (TPU) v3, kindly provided by the GoogleTM TPU Research Cloud. We adopt the Adam optimizer. We employ a cyclical triangular learning rate [300] to avoid falling into local minima of the parameter space. We adopt a batch size equal to $128 \times 8 = 1024$ to fully exploit the computational capabilities of the TPU. We train the model for 72 epochs, for a total of 9 triangular learning rate cycles, with learning rate ranging from 10^{-6} to 10^{-3} . As already anticipated, we have generated 1 million maps, 90% of which have been employed in the training and 10% for validation.

To test the convergence of the model we evaluate the mean square error (MSE) defined as $MSE = \sum_{i=1}^N (y_i^{\text{true}} - y_i^{\text{pred}})^2$. In Figure 6.10 we plot the MSE as a function of the training epoch for the training and validation dataset. We can see that, starting from epoch 10, the validation and training lines are close to each other, which means that our model is not overfitting the training dataset. We observe that the MSE reaches a plateau at around epoch 60, meaning that further training of our model would not lead to improvement in performance.

Once the CNN is fully trained, we extensively check the convergence of the reconstructed dN/dS and F_{iso} against their inputs, by using the maps of the validation set. Figure 6.11 shows a few illustrative examples of dN/dS , randomly generated according to the priors and parameters intervals listed in Table 6.2, together with their reconstructions. The dashed black lines show the input dN/dS and the black solid lines denote the corresponding distributions reconstructed by

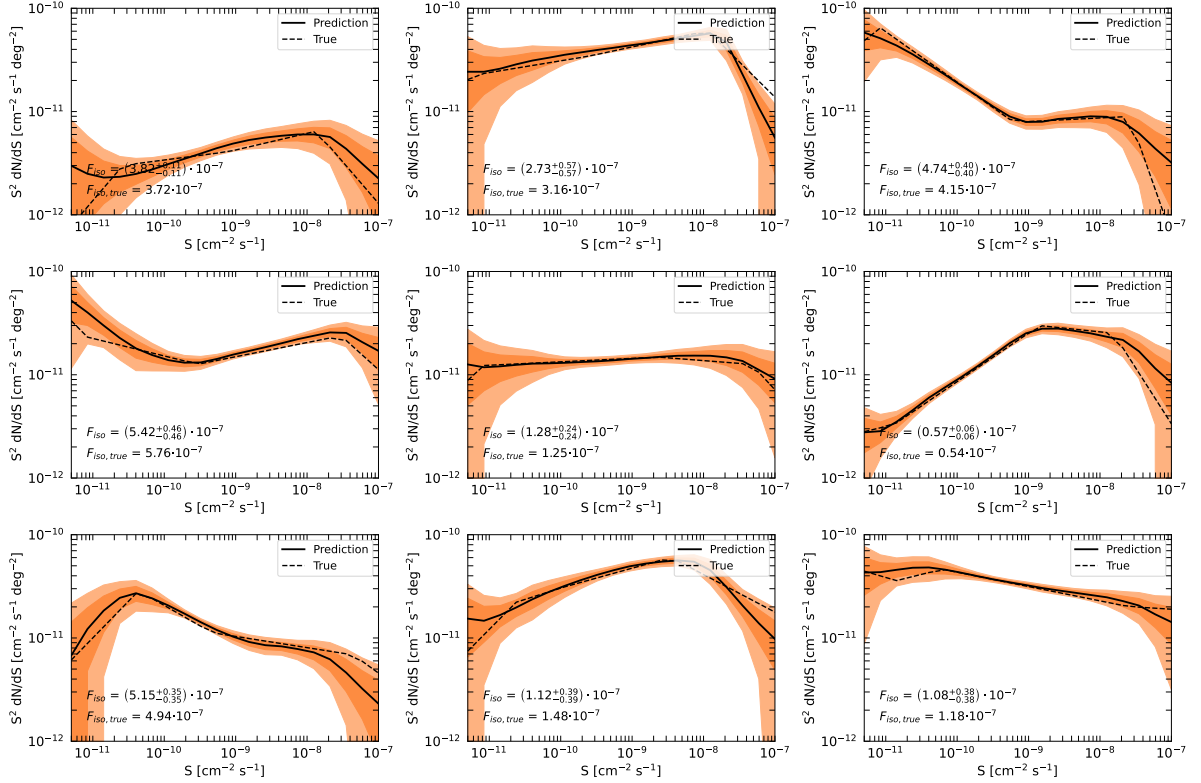


Fig. 6.11. Some example dN/dS from the validation dataset. Each panel shows the input dN/dS and F_{iso} and the corresponding quantities reconstructed by the CNN. The colored bands indicate 1-2 σ uncertainty regions. See text for more details.

the CNN. The input and reconstructed values of F_{iso} are also quoted. The colored bands refer to the 1σ and 2σ confidence intervals as estimated by the CNN. The CNN recovers a dN/dS which is compatible with the input. The error bands are large for high fluxes, due to the fact that the number of sources at those values of S is low and therefore a large statistical uncertainty is expected. The error band becomes large again at low fluxes, indicating that the CNN reaches a confusion limit: even though here the number of sources is very large, their faintness makes progressively harder to identify their number and a confusion between faint sources and the average isotropic component F_{iso} arises, the lower the value of S becomes. This same behaviour also occurs in the 1point-PDF analysis of [274]. Concerning the value of F_{iso} , the reconstruction is very good, compatible with the input and with a small uncertainty.

6.5.3.6 Frequentist error estimation and Bias

Besides the Bayesian error automatically provided as output by the CNN, we also estimate the error in an alternative frequentist-like way, in order to provide a cross-check. This second method gives as a further advantage also the possibility to estimate the amount of bias on the reconstructed dN/dS . To this aim, we have built, for each of the 20 bins in flux and for F_{iso} , the histogram of the deviations between the true and reconstructed dN/dS and F_{iso} . The results for 14 values of S across its interval of interest, and for F_{iso} , are shown in Figure 6.12 (the remaining histograms are similar in content and are not shown here for economy of space). The plot shows the distribution of models which exhibit a specific relative deviation (expressed as percentage of

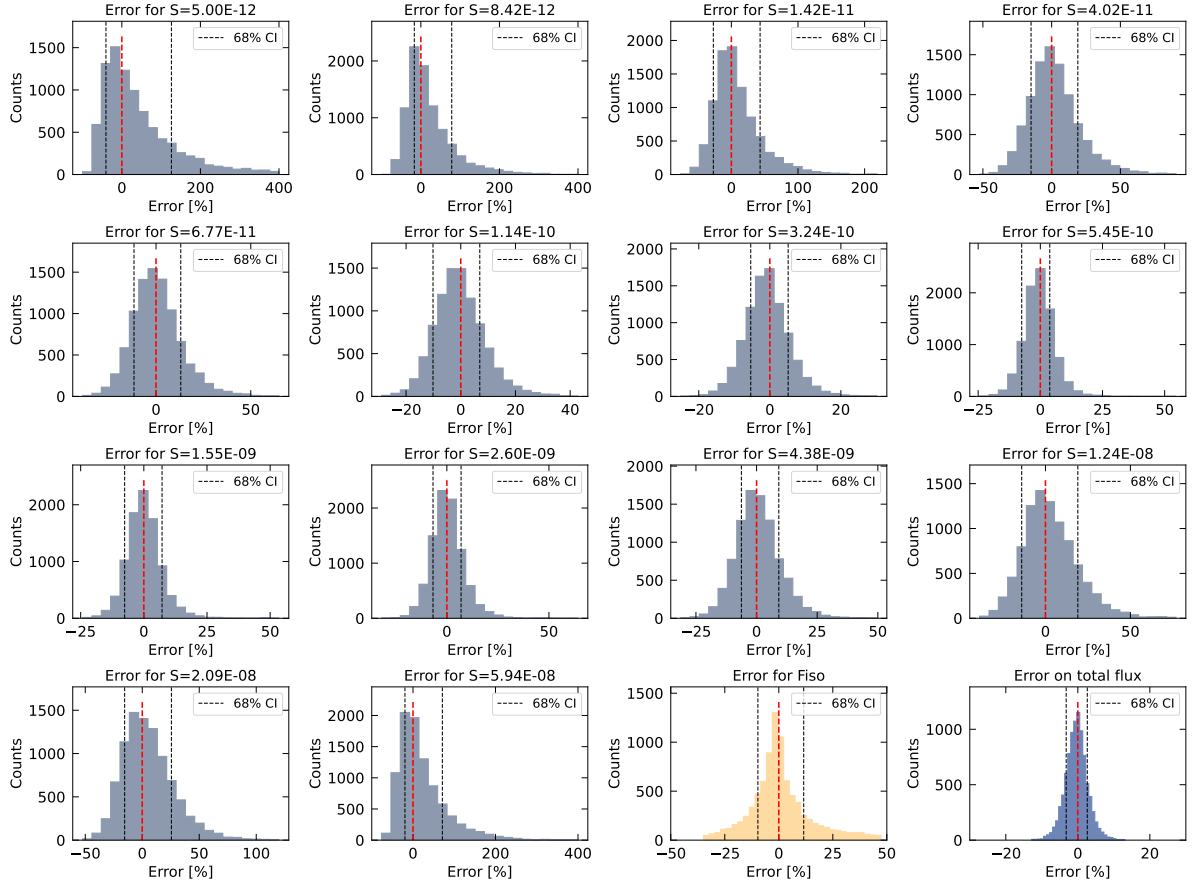


Fig. 6.12. Histograms of the percentage deviations between the true and reconstructed dN/dS for 14 values of S across its interval of interest, and for F_{iso} . The bottom right panel shows the same, but for a derived quantity, namely the total flux. The dotted vertical black lines indicate the 68% confidence region.

the true value). By using these distributions of the deviations (normalized to unity), we derived the 1σ and 2σ confidence level intervals on the reconstructed dN/dS and F_{iso} . As anticipated in Section 6.5.3.5, at low and large fluxes the distributions are wide, implying a poorer ability to reconstruct the underlying physical models, while in the flux interval $(1 \cdot 10^{-11}, 2 \cdot 10^{-8}) \text{ cm}^{-2} \text{ s}^{-1}$ the 1σ error is below 30%, with an error below 20% for S in $(1 \cdot 10^{-10}, 1 \cdot 10^{-8}) \text{ cm}^{-2} \text{ s}^{-1}$. The 1σ error on F_{iso} is also of the order of 20%. The last panel of Figure 6.12 shows the histogram for the reconstructed value of the total photon flux: this is a sanity check and shows that the CNN correctly reproduces the normalization of the maps in terms of total counting rate. The uncertainty on the total flux is 5%. The 1σ (non symmetric) intervals for all the 20 flux bins are also reported in Figure 6.13. A more direct comparison of the Bayesian and frequentist errors is provided in the results section.

The same histograms of Figure 6.12 allow us also to check whether the reconstructed values are biased. We define the bias as the deviation between the median of the distribution and the input value. The values of the bias are shown, for each of the 20 values of flux, in Figure 6.13 as squares. We see that for all fluxes, the bias is very close to zero, which implies that the reconstructed values not only are determined with good precision (since the error bar is small),

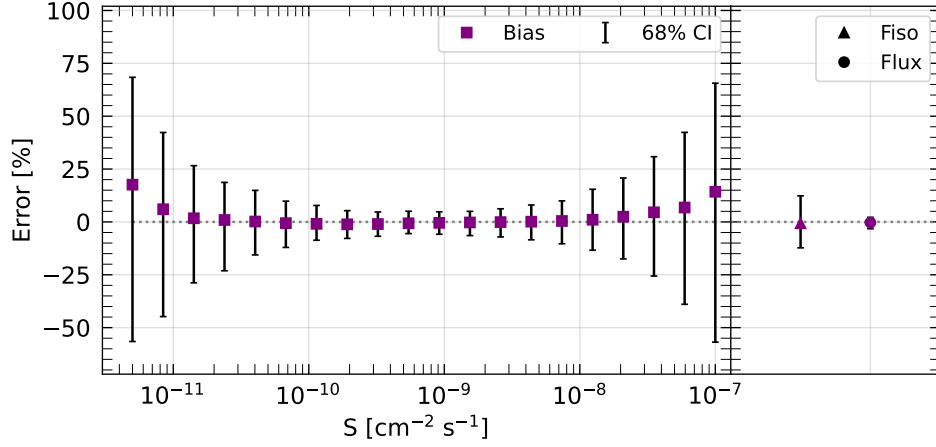


Fig. 6.13. 68% error for the reconstructed dN/dS as function of S , for F_{iso} and for the total flux. Also shown is the bias given by the median of the histogram for each S bin (squares).

but also without a bias toward larger or lower values. The only exception is a positive bias for very low fluxes, where the CNN meets its confusion limit, and for very large fluxes, where the low source-counts statistics produces a small overestimate of the underlying dN/dS . In both cases, however, the bias is small compared to the overall error.

6.5.4 Results

After the training and validation processes described in the previous Section, we have applied the fully trained CNN to the 14 years *Fermi*-LAT data. The result of our baseline analysis, which adopts the `gll_iem_v07` galactic foreground and a latitude cut of $|b| < 30^\circ$, is shown in Figure 6.14. The blue line is the reconstructed dN/dS and the error bands are the Bayesian errors obtained as described in Section 6.5.3.4. The blue points show the dN/dS of the resolved sources, obtained from the 4FGL-DR3 catalog. We can see that in the resolved limit, the CNN reconstructs a dN/dS fully compatible with the one derived from the catalog, but then extends it to the unresolved regime with a behaviour compatible with $dN/dS \sim S^{-2}$ down to the smallest flux considered of $S = 5 \cdot 10^{-12} \text{ cm}^{-2} \text{ s}^{-1}$. Below $S \sim 10^{-10} \text{ cm}^{-2} \text{ s}^{-1}$ the dN/dS exhibits a slight turn-up, although the feature has a very low statistical significance.

The result obtained here is also compatible with the one obtained in [274], as shown in Figure 6.15. The difference between the two analyses, apart from the technique adopted to extract the dN/dS , stands in the fact that here we update the data set to 14 years of *Fermi*-LAT data collection and to improved detector response functions, as compared to [274]. Another difference is that in [274] the dN/dS was reconstructed with a broken power-law, while here we are allowing for a more flexible functional dependence (we are using 20 points in the flux interval shown in Figure 6.14). The result is nevertheless quite compatible with a power law behaviour with $dN/dS \sim S^{-2}$ for mid-values of S , $dN/dS \sim S^{-3}$ for $S > 10^{-8} \text{ cm}^{-2} \text{ s}^{-1}$ and a slight decrease in the power-law index for fluxes $S < 10^{-10} \text{ cm}^{-2} \text{ s}^{-1}$.

In Figure 6.16 we show the measurement for the Fermi map employing the frequentist estimation of the error. It can be seen that the error is very similar to the Bayesian one, with some minimal difference at low fluxes. This provides confidence that the estimated error is robust.

The CNN provides also the value $F_{\text{iso}} = 4.90 \pm 0.43 \cdot 10^{-7} \text{ cm}^{-2} \text{ s}^{-1} \text{ sr}^{-1}$, which can be compared with the value $F'_{\text{iso}} = 4.91 \pm 0.04 \cdot 10^{-7} \text{ cm}^{-2} \text{ s}^{-1} \text{ sr}^{-1}$ reported in Table 6.3. We remind that F'_{iso}

is the sum of F_{iso} and of the contribution of sources below the 4FGL catalog threshold. Indeed, if we calculate the contribution of sources using our best fit dN/dS integrated in the range $[5 \cdot 10^{-12}, 1 \cdot 10^{-10}] \text{ cm}^{-2} \text{ s}^{-1}$, the upper limit being an approximate value for the catalog threshold, and we add F_{iso} we obtain a value of $(6.41 \pm 0.61) \cdot 10^{-7} \text{ cm}^{-2} \text{ s}^{-1} \text{ sr}^{-1}$. This is in slight tension with F'_{iso} , although the two agree at the $\sim 2\sigma$ level.

In order to test the stability of the results, we repeated the analysis by using the same `gll_iem_v07` galactic foreground but with two different latitude cuts: $|b| < 40^\circ$ and $|b| < 50^\circ$. The positive consequence of a higher latitude cut is a lower residual foreground contamination. However, the amount of available data is reduced. The CNN has been fully re-trained for both of these situations, and the ensuing results when applied to the *Fermi*-LAT data are shown in Figure 6.17. We notice that the results, both in terms of behaviour and uncertainty estimate, are fully compatible with the results of the baseline analysis. The dN/dS exhibits a slight decrease toward lower fluxes as compared to the baseline case, although the size of this effect is not statistically significant. The values of F_{iso} are also fully compatible among them.

A second test of stability has been performed in order to check the impact of galactic foreground modeling. We have performed a new training of the CNN with the alternative template `gll_iem_v05`, and with the $|b| < 30^\circ$ latitude cut. The result is shown in Figure 6.18. The reconstructed dN/dS matches to a large degree the source-count distributions obtained with `gll_iem_v07`. This reassures us that, even though the galactic foreground modeling introduces a degree of uncertainty, nevertheless the results of the CNN for dN/dS are remarkably stable. The value of F_{iso} is smaller in this case, equal to $F_{\text{iso}} = 1.82 \pm 0.15 \cdot 10^{-7} \text{ cm}^{-2} \text{ s}^{-1} \text{ sr}^{-1}$, which can be compared with the value $F'_{\text{iso}} = 2.57 \pm 0.06 \cdot 10^{-7} \text{ cm}^{-2} \text{ s}^{-1} \text{ sr}^{-1}$ reported in table 6.3. Adding to F_{iso} the integral of the contribution of sources in the interval in the range $[5 \cdot 10^{-12}, 1 \cdot 10^{-10}] \text{ cm}^{-2} \text{ s}^{-1}$ using the obtained dN/dS , we obtain a total flux of $(2.92 \pm 0.54) \cdot 10^{-7} \text{ cm}^{-2} \text{ s}^{-1} \text{ sr}^{-1}$, which is in very good agreement with F'_{iso} .

In Appendix 6..6, we also discuss the results of further tests that we performed to cross-check the stability of the dN/dS derived in this section, and that confirm the robustness of the result.

6.5.5 Conclusions

In this paper we have studied the differential source-count distribution of gamma-ray sources below the *Fermi*-LAT threshold for source identification. We adopted a convolutional neural network methodology devised to deal with spherical sky-maps and able to extract information related to small scale fluctuations. The technique is based on the EfficientNet V2M model, applied to a patched version of the 2-dimensional spherical sky-map projection of the gamma-ray emission. The neural network has been trained on 900k synthetic maps and validated on 100k additional maps. These synthetic maps have been generated from the flux of random selections of source-counts distributions, to which galactic foreground models and an isotropic component have been added. After training and validation, the neural network has been applied to the *Fermi*-LAT sky-map.

We concentrated our analysis on the photon energy interval (1,10) GeV and to 14 years of data. The main result is shown in Figure 6.14: the reconstructed source count distribution exhibits a $dN/dS \sim S^{-2}$ behaviour over almost four orders of magnitude in flux in the range $[5 \cdot 10^{-12}, 1 \cdot 10^{-8}] \text{ cm}^{-2} \text{ s}^{-1}$, which includes the unresolved regime range $[5 \cdot 10^{-12}, 2 \cdot 10^{-10}]$. In the regime where *Fermi*-LAT has sensitivity to individually resolve gamma-ray sources, Figure 6.14 shows that the neural network fairly reproduces the observations, giving confidence on its reliability and strength. While in the unresolved regime the result is in agreement with previous studies performed with different methodologies. We have also further validated the

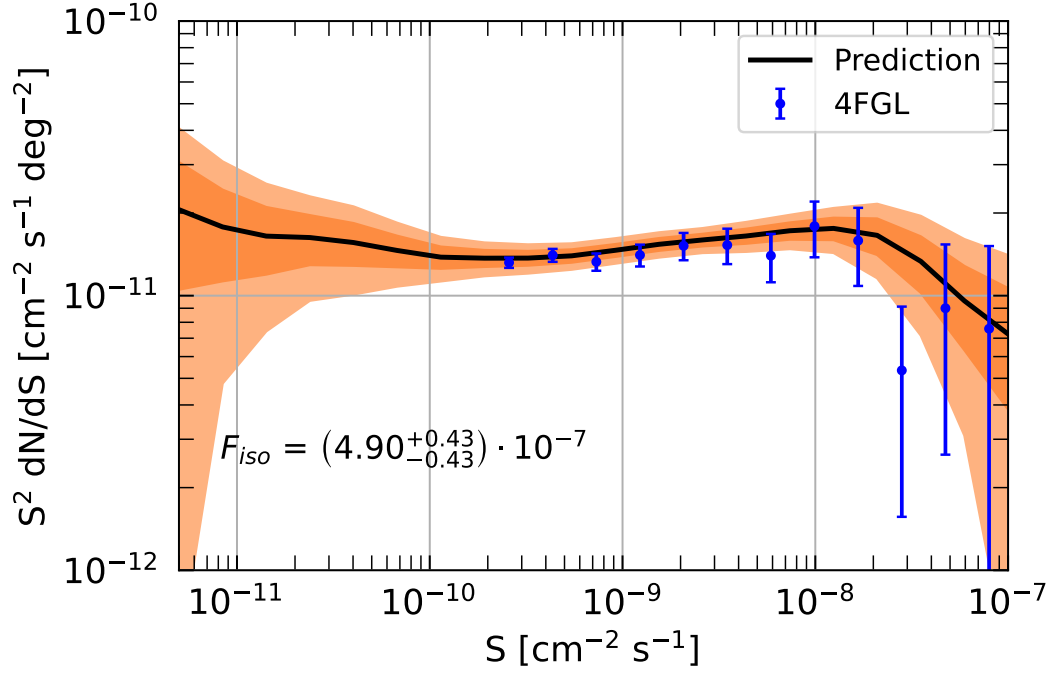


Fig. 6.14. Reconstructed dN/dS and F_{iso} and their Bayesian errors, when the trained CNN is applied to the *Fermi*-LAT map. This is the main baseline result of the analysis.

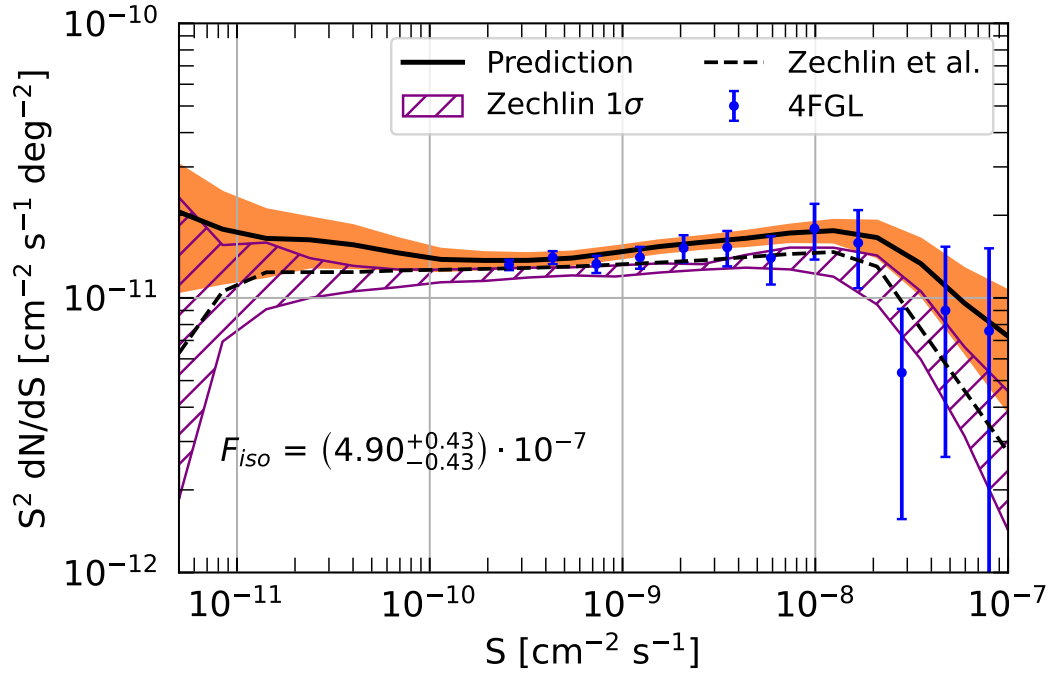


Fig. 6.15. Same as Figure 6.14, including a comparison with the result of Ref. [274].

result performing several test of stability which have confirmed that our findings is stable and

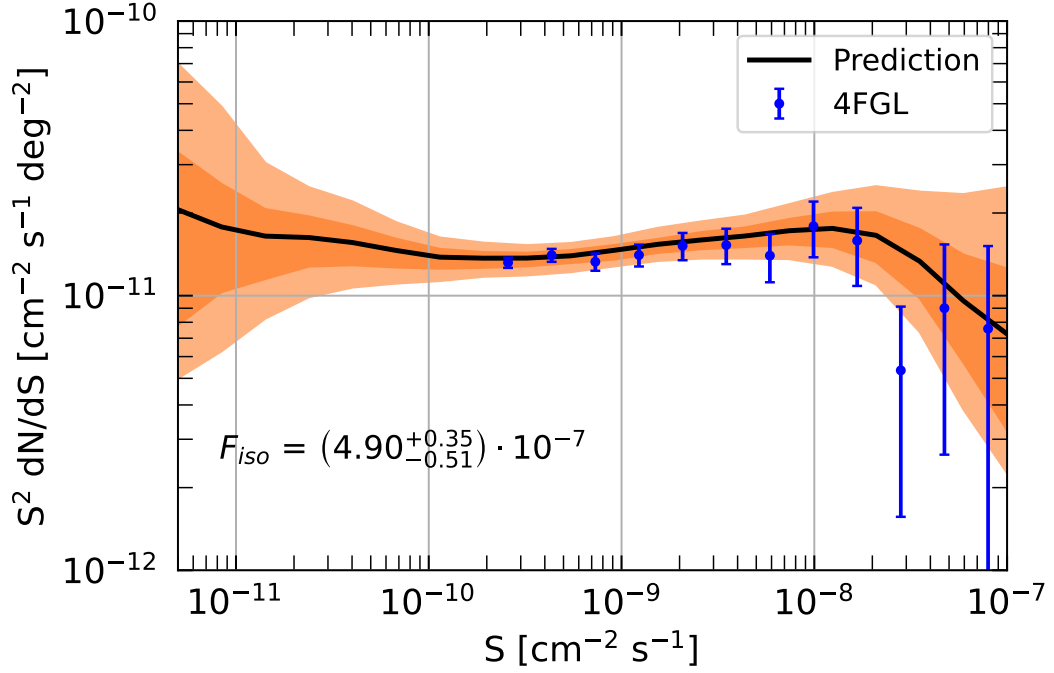


Fig. 6.16. Same as in Figure 6.14, but with errors estimated through a frequentist approach.

robust.

The methodology presented here provides a proof of principle for the adoption of a CNN to the reconstruction of the source-count distribution of the extra-galactic sky. With respect to other methods to extract the source-count distribution in the unresolved regime, like the 1-point PDF one, the use of a CNN avoids the need to calculate complicated and numerically demanding likelihoods. Possible future applications includes the extension to multiple energy ranges and energy correlations, and the investigation of features in the dN/dS which might indicate the presence of exotic components like dark matter. A further aspect which would be interesting to investigate is to compare different approaches to machine learning on spherical domains. In particular, it might be interesting to employ a graph convolutional neural network as suggested in [150], in order to preserve spatial relations between distant pixels, and rotational invariance. Finally, recent developments in the field of simulation based inference, such as our case, suggest that a promising approach to estimate the complex posterior distribution of the objective parameters are the so-called neural likelihood ratio estimation techniques [301], and in particular Truncated Marginal Neural Ratio Estimation [134, 302, 303]. Since these techniques are tailored for simulation based problems, we hypothesise that it would be possible to achieve similar performance to our current neural network architecture with an even simpler structure and possibly less training samples.

6.6 Further Tests

In this appendix we describe further tests that we have performed to cross-check the stability and robustness of the dN/dS derived in the main text.

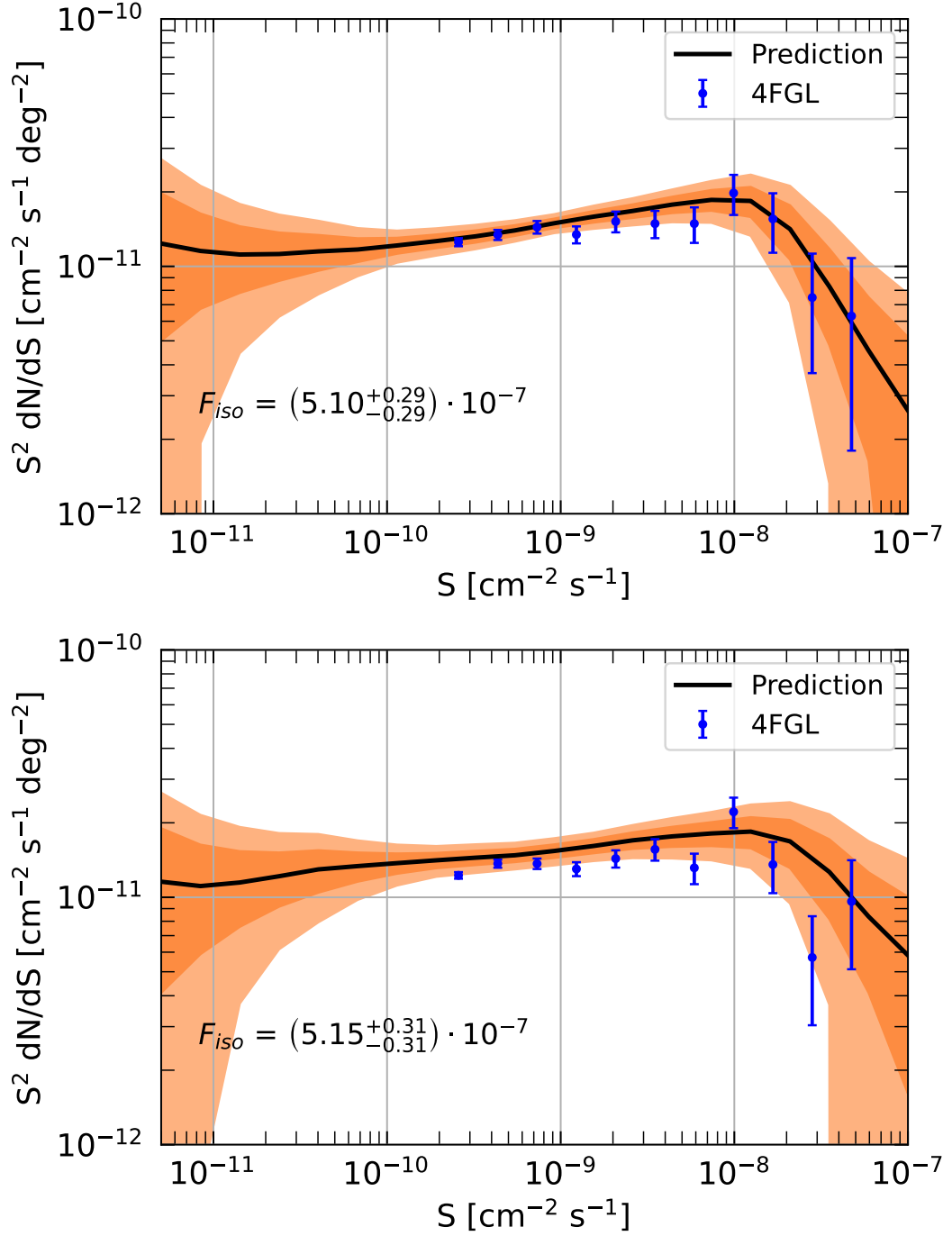


Fig. 6.17. Same as Figure 6.14 but for a latitude cut of 40° and 50° . The blue datapoints of the dN/dS of the 4FGL sources have been updated using only the source in the given region of interested.

6.6.1 Flat $S^2 \frac{dN}{dS}$

Figure 6.19 shows the stability of the CNN output for a specific repeated dN/dS input. In this case, the CNN, trained on the wide variability of models of Table 6.2, is applied to a set of maps,

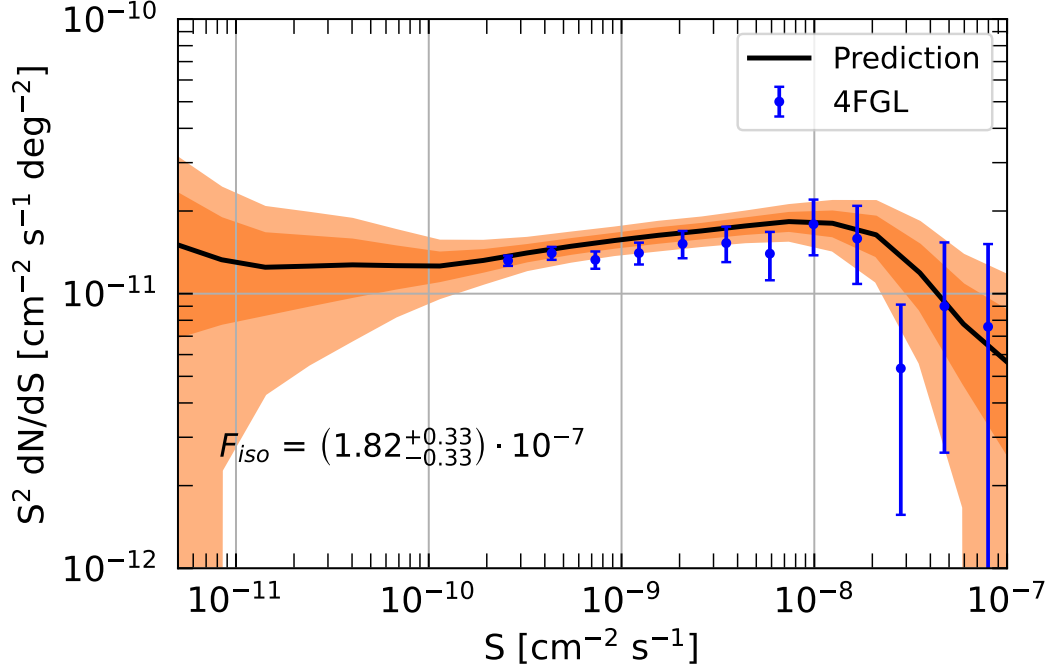


Fig. 6.18. Same as Figure 6.14 but using a CNN trained using the foreground model v05.

all generated with a specific $dN/dS \sim S^{-2}$. This specific choice of input dN/dS is meant to test a case which is expected to be similar to the real *Fermi*-LAT case, as was found in [274] for mid-values of S . The plots show that the CNN is remarkably consistent in reproducing the correct behaviour. The drop at high fluxes is mostly due to the fact that the CNN has been instructed with maps that contain a dropping dN/dS at high fluxes (see Table 6.2 - the reason for this choice of prior is that sources in the 4FGL catalog exhibit this drop in the source count). For mid-values of S , the reconstructed $S^2 dN/dS$ is quite flat, with a small error band. At low S , well below the *Fermi*-LAT threshold and when approaching the confusion limit, some mild deviation occurs, but the true dN/dS lies always within the uncertainties. These tests gives us confidence that the CNN should be able to faithfully reconstruct the source distribution of the *Fermi*-LAT sky, when applied to the real data in Section 6.5.4.

As a further test, we show in Figure 6.20 the case where we apply our method to maps which contain a galactic foreground model different from the one used in the training. The CNN has been built (like in our baseline case) with maps generated with the galactic template model `gll_iem_v07`. The same model is used in the foreground subtraction performed in the analysis. On the contrary, for this test we have generated maps using the alternative `gll_iem_v05` foreground model, and let the above `gll_iem_v07`-trained CNN to analyze them. This has been done in order to verify the resilience of the method on the imperfect knowledge of the galactic foreground. Figure 6.20 shows that the reconstructed dN/dS has a level of agreement with the input model comparable to the analogous case of Figure 6.19. Some larger deviation is present at very low fluxes, even though the reconstructed and input models are compatible within 1σ .

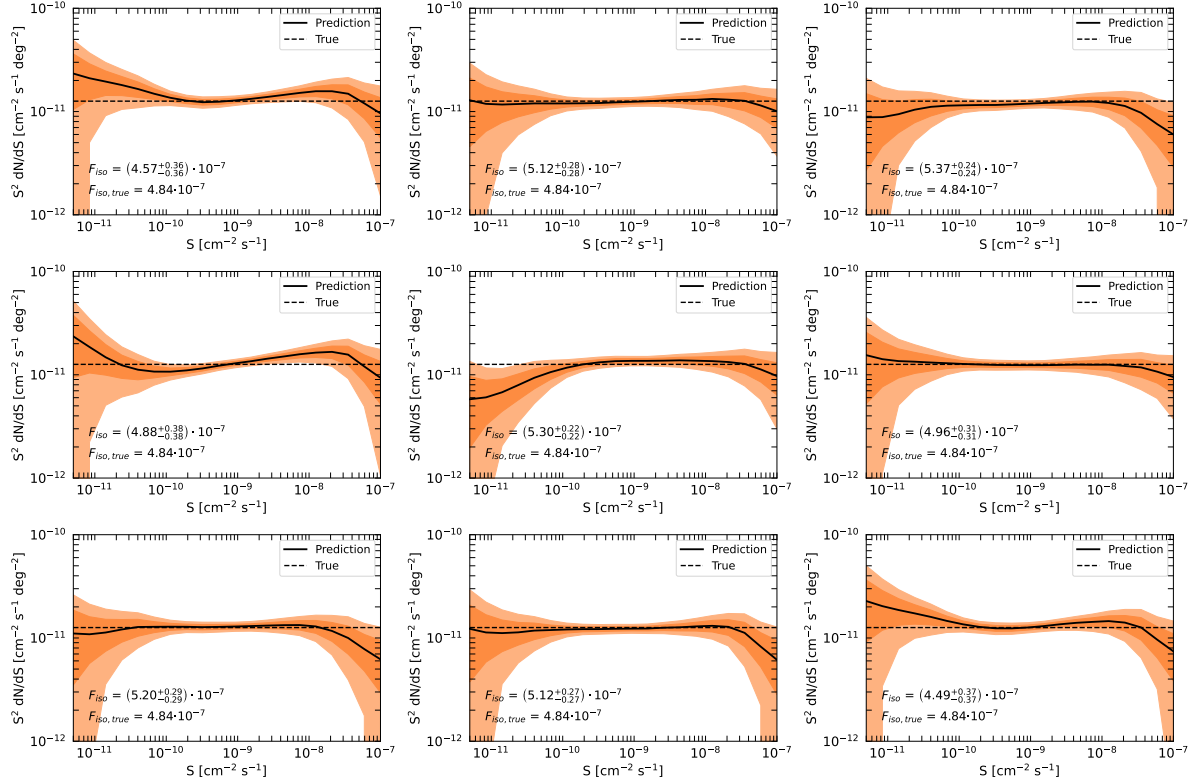


Fig. 6.19. Same as Figure 6.11 but for 6 different random realization of the same flat $S^2 dN/dS$ and same $F_{iso,true} = 4.84 \cdot 10^{-7} \text{ cm}^{-2} \text{ s}^{-1} \text{ sr}^{-1}$.

6.6.2 UltraCleanVeto Selection

In this section we further test the robustness of the results to the *Fermi*-LAT data selection. In particular, while for the main results we used the SOURCEVETO event class selection, here we test the ULTRACLEANVETO selection. This event class has more stringent cuts with respect to the SOURCEVETO class, and it thus contains a lower residual charged cosmic-ray background, at the prize of a lower effective area (by about 15%). Except for the different data selection all the rest of the analysis is performed in the same identical way as for the SOURCEVETO case. The resulting dN/dS is shown in Figure 6.21 and it can be seen that it is compatible with the main results, while, as expected, the value of F_{iso} is a bit lower.

6.6.3 Multipole Analysis

Since foreground modeling can be an important source of systematic uncertainty, we investigate this potential issue performing a further test. As discussed above, the signal we try to extract from the gamma ray maps, is a small-scale effect, since it is due to a distribution of point sources. To a large degree, this point sources are isotropically distributed in the sky and contribute to the fluctuations of this isotropic field at small angular scales. On the contrary, the galactic foreground is quite diffuse, as can be seen in Figure 6.6, and therefore contributes mainly to large-scale anisotropies. In order to further remove from the data the possible large-scale residual component of the galactic foreground, which might still be present after imperfect foreground removal performed with the use of the templates, we adopt a method based on the transformation of the flux to harmonic space and the removal from the maps of the low-multipoles (i.e., large

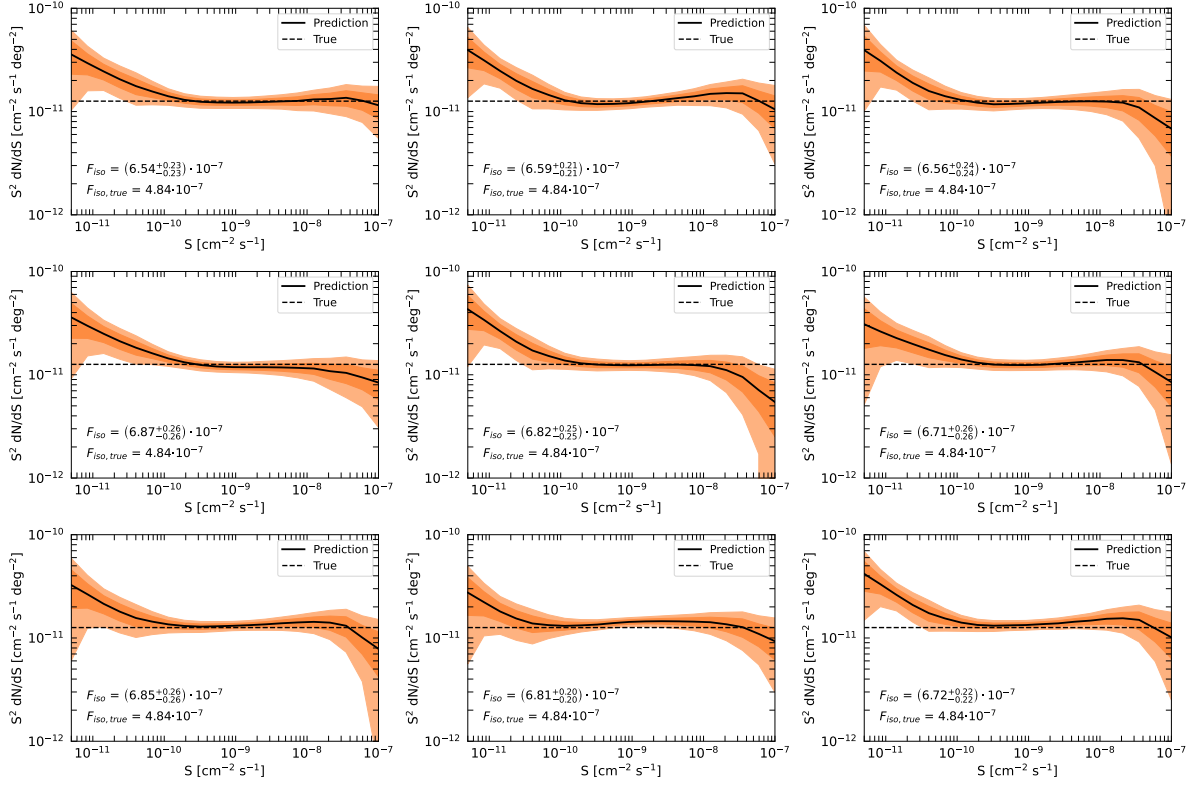


Fig. 6.20. Same as Figure 6.19 but for synthetic maps generated using version v05 of the foreground model.

angular scales) contribution. We stress that this procedure is applied only to the *Fermi*-LAT data map, and not to the synthetic maps. For the latter, the foreground subtracted is the same used to generate the maps, so foreground subtraction is, by definition, “perfect”. The procedure follows these steps:

- We start with the foreground-subtracted flux map, to which we apply the $|b| < 30^\circ$ latitude cut and a 2° mask around each of the bright sources of the 4FGL catalog (which would otherwise dominate the angular power spectrum). We first determine and remove the monopole ($l = 0$) and dipole ($l = 1$) components from the masked map (for this, we use the `remove_dipole` routine from the `Healpy` library). In this way, after the subtraction procedure outlined below, the monopole (i.e. the total flux) and dipole information of the *Fermi*-LAT map are retained.
- This map is then subject to harmonic space decomposition:

$$\mathcal{M}_{\text{masked}}(\theta, \phi) = \sum_{l=0}^{\infty} \sum_{m=-l}^l a_{lm} Y_{lm}(\theta, \phi) \quad (6.6)$$

where θ and ϕ are angles on the sphere, $Y_{lm}(\theta, \phi)$ the spherical harmonics and the harmonic coefficients a_{lm} completely encode the same information present in the map $\mathcal{M}_{\text{masked}}$ (notice that, even though in Eq. (6.6) we formally decompose over all multipoles, the monopole and dipole amplitude are vanishing, since they have been removed from $\mathcal{M}_{\text{masked}}$);

- The harmonic coefficients a_{lm} are determined with the `map2alm` Healpy function. The ones that are retained for the foreground cleaning are those referring to multipole values $l < l_{\max}$, where $l_{\max}$ will be varied and the dependence of the results with $l_{\max}$ will be studied (at this level, the monopole and dipole are not present, because of the above point);
- By using the a_{lm} corresponding to multipoles $l < l_{\max}$, we use Eq. (6.6) to construct a map $\mathcal{M}_{\text{residual}}$ which contains only the large-scale component. This is performed with `alm2map` Healpy function.
- We subtract the residual map from the original map, thus obtaining a map where the large scale residual components have been removed. The size of the scale is determined by $l < l_{\max}$.

With this algorithm we obtain maps where the large scale residual features associated to imperfect galactic foreground subtraction are further removed. The results we obtain on the reconstructed dN/dS for the *Fermi*-LAT data (after this “multipole cleaning”) are shown in Figure 6.22, for different values of $l_{\max}$ and for the `gll_iem_v07` template. By comparing Figure 6.22 with Figure 6.14, which refers to the same foreground template and latitude cut, we can see that the results are extremely stable up to $l_{\max} \sim 100$. This means that our baseline analysis is not affected by incomplete foreground subtraction, and can be considered reliable. The fall-off of the dN/dS when $l_{\max} > 100$ is expected since $l_{\max} = 100$ corresponds to an angular scale of the order of 2° , which starts to be compatible with the size of (PSF-smeared) point sources. Therefore, a cleaning with $l_{\max} > 100$ starts to remove point sources instead of galactic foregrounds.

The same analysis has been performed on a pipeline fully based on `gll_iem_v05`: the result is shown in Figure 6.23, to be compared with Figure 6.18. Also in this case, the results are extremely stable up to $l_{\max} \sim 100$, and the same above considerations apply.

Finally, to check the strength of the method against the imperfect knowledge of the background, we performed the same analysis of multipole cleaning by adopting our baseline pipeline (based on `gll_iem_v07`) on maps generated with `gll_iem_v05` (similarly to what has been done for Figure 6.20). The results are shown in Figure 6.24, where again we see that the results are pretty consistent, up to $l \sim 100$. It can be seen, moreover, that the procedure of multipole cleaning helps in removing the, nonetheless small, positive bias which is present toward low fluxes in this case.

In conclusion, these tests give confidence on the reliability of the method and that the main result of Figure 6.14 corresponds to a fair representation of the source-count distribution of the unresolved gamma-ray sources.

6.6.4 Changing A_{gal}

In order to test the stability of the results against changes in the values of A_{gal} , we have provided, as input to our trained CNN, *Fermi*-LAT maps cleaned with different values of A_{gal} , ranging from 0.78 to 1.0 (our fiducial value is 0.888 ± 0.005 , thus the chosen range encompass many standard deviations around its central value). The results are shown in Figure 6.25 below. It can be seen that the dN/dS is considerably robust and stable to different values of A_{gal} , despite the large range explored. Instead, different values of A_{gal} seem to imply a significant variation in the reconstructed value of F_{iso} , as expected since at the large Galactic latitudes explored in our analysis ($|b| > 30^\circ$) there is a certain amount of degeneracy between a purely isotropic emission and the Galactic foreground emission itself. However, this does not affect the reconstructed dN/dS .

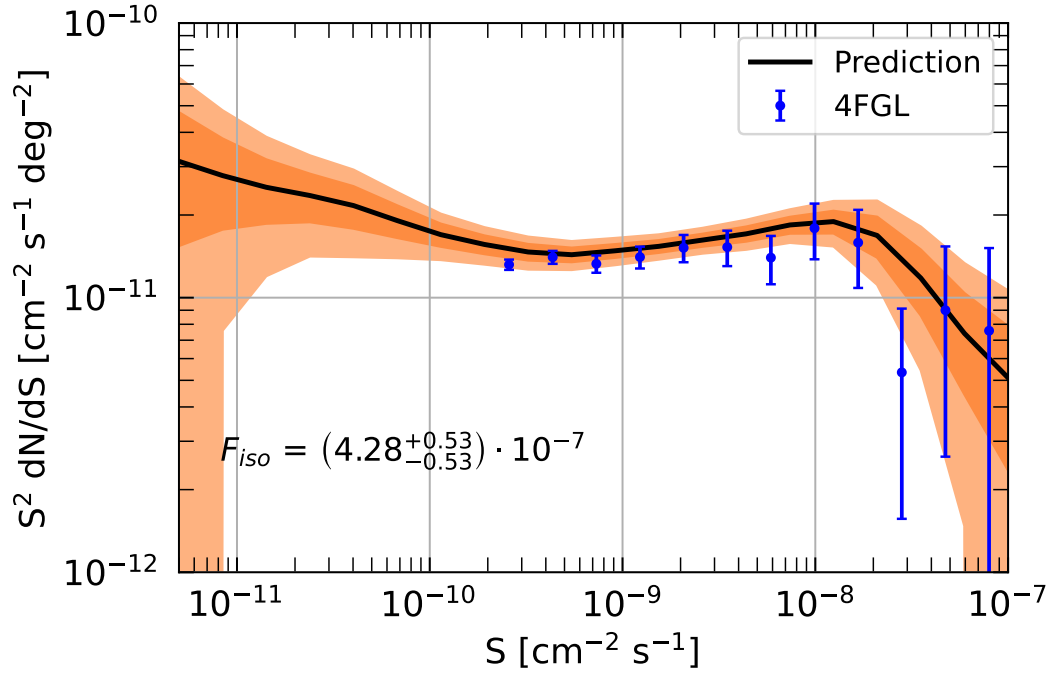


Fig. 6.21. Same as Figure 6.14 but for the ULTRACLEANVETO *Fermi*-LAT data selection.

6.6.5 Fully spherical CNN

In this section we show a comparison of the results obtained using a fully spherical CNN, instead of the map2patch algorithm used in the main analysis. In particular, we adopt here the NNHealpix [152] algorithm.

Figure 6.26 shows the reconstructed dN/dS , F_{iso} and their frequentist errors, using the spherical CNN applied to the *Fermi*-LAT map. The results are fully compatible with the ones obtained with the map2patch algorithm, both for the mean value and the errors, as can be seen by confronting with Figure 6.16.

Figure 6.27 instead, shows some examples of random input dN/dS and F_{iso} from the validation dataset of the Spherical CNN, and the corresponding quantities and their errors reconstructed by the CNN. The spherical CNN has similar performances as the CNN that uses map2patch, as can be seen comparing with Figure 6.11. The same occurs also in Figure 6.28, which refers to a flat $S^2 dN/dS$, to be compared with Figure 6.19 obtained with map2patch.

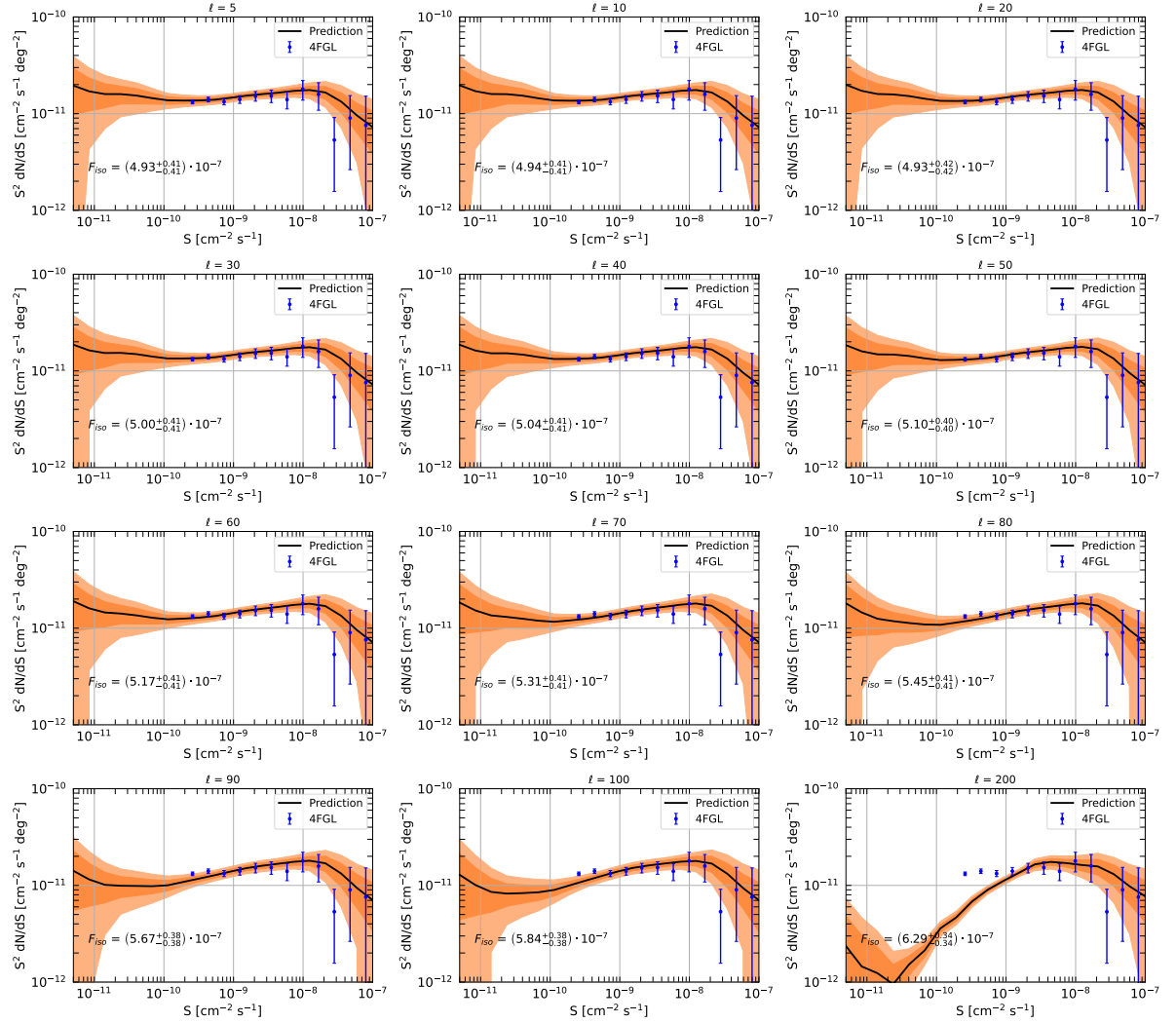


Fig. 6.22. Same as Figure 6.14 but where the *Fermi*-LAT map given as input to the CNN has been processed with the improved foreground cleaning procedure based on multipole decomposition described in the text. Each panel refer to a different maximal multipole $\ell_{\max}$ used for the cleaning.

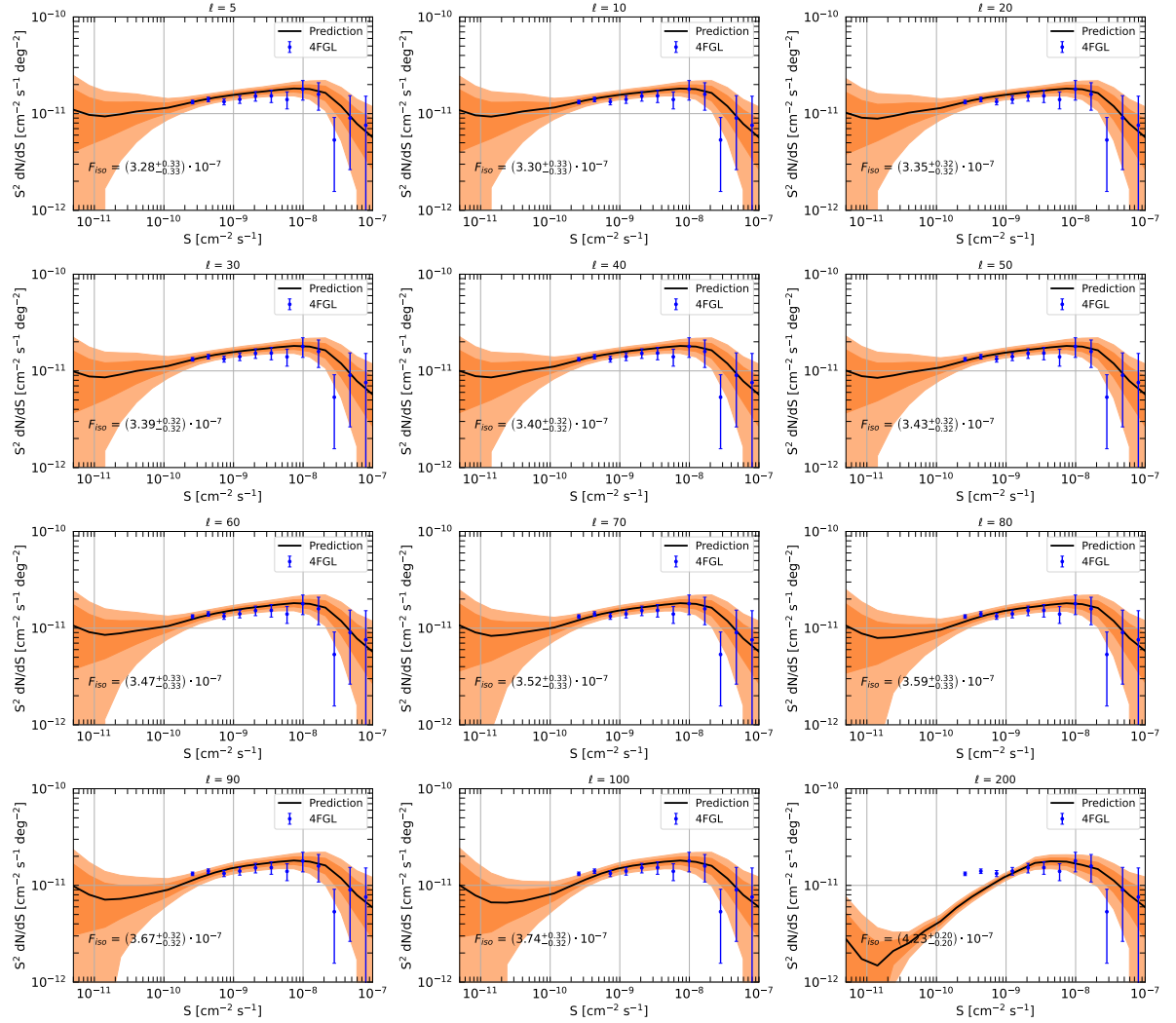


Fig. 6.23. Same as Figure 6.22 but using a CNN trained with the v05 version of the foreground model, as well as using as input a *Fermi*-LAT count map processed with the same foreground model.

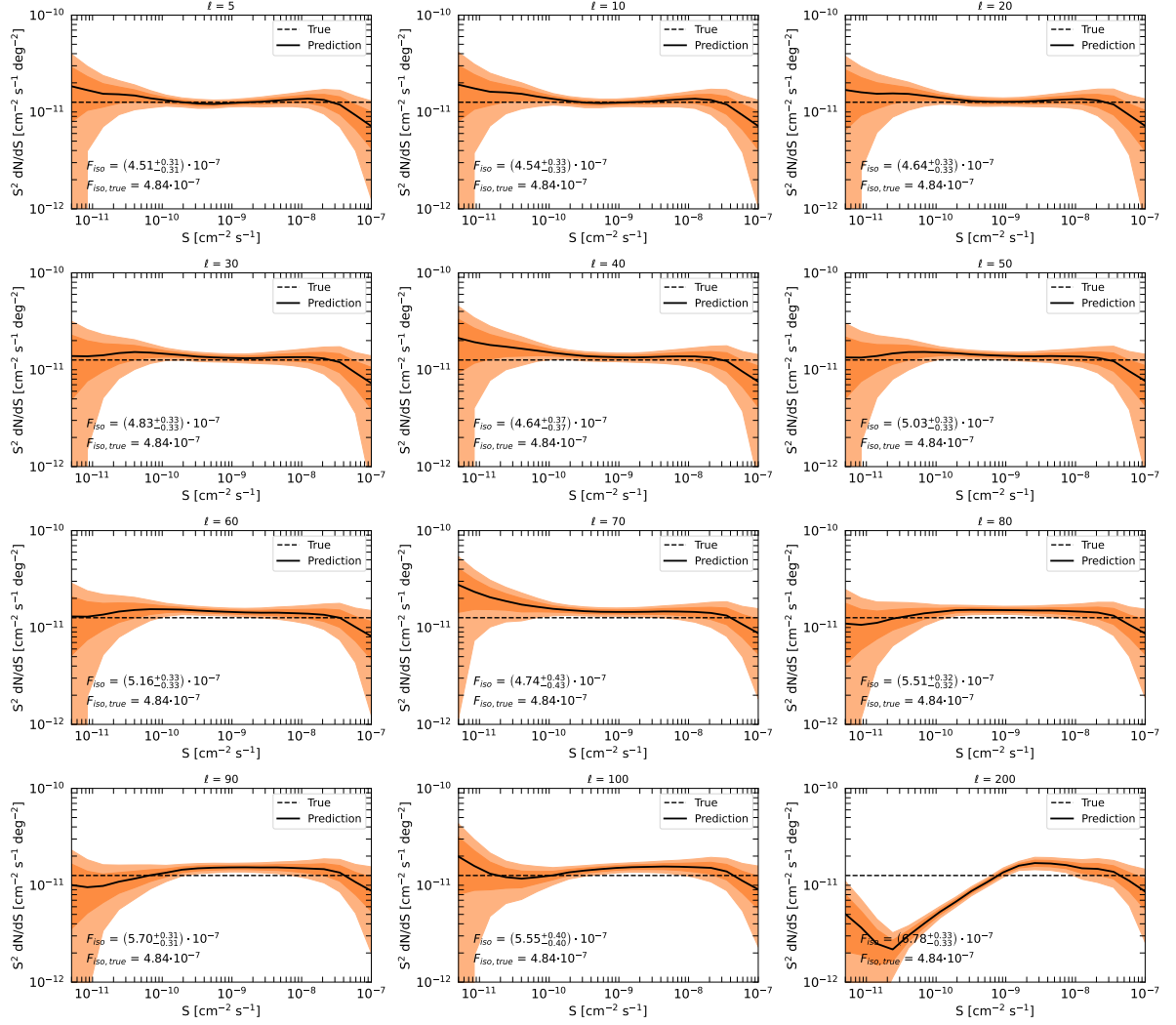


Fig. 6.24. Same as Figure 6.22 but using as input a synthetic map generated with a flat $S^2 \frac{dN}{dS}$ and with the v05 version of the foreground model.

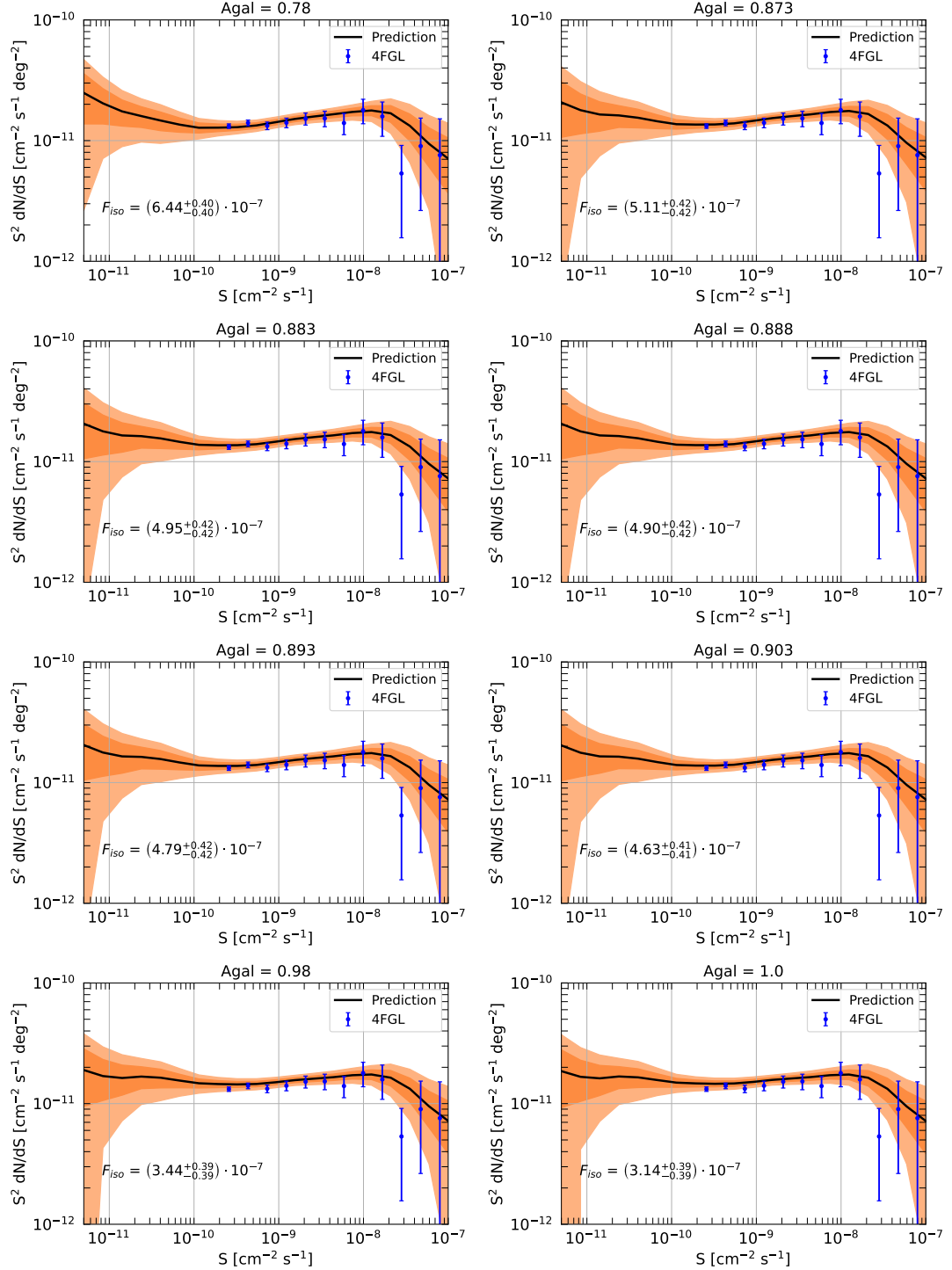


Fig. 6.25. Reconstructed dN/dS and F_{iso} and their Bayesian errors, when the trained CNN is applied to the *Fermi*-LAT map using different values of A_{gal} as indicated in the panel labels.

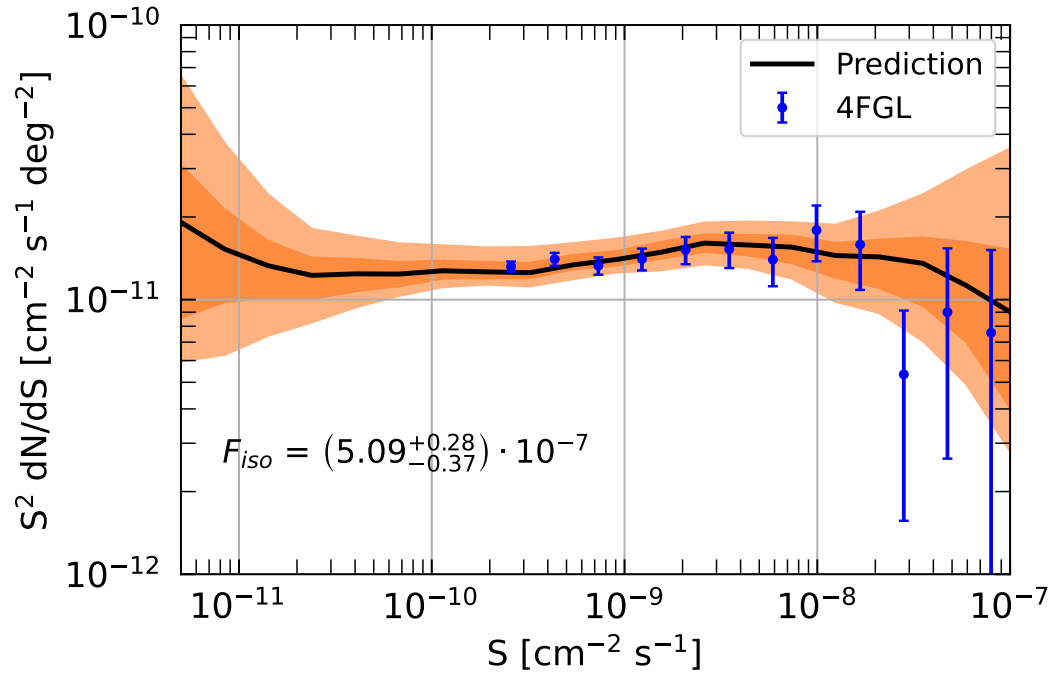


Fig. 6.26. Reconstructed dN/dS and F_{iso} and their errors, using a fully spherical CNN, trained on simulated maps and then applied to the *Fermi*-LAT map.

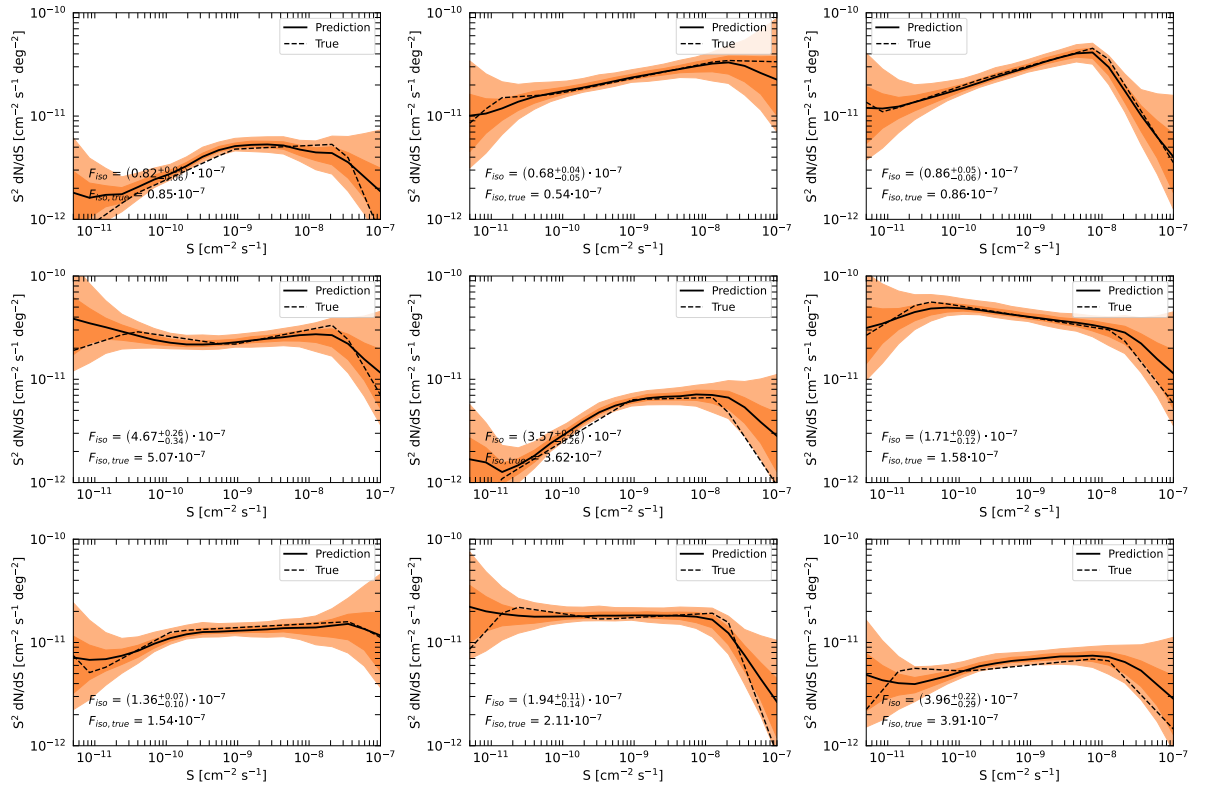


Fig. 6.27. Some example dN/dS from the validation dataset of the spherical CNN. Each panel shows the input dN/dS and F_{iso} and the corresponding quantities reconstructed by the CNN. The colored areas indicate the 1σ and 2σ uncertainty bands.

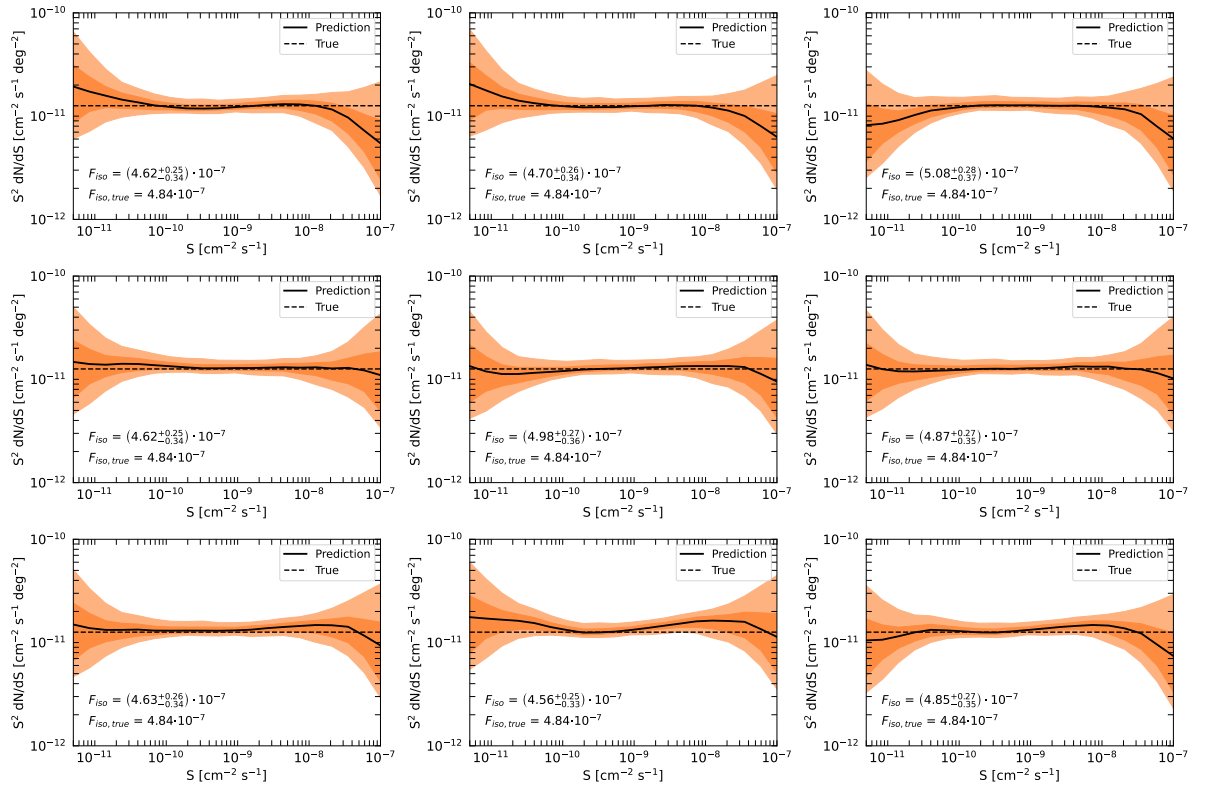


Fig. 6.28. Same as Figure 6.27 but for 6 different random realization of a flat $S^2 dN/dS$ and same F_{iso} .

PROBABILISTIC CATALOGING

The previous chapter demonstrated that the source-count distribution of faint gamma-ray emitters can be recovered from the photon-count statistics of the *Fermi*-LAT sky using simulation-based inference and deep learning. The resulting dN/dS extends as $\sim S^{-2}$ down to fluxes roughly 50 times below the nominal detection threshold, capturing a population of sub-threshold point sources — blazars, star-forming galaxies, misaligned active galactic nuclei — that contribute to the isotropic diffuse emission but remain individually undetected. Yet the dN/dS is a one-dimensional flux distribution: it tells us *how many* sources exist at each flux, but it says nothing about *where* any specific source is located. The challenge, then, is whether this population-level statistical knowledge can be turned into spatial information.

This chapter takes the next step by complementing the source-count distribution with positional identifications. The core idea is to use the recovered dN/dS as a generative model for ensembles of synthetic skies, enabling a frequentist statistical comparison that identifies candidate source directions below the detection threshold.

Section 7.1 examines why the standard fixed-threshold cataloging paradigm discards useful information and produces a spatially inconsistent detection scale. Section 7.2 describes the conceptual framework for converting the dN/dS into spatial source candidates through simulated-sky comparison, and introduces the frequentist meta-parameters that control the depth and purity of the resulting catalog. The remainder of the chapter presents the original research paper, which applies this framework to fourteen years of *Fermi*-LAT data and delivers a publicly available probabilistic catalog. Together with the dN/dS of Chapter 6, this probabilistic catalog completes the characterization of the astrophysical point-source foreground against which any residual dark matter contribution must be identified, as pursued in Chapter 8.

7.1 The Limits of Fixed-Threshold Cataloging

Before describing how the source-count distribution can be used to build a probabilistic catalog, we first examine the limitations of the standard approach. The fixed-threshold paradigm behind the *Fermi*-LAT catalogs has been in use for over a decade, but it comes at two costs: a spatially

non-uniform detection significance, and the loss of all sub-threshold information. We take these in order.

7.1.1 The Standard Catalog-Construction Paradigm

The *Fermi*-LAT fourth source catalog (4FGL) is the most comprehensive catalog produced from the instrument described in Chapter 2, incorporating over a decade of survey-mode observations in the gamma-ray band [121]. Point-source detection in this catalog follows a well-defined statistical procedure. In each region of interest (RoI) partitioning the sky, the analysis evaluates a likelihood-ratio test statistic defined as $TS = 2\ln(\mathcal{L}/\mathcal{L}_0)$, where $\mathcal{L}$ is the maximum likelihood including the source model and $\mathcal{L}_0$ is the maximum likelihood for the background-only hypothesis [121]. Sources are retained in the catalog if they exceed a threshold of $TS > 25$, corresponding to a significance of approximately 4σ when evaluated against a χ^2 distribution with four degrees of freedom (two positional and two spectral parameters for a power-law model) [121]. Any candidate that falls below this threshold is removed from the model, and the fit in that RoI is readjusted accordingly.

This detection procedure is inherently local. The sky is divided into roughly 1750 RoIs, each containing a core region of actively fitted sources and an outer ring whose width adapts to the energy-dependent point-spread function [121]. Within each RoI, the normalization and spectral tilt of the Galactic diffuse emission template and the normalization of the isotropic component are fitted independently, and the log-likelihood is further split into components defined by energy intervals and event types before being summed for optimization [121]. Since the background model parameters are determined locally, the same numerical TS value carries different statistical weight at different sky positions. A source yielding $TS = 30$ in a sparse high-latitude field sits on a very different background than one with $TS = 30$ in the crowded Galactic plane. The resulting detection scale is therefore spatially inhomogeneous: the TS encodes the local excess above the locally fitted background, not a globally calibrated measure of source significance.

The fixed detection threshold imposes a binary classification on the sky. Every direction is either “in” the catalog ($TS \geq 25$) or “out,” and the information carried by pixels with TS values just below the cut — pixels that may well host genuine astrophysical sources — is entirely discarded. The construction of each successive data release inherits this binary structure. While such a threshold is operationally necessary to maintain a controlled false-positive rate across thousands of candidate seeds, it discards a range of TS values that, as we show below, can be recovered by comparing observed pixels against simulated sky ensembles.

7.1.2 The Information Below Threshold

The fixed detection threshold does not only exclude noise. Genuine astrophysical sources exist below $TS = 25$ — the dN/dS recovered in Chapter 6 confirms this directly — and their collective emission contributes to the high-latitude gamma-ray sky at a level that depends on the population model assumed. As discussed in Section 6.2, blazars, misaligned active galactic nuclei, star-forming galaxies, and millisecond pulsars all contribute to this unresolved emission, and their aggregate flux may account for a large fraction of the unresolved gamma-ray background [104, 111], though the individual budget shares remain model-dependent.

The source-count distribution recovered in Chapter 6 (cf. Section 6.2) quantifies this sub-threshold population directly. The dN/dS measured from high-latitude photon counts extends as $\sim S^{-2}$ down to fluxes of approximately $5 \times 10^{-12} \text{ cm}^{-2} \text{ s}^{-1}$, roughly 50 times below the nominal catalog sensitivity in the 1–10 GeV band [304] (cf. Chapter 6). This faint extension implies that many sources whose individual signals fall just below the $TS = 25$ cut-off carry real astrophysical

information yet are systematically excluded from every standard catalog. For science cases that depend on sample size rather than per-source purity, this information loss is particularly costly. Cross-correlation studies between gamma-ray maps and galaxy catalogs, for instance, gain statistical power from a larger sample of source directions, even at the price of a moderate false-positive fraction (cf. Chapter 8). Similarly, multi-wavelength source identification campaigns benefit from having a larger set of candidate positions to follow up.

The practical consequences of the fixed threshold become visible in the incremental catalog releases. In the 4FGL-DR3, which extends the observation baseline from eight to twelve years, 112 sources originally included in the first data release dropped below $TS = 25$, fifteen of which fell below $TS = 16$, with the lowest reaching $TS = 9.8$ [228]. An additional 69 sources from the second data release also fell below threshold [228]. These sources were retained for continuity and explicitly flagged, but the episode exposes a structural problem: highly variable objects such as flat-spectrum radio quasars can be bright enough to enter the catalog during one observation window and fade below threshold in a subsequent accumulation [136]. Lowering the threshold does not help, since it would increase false detections from background fluctuations [136]. What is needed is a framework that replaces the binary in-or-out decision with a simulation-calibrated probability of source presence, informed by the sub-threshold population measured through the dN/dS .

7.2 From Population Statistics to Spatial Information

The previous section showed that a fixed TS threshold discards sub-threshold sources whose aggregate properties — including the dN/dS — are measurable from photon-count statistics [273, 274]. The dN/dS is, however, a purely statistical quantity: it tells us how many sources exist at each flux level, but nothing about where they are. This section describes how the dN/dS can be complemented with spatial information by comparing ensembles of simulated skies against the real data.

The idea, developed in full in the paper body (Section 7.3), is to use the recovered dN/dS as a generative model for synthetic skies and then compare their pixel-level TS distributions with the real *Fermi*-LAT sky using a frequentist test. Section 7.2.1 describes the simulated-sky comparison at a qualitative level; Section 7.2.2 introduces the statistical machinery — a per-pixel TS , a KS test, and a Quality Factor — that makes the comparison quantitative. Full details appear in the paper body.

7.2.1 The Core Idea: Simulated-Sky Comparison

As established in Section 7.1, the dN/dS recovered in Chapter 6 extends to fluxes roughly 50 times below the *Fermi*-LAT detection threshold (cf. Section 6.2), yet it carries no spatial content whatsoever. The question we now address is whether this population-level flux distribution can be leveraged to identify specific sky directions that are likely to host sub-threshold point sources.

The idea is straightforward. If the dN/dS is known, one can use it as a generative model to produce realistic synthetic skies. For each realization, source fluxes are drawn from the dN/dS , with the expected number of sources per flux bin given by the integral of dN/dS over that bin and the actual number drawn from a Poisson distribution centered on this expectation. Since the dN/dS carries no positional information, these synthetic sources are placed at uniformly random positions on the sphere. Each synthetic sky is then processed through the same pipeline applied to the real data: the point sources are convolved with the *Fermi*-LAT PSF, multiplied by the exposure map, and converted into a pixelized photon-count map on top of a modeled diffuse

background (see Section 7.3 for full details). By repeating this procedure 5000 times — each time sampling a slightly different dN/dS realization within the uncertainties estimated via Gaussian Process interpolation — one obtains an ensemble of synthetic skies that collectively encode what the gamma-ray sky should look like if the recovered dN/dS is correct.

The comparison between synthetic and real skies cannot be performed on a pixel-by-pixel basis, because the synthetic sources are randomly positioned and their spatial pattern never matches the real sky. Instead, the relevant comparison is statistical: one computes a pixel-wise test statistic (TS) for every pixel in both the real and synthetic maps and then compares the resulting TS distributions. At high TS, where bright sources dominate, the synthetic and real distributions agree well. As one moves to lower TS values, the simulated distributions eventually diverge from the real sky, because background modeling systematics become increasingly important at faint fluxes. The threshold at which this divergence becomes statistically significant determines how deep below the nominal catalog limit the method can reliably probe. Pixels in the real sky whose TS exceeds this threshold are labeled “firing pixels” — candidate directions that are likely to host a genuine point source.

[AA: idk if I want to keep this part] This approach differs from earlier efforts at probabilistic gamma-ray cataloging. The Bayesian framework PCAT [305] samples posterior distributions of full catalogs — including source positions, fluxes, and spectra — using transdimensional Markov Chain Monte Carlo. While PCAT provides a complete probabilistic description of the source population in a given region of interest, its computational cost scales roughly as the square of the number of modeled sources, which limits its applicability to restricted sky areas. In its original implementation, PCAT was applied to a $40^\circ \times 40^\circ$ patch around the North Galactic Pole, requiring approximately 250 CPU hours for a single inference run [305]. Machine learning approaches to probabilistic cataloging have also been explored [136], though these too have been applied to localized regions rather than to the full high-latitude sky.

The simulated-sky comparison framework adopted here takes a different route. Rather than sampling the full catalog posterior, it reduces the problem to a comparison of one-dimensional TS distributions, which is computationally inexpensive and naturally scalable to the entire high-latitude sky. The price for this simplification is that the method does not return individual source parameters (flux, spectrum, position uncertainty); its output is a map of candidate firing pixels at a chosen pixel resolution. This is sufficient for the downstream application we have in mind — cross-correlation studies (Chapter 8) — where the relevant quantity is the number and angular distribution of candidate source directions, not the properties of individual sources. We now turn to the statistical framework that makes this comparison quantitative.

7.2.2 A Frequentist Framework: TS, KS Test, and Quality Factor

The framework outlined in the previous section is entirely frequentist. The dN/dS enters as a generative model for the synthetic skies, not as a Bayesian prior to be updated. Three quantities define the statistical procedure.

A per-pixel test statistic quantifies how much the observed photon count in each pixel deviates from the background-only expectation. Unlike the TS used by the *Fermi*-LAT collaboration, which is computed locally in each region of interest and therefore carries spatially varying significance, the TS adopted here is a single, globally coherent quantity defined identically for every pixel. It is treated as a heuristic “signal interest label” whose probabilistic meaning comes entirely from the comparison with simulated skies, not from an assumed analytic distribution.

A Kolmogorov–Smirnov test then compares the cumulative TS distributions of the real sky and each synthetic realization above a minimum threshold TS^* . At high TS^* , where bright

sources dominate, the distributions agree; as TS^* is lowered, background systematics cause them to diverge, setting the depth to which the method can probe.

[AA: check again the definition of the QF] Finally, a Quality Factor (QF) — the fraction of the 5000 synthetic realizations passing the KS test at a given TS^* and significance level α — controls the purity of the selection. A high QF yields a conservative set of firing pixels; a lower QF extends coverage deeper into the sub-threshold regime at the cost of more spurious directions. The pair (α, QF) sets the operating point, and both can be tuned to the downstream application.

Two features of this framework are worth noting. First, the globally homogeneous TS scale — in contrast to the *Fermi*-LAT catalog’s locally computed TS — is particularly advantageous for statistical analyses that combine information across large sky areas, such as the cross-correlation studies discussed in Chapter 8. Second, at the chosen pixel resolution ($N_{\text{side}} = 512$, corresponding to $\sim 0.12^\circ$), a single astrophysical source may produce multiple adjacent firing pixels due to PSF spreading, and conversely, multiple faint sources may contribute to a single pixel. The method’s output is therefore a map of firing pixels, not a traditional source catalog.

The remainder of this chapter presents the full analysis: the data selection, the detailed statistical procedure, the results for different (α, QF) operating points, and the validation against the 4FGL-DR3 catalog.

7.3 Deepening gamma-ray point-source catalogues with sub-threshold information

The goal of this analysis is to probabilistically extend the *Fermi*-LAT source catalogue exploiting the dN/dS recovered in Chapter 6. We do so by proposing a new methodology applied to the *Fermi*-LAT data which consists in a) proposing a test statistic (TS) for the gamma-ray counts in the sky, whose distributions across spatial pixels corresponding to the *Fermi*-LAT measured map and those coming from the simulations can be compared; and b) identifying the pixels in the measured map whose TS values are large enough depending on the previous comparison, therefore overcoming the limitation of the dN/dS analysis of not being able to provide spatial information.

In Section 7.3.1 we describe the data selection, while Section 7.3.2 introduces the null- and alternative-hypothesis models used in the analysis. Section 7.3.3 is devoted to the quantitative setting of the problem and the statistical procedure followed. In Section 7.3.4 we present the results, which are also made publicly available at <https://doi.org/10.5281/zenodo.8070852>. In Section 7.3.5 we outline some perspectives and present our conclusions.

7.3.1 Data selection

We consider the updated 14 years *Fermi*-LAT data set (from week 9 to week 745) in the (1, 10) GeV energy range¹ and Pass 8 event selection [118, 119]. This ensures a good balance between high statistics and good angular resolution of the detector. We employ the Fermi Science Tools suite version 2.0.8 [281] to analyse *Fermi*-LAT data with the following settings: We adopt the P8R3_SOURCEVETO_V3 instrument response functions (IRF), event class (EVCLASS) 2048 (SOURCEVETO) and event type (EVTYPE) 1 (FRONT). We used standard quality selection criteria, i.e. DATA_QUAL==1 and LAT_CONFIG==1. Atmospheric γ rays from the Earth limb emission are removed by adopting a cut on the maximum zenith angle ZMAX of 90 degrees. We consider FRONT events in order to have an optimal angular resolution, and the SOURCEVETO class of events, in

¹The energy range is chosen for consistency with the dN/dS determination used below.

Healpix order	9, 10
Weeks	9 – 745
E _{min}	1 GeV
E _{max}	10 GeV
Instrument Response Functions (IRFs)	P8R3_SOURCEVETO_V3
EVCLASS	2048 (Source Veto)
EVTYPE	1 (Front)
ZMAX	90

Table 7.1. Fermi Science Tools settings used for the 14-year data set analysis.

order to have a good suppression of the charged cosmic-ray background while still retaining a large event statistics. In table 7.1, we provide the reader with a summary of the settings.

We employ the Fermi Science Tools to compute photon counts map, exposure map and point spread function (PSF) of the LAT. The maps are binned in N_{pix} pixels with equal area relying on HEALPix.² The chosen pixelization is expressed in terms of the “resolution parameter” N_{side} controlling the number of subdivisions of a great circle on the sphere, related to N_{pix} via $N_{\text{pix}} = 12N_{\text{side}}^2$. As we motivate in section 7.3.4, we simulate the sky with $N_{\text{side}} = 1024$, but run the pixel analysis with $N_{\text{side}} = 512$.

We perform an energy-integrated analysis in the (1, 10) GeV energy range, but our procedure can be extended to multiple energy bins.

7.3.2 Null- and alternative-hypothesis models

Our goal is to devise an algorithm able to detect point sources as γ -ray excesses on top of a Poisson-distributed model for the background diffuse emission components. In this respect, our null-hypothesis model only consists of diffuse emission components, while the alternative hypothesis model also contains point sources.

We construct our background-only map $\mathcal{B}$ including two components:

$$\mathcal{B} = A_{\text{gal}} \cdot \mathcal{G} + F_0, \quad (7.3.1)$$

where

- F_0 is an isotropic (usually assumed extragalactic) component accounting for the average flux of all the unresolved sources as well as possible diffuse emission mechanisms.
- $\mathcal{G}$ is the gll_iem_v07 template morphology of the diffuse γ -ray emission from the Milky Way provided by the *Fermi*-LAT collaboration³, with normalization A_{gal} .

The parameters A_{gal} and F_0 are determined via a fit to the *Fermi*-LAT photon counts map, as explained in section 7.3.3. In order to obtain a background map in units of counts in each of the N_{pix} pixels, we multiply by the exposure map $\mathcal{E}$ and the steradian-to-pixel conversion factor $4\pi/N_{\text{pix}}$:

$$C_{\mathcal{B}} = \frac{4\pi}{N_{\text{pix}}} \int_{1\text{ GeV}}^{10\text{ GeV}} \mathcal{B}(E) \mathcal{E}(E) dE. \quad (7.3.2)$$

The exposure map is extracted using the Fermi Science Tools for 10 logarithmic bins in the (1, 10) GeV range (and treated as the corresponding piecewise function when performing the integral), in order to be consistent with the binning of the Galactic foreground model. We also

²<https://healpix.sourceforge.io>

³<https://fermi.gsfc.nasa.gov/ssc/data/access/lat/BackgroundModels.html>

account for the PSF, responsible for an angular smoothing of the map, via the Fermi Science Tools. Similarly to the map $\mathcal{E}$, we average over the (1, 10) GeV range with a $E^{-2.4}$ weight, according to the overall energy dependence of the data at high Galactic latitudes [283].

On the other hand, our first alternative hypothesis model is based on the 4FGL-DR3, and will be used to validate our procedure and TS analysis on the official *Fermi* catalogue. We generate a new map

$$\mathcal{K} = \mathcal{C} + A_{\text{gal}} \cdot \mathcal{G} + F_0, \quad (7.3.3)$$

including the same background model as in eq. (7.3.1) (i.e. same parameters for A_{gal} and F_0), and adding a synthetic map associated to the list of sources in the 4FGL-DR3 catalogue with $|b| \geq 30^\circ$, $\mathcal{C}$. The map is appropriately smoothed via the PSF and convoluted with the exposure map, and converted into a pixelized count map, as previously described.

Finally, as our second alternative hypothesis we consider a model for point sources based on the source-count distribution derived in [272]. Analogously to the catalogue, we include an additional source component, $\mathcal{S}$, on top of the background-only model:

$$\mathcal{M} = \mathcal{S} + A_{\text{gal}} \cdot \mathcal{G} + F_{\text{iso}} \quad (7.3.4)$$

The map $\mathcal{S}$ is constituted of point sources randomly placed on the sphere, drawn according to the dN/dS inferred in [272]. We produced 5000 realisations of this source model, by considering variations of the dN/dS parameters inferred in [272]. In particular, we adopt the model related to the `gll_iem_v07` foreground template ⁴ and $|b| < 30^\circ$ Galactic plane cut, and we vary the dN/dS within the estimated uncertainties by fitting a Gaussian Process (c.f. [306]) to the dN/dS output of the neural network ⁵. For each parameter configuration of the dN/dS , the F_{iso} is consistently determined by the procedure proposed in [272]. Note that the residual Poisson-distributed isotropic component F_{iso} is now lower than F_0 in eq. (7.3.1), since part of the flux F_0 is accounted for by the discrete sources.

We note that both the $\mathcal{K}$ and $\mathcal{M}$ maps are simulated, i.e. are “theoretical skies”, not the “measured sky”. The map $\mathcal{K}$ is deterministic, meaning we simulate it by accounting for the Fermi PSF for the point-like sources whose fluxes and positions are given by the 4FGL catalogue, on the top of the assumed smooth background (the remaining contributions to the $\mathcal{K}$ map in eq. 2.3). In the case of the map $\mathcal{M}$, we rely on statistical information, in which the dN/dS does not describe the position of any gamma-ray source, nor does it tell us their exact number. Rather, the integral of dN/dS in a flux bin tells us the expected number of sources in that bin, while the actual number of sources is only defined up to Poisson fluctuations. Once such a number of sources per flux bin has been determined, we place them at uniformly random positions in the sky, accounting for the $\mathcal{S}$ term in Eq. (7.3.4). In summary, we run the simulator developed in [272] in order to build the maps $\mathcal{M}$.

7.3.3 Problem setting and statistical framework

Our first step is the determination on the null-hypothesis model, consisting solely of Poisson-distributed background emission. To this end, we apply a mask of $|b| < 30^\circ$ around the Galactic plane, and remove the emission of all sources in the 4FGL-DR3 catalogue, further masking a 1°

⁴This is the latest foreground model provided by Fermi-LAT.

⁵The reason for considering a Gaussian Process here is two-fold: to interpolate the dN/dS to values of the flux not included in the neural net output, as well as to have an estimation of the uncertainty band of dN/dS at such interpolated values.

disk around each source centroid. This cut roughly corresponds to the 90% containment angle of the PSF and represents a good compromise between removing the bulk of the pointlike emission while avoiding a too large removal of the diffuse emission area. We checked that the results are not particularly sensitive to the exact cut used in the $1^\circ - 2^\circ$ range. We then compute the expected photon counts λ_i of model $\mathcal{B}$ (depending on A_{gal} and F_0) in the pixel i and, given the actual counts k_i , define the Poissonian likelihood

$$\mathcal{L}_0 = \prod_{i=1}^{N_{\text{pix}}} \frac{\lambda_i^{k_i} e^{-\lambda_i}}{k_i!}. \quad (7.3.5)$$

The adopted parameters A_{gal} and F_0 are those maximising the likelihood above. For $N_{\text{side}} = 512$, we find $A_{\text{gal}} = 0.914$ and $F_0 = 4.81 \cdot 10^{-7} \text{ cm}^{-2} \text{ s}^{-1} \text{ sr}^{-1}$.

Our second step is to quantify the significance of additional point sources on top of the null-hypothesis model. For this purpose, we define a test statistic (TS) function in each pixel, quantifying how well the observed photon counts agree with the background-only hypothesis:

$$TS_i \equiv \frac{(x_i - \lambda_i)^2}{\lambda_i}, \quad (7.3.6)$$

where we indicate with λ_i the best-fit background-only model counts in the pixel i , computed according to eq. (7.3.1); x_i can be either the observed photon counts (when applied to the actual photon count map of data, leading to the single set TS_i^{data}) or the photon counts in pixel i generated via synthetic maps. If the maps are generated according to the hypothesis eq. (7.3.3), the procedure yields the single set TS_i^{cat} ; if the maps are randomly taken from the model of eq. (7.3.4), we obtain a number of sets TS_i^{sim} equal to the number of realisations. The TS choice of eq. (7.3.6) is inspired by Pearson's χ^2 test statistic for Poisson distributed counts, whose variance $\sigma_i^2 = \lambda_i$. This choice also loosely mimics the one adopted by the *Fermi*-LAT collaboration, albeit no exact correspondence can be expected. In fact, *Fermi*-LAT uses a recursive procedure where, after a first step similar to the above, energy-dependent renormalisations of the background are allowed independently in each “region of interest”. We prefer to use a simple and coherent definition valid for all pixels in the sky, also because we do not assume a particular probability distribution for the TS values. Rather, we use it as a “signal interest label”, and derive a probabilistic interpretation for it from simulations of synthetic maps (see below).

Obviously, since in our simulations the sources are uniformly distributed across the sky, the spatial distribution of TS_i^{sim} never coincides with the one obtained from the actual data, TS_i^{data} , nor with the one from the 4FGL-DR3 catalogue, TS_i^{cat} . However *the histogram of the TS_i^{sim} values should statistically match the actual one TS_i^{data} , if the functional form and inferred parameters of the underlying dN/dS are correct*. Since the determination of dN/dS exploits information below the nominal catalogue's threshold, the statistical agreement between the actual map and the synthetic maps should hold at values below this threshold and inform us about new, faint sources. This is the gist of the argument that we exploit to extend *statistically* the *Fermi*-LAT catalogue below threshold. This expectation is illustrated in figure 7.1, where we report the descending cumulative TS distribution for real data (black), the 4FGL-DR3 catalogue source model, $\mathcal{K}$ (gray), and for several realisations of the dN/dS source model, $\mathcal{M}$ (blue). While the distributions match rather well at high TS , the one associated to the catalogue departs from the data one well before the simulation's ones do.

As a next (and third) step, we attribute a statistical meaning to the TS distributions. At large TS , we expect the observed map and the simulated ones to agree well: Visible (but insignificant)

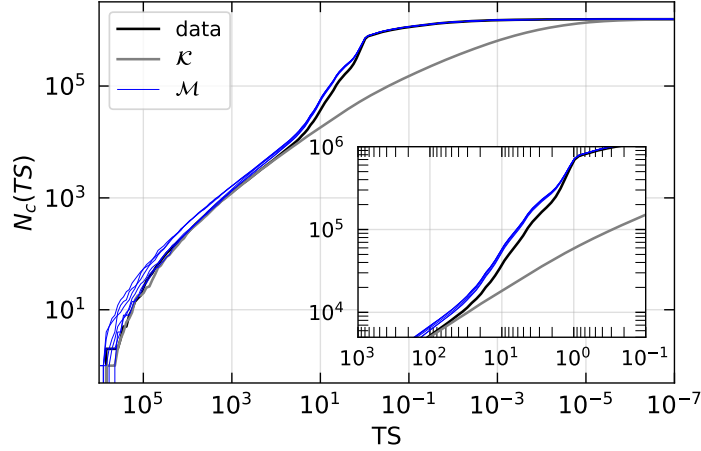


Fig. 7.1. Descending cumulative TS distribution for the maps corresponding to: real data (black), 4FGL-DR3 catalogue (gray) and a few realisations of the dN/dS source model (blue).

discrepancies are expected at the highest values of TS , where Poisson fluctuations are important. At low TS we expect the deviations from the two distributions to eventually become statistically significant. This is due to the interplay of the weakness of the sources and the systematic errors in the background template, that are more and more relevant at low fluxes.

To assess how well the two distributions are in agreement, we apply a two-sample Kolmogorov-Smirnov (KS) test [128] to the normalised cumulative distribution functions of real data and simulated sky (or catalogue sky), both cut at a minimum TS_* , which we vary on a grid. Their compatibility depends on the confidence level $1 - \alpha$ at which we set the test. This is strictly speaking arbitrary, although conventionally α is set to 0.05 in most applications. We will actually consider three values of α : 0.01, 0.05 and 0.1, to assess the dependence of the results on this meta-parameter. In general, increasing α yields more conservative constraints, as the criterion to reject the null hypothesis becomes more restrictive (i.e. the distributions should agree more closely).

For the single catalogue model in eq. (7.3.3), we present the results in terms of the TS_* leading to agreement with real data as a function of α .

In the case of the dN/dS model, since we have multiple realisations, we quantify what is the fraction of simulations which passes the test against the true sky distribution as a function of TS_* (and α). This provides a *quality factor* QF that we associate to the considered cut. Lacking a better criterion, a reasonable choice may be to require a majority of realisations to lead to an agreement (i.e. $QF > 0.5$), but it is strictly speaking another meta-parameter which depends on the criteria of the user. We will also show how results depend on QF , with QF 's closer to 1 associated to more restrictive requirements. Note that we will “naively” interpret the QF as probability of the TS being associated to a source, by considering the whole sample of simulations as equally viable (i.e. only the prior information based on the pixel statistics is used). In principle, one could refine this indicator by restricting the sample of simulations to the ones which reproduce the high- TS distribution. We expect the actual probability to be somewhat higher than the one estimated via the QF . Also note that nowhere in our procedure we have relied on the 4FGL catalogue as input.⁶

⁶Technically, we mask the catalogue position to fit the background model, but this is done only to lead to a better

The final step is how to identify the “firing pixels” in the true sky which are going to constitute the main deliverable of our work. For a fixed KS test significance α and a quality factor QF , we obtain a value TS_* above which a fraction QF of simulations is compatible with the measured sky. Those pixels i in the latter having $TS_i > TS_*$ will simply constitute of candidate pixels.

We conclude this section with a general remark. We note that our procedure treats every pixel independently, assigning a TS value to each of them, regardless of the TS values of the neighboring pixels. In this respect, it is useful to distinguish between a source and a “firing pixel”: Multiple pixels may be firing (i.e. having a sufficiently large TS) in association to a single source, depending on the choice of N_{pix} , the intensity of the source and the PSF of the instrument (indirectly, thus, also on the spectrum of the source). This instance is more and more likely the smaller the pixel size, i.e. the larger N_{pix} , is. Conversely, multiple sources may be associated to a single pixel, notably for choices of N_{pix} which are too low. Clearly, for the exercise proposed here to be useful, we must require that the choice of N_{pix} leads to a number of pixels in the non-masked sky sufficiently larger than the current number of sources in the catalogue in the same region. At the same time, a pixel size much smaller than the angular resolution would be meaningless. Keeping this in mind, for the sake of simulations, we fix $N_{\text{side}} = 1024$, which corresponds to an angular resolution of approximately 0.06° , in order to match the angular resolution of the Galactic foreground template. However, since the *Fermi*-LAT angular resolution in the (1,10) GeV energy band is at best of the order of 0.15° ⁷, for the sake of our pixel analysis, we will downgrade our simulations to $N_{\text{side}} = 512$, which corresponds to an angular resolution of approximately 0.12° . The choice of this value for N_{side} ensures that there are considerably more pixels than the number of resolved sources (approximately 500 times the number of resolved sources in the region $|b| > 30^\circ$) and that the pixel size of our analysis roughly matches the best resolution of the LAT at the energies of interest. The TS analysis is therefore performed with synthetic and real data maps pixelized with $N_{\text{side}} = 512$.

7.3.4 Results

We comment here on some summary statistics and synthetic analysis of our main results. In table 7.2 we report, for different values of α , the TS_* above which the 4FGL-DR3 catalogue source model and real data pass the KS test (second column, TS_*^{cat}), and the corresponding TS_* where the dN/dS simulated sky maps do the same (TS_*^{sim}), at different values of QF .

α	TS_*^{cat}	$TS_*^{\text{sim}}(QF = 0.9)$	$TS_*^{\text{sim}}(QF = 0.8)$	$TS_*^{\text{sim}}(QF = 0.5)$
0.01	53	57	42	33
0.05	62	147	48	36
0.10	63	307	74	37

Table 7.2. Required TS_* for three values of α (KS test confidence level) for the matching between 4FGL-DR3 catalogue source model and real data (second column, TS_*^{cat}), and the corresponding TS_* for the dN/dS simulated sky maps vs real data (rest of columns, TS_*^{sim}), at different values of QF .

background model and thus a more meaningful TS scale. In principle, we could fit the background model without masking the catalogue source positions. That would alter our TS scale, but the same bias will affect the simulated sky maps that we use to attribute a statistical meaning to the TS .

⁷<https://fermi.gsfc.nasa.gov/science/instruments/table1-1.html>

α	$\widetilde{TS}_*^{\text{sim}}(QF = 0.9)$	$\widetilde{TS}_*^{\text{sim}}(QF = 0.8)$	$\widetilde{TS}_*^{\text{sim}}(QF = 0.5)$
0.01	181	161	104
0.05	301	180	148
0.10	400	200	160

Table 7.3. Required TS_* for three values of α (KS test confidence level), for the dN/dS simulated maps to match the 4FGL catalogue model map, at different values of QF . Note that the KS test is different from the one in table 7.2, where the matching was with respect to the *Fermi*-LAT data map.

The first conclusion is that, provided that the choice of QF (and/or of α) is not too stringent, our model maps $\mathcal{M}$ agree with the data better than the catalogue map does, i.e. the agreement extends to lower TS_* . Compare in particular the second with the last column in table 7.2. This is just a manifestation of the fact that, at least for a majority of realisations, including sub-threshold sources provides a much better description of *Fermi*-LAT sky map. On the other hand, requiring a too large QF may be counter-productive: Even at high- TS , there is a non-negligible fraction of the simulations that does not agree with the data as measured by the KS test. This manifests in the high values of TS_* found in the $QF = 0.9$ column and to some extent for $QF = 0.8$ as well.

A second sanity check is that the dN/dS model is better in agreement with the data than with the catalogue: This manifests in the values of $\widetilde{TS}_*$ (see table 7.3) for which the dN/dS model yields maps in agreement with the catalogue map $\mathcal{K}$, hence indicated with a tilde. As we can see from the table, the $\widetilde{TS}_*$ values are significantly higher than the corresponding TS_* reported in table 7.2.

A further natural question to ask oneself is if the statistical procedure leads to a competitive sample of candidate source positions in the sky, when compared to the *Fermi*-LAT catalogue analysis. After all, requiring an agreement of the TS histograms in the KS-sense is a *qualitatively* different procedure than the one used by the collaboration to infer their catalogue. A direct comparison is impossible, if anything since our discretisation of the sky only allows us to identify firing pixels, which are not in a one-to-one correspondence with sources, as discussed in the previous section. However, for a reasonable choice of the pixel size as the one adopted here (see discussion at the end of section 7.3.3) we expect the number of firing pixels to be a reasonable proxy of the number of underlying candidate sources. Our results are reported in table 7.4.

α	N^{cat}	$N^{\text{sim}}(QF = 0.9)$	$N^{\text{sim}}(QF = 0.8)$	$N^{\text{sim}}(QF = 0.5)$
0.01	7412	7050	8543	10068
0.05	6707	3975	7891	9636
0.10	6580	2492	5965	9409

Table 7.4. Number of firing pixels associated to the same cases of table 7.2.

A comment on this result is that the number of pixels inferred from the catalogue map is in the ballpark of the sources reported in the *Fermi*-LAT catalogue. This result was not obvious a priori. We may tentatively conclude that the power of the procedure we follow in identifying a firing pixel is at least roughly comparable to the criterion the *Fermi*-LAT collaboration follows in identifying a source. A second interesting conclusion is that the number of firing pixels can be significantly higher (up to a factor 1.5) if using the dN/dS model maps, at least if sufficiently

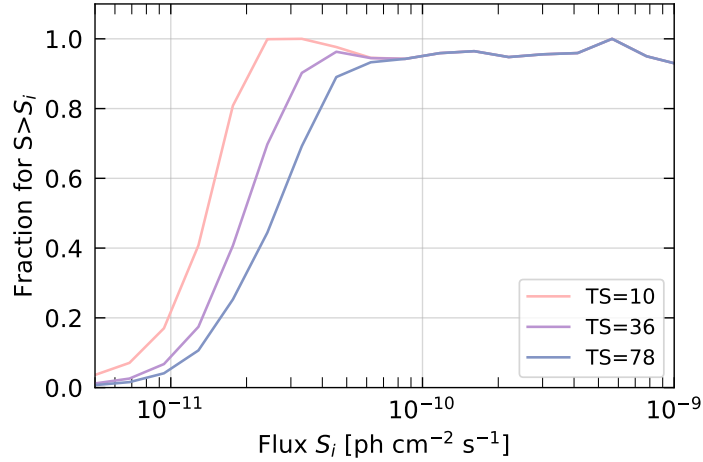


Fig. 7.2. Fraction of sources in the model $\mathcal{K}$ map, whose position corresponds to a firing pixel for different values of TS . See text for details.

loose criteria are used for QF and/or the level of KS agreement, i.e. α . We can conclude that our procedure can be used to increase the number of candidate pixels in the sky that we can probabilistically attribute to sources, with respect to the ones one would deduce by merely using the existing catalogue.

A question arises though: Are the identified pixels in the sky truly associated to sources? Of course we cannot answer this question without lengthy and careful follow-up studies and extensions of the present approach. We discuss some interesting directions in the next section. However, an interesting sanity check is that the bulk of the positions of *Fermi*-LAT sources should match part of the recovered firing pixel positions.

In figure 7.2 we compute, as a function of the flux, the fraction of all the pixels in the catalogue map $\mathcal{K}$, having a flux $S > S_i$, which coincide with the pixels of our firing pixels map obeying the same flux condition, for different TS cuts. The pixel coincidence is actually relaxed to a distance corresponding to the 68% containment angle of the PSF, which naively accounts for the uncertainty of the quoted source centroids in the 4FGL catalogue. In figure 7.2 we compute the fraction of all the pixels in the catalogue map $\mathcal{K}$ with flux $S > S_i$, whose positions are found in spatial coincidence with pixels in our firing-pixels map obeying the same flux condition.⁸ We display the fraction as a function of the flux and for different TS cuts.

Roughly⁹, *Fermi*-LAT starts to loose sensitivity to sources below $2 \times 10^{-10} \text{ cm}^{-2} \text{ s}^{-1}$ [269, 272, 274, 307], above the rising region of the sigmoid-like curves of figure 7.2. As expected, modulo fluctuations, the fraction tends towards unity at high fluxes. Note that we do not expect perfect recovering of the sources in the *Fermi*-LAT catalogue, anyway, either due to some simplifications (e.g. we use energy-integrated quantities for our analysis, as opposed to the differential energy analysis by the LAT collaboration) or methodological differences, such as the pixelisation procedure or the iterative background estimate procedure in *Fermi*. But it is clearly the case that we do recover the largest majority of the “bona fide” sources that we should.

Figure 7.3 gives us a more detailed look at the information contained in figure 7.2. In this

⁸The pixel-wise spatial coincidence is actually relaxed to a distance corresponding to the 68% containment angle of the PSF, which naively accounts for the uncertainty of the quoted source centroids in the 4FGL catalogue.

⁹More quantitative considerations require energy- and position-dependent assessments.

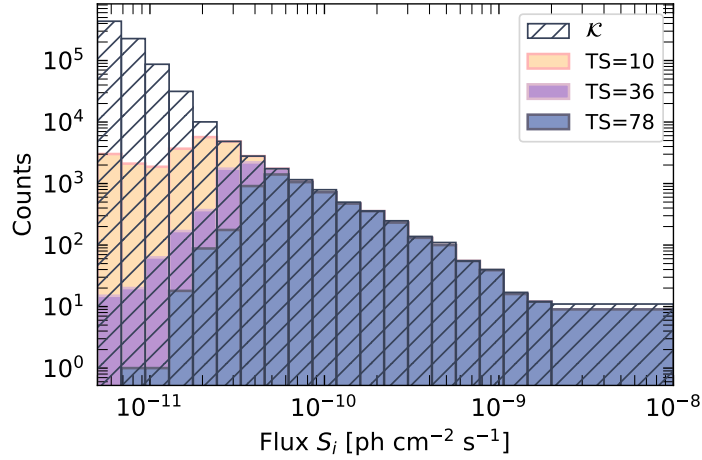


Fig. 7.3. Analogous to figure 7.2, but for differential flux; i.e. number of pixels for a given flux bin. White histograms correspond to the whole catalogue map $\mathcal{K}$, while colored histograms are only those pixels in $\mathcal{K}$ coinciding (within the 68% PSF containment angle) with the firing pixels of our dN/dS model map.

case we do not consider the “cumulative” number of pixels, but only those lying in a certain flux bin. The colored histograms correspond to the pixels in map $\mathcal{K}$ matching with the firing pixels of our dN/dS model map (related to the numerator of the fraction computed in figure 7.2). The white histogram instead correspond to all the pixels in map $\mathcal{K}$ (related to the denominator of the aforementioned fraction). The sigmoid curve in figure 7.2 grows close to unity at large S since—barring fluctuations—the quasi-totality of sources in the catalogue are found within the adopted cuts. When lowering the cut TS , the peak of the colored histogram shifts to lower S , while still preferentially populating the high- S tail (compare case $TS = 36$ with case $TS = 78$). If the TS is lowered too much, as illustrated by the case $TS = 10$, the matching starts developing a “flat tail” at lower S , hinting at a growing random match component. Eventually, the curves in figure 7.2 drop to zero at low S not because of the scarcity of matchings (i.e., the numerator), but due to the faster growing number of pixels having flux above the indicated value (i.e. the denominator).

As a further check, we assess how robust the selected catalogue is to variations of the background model. We compare the results obtained with the `gll_iem_v07` template with the previous incarnation of the galactic foreground model developed by the Fermi collaboration: `gll_iem_v05_rev1`. Since the TS^* scale and related QF are quantitatively different if the different backgrounds are used, we proceed by comparing the two results *at equal number of selected sources*. We select TS^* cuts delivering the same number of selected sources for the two cases and evaluate the exact overlap of pixel candidates. The results are reported in Table 7.5. As can be seen, within a few percent, the catalogues obtained agree.

The main output of our work consists of sky maps of “firing” pixels (i.e. source candidates). These pixels have a significant TS value among all pixels for which there is a statistical compatibility between simulations and real data, according to KS tests. The firing pixels depend on the significance α of the KS test, as well as on a “quality factor” QF , accounting for the fraction of simulated maps passing the KS test. We provide our results electronically in the form of a Python package called `gPCS` (for gamma-ray Photon Count Statistics), as well as a summary FITS, both

N	coincidence
3000	98.4%
5000	97.8%
9000	97.8%

Table 7.5. The fraction of firing pixels in common between the selections derived for the two galactic foreground models. The comparison is performed at equal number of selected sources for the two cases.

available on Zenodo: <https://doi.org/10.5281/zenodo.8070852>. The Python package can be used to determine the firing pixels given either a TS_* or a quality factor and α parameter. The Python package also includes a convenience function to export the firing pixels in the form of a FITS file, and we provide an example on how to compute the summary table provided with the package. The gPCS Python package can be easily installed using pip, with the command:

```
pip install gPCS
```

In order to get the firing pixels, given either TS_* or a quality factor and α , we can use the following snippet of code:

```
import numpy as np
from gPCS import gPCS

# specify manually a  $TS_{\text{star}}$ 
TS_star = 36
pixel_firing = gPCS.get_firing_pixels(TS_star, filter=False)
print(len(pixel_firing))

# Compute the  $TS_{\text{star}}$  from a chosen QF and alpha
# [alpha must be 0.01, 0.05 or 0.1]
QF = 0.5
alpha = 0.05
TS_star = gPCS.get_TS_from_QF(QF, alpha=alpha)
pixel_firing = gPCS.get_firing_pixels(TS_star, filter=False)
print(len(pixel_firing))
```

It is possible to export a FITS table using the `export_fits_table` function. For example to reproduce the table provided with the library we can use:

```
gPCS.export_fits_table(filename="firing_pixels.fits", QF=0.50, alpha=[0.01, 0.05, 0.1])
```

The FITS table has the following fields:

```
pixel      : pixel index
TS         : TS value for each pixel
QF_best    : QF value obtained by considering all the simulations
QF_min     : lower bound of the QF range
QF_max     : upper bound of the QF range
```

For further details on the package usage, please refer to the documentation and examples: <https://github.com/aurelio-amerio/gPCS>.

7.3.5 Discussion and conclusions

We have presented an approach to statistically push the *Fermi*-LAT sensitivity to point-like sources at high latitudes below the current threshold for detection, leveraging on the fact that the underlying source count distribution function is constrained even for lower test statistics (TS). This function has been recently re-derived using machine learning methods, yielding results in accordance with existing pixel statistics analyses. This information, not accounted for in the current catalogue construction procedure, allowed us to provide a catalogue of directions (i.e. firing pixels) in the sky likely to be associated to sources, albeit only in a probabilistic sense, that we assessed via a Kolmogorov-Smirnov (KS) test. The actual catalogue of directions depends on meta-parameters such as the confidence level $1 - \alpha$ of the KS test, or the fraction QF of simulated skies for which the simulation agrees with data above a certain TS level. For reasonable choices of these meta-parameters, we found a number of “firing pixels” $\sim 50\%$ higher than what one would infer from a catalogue only including sources listed in the latest incarnation of the *Fermi*-LAT catalogue. Our results also pass some sanity checks, like the fact that *Fermi*-LAT catalogue sources which are luminous enough are found within the directions inferred by our method, within the angular resolution of the instrument. If compared with the “local” iterative procedure followed by *Fermi*-LAT to establish their catalogue, another possible advantage of our catalogue of directions in the sky is that its TS scale has a homogeneous meaning, and may be more suitable for global statistical analyses.

We provide our results in digital form, allowing the user to select the meta-parameters QF and α to be more or less restrictive in their selection criteria, hopefully adapting to a wide range of applications. The most obvious one we can think of consists of cross-correlation studies, either multi-wavelength or multi-messenger, where the statistical advantage of a significantly larger sample more than compensates having a (sub-leading) fraction of spurious directions. Another straightforward use of our results could be a guided source search and identification program via multi-wavelength studies, which would also help transforming some of these candidate source directions into *bona fide* sources. In turn, these studies may further benefit of machine learning techniques that have been recently proposed to ease the probabilistic classification of unassociated sources [136].

We stress that, although our results are based on the source-count distribution as inferred from a specific machine learning analysis of the photon counts, the method developed here to identify “firing pixels” can be applied to any other determination of the dN/dS of the high-latitude sky, and it is, in this respect, independent on the choice of the dN/dS as long as the latter correctly describes point sources in the faint regime.

Since our work had primarily a methodological proof-of-principle motivation, we have resorted to a couple of technical simplifications: We worked with an energy-integrated spectrum and we limited ourselves to high Galactic latitudes. We could rather naturally lift the former approximation by a brute force approach. We expect the main complication to be computational, but the problem should remain affordable in a reasonable time. Also the latter limitation could be in principle lifted, extending the analysis to low Galactic latitudes. The main problem we can anticipate is that the existing background models fail to perform as well at low latitudes as they do at high latitudes. One also needs to include multiple populations contributing to both resolved and unresolved emission, see e.g. [113] and sec. 6.2 in [122] for the role played by Galactic millisecond pulsars in this context. In the worst case, the method may not allow one to gain much more with respect to more traditional techniques. Depending on the outcome, one may however think of multi-zone models for the background or the like.

A third direction for future progress is to attach a more realistic probability scale to the TS maps. We have constructed our synthetic skies only based on the statistical properties for the sources previously deduced via the pixel analysis. However, only a subset of these simulations is consistent with the properties of the actual sky: Notably at high flux where only a few sources exist, the associated TS distribution significantly depends on the distribution of the sources in the sky randomly falling in a lower or higher background region (for an illustration, see the large variability at high TS in figure 7.1). As briefly discussed in the article, a promising approach would be to narrow the sample of admissible synthetic skies used to compute QF to the sole simulations which are statistically consistent with the TS pattern of the most luminous sources. Such ‘constrained’ simulations should then improve the reliability of the QF scale. Another working direction with the same goal could be to improve the description of background and/or the parameterization for sub-threshold sources in the pixel analyses beyond what considered in [272]. This may bring the simulation distribution illustrated in figure 7.1 in even better agreement with the TS histogram. Finally, exploring alternative metrics to the agreement in the KS sense is also a direction potentially worth exploring.

CROSS-CORRELATIONS AND FUTURE PROSPECTS

The preceding chapters pursued dark matter through its most localized manifestations: the concentrated gamma-ray emission expected from the Galactic Center — where the dark matter density reaches its peak — and from individual subhalos that remain unresolved among the unassociated sources detected by the *Fermi* Large Area Telescope. Neither strategy returned a definitive signal. The Galactic Center excess persists, but a decade of increasingly sophisticated analyses has not resolved the fundamental degeneracy between a dark matter annihilation origin and a population of unresolved millisecond pulsars [62, 170, 171, 186]. The search for dark matter subhalos among unassociated sources [220] found no significant evidence for an anomalous population at current sensitivity levels, consistent with theoretical expectations — a result that establishes competitive constraints but does not close the door. At the population level, the source-count distribution and probabilistic catalog constructed in Chapters 6 and 7 mapped the unresolved gamma-ray sky to fluxes roughly 50 times below the detection threshold [272, 274, 308], characterizing the astrophysical landscape in which any dark matter contribution must reside, but stopping short of isolating it.

This chapter opens Part IV by stepping back from individual sources and asking a different question: if dark matter annihilates or decays, it does so inside every gravitationally bound halo and subhalo across all cosmic epochs. The accumulated, unresolved emission from this entire population contributes to the unresolved gamma-ray background (UGRB). Separating a dark matter component from the astrophysical foreground — blazars, star-forming galaxies, misaligned active galactic nuclei — requires more than spectral information alone, since distinct source classes partially overlap in energy. What discriminates dark matter is its unique spatial distribution: the large-scale structure of the universe, traced by dark matter halos from galaxy clusters down to sub-Earth mass clumps, imprints specific angular anisotropy patterns on the gamma-ray sky that differ systematically from those of astrophysical backgrounds [104, 162, 163]. The angular cross-power spectrum between gamma-ray maps and a gravitational tracer — such as a galaxy catalog or a cosmic shear field — provides a statistically powerful handle on this signal, because it vanishes for astrophysical source populations at redshifts incompatible with the tracer, but is maximal for dark matter emission that clusters with local structure.

We now examine this cross-correlation approach in detail from three directions. Section 8.1

places the technique in the context of the thesis as a whole, tracing the conceptual arc from resolved Galactic targets to the cosmic web and establishing why a shift in scale is both necessary and well-motivated given the null results of the previous searches. Section 8.2 develops the physical and mathematical principles of the cross-correlation method — the role of window functions and tomographic redshift selectivity, the noise cancellation advantage over auto-correlations, and the identification of low-redshift galaxy catalogs as the optimal gravitational tracer for dark matter annihilation. Section 8.3 introduces the Cherenkov Telescope Array Observatory (CTAO), which extends the accessible energy range an order of magnitude above the *Fermi*-LAT ceiling and brings effective collecting areas six orders of magnitude larger, opening the TeV mass window to cross-correlation searches for the first time. The paper that follows then presents a full sensitivity forecast for detecting dark matter signals through cross-correlations between the CTAO extragalactic survey and the 2MASS galaxy catalog [309].

8.1 From Resolved Sources to the Cosmic Web

Dark matter annihilates wherever its density is high enough — in every gravitationally bound structure from the Milky Way’s central parsecs to the faintest galaxy-cluster subhalos a billion light-years away. The preceding chapters have probed this emission one scale at a time: the Galactic Center at sub-degree resolution, individual subhalos as point sources, and the statistical aggregate of the sub-threshold population. Each approach identified a genuine signal channel and, in doing so, also identified what it could not access. This section steps back to map that landscape explicitly, and then to introduce the complementary strategy that forms the core of Part IV: exploiting the spatial cross-correlations between the unresolved gamma-ray sky and the large-scale structure of the universe as traced by galaxy catalogs.

Section 8.1.1 reviews what Parts II and III established and where the limiting systematics lie. Section 8.1.2 then develops the physical argument for why the dark matter-large-scale-structure cross-correlation provides orthogonal sensitivity: it selects a low-redshift emission window where dark matter dominates astrophysical backgrounds, and it suppresses noise from source populations that do not cluster with local galaxies.

~~8.1.1 The Thesis So Far: Scales and Strategies~~

The work collected in Parts II and III progressively refined both the targets and the statistical methods of dark matter searches in gamma rays. Each stage addressed a specific astrophysical scale and, in doing so, identified the limitations that motivated the next approach.

The Galactic Center excess (GCE), analyzed in Chapter 4, remains the strongest expected annihilation signal in the Milky Way: an extended, approximately spherically symmetric excess of GeV photons whose morphology and spectrum are consistent with a $\sim 35\text{--}50$ GeV WIMP annihilating into $b\bar{b}$ with a cross-section near the thermal relic value [62]. A decade of increasingly refined analyses has not resolved whether the excess originates from dark matter or from an unresolved population of millisecond pulsars in the Galactic bulge [170]; the two hypotheses produce nearly indistinguishable observational signatures at the current sensitivity of the data.

The search for individual dark matter subhalos among *Fermi*-LAT unassociated sources (Chapter 5) approached the problem from a different angle: rather than modeling a known extended excess, it asked whether any high-latitude unassociated source belongs to the subhalo population predicted by Λ CDM [220]. No significant excess beyond astrophysical components was found, and competitive upper limits on $\langle\sigma v\rangle$ were derived for masses from 10 GeV to 1 TeV.

Chapters 6 and 7 shifted scope from individual source identification to the aggregate statistical properties of the gamma-ray source population. The source count distribution dN/dS was recovered well below the detection limit using simulation-based inference [272], and a probabilistic catalog extended the census of candidate sources beyond the standard 4FGL catalog [308]. Together, these analyses characterized the astrophysical landscape in which any dark matter contribution must reside, but they did not target dark matter specifically.

The conceptual thread connecting Parts II and III is one of increasing scale and decreasing signal strength. The GCE offers the highest expected dark matter surface brightness but is contaminated by Galactic systematics and thermal emission degeneracies. Subhalo searches target individual objects at predictable fluxes but face the sparse signal problem: the expected number of detectable dark matter subhalos at current sensitivity is at most of order a few dozen, and none was observed. Population statistics describe the aggregate unresolved sky but do not isolate a dark matter component. The signal we have not yet exploited is one that is distributed across all angular scales: the spatial correlation pattern that the hierarchical distribution of dark matter halos — from the nearest cluster down to the smallest subhalo — imprints on the gamma-ray sky at cosmological distances.

8.1.2 Dark Matter in the Cosmic Web

If dark matter is a particle that annihilates, it does so everywhere it is dense enough for pairs to meet — not just in the Galactic Center or in individual subhalos, but inside every gravitationally bound structure across the observable universe. Understanding how much gamma-ray emission this process produces at cosmological distances, and how that emission is distributed across the sky, requires first understanding how dark matter structures are arranged in space.

In the standard Λ CDM cosmological framework, structures form hierarchically from the bottom up. Small-amplitude density perturbations in the early universe collapse first into low-mass dark matter halos; these merge and accrete material to build progressively more massive structures over cosmic time. The result is a cosmic hierarchy: galaxy clusters at the top, hosting individual galaxy halos, which in turn host populations of subhalos down to the minimum halo mass set by the free-streaming length of the dark matter particle — roughly Earth-mass ($\sim 10^{-6} M_\odot$) for a typical ~ 100 GeV WIMP [104]. This hierarchical web of halos and subhalos, distributed across the universe, is what we call the cosmic web of dark matter.

The gamma-ray flux from dark matter annihilation is proportional to the square of the local dark matter density. Dense, compact structures are therefore the dominant contributors to the total signal, and the cumulative emission from all unresolved halos and subhalos across all redshifts produces a diffuse, approximately isotropic background. This is the dark matter component of the unresolved gamma-ray background (UGRB) — also called the isotropic diffuse gamma-ray background (IGRB) — whose total energy spectrum between 100 MeV and 820 GeV has been measured by the *Fermi*-LAT [111]. Well-established astrophysical source populations contribute their own guaranteed fraction to the UGRB: unresolved blazars (HSPs and LSPs) dominate at high energies and account for perhaps 20–30% of the total intensity, while star-forming galaxies, misaligned AGNs, and millisecond pulsars add further contributions across the full energy range [104]. Because the exact composition of the UGRB is unknown, any dark matter signal must compete with and be separated from these astrophysical backgrounds.

The key challenge is spectral degeneracy. At energies accessible to *Fermi*-LAT (0.1–300 GeV), the spectra of heavy WIMPs and certain blazar populations overlap enough that energy information alone cannot distinguish them. The annihilation flux is additionally sensitive to highly uncertain quantities: the concentration-mass relation of dark matter halos extrapolated to

~~sub-resolution masses~~, the abundance of substructure, and the choice of minimum halo mass — all of which enter through the so-called boost factor $B(M, z)$ that encodes how much the clumpy distribution of subhalos enhances the signal above a smooth halo **assumption** [104]. The boost factor itself can vary by orders of magnitude depending on these assumptions. A purely spectral analysis of the UGRB therefore constrains combinations of particle physics parameters and astrophysical modeling choices that are difficult to disentangle.

Spatial anisotropies provide the **orthogonal handle**. The UGRB is not perfectly isotropic: each contributing population imprints spatial fluctuations ~~whose statistical properties — the angular power spectrum C_ℓ — are determined~~ by the clustering properties of the sources and their redshift distribution. Dark matter annihilation traces the squared density field of the cosmic web, which is more clustered on small angular scales and at lower redshifts than the blazars that constitute most of the astrophysical background [165, 310]. The *Fermi*-LAT Collaboration provided the first measurement of the UGRB angular power spectrum, detecting an approximately scale-invariant (Poisson) component consistent with unresolved point sources, but without clear evidence for a dark matter contribution [310]. The auto-correlation power spectrum, however, suffers from a fundamental limitation: the signal from dark matter is always mixed with the shot noise and correlated fluctuations of astrophysical sources that contribute at the same angular scales. Furthermore, the *measured* angular power spectrum — a single number per multipole ℓ constructed from the 2D projected gamma-ray sky — is a line-of-sight integral over all redshifts and therefore carries no redshift resolution. While the theoretical model for C_ℓ is built from window functions $W_\gamma(z)$ that encode the redshift distribution of each source class (as developed in Section 3.5), this information is observationally inaccessible from gamma-ray data alone: **there is no way to decompose the measured C_ℓ into contributions from different redshifts.** Cross-correlating the gamma-ray map with an external gravitational tracer resolves **both problems**. A galaxy catalog provides a redshift-dependent map of large-scale structure that correlates with the dark matter distribution at specific cosmic distances. **The cross-power spectrum between the gamma-ray sky and the galaxy density field vanishes by construction for any source population that does not cluster with the selected galaxies — this is the noise-cancellation advantage that makes cross-correlation a qualitatively more powerful probe than auto-correlation [162, 163].** The dark matter annihilation signal peaks at low redshift (because the flux is proportional to ρ^2 and thus weights dense, nearby structures more heavily), precisely where a catalog of local galaxies has the highest completeness. By choosing a catalog with $z \lesssim 0.1$, ~~one can arrange for~~ the dark matter window function ~~to~~ overlap substantially with the galaxy window function while the **blazar window function** — whose redshift distribution peaks at $z \sim 0.3\text{--}0.4$ — mostly does not. The angular cross-power spectrum developed formally in Section 3.5 is the mathematical realization of this strategy.

Figure 8.4 illustrates the core idea. The dark matter window function $W_\gamma^{\text{DM}}(z)$, which describes the redshift distribution of the gamma-ray emission, is sharply peaked at $z < 0.1$ due to the ρ^2 density **weighting** of the annihilation rate. Local galaxy catalogs such as 2MASS, covering $z \lesssim 0.1\text{--}0.2$, have window functions that **closely follow this peak**. In contrast, blazars at energies of 50 GeV are distributed over $z \sim 0.1\text{--}0.4$, and at 1 TeV they are further suppressed at high redshift by extragalactic background light (EBL) absorption, but still do not overlap strongly with the very local galaxy distribution. **The cross-correlation signal between the gamma-ray sky and local galaxies is therefore dominated by whatever gamma-ray emission originates in the same volume as the galaxies — precisely the regime where dark matter is the dominant cosmological source.**

8.2 The Cross-Correlation Technique

Section 3.5 laid out the mathematical formulation of the cross-correlation technique. What remains is to understand why it outperforms auto-correlation and spectral analyses in practice. Three properties are key: redshift selectivity from overlapping window functions (Section 8.2.1), the statistical gain from combining two independent observables (Section 8.2.2), and the choice of low-redshift catalogs that reinforces both (Section 8.2.3).

8.2.1 Tomographic Redshift Slicing

The angular cross-power spectrum $C_\ell^{\gamma g}$ introduced in Section 3.5 measures how the gamma-ray intensity field and the galaxy number counts fluctuate together as a function of angular scale. Under the Limber approximation (Eq. 3.5.4), this is a single integral over comoving distance, weighted by the product of the gamma-ray and galaxy window functions $W_\gamma(\chi)$ and $W_g(\chi)$. The key observation is that $C_\ell^{\gamma g}$ receives contributions only from the redshift range where both window functions are simultaneously nonzero. Choosing a galaxy catalog with a narrow, well-understood redshift distribution therefore selects a specific slice of the universe, filtering out emission from all other distances. This is the essence of the tomographic approach [104, 311].

The window function $W_\gamma(z)$ encodes the physical character of the gamma-ray emission. For unresolved astrophysical sources such as blazars, W_γ depends on the luminosity function and the spectral energy distribution of the source class, modulated by EBL absorption at high energies. For dark matter annihilation, it is proportional to $\langle \sigma v \rangle \Delta^2(z) (\Omega_{\text{DM}} \rho_c / m_\chi)^2 (1+z)^3 e^{-\tau(E,z)}$, where $\Delta^2(z)$ is the flux multiplier that captures the enhancement from DM clustering and substructure (see Section 1.4). Because the annihilation rate goes as ρ^2 , it is highly sensitive to the dense central regions of halos and is effectively confined to the local universe: the DM window function peaks sharply at $z \lesssim 0.1$ and decays rapidly thereafter, regardless of the DM mass or annihilation channel.

Astrophysical backgrounds present a very different picture. Unresolved blazars¹ have their emission peak at $z \sim 0.3$ – 0.5 at 50 GeV. At higher energies, EBL absorption progressively shrinks the gamma-ray horizon: above ~ 1 TeV, the observable volume contracts to $z \lesssim 0.1$ – 0.2 , irrespective of source type. Star-forming galaxies trace the cosmic star-formation rate and dominate at much higher redshifts, $z \sim 1$ – 3 . These populations are effectively invisible to a galaxy catalog covering the local universe. Figure 8.4 illustrates this separation quantitatively for the benchmark scenario considered in the analysis of this chapter: the DM window functions are confined to low redshift, while blazar contributions extend to $z \sim 0.4$ – 0.5 at 50 GeV.

The galaxy catalog window function $W_g(z) = dN_g/dz$, normalized to unity, is the redshift distribution of the galaxies in the sample. A dense, shallow catalog covering $z \lesssim 0.1$ such as the Two Micron All-Sky Survey (2MASS) aligns almost perfectly with the DM annihilation window function. The cross-correlation signal is then proportional to the overlap integral $\int W_\gamma W_g d\chi$. Because the blazar contribution to W_γ lies at much higher redshift than W_g , it does not contribute to the signal — the two window functions simply do not overlap. The DM signal, by contrast, overlaps strongly, and the cross-correlation becomes a redshift-selective probe of the local DM distribution.

This tomographic selectivity is the principal advantage of the cross-correlation over the auto-correlation of the gamma-ray sky alone. Even if a DM signal were present in the total

¹High-synchrotron-peaked BL Lacertae objects are often denoted HSP, while low and intermediate-synchrotron-peaked BL Lacs are often grouped together and denoted LISP.

intensity, the power-law continuum of unresolved blazars would prevent its isolation without additional redshift information. Cross-correlation provides exactly that handle: by choosing a catalog confined to the local universe, one selects for the DM component while astrophysical backgrounds at higher redshifts cancel from the observable.

8.2.2 Why Cross-Correlation Outperforms Auto-Correlation

The statistical properties of the cross-correlation estimator reveal why this approach is fundamentally more powerful than auto-correlation for dark matter searches. The variance of the cross-power spectrum (Eq. 3.5.8) contains the product of the gamma-ray noise term ($C_\ell^{\gamma\gamma} + C_N/B_\ell^2$) and the galaxy noise term ($C_\ell^{gg} + C_{N_g}$). The photon noise C_N in the gamma-ray map and the galaxy shot noise C_{N_g} are physically independent: their cross-noise vanishes on average. The noise therefore inflates the error bars on $C_\ell^{\gamma g}$ but does not add a systematic offset to the signal itself.

The auto-correlation variance (Eq. 3.5.9) is qualitatively different. Here the photon noise enters quadratically and adds directly to the gamma-ray auto-power spectrum. For a ground-based instrument like CTAO, the cosmic-ray rejection rate is never perfect, so C_N is dominated by the irreducible cosmic-ray background rather than by photon statistics alone, making it very large. Any attempt to extract a DM signal from $C_\ell^{\gamma\gamma}$ must therefore contend with subtracting this large noise term with high precision. By contrast, C_N in the cross-correlation variance multiplies the galaxy auto-spectrum $C_\ell^{gg} + C_{N_g}$, which is well measured independently; the DM signal is not hidden beneath the noise but simply requires accumulating sufficient signal-to-noise over multipoles and energy bins.

The second advantage of cross-correlation concerns the astrophysical foreground itself. The gamma-ray auto-correlation power spectrum $C_\ell^{\gamma\gamma}$ is dominated by the Poisson shot noise of unresolved blazars at essentially all multipoles of interest [104, 162]. These are rare, bright point sources whose auto-correlation is a flat (Poissonian) C_ℓ from the 1-halo term, covering any diffuse DM component. The detectability of astrophysical sources in auto-correlation therefore depends on how deeply the instrument can resolve individual blazars: the better the point-source sensitivity, the lower the residual unresolved blazar background and the smaller $C_\ell^{\gamma\gamma}$. For the DM signal, which is effectively diffuse and unresolved, the auto-correlation sensitivity therefore degrades as a function of the Poisson blazar floor.

Cross-correlation with a galaxy catalog suppresses this foreground through the window function mismatch described in Section 8.2.1. The high-redshift blazar population does not correlate with a local galaxy catalog, so their 1-halo Poisson term does not appear in $C_\ell^{\gamma g}$. The cross-correlation SNR for the astrophysical signal depends on the point-source sensitivity only weakly, through the window function of the resolved fraction, whereas the auto-correlation SNR is strongly sensitive to it because the detection threshold enters both the window function and the three-dimensional power spectrum. For dark matter, the situation is even more favourable: the DM emission is intrinsically diffuse — no individual extragalactic DM halo has been resolved as a gamma-ray point source to date — so the DM window function does not depend on the detection threshold at all. The cross-correlation sensitivity to a DM signal is therefore essentially immune to the blazar detection threshold. The quantitative impact of this suppression on the dark matter sensitivity is derived in the paper that follows, which compares the cross-correlation and auto-correlation constraints directly.

Beyond the statistical advantages described above, the cross-correlation power spectrum carries richer physical information than either the energy spectrum or the auto-correlation alone. The angular dependence of $C_\ell^{\gamma g}$ can be decomposed into a 1-halo and a 2-halo contribution,

following the halo-model formalism introduced in Section 3.5 and detailed in [104, 163]. The 1-halo term captures correlations between pairs of points within the same halo — for DM this traces the density profile within individual halos and dominates at small angular scales ($\ell \gtrsim 200$). The 2-halo term describes the large-scale clustering between different halos, following the linear matter power spectrum weighted by a bias factor. Measuring $C_\ell^{\gamma g}$ across multipoles therefore allows one to probe both the internal structure of DM halos and their clustering properties simultaneously, a richer physical picture than the energy spectrum or the auto-correlation alone can provide.

8.2.3 The “Golden Channel”: Low-Redshift Galaxy Catalogs

The redshift overlap argument of Section 8.2.1 points to a specific class of galaxy catalogs as optimal for DM annihilation searches: dense, all-sky surveys covering the local universe at $z \lesssim 0.1$ – 0.2 . Two datasets stand out in this role. The Two Micron All-Sky Survey (2MASS) Extended Source Catalog contains approximately one million galaxies and has a redshift distribution peaking at $z \approx 0.072$, making it one of the shallowest but most complete all-sky galaxy surveys available [104]. The 2MASS Redshift Survey (2MRS) is its spectroscopic subsample, extending to $z \sim 0.1$ with full spectroscopic redshifts, providing both a precise dN_g/dz for modeling and higher galaxy density per solid angle in the local volume. Both catalogs are nominally all-sky; however, in the cross-correlation analysis the low-latitude region around the Galactic plane (approximately $|b| < 30^\circ$) is masked in the gamma-ray maps to suppress Galactic diffuse emission [104, 311]. This masked region coincides with the region where dust obscuration already limits the completeness of infrared galaxy catalogs, making the masking consistent across both datasets.

The practical advantage of 2MASS and 2MRS is that the blazar contribution to the UGRB in the local volume is strongly suppressed. Although unresolved blazars may account for roughly 20–30% of the total UGRB intensity integrated over all redshifts [114], this contribution arises predominantly from sources at $z \gtrsim 0.3$, where the large luminosity distance pushes their apparent flux below the Fermi-LAT detection threshold. In the local universe ($z \lesssim 0.1$), by contrast, even intrinsically faint blazars are close enough that their flux exceeds the point-source sensitivity of the Fermi-LAT [104]; they are therefore individually detected, cataloged, and removed from the diffuse maps. As a result, the unresolved blazar window function drops sharply at low redshift, and unresolved blazars in the volume probed by 2MASS and 2MRS are a negligible component of the UGRB at Fermi-LAT energies [104, 312]. The cross-correlation of the UGRB with these catalogs is thus free from the dominant astrophysical contaminant that plagues both the gamma-ray auto-correlation and searches at higher redshift. The signal remaining in $C_\ell^{\gamma g}$ is a mixture of star-forming galaxies (SFGs), misaligned active galactic nuclei (mAGN), and any DM contribution — precisely the populations one expects to find in the local large-scale structure.

This expectation was confirmed observationally with 60 months of Fermi-LAT data: Xia et al. [311] detected a statistically significant cross-correlation between the UGRB and several galaxy catalogs at angular scales below one degree, finding that the signal is dominated by unresolved star-forming galaxies and misaligned AGNs, with blazars contributing less than 20%. Cuoco et al. [312] then performed a joint fit to the measured cross-correlation functions, decomposing the signal into DM, blazars, SFGs, and mAGN components. A non-zero DM contribution was allowed by the data but degenerate with the astrophysical components; the resulting upper limits on the annihilation cross section and decay lifetime were competitive with contemporary constraints from dwarf spheroidal galaxies and galaxy clusters, while arising from an independent observable — the spatial coherence of DM emission with the cosmic web [313].

Beyond galaxy number counts, gravitational lensing maps offer an alternative tracer with independent systematics. Cross-correlating the UGRB with weak-lensing convergence maps, Shirasaki et al. [314] found no significant signal and used the null result to set upper limits on $\langle\sigma v\rangle$, demonstrating that cosmic shear is useful for DM searches even in the absence of a detection. Camera et al. [163] showed theoretically that the shear window function overlaps well with the DM annihilation window function, ~~making this combination particularly clean from an astrophysical foreground perspective.~~ The main practical challenge is the much larger noise of current weak-lensing surveys compared with galaxy number counts.

Looking ahead to CTAO, the choice of 2MASS and 2MRS as the reference catalogs is well motivated: their redshift distributions ~~maximize the overlap with the DM window function in the energy range where CTAO is most sensitive ($\gtrsim 50$ GeV, where EBL absorption shrinks the gamma-ray horizon of astrophysical sources to $z \lesssim 0.2$).~~ As the EBL horizon effect progressively suppresses ~~gamma-ray~~ emission at higher energies, the auto-correlation background decreases while the DM window function remains sharply peaked at low redshift. This energy-dependent competition makes the cross-correlation with local catalogs an even more powerful tool at TeV energies ~~than at GeV energies~~, a feature that CTAO is uniquely positioned to exploit.

8.3 The Cherenkov Telescope Array Observatory

~~The cross-correlation technique described in the preceding section reaches its full potential when paired with an instrument whose energy coverage, sky coverage, and angular resolution are matched to the physical scales of the DM signal.~~ The Cherenkov Telescope Array Observatory (CTAO) satisfies all three requirements. Its energy range, extending from 20 GeV to 300 TeV, spans precisely the regime where heavy WIMP candidates ~~produce their signal~~; its planned extragalactic survey covers a quarter of the sky with approximately uniform exposure; and its angular resolution at the relevant energies exceeds that of Fermi-LAT by an order of magnitude. The following subsections introduce CTAO's key characteristics and its extragalactic survey strategy ~~before connecting them to the sensitivity forecast that constitutes the paper in this chapter.~~

8.3.1 From Fermi-LAT to CTAO

The analyses in Chapters 4–7 were built on data from the Fermi Large Area Telescope (Fermi-LAT), which has operated as the premier instrument for indirect dark matter searches in the GeV band for over 15 years (see Section 2.3.1). As a space-based pair-conversion telescope, the LAT detects gamma rays from 20 MeV to beyond 300 GeV with a field of view of approximately 2.4 sr, surveying the entire sky every ~~two orbits~~ (~ 3 hours) [117]. Its effective area, however, is constrained by spacecraft payload limitations to roughly 1 m^2 at GeV energies, ~~a bound common to all satellite-borne instruments.~~ This ceiling in collecting area, combined with the steeply falling flux of gamma rays at high energies, makes Fermi-LAT poorly suited for probing WIMP candidates with masses ~~well~~ above 1 TeV.

The Cherenkov Telescope Array Observatory (CTAO) is ~~designed~~ to overcome ~~exactly~~ this limitation [315]. Rather than detecting photons directly in space, ground-based imaging atmospheric Cherenkov telescopes (IACTs) reconstruct very-high-energy gamma rays from the brief flashes of Cherenkov light produced when an incident photon strikes the upper atmosphere and initiates an electromagnetic shower. By ~~sampling~~ this shower at ground level, CTAO effectively uses the atmosphere as a calorimeter ~~of continental scale~~, achieving effective areas orders of magnitude larger than any space-based detector. The observatory will consist of two arrays, one

in the northern hemisphere and one in the south, built from three telescope types tailored to different energy regimes: the Large-Sized Telescopes (LSTs, sensitive at lower energies, field of view 4.3°), the Medium-Sized Telescopes (MSTs, field of view $7.5^\circ\text{--}7.7^\circ$), and the Small-Sized Telescopes (SSTs, fields of view 8.8° , deployed only in the south where Galactic observations are prioritized). Across the three telescope classes, CTAO covers a broad energy range from 20 GeV up to 300 TeV, with an angular resolution and sensitivity that improve by an order of magnitude over current-generation IACTs such as H.E.S.S., MAGIC, and VERITAS [315].

The extension to TeV energies opens a qualitatively new window for dark matter searches. Heavy WIMP candidates with masses $m_\chi \gtrsim 1$ TeV produce gamma rays predominantly in the TeV band, a range where Fermi-LAT has negligible sensitivity. More importantly for cross-correlation studies, EBL absorption becomes significant above ~ 1 TeV, attenuating photons from sources at $z \gtrsim 0.1\text{--}0.2$ and effectively cutting off the astrophysical backgrounds. The LST and MST telescopes, built in both hemispheres, are optimised for extragalactic observations, where the unresolved emission receives its dominant DM contribution from structures at $z \lesssim 0.1$. At energies above 20 GeV and increasingly toward the TeV regime, star-forming galaxies and misaligned AGN contribute negligibly to the unresolved background; only blazars, both High-Synchrotron-Peaked (HSP) and Low-to-Intermediate-Synchrotron-Peaked (LISP), remain as the principal astrophysical foreground. Their window functions peak at $z \sim 0.3\text{--}0.4$ at 50 GeV and are pushed down to $z \sim 0.1\text{--}0.2$ above 1 TeV by EBL attenuation — a range that still sits above the DM-dominated window, which we exploit through the choice of the 2MASS/2MRS galaxy catalogs (Section 8.2.3). This residual separation between the DM and blazar window functions is the physical lever arm that makes the cross-correlation technique viable at TeV energies with CTAO.

8.3.2 The Extragalactic Survey

Cross-correlation and anisotropy analyses require large sky maps with approximately uniform exposure. A pointed instrument with irregular coverage would introduce instrumental variations that could mimic or obscure genuine cosmological angular correlations. For CTAO, this requirement is met by the Extragalactic Survey (EGAL), one of the observatory’s designated Key Science Projects [315].

The EGAL survey targets a quarter of the sky, covering the region defined by $-90^\circ < l < 90^\circ$ and $|b| > 5^\circ$ in Galactic coordinates. The survey is planned over three years of operations, split between the two array sites: the Southern array will observe 15% of the sky with 400 hours, while the Northern array will cover the remaining 10% with 600 hours, for a total observation time of approximately 1000 hours [315]. The default observing strategy is a grid of telescope pointings spaced by approximately 3° across the survey area. Individual pointings are observed for 0.51 hours in the south and 1.11 hours in the north, yielding an effective average exposure of approximately 3 hours per point within the survey boundary (see Section 8.4.1). The wide fields of view of the individual telescopes — up to 8.8° for the SSTs and $7.5^\circ\text{--}7.7^\circ$ for the MSTs — create deliberate overlaps between adjacent pointings. These overlaps smooth out the exposure variations that inevitably arise when stitching together individual observations, keeping the relative fluctuations in the exposure map to approximately 10% within the survey region (see Section 8.4.1).

A second observation scenario considered in the paper appears from the accumulated off-source data collected through years of CTAO pointed observations across the extragalactic sky. In the most optimistic projection, these off-source data sum to an effective exposure of approximately 50 hours per sky location — roughly 25 times larger than the nominal EGAL exposure — yielding a factor of ~ 4 improvement in sensitivity compared to the 3-hour scenario (see Appendix 8.A).

This scenario does not require a dedicated survey and could in principle be reached earlier in the observatory’s operational lifetime, making it a practically relevant complement to the EGAL Key Science Project.

Uniform sky coverage is a technical requirement, not just an observational convenience, for the cross-correlation formalism. The angular power spectrum C_ℓ is computed from a masked, pixelised sky map, and anisotropies in the instrumental exposure cannot be distinguished from intrinsic anisotropies in the gamma-ray background without careful masking and exposure correction. At the multipoles $\ell \sim 10\text{--}100$ relevant for the DM cross-correlation signal, any residual exposure gradient couples into the signal through the survey window function and biases the recovered power spectra. The EGAL strategy’s uniform grid, combined with the broad telescope fields of view, is designed to minimise this systematic effect, providing the map quality required for the sensitivity forecast presented in the following paper.

The combination of the EGAL survey’s uniform sky coverage, CTAO’s order of magnitude gains in angular resolution and collecting area, and the cross-correlation formalism developed in Section 8.2 provides the ingredients for a quantitative assessment of CTAO’s reach. The section that follows – based on [309] – applies this framework, forecasting the sensitivity to dark matter annihilation and decay through cross-correlations between the CTAO extragalactic survey and the 2MASS galaxy catalog, and benchmarking the results against complementary search strategies.

[AA: follows the paper]

8.4 Dark Matter and Galaxy Cross-Correlations with the Cherenkov Telescope Array Observatory

The Cherenkov Telescope Array Observatory (CTAO) will be a ground-based Cherenkov telescope performing wide-sky surveys, ideal for anisotropy studies such as cross-correlations with tracers of the cosmic large-scale structure. Cross-correlations can shed light on high-energy γ -ray sources and potentially reveal exotic signals from particle dark matter. In this work, we investigate CTAO sensitivity to cross-correlation signals between γ -ray emission and galaxy distributions. We find that by using dense, low-redshift catalogs like 2MASS, and for integration times around 50 hours, this technique achieves sensitivities to both annihilating and decaying dark matter signals that are competitive with those from dwarf galaxy and cluster analyses.

In this paper, we present a sensitivity forecast for detecting dark matter signals through cross-correlations between the CTAO extragalactic survey and the 2MASS galaxy catalog. Section 8.4.1 describes the CTAO instrument configuration and survey strategy. Section 8.4.2 develops the cross-correlation signal model, including the window functions and angular power spectra for both astrophysical sources and dark matter. Section 8.4.3 presents the sensitivity forecast results for dark matter annihilation and decay. We conclude in Section 8.4.4.

8.4.1 The Cherenkov Telescope Array

The CTAO will represent the next generation of imaging atmospheric Cherenkov telescopes (IACTs), improving the technology of the current ground-based γ -ray detectors, such as H.E.S.S., MAGIC, and VERITAS. It is designed to surpass the sensitivity of current IACTs by an order of magnitude in the TeV regime, to improve the angular resolution, as well as to enhance the surveying and monitoring capabilities. CTAO will consist of two IACT arrays: one located in the Southern hemisphere (CTAO-South) and the other in the Northern hemisphere (CTAO-North) [315].

The configuration will use three different types of telescope: large-, medium-, and small-sized telescopes (LST, MST, and SST, respectively), detecting photons over a broad energy range from 20 GeV up to 300 TeV. The LSTs will be most sensitive at lower energies, while the SSTs will cover the highest energies. The SSTs will be dedicated primarily to the observation of phenomena occurring within our Galaxy and will therefore be deployed only at the southern site from which the central/inner part of the Galaxy is most visible. LSTs and MSTs will instead be built both in the Northern and Southern locations and will focus on observations of extragalactic targets.

The telescopes will be equipped with cameras featuring wide fields of view: 4.3° for LSTs, $7.5\text{--}7.7^\circ$ for MSTs, and 8.8° for SSTs. A wide field of view is important for observing extended sources and diffuse emission, as well as for reducing the systematic errors of the measurements when requiring a uniform response over regions much larger than the telescope point-spread function (PSF). Several surveys covering large portions of the sky are planned as part of CTAO's observational strategy, and will be performed as Key Science Projects (KSPs) of the observatory early in the observatory's operation [315]. Thanks to its improved sensitivity and wider field of view, CTAO will surpass the surveying capabilities of existing IACTs. In particular, the Extra-Galactic survey (detailed below) will be the largest survey ever undertaken by IACTs.

At the same time, CTAO will carry out a large number of traditional, dedicated pointings across the sky. The off-source data collected through these observations will cumulatively cover a substantial sky area with a significant average exposure. In this work, we consider extragalactic data obtained from both observational strategies, as detailed in two subsections below.

8.4.1.1 The Extra-Galactic survey

To study the auto- and cross-correlation signals with gamma-rays, it is essential to use maps of large sky regions with approximately uniform coverage. Among CTAO Key Science Projects, the dedicated survey of the extragalactic sky—referred to as the extra-galactic (EGAL) survey - presents the required characteristics.

As described in [315] (Sect. 8), the EGAL survey will cover a quarter of the sky, targeting the region defined by Galactic coordinates $-90^\circ < l < 90^\circ$ and $b > 5^\circ$. The survey is planned to reach an integral sensitivity ~ 6 mCrab above 125 GeV. One of its main objectives is to provide a high-resolution map of the extragalactic sky for photons with energies at least up to 10 TeV and to compile a catalog of very high energy (VHE) extragalactic sources.

The EGAL observations are planned to be carried out over three years of observations, using the full CTAO Observatory (both the Southern and Northern arrays) to achieve nearly uniform sensitivity over the survey region. It is proposed that 15% of the sky will be covered by the Southern array using 400 h and 10% by the Northern array using 600 h, resulting in a total observation time of ~ 1000 h. The default observing strategy consists of a grid spaced by $\sim 3^\circ$ across the survey area, with each pointing observed for 0.51 h in the South and 1.11 h in the North. This configuration is expected to yield an effective exposure of approximately 3 h per point within the EGAL survey boundary.

8.4.1.2 Simulating survey observations

To simulate the planned EGAL survey observations, we use the software package `ctools`,² developed for a scientific analysis of CTA data. We used the instrument response functions

²<http://CTA.irap.omp.eu/ctools/>.

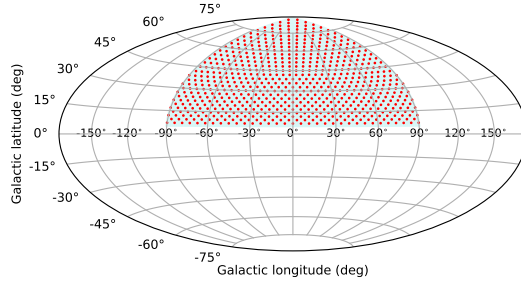


Fig. 8.1. Pointing directions for the region of the sky planned to be covered by the extragalactic survey. The pointing directions are shown using a HEALPix pixelization with NSIDE = 16.

(IRFs) for the observatory,³ labeled prod3b-v2, defining the so-called “omega” configuration. The respective IRF data files are publicly available at Ref. [316].

To enclose the region of the sky covered by the EGAL survey, we simulate observations using telescope-pointing directions corresponding to a HealPix realization with NSIDE = 16, resulting in 694 (3072) pointings within the EGAL survey bounds (over the whole sky). With this setup, the typical angular separation between adjacent pointings is $\sim 3.7^\circ$. For definiteness, the radius of the field of view (FoV) of each pointing is set to 6° . The pointing directions used for the simulation of the EGAL survey within the region of interest (ROI) are shown in Fig. 8.1.

The overlap of the FoV from different pointings contributes to a more uniform coverage of the survey’s ROI, minimizing the fluctuations in the exposure that inevitably occur when smaller FoV observations are used to cover a large survey area. Depending on the selected HealPix grid size (i.e., the values of the NSIDE parameter) and the pointing radius for individual observations, the overlap between nearby pointings can vary in extent. The overlaps between pointings are schematically shown in Fig. 8.2, where the upper panel presents a zoomed-in scheme of the pointing strategy from Fig. 8.1, and the bottom panel shows the simulated exposure of the same region. These images demonstrate that the overlap of nearby pointings creates an anisotropy pattern in the corresponding observation exposure, with variations of approximately $\sim 10\%$ relative to the average exposure within the plotted region.

The IRFs we use are optimized for observations under different zenith angles. For each pointing direction, we assign one of the three available IRF optimizations based on the minimum zenith angle at which that direction can be observed. To estimate the minimal zenith angle, we use the Earth coordinates of both array positions, i.e., $(l_N, b_N) = (17.8920, 28.7622)$ and $(l_S, b_S) = (79.4041, -24.6272)$ for the Northern and Southern array, respectively. The zenith angle is calculated as the angular distance between the array location and the pointing direction when they are closest. The resulting zenith angles for the Northern and Southern arrays are shown in the left panel of Fig. 8.3. The strategy for assigning the IRFs based on zenith angle is adapted from Ref. [317]. For zenith angles smaller than 30° , we use the IRF optimized for 20° angles; for zenith angles between 30° and 50° we use the IRF optimized for 40° angles, and for zenith angles larger than 50° we use IRF optimized for 60° angles, as shown in the right panel of Fig. 8.3. All IRFs used in this work are optimized for 5 hours of observation time, independently of the array or the zenith angle.

The observational strategy presented in Section 8 of Ref. [315] divides the extragalactic survey area into two parts: roughly $\sim 15\%$ of the sky to be observed using the Southern array and

³See <http://www.CTA-observatory.org/science/CTA-performance/> for more details.

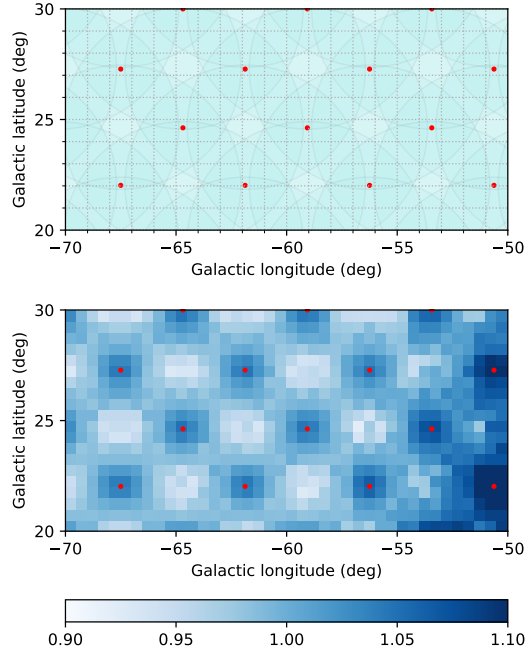


Fig. 8.2. Overlapping areas of different pointings. *Top:* A zoomed-in scheme of the pointing strategy shown in Fig. 8.1 (red), together with the overlapping fields of view of individual pointings (turquoise). *Bottom:* Simulated exposure map showing variations relative to the average exposure, highlighting the anisotropic pattern created by overlapping pointing regions.

$\sim 10\%$ with the Northern array. However, the precise division of the survey observations between the two arrays has not yet been finalized. The planned strategy remains under discussion and is still subject to adjustments. For simplicity, we simulate observations of each pointing direction within the survey boundary using both Southern and Northern arrays with the corresponding zenith angle optimized IRFs. We use 1.11 hours and 0.51 hours per pointing direction with the Northern and Southern arrays, respectively. This strategy provides a reasonable approximation for simulating the survey when combined with zenith-angle-optimized IRFs, as regions observed at higher zenith angles will yield fewer detected events, and vice versa. As a result, the sky coverage of both arrays will be complementary, as illustrated in the right panel of Fig. 8.3.

The survey observations are simulated for energies between 30 GeV and 30 TeV, using 30 log-spaced energy bins to ensure proper sampling of the IRFs across the energy range. For the purpose of estimating the background events and exposure over the whole extragalactic survey, we simulate the entire survey ROI, using coarser spatial binning of $1.0^\circ \times 1.0^\circ$ per pixel. For the sensitivity analysis, we instead simulate templates covering an ROI of $6.0^\circ \times 6.0^\circ$, using a finer spatial binning of $0.02^\circ \times 0.02^\circ$ per pixel.

8.4.1.3 Off-source data scenario

In addition to performing extensive surveys of the γ -ray sky, CTAO will be operated as an open, proposal-driven observatory. The observation time awarded to the wider research community can be added to the dedicated survey observation times of CTAO, and increasing the overall exposure that will be achieved over the first 10 years of telescope operation. An educated guess based on the performance of current IACTs can be made for CTAO observations performed through

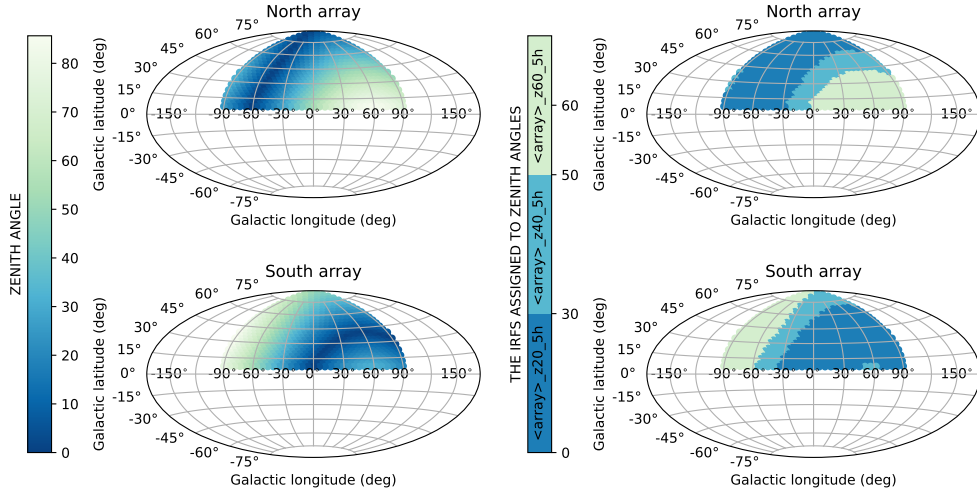


Fig. 8.3. Strategy for assigning Instrument Response Functions (IRFs) across the extragalactic survey region as a function of zenith angle. In the left panel, the color scale shows, for each sky position, the zenith angle as measured in the observatory reference frame during observation (the values of the zenith angles and their color coding are reported on the vertical bar on the left). Since the IRFs are derived in three zenith angle bins (indicated by the color bar on the side of the right panel), the right panel shows which IRF is assigned to each region of the sky. The three IRFs shown here refer to 5 hrs of observation and to zenith angles below 30 deg (dark blue), between 30 and 50 deg (light blue) and above 60 deg (green).

the proposal-driven program. In Ref. [318], the authors calculate the sky coverage and resulting observation time of the CTAO proposal program, assuming similarity with the actual operations of the MAGIC telescope. They extrapolate the 6.5 years of data obtained with the MAGIC array to 10 years of observation time using the two CTAO arrays. We adopt a similar approach to estimate the most optimistic case scenario for the observations of the extragalactic sky with CTAO, assuming an effective observation time of ~ 50 hours per point and the same sky coverage as in the EGAL survey, consistent with the findings in Ref. [318]. In the sensitivity analysis described below, we use a corresponding re-scaling factor to estimate the point source sensitivity of CTAO under this scenario. Details on how we derive the effective observation time for this scenario can be found in Section 8.A.

8.4.2 The cross-correlation signal

The cross-correlation between a map of the UGRB, $\gamma(\hat{n})$, and fluctuations in galaxy number counts, $g(\hat{n})$, can be measured through the cross-correlation angular power spectrum (APS) [162, 163, 319]. In our analysis, the field g will be the distribution of galaxies, which is a biased tracer of the underlying dark matter distribution. In the Limber approximation [164, 320, 321], the APS can be written as [162, 163, 319]

$$C_{\ell}^{\gamma g} = \int \frac{d\chi}{\chi^2} W_{\gamma}(\chi) W_g(\chi) P_{\gamma g}(k = \ell/\chi, \chi), \quad (8.4.1)$$

where ℓ denotes the multipole, k the wavenumber, $\chi(z)$ is the radial comoving distance, related to redshift z in a flat cosmology through the relation $d\chi = c dz/H(z)$ with $H(z)$ being the expansion rate of the Universe. The term $P_{\gamma g}(k, z)$ represents the 3D cross-correlation power spectrum (PS), embedding the information on the fluctuations of the two correlated fields. The function $W_{i=\{\gamma, g\}}$ denotes the window functions of the gamma-ray intensity field, γ , and the galaxy number counts fluctuations, g , respectively, and describes the redshift distribution of each field (see Appendix 8.B for full details).

As astrophysical γ -ray sources we consider only BL Lacs, subdivided into the two sub-categories of high-synchrotron-peaked (HSP) blazars and low and intermediate synchrotron-peaked (LISP) blazars [322]. We model the spectral energy distribution (SED) of these sources as power-laws with index Γ and exponential cutoffs E_{cut} , whose specific parameter values are reported in Table 8.1, together with the minimum and maximum luminosities of the population. Since we focus on CTAO and thus on energies above approximately 20 GeV, other typical γ -ray emitters like misaligned AGN, star-forming galaxies, and flat-spectrum radio quasars can be neglected, since their contribution are dominant at lower energies [104]. Full details on the modeling of the astrophysical sources are provided in Appendix 8.B, together with the expressions for their window functions and 3D cross-correlation power spectra, obtained within the halo model formalism.

Let us just recall here that the window function of the γ -ray emission of astrophysical sources depends on the γ -ray luminosity function (GLF) of each specific source-type and on their SED, and is integrated over the luminosity of the sources up to a threshold luminosity, which corresponds to the minimal luminosity that a source must possess in order to be resolved by the detector (in this way, the window function which is then obtained pertains to the *unresolved* sources, those contributing to the UGRB). The full expression and the detailed modeling can be found in Appendix 8.B.2.

For the DM γ -ray emission, we consider both annihilating and decaying DM. In the case of annihilation, the relevant particle physics parameters are the DM mass m_χ and the thermally-averaged annihilation cross-section $\langle\sigma v\rangle$. For decaying DM, the key parameters are the mass and the lifetime τ . These will be the parameters for which we provide the forecast sensitivity in the cross-correlation analysis. The expressions for the window functions and the 3D cross-correlation power spectra are reported in Appendix 8.B. Here we recall that the window functions are proportional to $\langle\sigma v\rangle$ for annihilating DM and inversely proportional to τ in the case of decaying DM, and proportional to m_χ^{-2} and m_χ^{-1} for annihilating and decaying DM, respectively. The window function of the gamma-ray emission from DM annihilation or decay also depends on the average dark matter content of the Universe and on the energy spectrum produced by dark matter annihilation or decay, which in turn depends on the specific annihilation/decay channel, for which we will consider $b\bar{b}$ as the representative case and provide results also for the channels e^+e^- , $\tau^+\tau^-$ and W^+W^- . In the case of annihilating dark matter, the window function also contains a so-called flux-multiplier factor, which accounts for the enhancement of the flux due to the clumping of DM and the presence of sub-structures within the DM halos. The full expressions and the detailed modeling of the DM window functions can be found in Appendix 8.B.2. A few examples of the redshift behavior of the γ -ray window functions for DM annihilation are shown in Figure 8.4, for two photon energies and two DM masses. For comparison, we also show the window functions of unresolved LISP and HSP astrophysical source classes, assuming an observation time of 3 hours. The observation time affects the window functions of astrophysical sources, since it determines the minimum luminosity of the sources that can be resolved by the instrument (and therefore do not contribute to the unresolved component considered in our analysis). The case

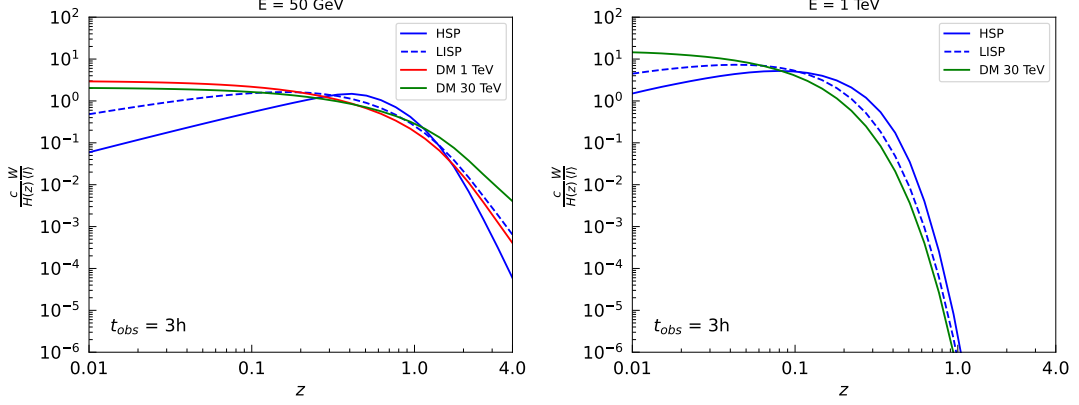


Fig. 8.4. The γ -ray window functions, normalized to the mean γ -ray intensity and divided by the Hubble parameter, as a function of redshift. Blue solid and dashed lines show HSP and LISP astrophysical sources for 3 hours of observation, respectively. Red and green lines correspond to DM annihilation into $b\bar{b}$, $\langle\sigma v\rangle = 3 \times 10^{-26} \text{cm}^3 \text{s}^{-1}$ for DM masses of 1 TeV (red) and 30 TeV (green). Upper panel: $E_\gamma = 50 \text{ GeV}$; lower panel: $E_\gamma = 1 \text{ TeV}$ (only 30 TeV DM case shown).

for 50 hours of observation is shown in Figure 8.15 in Appendix 8.B.

For the present analysis, which primarily focuses on the sensitivity to a DM signal, we adopt as galaxy catalogs the Two Micron All-Sky Survey (2MASS) [323, 324] and the 2MASS Redshift Survey (2MRS) [325, 326] catalogs, since they cover the low redshift range $z \lesssim 0.1$ where most of the DM γ -ray emission originates. Astrophysical sources, on the other hand, have a peak in emission at redshifts $z \sim 0.3\text{--}0.4$ at $E \sim 50 \text{ GeV}$, and $z \sim 0.1\text{--}0.2$ at $E \gtrsim 1 \text{ TeV}$ as shown in Figure 8.4 and in Figure 8.15 in Appendix 8.B. The latter range is a consequence of strong absorption of γ -rays above $\sim 1 \text{ TeV}$ due to interactions with the extragalactic background light (EBL), which limits the γ -ray horizon to $z \sim 0.1\text{--}0.2$ independently of the type of source.

The present analysis is thus optimized for DM searches, while for astrophysical sources it will be beneficial to consider galaxy catalogs covering higher redshifts, up to $z \sim 0.4\text{--}0.5$. We leave this extension to future studies. The window functions of galaxy number counts are given by the redshift distribution of the number of objects in the catalog. Full details on the window functions of the catalogs adopted here can be found in Appendix 8.B.2. The modeling of galaxy clustering within halos is based on the halo occupation distribution (HOD) formalism [166, 327, 328], which is also outlined in Appendix 8.B, together with the 3D cross-correlation power spectra.

Let us note that in our analysis, the window function $W_g(z)$ of the galaxy catalogs is normalized to unity: $\int dz W_g(z) = 1$. For the γ -ray window function $W_\gamma(\chi)$, we define it such that the (average) intensity energy spectrum is obtained as: $I_\gamma(E) = \int d\chi W_\gamma(E, \chi)$.

The cross-correlation APS signal has a variance $(\Delta C_\ell^{\gamma g})^2$ given by [162, 163, 319]

$$(\Delta C_\ell^{\gamma g})^2 (2\ell + 1) f_{\text{sky}} = (C_\ell^{\gamma g})^2 + \left[C_\ell^{\gamma\gamma} + \frac{C_N}{(B_\ell)^2} \right] (C_\ell^{gg} + C_{N_g}), \quad (8.4.2)$$

where $C_\ell^{\gamma\gamma}$ and C_ℓ^{gg} are the γ -ray and galaxy auto-correlation APS, C_N is the photon noise, C_{N_g} the galaxy catalog noise, and B_ℓ denotes the beam correction accounting for the finite angular resolution of the γ -ray instrument. No beam correction is applied to the galaxy term, since the angular resolution of galaxy catalogs (typically of the order of arcseconds) is negligible for the

	Γ	E_{cut} [GeV]	L_{min} [erg/s]	L_{max} [erg/s]
HSP	1.86	910^{+1100}_{-450}	10^{38}	$3 \cdot 10^{45}$
LISP	2.08	37^{+85}_{-20}	10^{38}	$3 \cdot 10^{46}$

Table 8.1. Parameters and uncertainties for the power-law with exponential cut-off SED models of HSP and LISP populations, including the minimum and maximum luminosity, adopted from Table 1 of Ref. [322].

purposes of this analysis. The parameter f_{sky} denotes the fraction of the sky covered by the analysis. All the relevant terms in Eq. 8.4.2 are outlined in Appendix 8.B.

For completeness, in the following analysis, we will also use the γ -ray auto-correlation APS, which is a measure of the intrinsic anisotropy of the γ -ray emission. It reads [162, 163, 319, 329]

$$C_{\ell}^{\gamma\gamma} = \int \frac{d\chi}{\chi^2} W_{\gamma}^2(\chi) P_{\gamma\gamma}(k = \ell/\chi, \chi), \quad (8.4.3)$$

where $P_{\gamma\gamma}$ is the 3D power spectrum of the γ -ray emission. The corresponding variance on $C_{\ell}^{\gamma\gamma}$ is [162, 163, 319]

$$(\Delta C_{\ell}^{\gamma\gamma})^2 = \frac{2}{(2\ell + 1)f_{\text{sky}}} \left[C_{\ell}^{\gamma\gamma} + \frac{C_N}{(B_{\ell})^2} \right]^2. \quad (8.4.4)$$

8.4.3 Sensitivity Forecast

8.4.3.1 Astrophysical sources

As a first step, we determine whether the anisotropy and the cross-correlation signals from unresolved astrophysical sources are detectable. As a statistical figure of merit, we adopt the signal-to-noise ratio (SNR), defined, for the cross-correlation, as

$$\text{SNR} = \sqrt{\sum_{\ell,i} \left(\frac{C_{\ell,i}^{\gamma g}}{\Delta C_{\ell,i}^{\gamma g}} \right)^2}, \quad (8.4.5)$$

where the sum runs over all the multipoles ℓ considered in the analysis and all energy bins i . For the auto-correlation, the formula is analogous but using $C_{\ell,i}^{\gamma\gamma}$ and $\Delta C_{\ell,i}^{\gamma\gamma}$.

To be more specific, the energy bins adopted in our analysis are 16, logarithmically spaced in the 30 GeV – 30 TeV range. The sum over multipoles extends from $\ell = 10$ up to $\ell = 1000$. The lower value $\ell = 10$ is adopted to avoid the largest angular scales (i.e., lowest multipoles), where the γ -ray Galactic foreground is the dominant emission, and, if not properly modeled and removed, can create some artifact in the reconstructed signal, especially in the auto-correlation. This issue is, however, mostly of concern for a real data analysis like the one in [330], where we refer the reader for further details. In our forecast, starting the sum from $\ell = 10$ or $\ell = 1$ has a negligible impact on the final results. At the same time, we do not consider very large multipoles, since the (energy-dependent) angular resolution of the γ -ray detector prevents us from accessing information on angular scales which are much smaller than its point-spread-function (PSF). The CTAO PSF has been determined through a Monte Carlo simulation of the detector and is shown in Figure 8.10 in Appendix 8.B, together with its harmonic space transform B_{ℓ} , which enters in the expression for the variance $\Delta C_{\ell,i}^{\gamma g}$ of Eq. (8.4.2). From Figure 8.10 we can see that for

SNR for 3 hrs	Auto-correlation	Cross with 2MASS
LISP	0.042	$8.2 \cdot 10^{-3}$
HSP	0.31	0.1
SNR for 50 hrs	Auto-correlation	Cross with 2MASS
LISP	0.18	0.022
HSP	0.9	0.16

Table 8.2. The predicted signal-to-noise ratios for 3-hours and 50-hours observations of LISP and HSP source classes, obtained for both γ -rays auto-correlation and cross-correlation with the 2MASS galaxy catalog.

SNR 50hrs low res	Auto-correlation	Cross with 2MASS
LISP	0.7	0.033
HSP	5.1	0.41

Table 8.3. Same as Table 8.2 but for 50hrs-observations, assuming a source detection threshold approximately three times worse than in Table 8.2. This illustrates the strong dependence of the auto-correlation signal on the point source detection threshold.

multipoles above about 1000 the beam function B_ℓ sharply decreases, thus inflating the variance and making those large multipoles irrelevant for the SNR.

The results for 3 hours and 50 hours of observation time are shown in Table 8.2, for both LISP and HSP source classes, and for both the auto-correlation and cross-correlation APS with the 2MASS galaxy catalog. We find that a purely astrophysical signal from unresolved sources can reach at most (for 50 hours of observation) only a marginal evidence of $\text{SNR} \simeq 0.9$ in the autocorrelation, and about 0.16 in the cross-correlation case. These values of SNR correspond to the HSP blazars, whose hard spectra are better suited to the CTAO energy range. The SNR is even smaller for LISPs, whose softer spectra result in a poorer overlap with the CTAO sensitivity.

As an aside comment, let us note that this is not necessarily a negative result: it simply reflects the fact that the expected sensitivity of CTAO to point-source detection would be good enough to resolve a large fraction of blazars, thereby reducing the auto- and cross-correlation APS of the *unresolved* population. In this case, it would be actually convenient and sufficient to perform auto- and cross-correlation APS analyses directly on the catalog of resolved sources in order to study their anisotropy properties.

However, we wish to stress that this result, i.e., the fact that CTAO sensitivity would be enough to resolve a significant fraction of blazars, strongly depends on the assumed point-source sensitivity of the detector, which for CTAO remains subject to considerable uncertainty (as discussed in Section 8.4.1). This is because the final configuration and specifications of CTAO have not yet been fully settled. For example, the point-source sensitivity differs significantly between CTAO-North and CTAO-South [315], and at the moment the relative share of North and South array observations to the survey is still undecided. To assess the capabilities of CTAO for the study of anisotropies of point-sources even in the case that the point-source sensitivity is worse than the nominal case derived and adopted in this paper, we repeat the analysis for point-sources by assuming a degraded point-source detection capability. Table 8.3 shows the

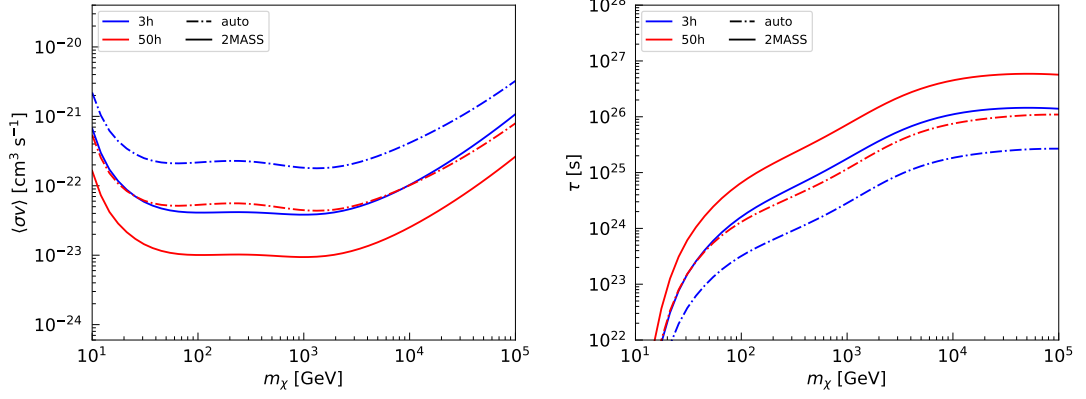


Fig. 8.5. Left: Predicted upper bounds on the dark matter cross-section $\langle\sigma v\rangle$, for annihilation into $b\bar{b}$, based on 3-hour and 50-hour of observations. The curves represent the bounds from γ -ray auto-correlation (dot-dashed) and cross-correlation with the 2MASS galaxy catalog (solid). Right: Same as the left panel, but for the lower bound on the DM particle lifetime τ .

SNR obtained for 50 hrs of observations, assuming a source detection threshold approximately three times worse than in the nominal case of Table 8.2. This has been achieved by using the 50 hrs exposure with a point-source sensitivity corresponding to 3 hrs of observations, which is approximately a factor of 3 worse than the nominal 50 hrs sensitivity (Note that, since we are in a strongly background-dominated regime, the uncertainty ΔC_ℓ does not depend on the C_ℓ itself, but only on the background and the noise, as discussed in Section 8.B). The result shows that the auto-correlation SNR for HSP indeed increases significantly, reaching $\text{SNR} \simeq 5$. The detectability of the cross-correlation APS signal also improves, but not so strongly as in the auto-correlation case. This is because the cross-correlation signal is less sensitive to the point-source detection threshold. As detailed in the Appendices, the point-source sensitivity impacts the cross-correlation signal only through the window function, whereas it affects both the window function and the 3D power spectrum in the auto-correlation case.

Regarding LISPs, on the other hand, the SNR remains very low in both cases, reaching at most 0.7 for the auto-correlation and 0.03 for the cross-correlation. This is expected, because LISPs have a SED cut-off around 40 GeV, resulting in minimal emission in the energy range where CTAO is most sensitive.

8.4.3.2 Dark matter

We now turn to assessing the CTAO sensitivity to an auto- or cross-correlation APS originating from DM annihilation or decay. To this end, we compare the null hypothesis, where only the contribution from astrophysical sources is present, with the alternative hypothesis scenario in which a DM γ -ray component is added on top of the astrophysical signal. From this comparison, we derive the minimum DM signal required to be statistically distinguishable from the purely astrophysical case.

For cross-correlations, we adopt the following test statistics:

$$\Delta\chi^2 = \sum_{\ell,i} \left(\frac{C_{\ell,i}^{\gamma_{\text{astro}} + \text{DM}g}}{\Delta C_{\ell,i}^{\gamma_{\text{astro}} + \text{DM}g}} \right)^2 - \sum_{\ell,i} \left(\frac{C_{\ell,i}^{\gamma_{\text{astro}}g}}{\Delta C_{\ell,i}^{\gamma_{\text{astro}}g}} \right)^2, \quad (8.4.6)$$

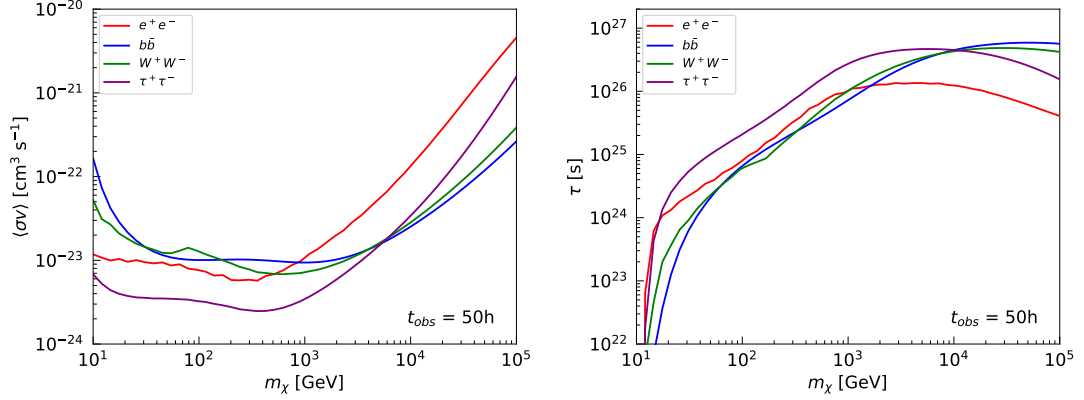


Fig. 8.6. Left: Predicted upper bounds on the dark matter cross-section $\langle\sigma v\rangle$ from the cross-correlation with 2MASS galaxies, based on 50 hours of observation, for various dark matter annihilation final states: e^+e^- , $b\bar{b}$, W^+W^- and $\tau^+\tau^-$. Right: Same as the left panel, but for the lower bound on the DM particle lifetime τ .

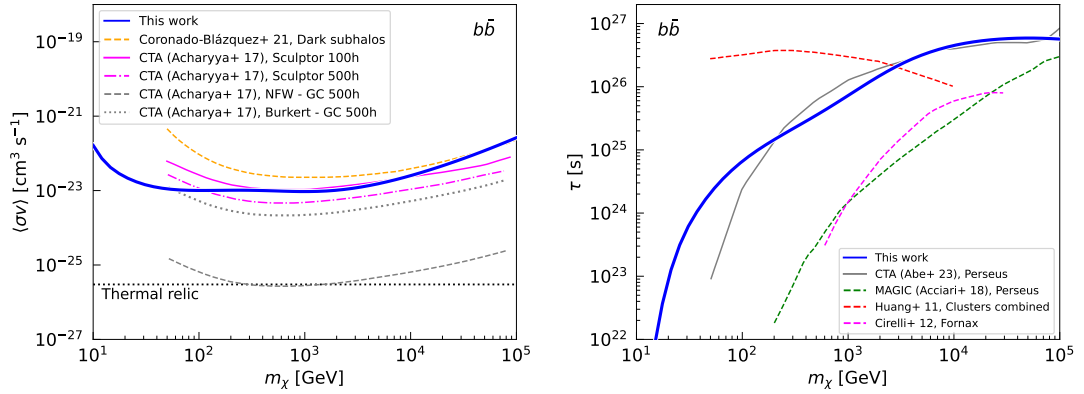


Fig. 8.7. Left: Comparison between the forecast on the upper bounds on the annihilation cross-section $\langle\sigma v\rangle$ obtained in this work with other bounds from the literature, for the $b\bar{b}$ annihilation channels: bounds from non-observation of dark sub-halos (dashed yellow) [318], from Sculptor dwarf galaxy [315] with 100 hours (solid magenta) and with 500 hours (dot-dashed magenta) of observations with CTAO. Sensitivities from the Galactic halo have large uncertainties due to the assumptions on the dark matter profile toward the Galactic Center, therefore we show two limiting cases of an NFW profile (dashed gray) and of a cored profile (dotted gray) profiles for 500h with CTAO [315]. Right: Comparison between the forecast on the lower bounds on the DM particle lifetime τ obtained in this work with other bounds from the literature, for the $b\bar{b}$ decay channel: bound from Perseus with CTAO [331] and MAGIC [332], from galaxy clusters [333] and from Fornax [334].

where $C_{\ell,i}^{\gamma_{\text{astro}}g}$ denotes the cross-correlation signal due to the presence of only astrophysical sources, while $C_{\ell,i}^{\gamma_{\text{astro}}+\text{DM}g}$ refers to the case where a DM-induced γ -ray emission is present. The corresponding standard deviations are $\Delta C_{\ell,i}^{\gamma_{\text{astro}}g}$ and $\Delta C_{\ell,i}^{\gamma_{\text{astro}}+\text{DM}g}$, respectively. Since the DM signal

depends on the DM mass m_χ and scales linearly with the annihilation rate $\langle\sigma v\rangle$ (or inversely with the lifetime τ in the case of decay), as outlined in Section 8.B, we derive bounds on the annihilation (or decay) rate by scanning over the DM mass m_χ . We then derive the 2σ bounds on the annihilation (or decay) rate by requiring $\Delta\chi^2 = 4$. A similar test statistic and procedure is employed for the auto-correlation case. We stress that the DM results are derived assuming the nominal case for the astrophysical sources scenario (i.e., HSP+LISP) and point-source sensitivity. Nonetheless, the LISP component has no impact on the DM results due to low-energy SED cut-off, as explained in the previous section. We have also verified that the point-source sensitivity uncertainty discussed in the previous section has a negligible impact on the DM results. To this aim, we explicitly derived DM sensitivities for the two cases of nominal and degraded point-source sensitivity, finding indistinguishable results.

Figure 8.5 shows the sensitivity to the DM annihilation and decay in the $b\bar{b}$ annihilation channel, derived from the auto-correlation and cross-correlation APS with the 2MASS catalog, for 3hrs and 5hrs of CTAO observations. The 5hrs case yields approximately a factor of 4 improvement in sensitivity compared to the 3hrs case, while the cross-correlation outperforms the auto-correlation by approximately a factor of 5. This is expected since in the auto-correlation, the DM signal competes with strong blazar auto-correlation, while in the cross-correlation, the blazar contribution is much lower, yielding a more favorable signal-to-background ratio. Sensitivities obtained from the cross-correlation with the 2MRS catalog are very similar and are shown Section 8.B. Note that in this case, the uncertainty on the point-source detection threshold does not play a significant role, since the DM signal is always diffuse.

Figure 8.6 shows the sensitivities of the cross-correlation technique for different annihilation channels, while Fig. 8.7 provides a comparison with existing results in the literature for alternative DM search strategies. In particular, we show bounds from observations of dwarf galaxies [315] and dark sub-halos [318] for DM annihilation, and from galaxy clusters for decaying DM [331–334]. Our results show that the cross-correlation technique is competitive, yielding constraints comparable to those from other methodologies. For comparison, the figure also includes the sensitivity for annihilation in the Galactic halo, for 500 hr of observations with CTAO [315]: in this case, the values of $\langle\sigma v\rangle$ that can be probed can be lower, but the reachable bounds are strongly dependent on the assumptions regarding the DM profile near the Galactic Center.

In Section 8.C we show a comprehensive set of plots for different hours of observations and DM channels.

8.4.4 Summary and conclusions

The CTAO will be the next generation of ground-based γ -ray Cherenkov detectors, offering a unique opportunity to study high-energy phenomena in the TeV range. In addition to targeted observations, CTAO is expected to carry out large-area sky surveys, making it particularly well-suited for anisotropy-based techniques such as auto-correlation and cross-correlation analyses. In this work, we have investigated the sensitivity of future CTAO data to constrain the properties of unresolved blazars and DM signals using these methods.

We found, in particular, that the auto-correlation APS method is better suited for probing the signal from unresolved blazars, with its sensitivity strongly dependent on the effective point-source detection threshold. On the other hand, the cross-correlation APS technique, when used in combination with galaxy catalogs, offers very promising sensitivity to potential DM annihilation or decay signal, reaching values around $10^{-23} \text{ cm}^{-3} \text{ s}^{-1}$ for annihilation cross-section $\langle\sigma v\rangle$ and 10^{27} s for decay. These sensitivities are comparable to and complementary with those obtained from existing strategies, such as observations of dwarf spheroidal galaxies or galaxy clusters.

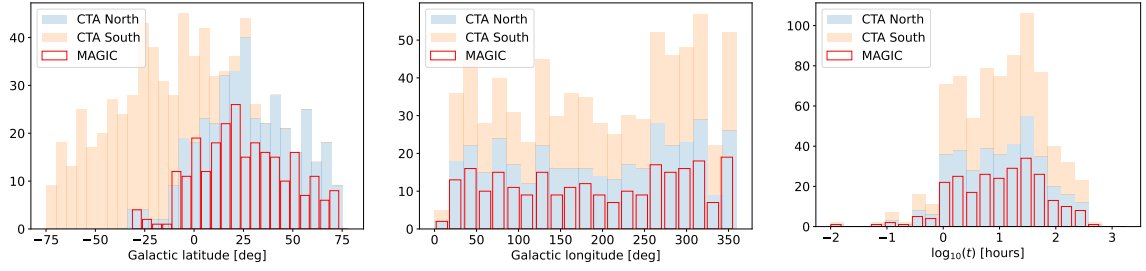


Fig. 8.8. Distributions of Galactic latitude (left), longitude (center), and observation time (right) for one realization of the predicted CTAO pointings over 10 years of observations. The stacked histograms show contributions from the Northern (blue) and Southern (orange) arrays. For comparison, the red outlined histograms indicate the corresponding distributions from 6.5 years of MAGIC observations.

In the future, these results can be further improved with upcoming, deeper, and more densely sampled galaxy catalogs, such as those from the European Space Agency’s *Euclid* satellite mission [335], as well as by incorporating cross-correlations with other large-scale-structure tracers, such as weak lensing.

Data Availability

The data that support the findings of this article are openly available on Zenodo [336].

8.A Off-source data scenario

As described in the main text, we follow the strategy outlined in Ref. [318] to extrapolate MAGIC observations to CTAO. The distribution of MAGIC pointings, that we consider when extrapolating the observation time and pointing directions of future CTAO measurements across the sky, is shown in Fig. 8.8, together with one realization of the predicted CTAO distributions of the same quantities, i.e. the Galactic latitude, longitude, and the observation time per pointing. The distribution of the cumulative pointing time over all pointings, obtained from 1000 different realizations of future CTAO observations, is shown in Fig. 8.9.

8.B Formalism and details of the modeling

The general cross-correlation formalism—the angular power spectrum, the Limber approximation, the variance expressions, and the halo model decomposition—is introduced in Section 3.5. This appendix provides the specific forms and modeling choices entering the CTAO analysis.

8.B.1 Angular Power Spectra

The angular power spectra for the γ -ray autocorrelation and for the cross-correlation between the unresolved γ -ray emission and a galaxy catalog g can be expressed as (cf. Eq. 3.5.4) [162, 163, 319]

$$C_{\ell}^{\gamma\gamma} = \int \frac{d\chi}{\chi^2} W_{\gamma}^2(\chi) P_{\gamma\gamma}(k = \ell/\chi, \chi) \quad (8.B.1)$$

$$C_{\ell}^{\gamma g} = \int \frac{d\chi}{\chi^2} W_{\gamma}(\chi) W_g(\chi) P_{\gamma g}(k = \ell/\chi, \chi), \quad (8.B.2)$$

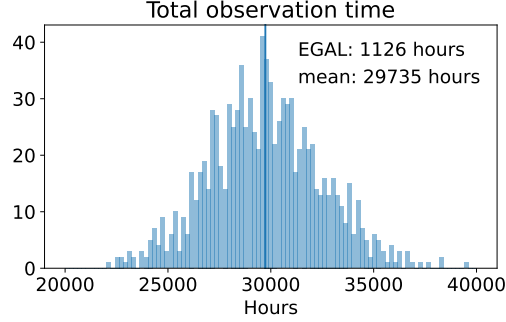


Fig. 8.9. Distribution of the total observation time from 1000 realizations of predicted CTAO pointings across the whole sky using both arrays. The blue vertical line marks the mean total observation time. The total time in the most optimistic scenario is about 25 times higher than that of the EGAL survey, effectively translating the nominal 3 hours per pointing into ~ 50 hours per sky location in this scenario.

Bin	$E_{\min}$	$E_{\max}$
Low	8 GeV	126 GeV
Mid	126 GeV	1.3 TeV
High	1.3 TeV	31.6 TeV

Table 8.4. Co-added energy bins used in the analysis for display purposes.

where ℓ denotes the multipole, χ is the comoving distance, $W_\gamma(\chi)$ and $W_g(\chi)$ are the window functions of the γ -ray emission and of the galaxy catalog, respectively, that is the functions describing how the observables are distributed in redshift. $P_{\gamma\gamma}$ and $P_{\gamma g}$ are the 3D power spectra of the auto- and cross- correlation, respectively, which quantify the fluctuations of the corresponding observables. All these components are discussed in the following sub-sections.

$C_\ell^{\gamma\gamma}$ and $C_\ell^{\gamma g}$ are both energy-dependent. In our approach, we compute the correlation functions within the specific energy bins for which the detector sensitivity is available. CTAO sensitivities are provided for 19 energy bins spanning the range (5 GeV, 501 TeV). Due to the sensitivity at low energies, we restrict our analysis to 16 bins, logarithmically spaced within the 30 GeV – 30 TeV range. For visualization purposes, the energy-dependent quantities are further coadded into 3 broader bins, as summarized in Table 8.4. These low-, medium-, and high-energy bins correspond to the CTAO bin numbers 2-7, 8-12, and 13-18, respectively. However, when computing the DM constraints, we make sure of all 19 energy bins available for CTAO (bins 1 to 19), thereby covering the full energy range from 5 GeV to 501 TeV.

The variance of the auto-correlation APS, in the Gaussian approximation, is given by Eq. (8.4.4).

The sky fraction observed by CTAO is $f_{\text{sky}} \sim 0.25$ and it is considered energy-independent in our analysis. The beam window functions B_ℓ account for the smearing of the angular signal due to the finite angular resolution of the instrument and can be defined in terms of the point spread function (PSF) as

$$B_\ell(E) = \int d\theta \text{PSF}(\theta, E) \sin\theta P_\ell(\cos\theta), \quad (8.B.3)$$

where $P_\ell(\cos\theta)$ are the Legendre Polynomials of order ℓ . The point-spread function $\text{PSF}(\theta, E)$ has been obtained through a Monte Carlo simulation of the CTAO detector, and is energy-dependent.

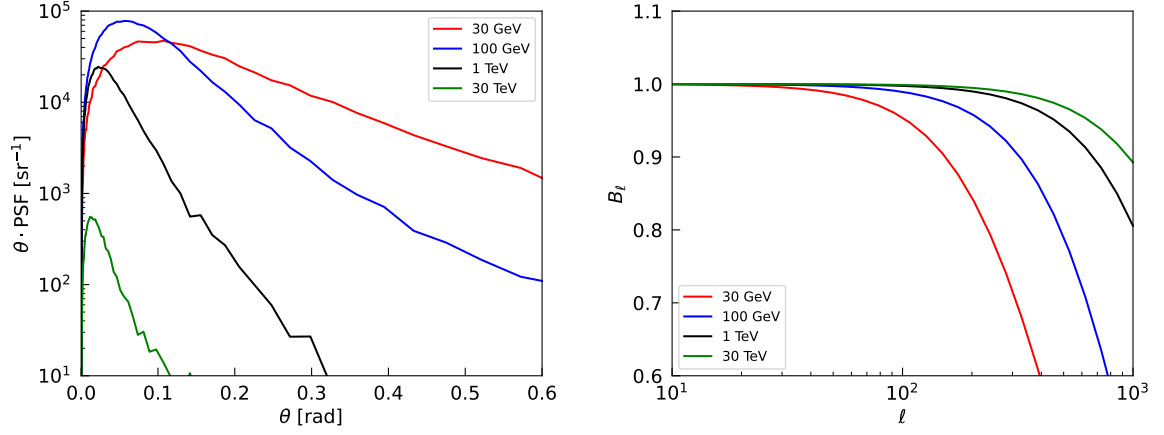


Fig. 8.10. Left: Point spread functions of CTAO, as a function of angle for different photon energies. Right: The corresponding CTAO beam window functions as a function of multipole.

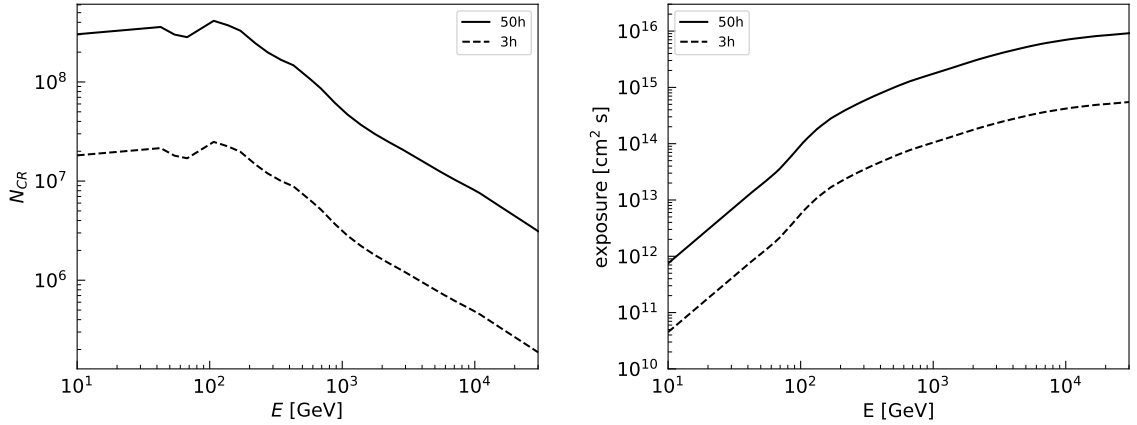


Fig. 8.11. Left: Cosmic-ray background N_{CR} as a function of energy. Right: CTAO exposure as a function of energy.

Fig. 8.10 shows the $\text{PSF}(\theta, E)$ as a function of the angle θ (left panel) and the corresponding beam window function as a function of the multipole ℓ (right panel) for different energies.

The beam window function in a specific energy bin is then obtained by averaging over the energy-dependent intensity flux dI/dE , namely

$$\langle B_\ell \rangle = \frac{1}{N} \int_{\text{bin}} dE B_\ell(E) \frac{dI}{dE}, \quad (8.B.4)$$

where $N = \int_{\text{bin}} dE dI/dE$. For definiteness, we use $dI/dE \propto E^{-2.3}$, which appropriately matches the observed behavior of the unresolved γ -ray intensity as measured by *Fermi*-LAT.

The photon noise C_N is also energy-dependent and is determined as

$$C_N = \frac{4\pi f_{\text{sky}}}{N_\gamma} \left(1 + \frac{N_{\text{CR}}}{N_\gamma} \right) \langle I \rangle^2, \quad (8.B.5)$$

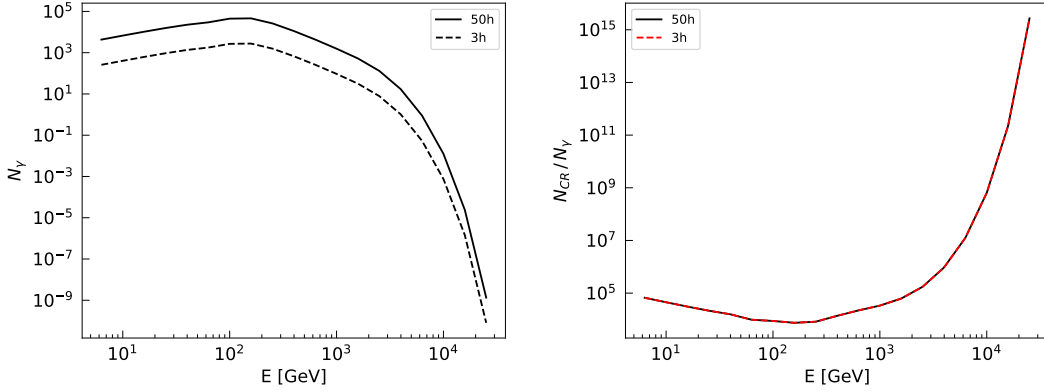


Fig. 8.12. Left: Number of expected γ -ray events N_γ in the CTAO survey as a function of energy. Right: Ratio of cosmic-ray events to expected γ -ray events (N_{CR}/N_γ) after quality cuts in the CTAO survey. The ratio worsens above 1 TeV due to significant γ -ray suppression from EBL attenuation. N_γ is determined for the HSP plus LISP source classes.

where N_{CR} represents the cosmic-ray irreducible background, N_γ is the number of γ -ray events, and $\langle I \rangle$ denotes the mean γ -ray intensity in the specific energy bin, which is determined from the window functions as: $\langle I \rangle = \int d\chi W(\chi)$. N_γ is therefore related to $\langle I \rangle$ via

$$N_\gamma = 4\pi f_{\text{sky}} \mathcal{E} \langle I \rangle, \quad (8.B.6)$$

where $\mathcal{E}$ represents the CTAO exposure in the given energy bin. The exposure and the counts associated with the cosmic-ray background are derived via Monte Carlo simulations of the CTAO array. The simulation has been performed for 1.62h of observation time, and the results were then rescaled to obtain the exposure for 3hrs and 50h. Fig. 8.11 displays the estimated cosmic-ray backgrounds N_{CR} (left panel) and exposures $\mathcal{E}$ (right panel) as a function of the energy.

Fig. 8.12 shows the the number of γ -ray events N_γ and the ratio N_{CR}/N_γ as a function of energy. Since $N_{CR}/N_\gamma \gg 1$ in all the energy bins considered in our analysis, the noise can be approximated as

$$C_N \simeq 4\pi f_{\text{sky}} \frac{N_{CR}}{N_\gamma^2} \langle I \rangle^2 \quad (8.B.7)$$

By substituting the expression of N_γ in Eq. (8.B.7), it becomes evident that C_N can be expressed only in terms of detector properties and configuration,

$$C_N \simeq \frac{N_{CR}}{4\pi f_{\text{sky}} \mathcal{E}^2} \quad (8.B.8)$$

The noise C_N is shown explicitly in Fig. 8.21, together with a selection of auto- and cross-correlation energy spectra.

In the case of cross-correlations, the variance is instead given by (cf. Eq. 3.5.8) [162, 163, 319]

$$(\Delta C_\ell^{\gamma g})^2 = \frac{1}{(2\ell + 1)f_{\text{sky}}} \left[(C_\ell^{\gamma g})^2 + \left(C_\ell^{\gamma\gamma} + \frac{C_N}{B_\ell^2} \right) (C_\ell^{gg} + C_{N_g}) \right]. \quad (8.B.9)$$

where C_ℓ^{gg} is the galaxy auto-correlation, C_N is the same as in the auto-correlation case, C_{N_g} is the galaxy catalog noise. There is no beam correction in the galaxy term, as the angular resolution

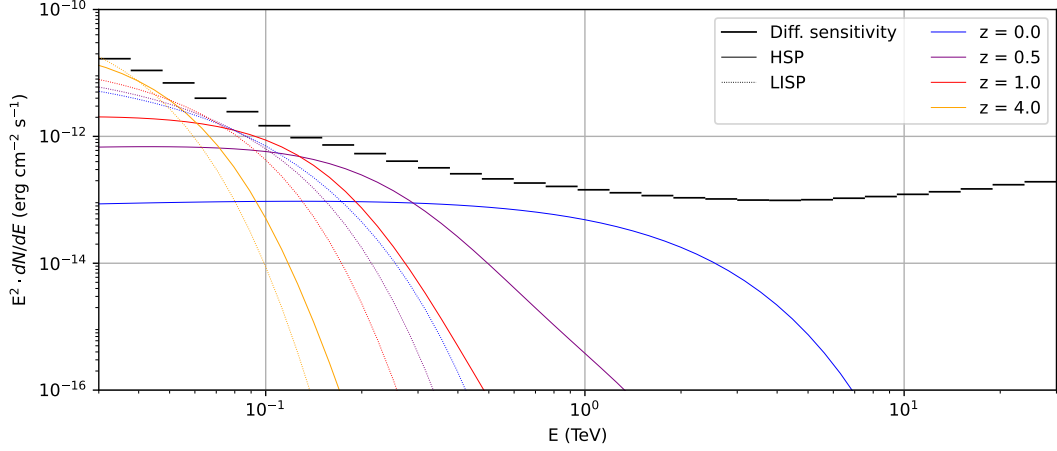


Fig. 8.13. Differential flux sensitivity for 5hrs of observation in the considered CTAO energy range (thick black horizontal dashes), compared with the energy spectra of HSP and LISP sources (solid and dotted lines, respectively) at four different redshifts. The spectra are normalized to the minimal flux detectable with the planned EGAL survey.

on the galaxy catalog is on the order of arcseconds, making it negligible in this analysis. The galaxy noise is given by

$$C_{N_g} = \frac{4\pi f_{\text{sky},g}}{N_g} \quad (8.B.10)$$

where $f_{\text{sky},g}$ represents the fraction of the sky covered by the catalog, and N_g is the number of galaxies in the catalog. Fig. 8.21 shows the quantity $\sqrt{C_N C_{N_g}}$ as a representative level for the cross-correlation noise, which approximately corresponds to the noise expected in a noise-dominated regime.

8.B.2 Window functions

Gamma-ray astrophysical sources The window function for the γ -ray emission of unresolved astrophysical sources can be written as

$$W(E, \chi) = \frac{dW}{dE d\chi} = \frac{d_L^2(z)}{(1+z)^2} \int_{L_{\min}}^{L_{\text{th}}(z)} dL \Phi(L, z) \frac{dF}{dE}(E, L, z) \exp[-\tau(E, z)] \quad (8.B.11)$$

where $d_L(z)$ is the luminosity distance, $\Phi(L, z)$ is the γ -ray luminosity function of the source population. In the present analysis, we use the luminosity functions of HSP and LISP as reported in [322]. Note that the above window function, which is used in Eq. 8.B.1-8.B.2 is differential in comoving distance, χ . The analogous quantity differential, differential in z , would be $W(E, z) = \frac{dW}{dE dz} = \frac{dW}{dE d\chi} \frac{d\chi}{dz} = W(E, \chi) \frac{d\chi}{dz}$ where $\frac{d\chi}{dz} = \frac{c}{H(z)}$. Also note that the above window is differential in energy. In Eq. 8.B.1-8.B.2, however, the corresponding quantity must be integrated over the energy bin considered. L is the γ -ray luminosity of the source, conventionally integrated over the interval (0.1, 100) GeV, that is

$$L = \int_{0.1 \text{ GeV}}^{100 \text{ GeV}} dE_r \frac{dL}{dE_r} \quad (8.B.12)$$

where E_r is the source rest-frame energy, related to the observed energy by $E_r = (1+z)E$, and dF/dE is the spectral energy distribution (SED) of the sources, which we model as a power law

with index Γ and an exponential cutoff,

$$\frac{dF}{dE} = A(L, z) E^{-\Gamma} \exp(-E/E_{\text{cut}}) \quad (8.B.13)$$

The values of Γ and E_{cut} for the two classes of astrophysical sources adopted in our models are reported in Table 8.1, together with the minimum luminosity of the source classes L_{min} . The parameter A , on the other hand, can be related to the luminosity L , by observing that (see Eq. 22 in [337])

$$L = \frac{4\pi d_L^2(z)}{(1+z)} \int_{0.1 \text{ GeV}}^{100 \text{ GeV}} dE_r E_r \frac{dF}{dE_r} = 4\pi d_L^2(z) \int_{0.1 \text{ GeV}/(1+z)}^{100 \text{ GeV}/(1+z)} dE E \frac{dF}{dE} \quad (8.B.14)$$

and therefore:

$$A(L, z) = \frac{L}{4\pi d_L(z)^2} \left[\int_{0.1 \text{ GeV}/(1+z)}^{100 \text{ GeV}/(1+z)} dE E^{(-\Gamma+1)} \exp(-E/E_{\text{cut}}) \right]^{-1} \quad (8.B.15)$$

The last term in Eq. (8.B.11) models the γ -ray absorption on extragalactic background light (EBL). For the optical depth $\tau(E, z)$, we adopt the results from Ref. [338]. The maximal luminosity $L_{\text{th}}(z)$ in Eq. (8.B.11), which refers to the threshold of detectability of resolved sources by the detector, depends on the CTAO sensitivity $F_{\text{sens}}^{\text{CTA}}(z)$ to point sources in the energy range of operation, which in our analysis is defined as (30 GeV, 30 TeV)

$$F_{\text{sens}}^{\text{CTA}}(z) = \int_{30 \text{ GeV}}^{30 \text{ TeV}} dE \frac{dF}{dE} \exp[-\tau(E, z)] = A_{\text{th}}(z) \int_{30 \text{ GeV}}^{30 \text{ TeV}} dE E^{-\Gamma} \exp(-E/E_{\text{cut}}) \exp[-\tau(E, z)] \quad (8.B.16)$$

From $F_{\text{sens}}^{\text{CTA}}(z)$, it is therefore possible to determine the value of $A_{\text{th}}(z)$, which allows us to determine the threshold luminosity by means of Eq. (8.B.14) as

$$L_{\text{th}}(z) = 4\pi d_L^2(z) F_{\text{sens}}^{\text{CTA}}(z) \frac{\int_{0.1 \text{ GeV}/(1+z)}^{100 \text{ GeV}/(1+z)} dE E^{-\Gamma+1} \exp(-E/E_{\text{cut}})}{\int_{30 \text{ GeV}}^{30 \text{ TeV}} dE E^{-\Gamma} \exp(-E/E_{\text{cut}}) \exp[-\tau(E, z)]} \quad (8.B.17)$$

The HSP and LISP point source sensitivity has been obtained through extensive Monte Carlo simulations [315, 339] and is shown in Fig. 8.13 for 3 hrs of observation across different energy bins. The same plot compares the differential sensitivity with the γ -ray spectra of LISP and HSP blazars normalized to the minimum detectable intensity. As shown, the spectra observed at Earth are redshift-dependent, as sources at different redshifts experience different levels of EBL absorption. Fig. 8.14 instead shows the integrated HSP and LISP point source sensitivity in the energy range 30 GeV to 30 TeV as a function of z , where the normalization of the SED in Eq. (8.B.13) has been fixed to its threshold value $A_{\text{th}}(z)$, obtained from Eq. (8.B.16) by employing the Monte Carlo modeling of $F_{\text{sens}}^{\text{CTA}}(z)$.

Gamma-ray emission from dark matter Window functions for dark matter are modeled as in Refs. [162, 312, 319], to which we refer for additional details. For decaying dark matter, we have

$$W(E, \chi) = \frac{dW}{dE d\chi} = \left(\frac{\Omega_{\text{DM}} \rho_c}{m_\chi} \right) \frac{\Gamma_d}{4\pi} \frac{dN}{dE} [(1+z)/E] \exp[-\tau(E, z)] \quad (8.B.18)$$

where Ω_{DM} is the dark matter density parameter, ρ_c is the critical density of the Universe, m_χ the DM particle mass, Γ_d the DM decay rate, dN/dE the γ -ray energy spectrum per decay event,

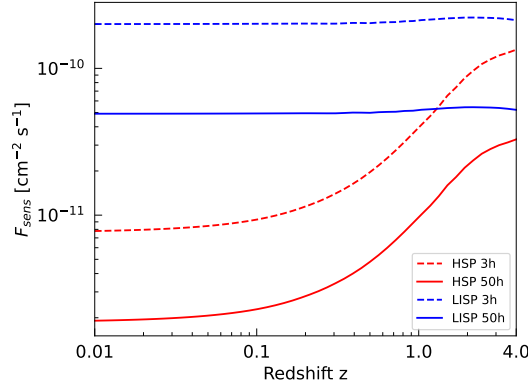


Fig. 8.14. CTA sensitivity $F_{\text{sens}}^{\text{CTA}}(z)$ for 3 hrs and 5 hrs of observation as a function of redshift, for the HSP and LISP γ -ray source classes.

which we take from [84] for decay into $b\bar{b}$, $\tau^+\tau^-$, W^+W^- . The last term in the equation refers again to γ -ray absorption, as discussed for the astrophysical sources.

For annihilating DM, the window function reads (cf. Eq. 3.5.5)

$$W(E, \chi) = \frac{dW}{dE d\chi} = \frac{1}{4\pi} \frac{\langle \sigma v \rangle}{2} \Delta^2(z) \left(\frac{\Omega_{\text{DM}} \rho_c}{m_\chi} \right)^2 (1+z)^3 \frac{dN}{dE} [(1+z)E] \exp[-\tau(E, z)], \quad (8.B.19)$$

where $\langle \sigma v \rangle$ is the velocity-averaged annihilation cross-section. Unlike DM decay, the annihilation window function also contains the flux multiplier factor $\Delta^2(z)$, which accounts for the enhancement of the flux due to the clumping of DM and the presence of sub-structures within the DM halos. Details on the way we model the flux multiplier can be found in Ref. [89, 340, 341]. Window functions for both DM and astrophysical sources are plotted in Figure 8.15 as a function of redshift for two different energies, and for 3 hrs and 5 hrs of observation time. It can be seen that the DM window always peaks at low z , while the astrophysical signal peaks at $z \sim 0.4 - 0.5$ at 50 GeV and $z \sim 0.1 - 0.2$ at 1 TeV, with the horizon shrinking due to the effect of EBL γ -ray absorption. Fig. 8.16, instead, displays the intensity $\langle I \rangle$ as a function of the energy for the two classes of astrophysical sources used in the analysis and for a few representative cases of DM annihilation.

Galaxy Catalogs The galaxy-catalog window functions are given by the redshift distribution of the number of objects in the catalog, namely

$$W_g(z) = \frac{dN_g}{dz}. \quad (8.B.20)$$

The window function is normalized such that $\int dz W_g(z) = 1$. Note again that $W_g(\chi)$ entering the cross-correlation APS is related to the redshift-dependent window function as $W_g(\chi) = W_g(z) \frac{dz}{d\chi} = W_g(z) H(z)/c$, where $H(z)$ is the Hubble parameter and c the speed of light. In this way we also have $\int d\chi W_g(\chi) = 1$. We use the dN_g/dz as given in [325, 326] for 2MRS and [323, 324] for 2MASS.

8.B.3 Galaxies Halo Occupation Distribution

To model the clustering of galaxies within the catalog, we use the Halo Occupation Distribution (HOD) formalism (see [166, 327, 328]).

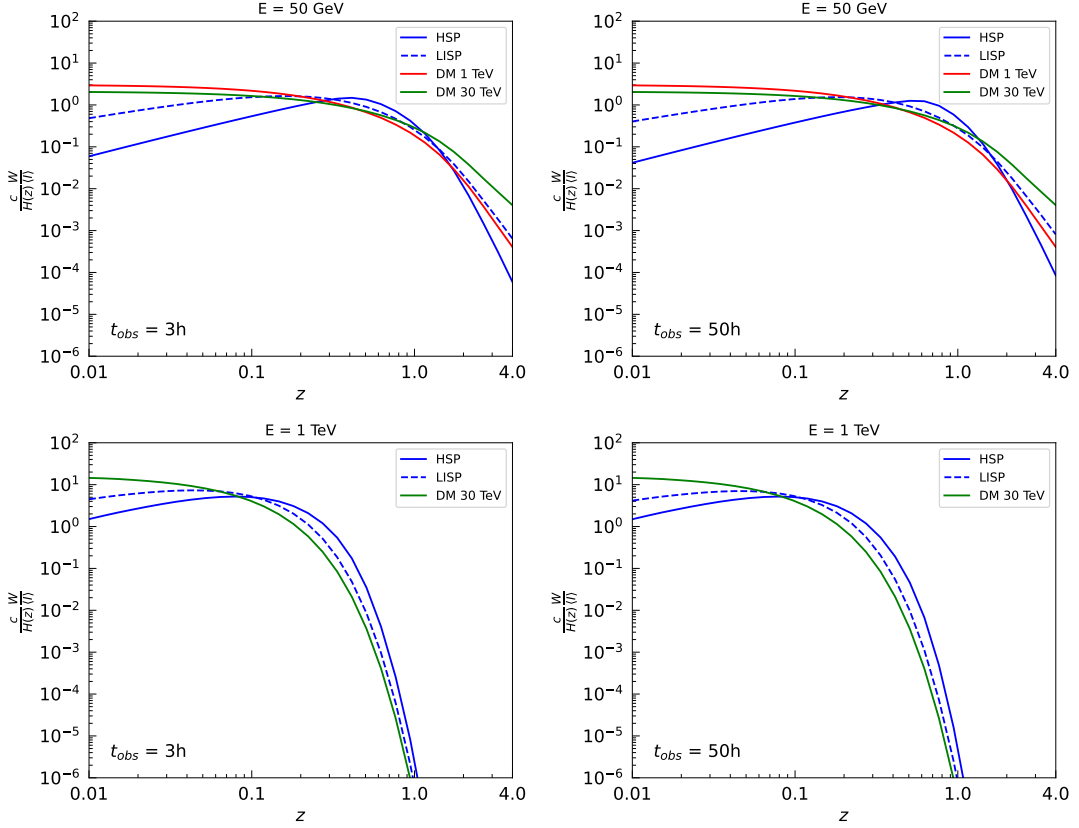


Fig. 8.15. γ -ray window functions, normalized to the mean γ -ray intensity, as a function of redshift (and divided by the Hubble parameter). Left panels show results for 3 hours of observation, and right panels for 50 hours. The blue solid and dashed lines represent the HSP and LISP astrophysical sources, respectively. The red and green solid lines refer to DM annihilation into $b\bar{b}$, $\langle\sigma v\rangle = 3 \times 10^{-26} \text{cm}^3 \text{s}^{-1}$ and masses of 1 TeV (red) and 30 TeV (green). The upper row is calculated for a γ -ray energy of 50 GeV, while the lower row is for 1 TeV (only the 30 TeV DM case is shown here). The 3-hour observation time cases are the same as shown in Figure 8.4.

The number of galaxies residing within a DM halo of mass M is parameterized as

$$\langle N_{\text{cen}}(M) \rangle = \frac{1}{2} \left[1 + \text{erf} \left(\frac{\log M - \log M_{\text{th}}}{\sigma_{\log M}} \right) \right], \quad (8.B.21)$$

$$\langle N_{\text{sat}}(M) \rangle = \left(\frac{M}{M_1} \right)^\alpha \exp \left(-\frac{M_{\text{cut}}}{M} \right), \quad (8.B.22)$$

where, for a halo of mass M , $\langle N_{\text{cen}}(M) \rangle$ is the average number of galaxies at the center of the halo and $\langle N_{\text{sat}}(M) \rangle$ represents the average number of satellite galaxies. The parameters M_{th} , $\sigma_{\log M}$, α , M_1 , M_{cut} characterize the HOD and we chose them according to [326] for 2MRS and [312] for 2MASS.

8.B.4 3D Power Spectra

The 3D Power Spectra $P(k)$ (PS) have been calculated by using the Halo Model formalism [166]. We have derived the autocorrelation 3D PS of decaying DM $P_\delta(k)$, annihilating DM $P_{\delta^2}(k)$ and

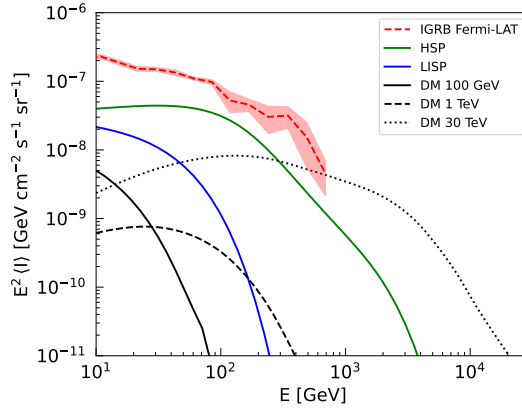


Fig. 8.16. γ -ray intensity in our benchmark model for HSP (solid green) and LISP (solid blue), along with three cases of dark matter annihilation for particle masses of 100 GeV (solid black), 1 TeV (dashed black), and 30 TeV (dotted black). The 100 GeV and 1 TeV lines correspond to an annihilation cross-section $\langle\sigma v\rangle = 3 \times 10^{-26} \text{cm}^3 \text{s}^{-1}$, while for 30 TeV we adopted $\langle\sigma v\rangle = 3 \times 10^{-23} \text{cm}^3 \text{s}^{-1}$. The figure also shows the measured IGRB (red dashed line and red band for its uncertainty) [111].

astrophysical sources $P_\gamma(k)$, and the 3D cross-correlation PS among the same three categories of tracers: $P_{g\delta}(k)$ for the cross-correlation between galaxies and decaying DM γ -ray emission, $P_{g\delta^2}(k)$ for the cross-correlation between galaxies and annihilating DM γ -ray emission, for the cross-correlation between galaxies and astrophysical sources γ -ray emission $P_{g\gamma}(k)$. The explicit formulas for the power spectra are reported in Ref. [312], to which we refer the reader for details.

Figures 8.17-8.18 report the plots of the various 3D power spectra adopted in the analysis. Figures 8.19-8.20 show a few samples of auto and cross-correlation APS as a function of the multipole ℓ . It can be seen that auto APS of astrophysical sources is constant in ℓ , since they are point-like, while the DM APS presents some structure due to the extension of the DM halos. Figure 8.21, instead, shows the APS as a function of energy at a fixed multipole $\ell = 100$. For comparison, the auto- and cross-correlation noise level is plotted, thus showing that the expected signal-to-noise ratio is typically low.

8.B.5 Poisson anisotropy

The auto-correlation angular power spectrum of blazars is dominated by the constant 1-halo term, and is therefore flat in ℓ . It is thus convenient to define the Poisson anisotropy as the normalization of this spectrum. Fig. 8.22 shows the Poisson anisotropy of unresolved blazars, divided by the square of the intensity as a function of energy for 3 hours and 50 hours of observation. As expected, the anisotropy decreases with increasing observation time, since a larger fraction of the brighter sources (which contribute more significantly to the anisotropy) become resolved. Across the entire energy range, the anisotropy remains at the level of percent or below.

The same figure also shows the CTA sensitivity to the Poisson anisotropy in three energy bins. To derive this sensitivity, we adopt the same methodology used for setting the DM limits. In this case, however, the null hypothesis corresponds to pure noise, that is, the absence of any source contribution, while the alternative hypothesis assumes the presence of a source class characterized by a Poisson anisotropy. We further assume that the energy dependence of the

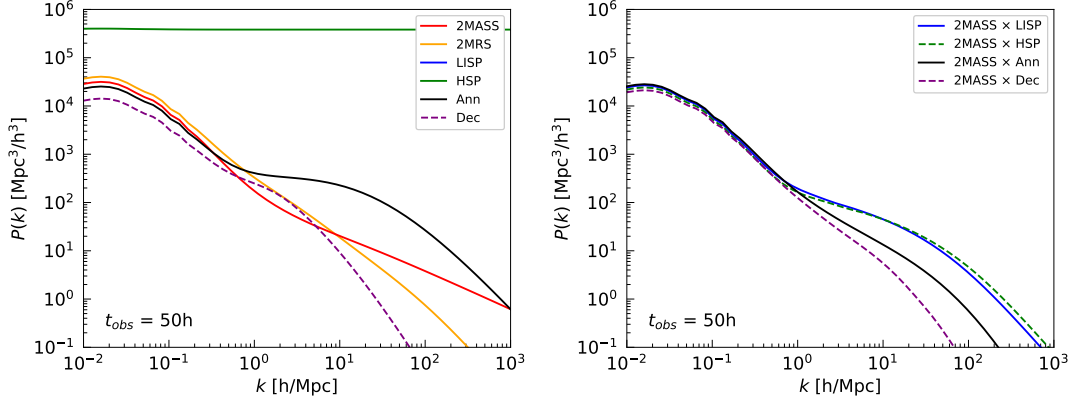


Fig. 8.17. Left: Auto-correlation 3D power spectra for 2MASS and 2MRS galaxies, LISP and HSP astrophysical sources, and annihilating and decaying dark matter. Right: Cross-correlation 3D power spectra for the cross-correlation between 2MASS galaxies and the various γ -ray source types considered in our analysis: LISP, HSP, annihilating and decaying dark matter.

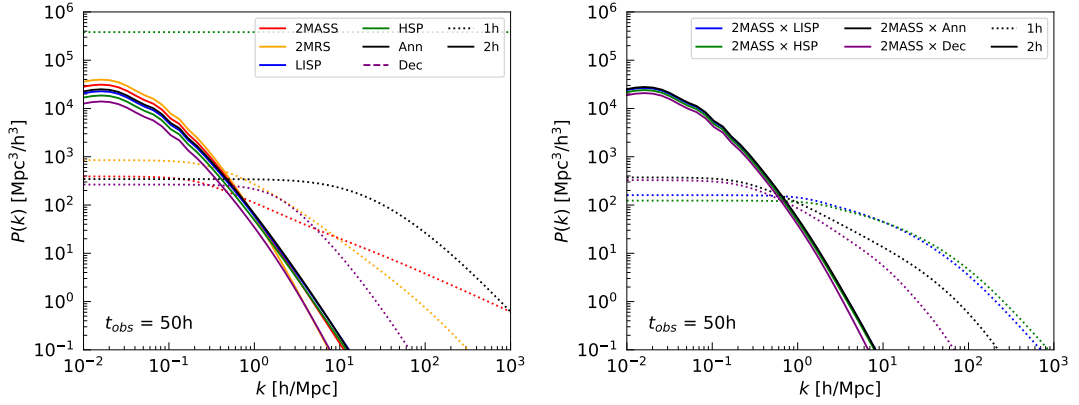


Fig. 8.18. The same as Figure 8.17 but with the 1-halo (dotted) and the 2-halo (solid) terms shown separately.

anisotropy of this source class is proportional to $\langle I \rangle^2$, i.e., it scales with the square of the observed intensity. As shown in the figure, the resulting sensitivity is at the level of approximately 10%.

8.C Further results

In Figures 8.23, 8.24, 8.25, 8.26, 8.27, 8.28, we show additional results on the constraints for the DM annihilation cross-section and the decay lifetime, exploring various annihilation and decay channels, as well as different observation times. We also provide further comparisons with existing bounds from the literature.

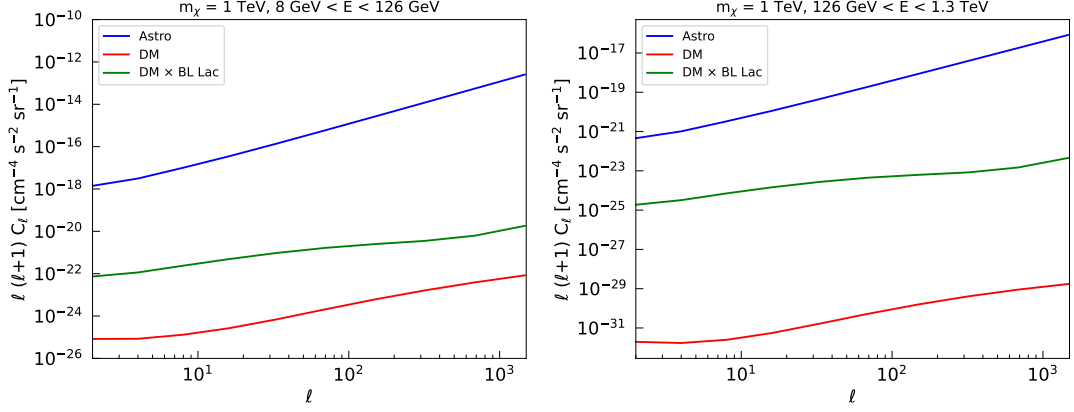


Fig. 8.19. Angular power spectra of the γ -ray auto-correlation as a function of the multipole, for two different energy bins. The upper (blue) curve refers to the auto-correlation among astrophysical sources, the median (green) and lower (red) curves to the correlation between sources and dark matter and the autocorrelation of dark matter, respectively, for a dark matter mass of 1 TeV and annihilation into $b\bar{b}$ with a cross-section $\langle\sigma v\rangle = 3 \times 10^{-26} \text{cm}^3 \text{s}^{-1}$.

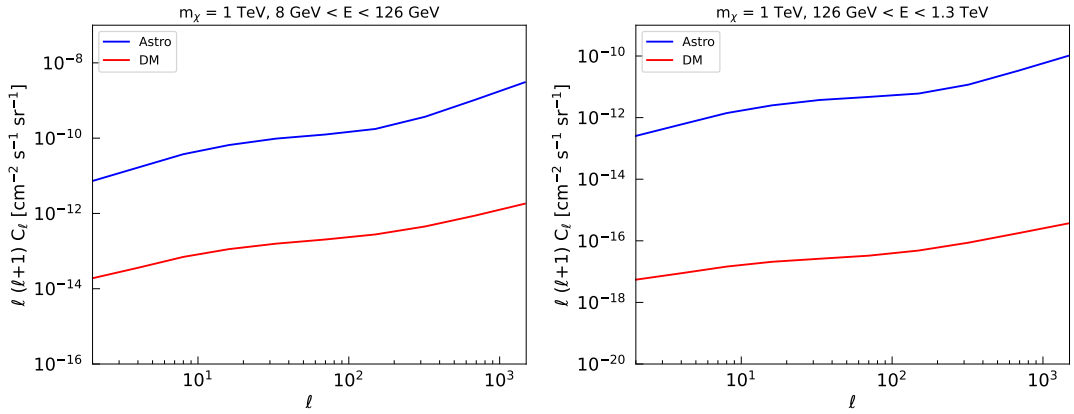


Fig. 8.20. Angular power spectra of the γ -ray cross-correlation with 2MASS galaxies, as a function of the multipole, for two different energy bins. The upper (blue) curve refers to the cross-correlation with astrophysical sources, while the lower (red) curves represent the correlation with a dark matter signal produced by a particle with mass of 1 TeV and annihilation cross-section into $b\bar{b}$ of $\langle\sigma v\rangle = 3 \times 10^{-26} \text{cm}^3 \text{s}^{-1}$.

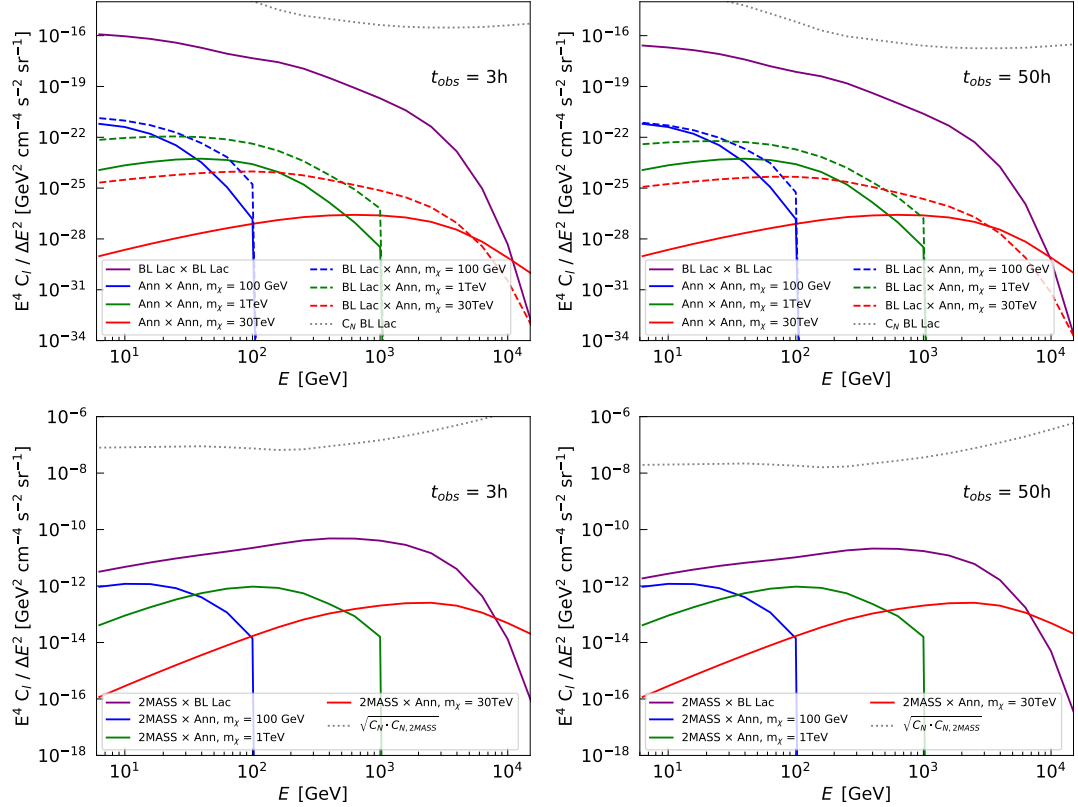


Fig. 8.21. Angular power spectra C_ℓ as a function of energy for $\ell = 100$. The left panels correspond to 3 hours of observation, while the right panels represent 50 hours. Dark matter signals are calculated for annihilation into $b\bar{b}$, for $\langle\sigma v\rangle = 3 \times 10^{-26} \text{cm}^3 \text{s}^{-1}$, and three representative masses: 100 GeV, 1 TeV, and 30 TeV. The solid and dashed lines show the auto-correlation and the cross-correlation C_ℓ for various combinations of γ -ray emitters and galaxy catalogs, as labelled. The black dotted lines in the topmost part of the panels represent the noise level for each category: for the upper row (γ -ray auto-correlations), the noise is dominated by the photon noise of astrophysical sources, and for the lower row (cross-correlation of γ -ray with galaxies), the geometric mean of the photon and galaxy noise is shown as a representative noise level.

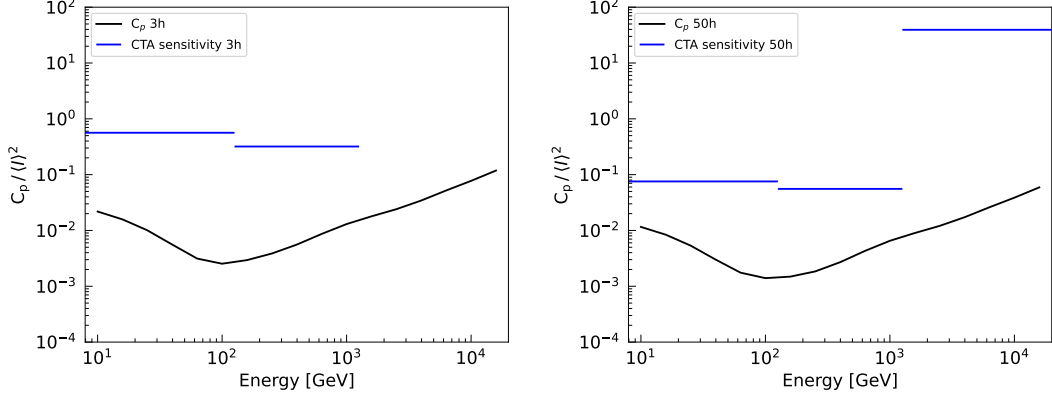


Fig. 8.22. CTA sensitivity to the Poisson anisotropy C_p (normalized to the square of the mean γ -ray intensity) in the low, mid and high energy bins, for 3 hours (left panel) and 50 hours (right panel) of observation. The predicted sensitivity is to the expected C_p from *unresolved* blazars (solid black lines). The reduced sensitivity at high energies is driven by the limited number of γ -ray events in this energy range.

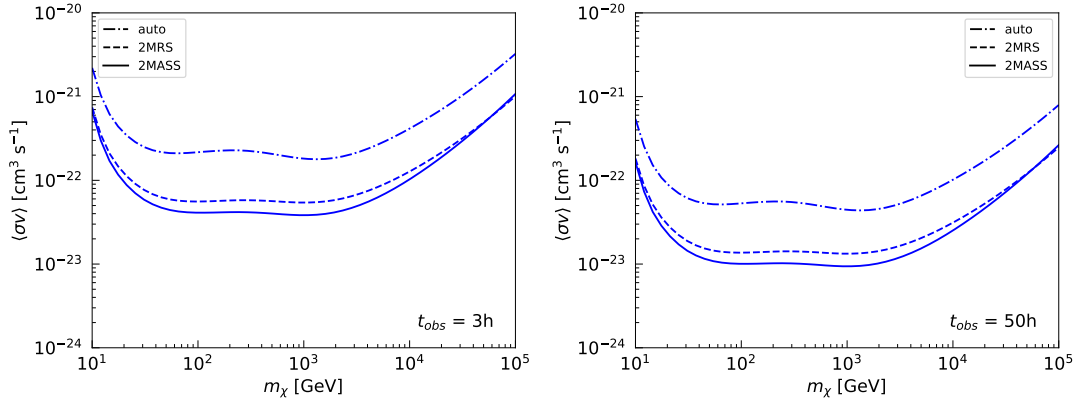


Fig. 8.23. Predicted upper bounds on the dark matter cross-section $\langle\sigma v\rangle$, for annihilation into $b\bar{b}$, for 3 hours (left panel) and 50 hours (right panel) of CTAO observations. The bounds are derived from γ -ray auto-correlation (dot-dashed), and cross-correlation with 2MRS (dashed lines) and 2MASS (solid) galaxy catalogs.

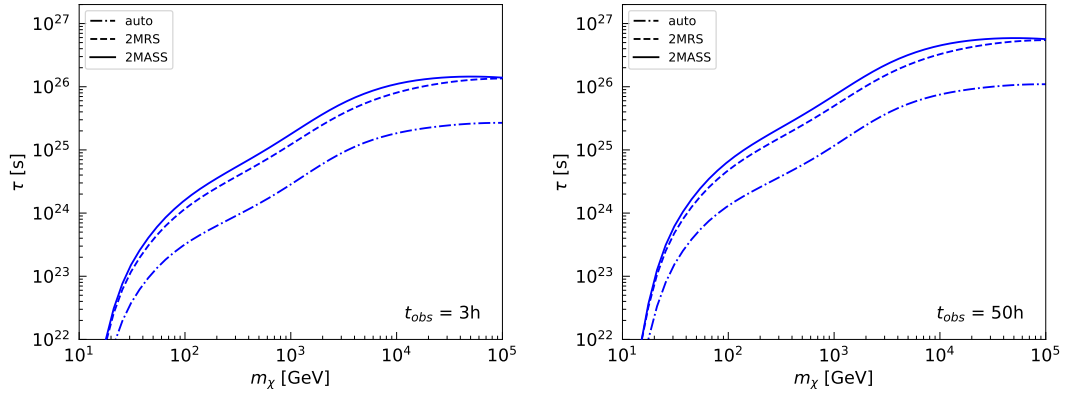


Fig. 8.24. Predicted lower bounds on the dark matter lifetime τ , for decay into $b\bar{b}$, for 3 hours (left panel) and 50 hours (right panel) of CTAO observations. The bounds are derived from γ -ray auto-correlation (dot-dashed), and cross-correlation with 2MRS (dashed lines) and 2MASS (solid) galaxy catalogs.

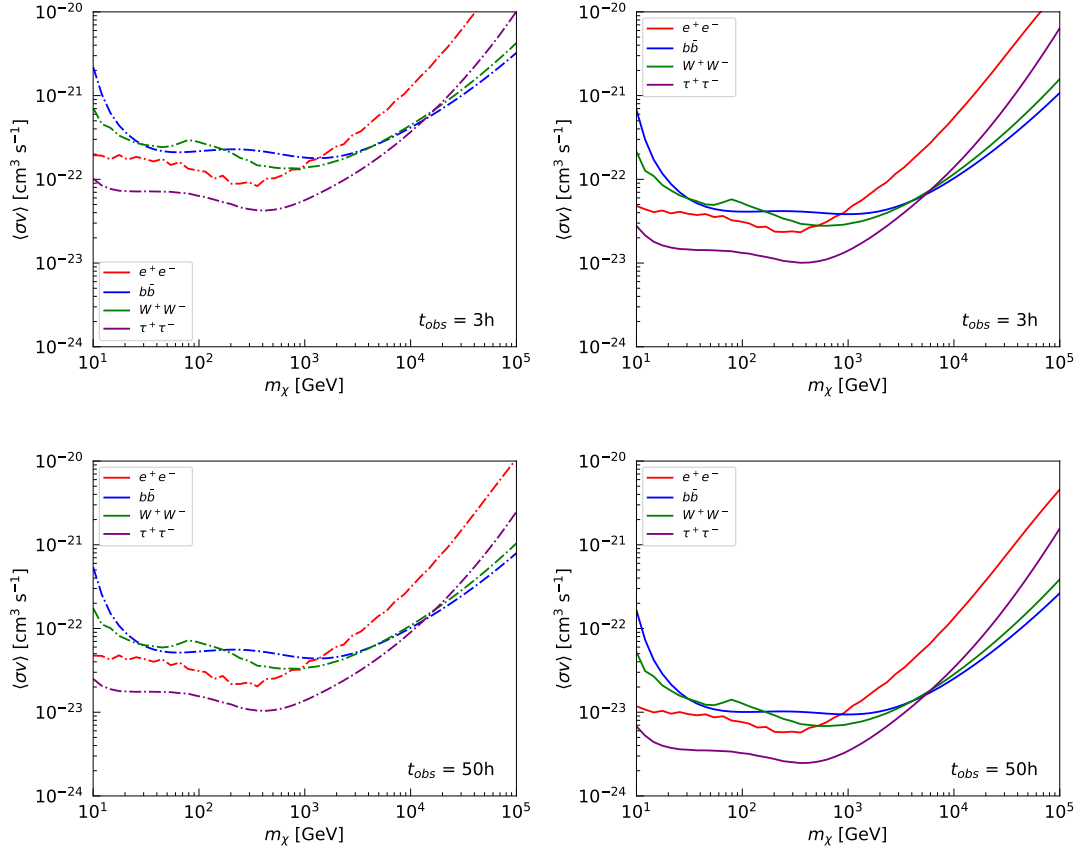


Fig. 8.25. Predicted upper bounds on the dark matter cross-section $\langle\sigma v\rangle$, for annihilation into various final states: e^+e^- , $b\bar{b}$, W^+W^- and $\tau^+\tau^-$. The upper row shows results for 3 hours of observation; the lower row for 50 hours. Left panels: bounds from γ -ray auto-correlation. Right panels: bounds from cross-correlation with 2MASS galaxies (the bottom-right panel reproduces the left panel of Fig. 8.6).

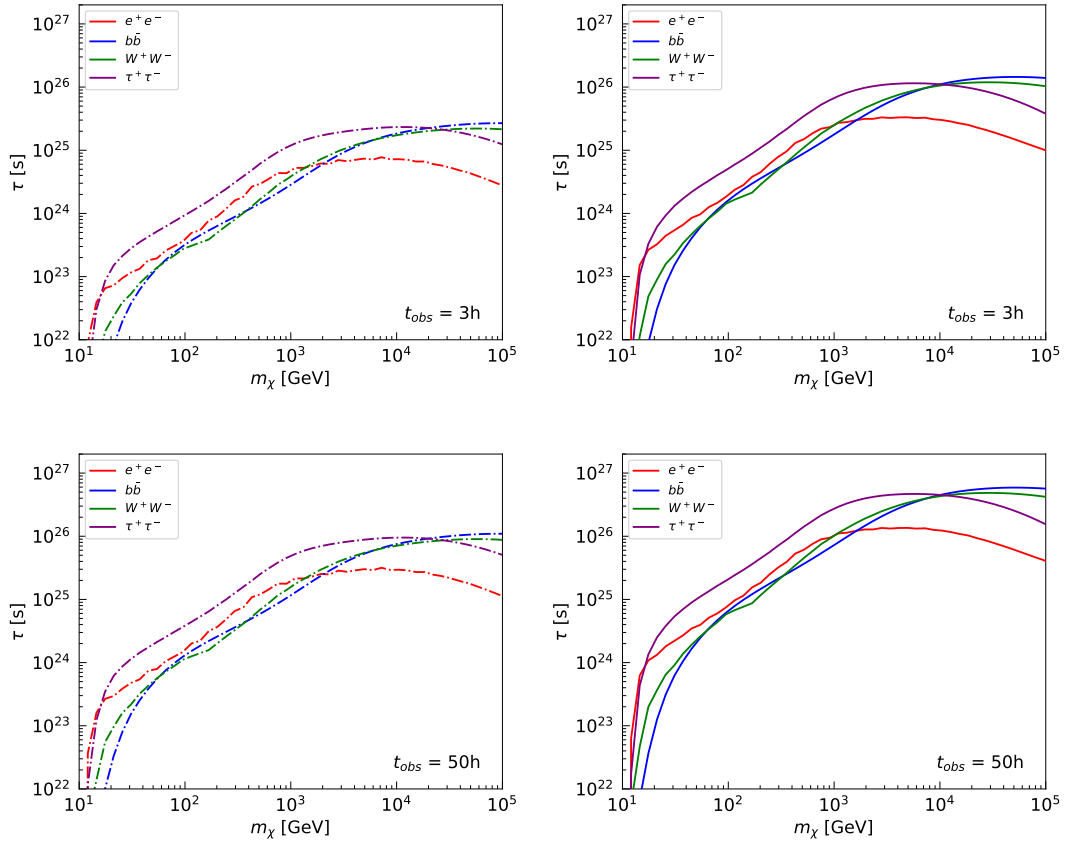


Fig. 8.26. Predicted lower bounds on the dark matter lifetime τ , for decay into various final states: e^+e^- , $b\bar{b}$, W^+W^- and $\tau^+\tau^-$. The upper row shows results for 3 hours of observation; the lower row for 50 hours. Left panels: bounds from the γ -ray auto-correlation. Right panels: bounds from cross-correlation with 2MASS galaxies (the bottom-right panel reproduces the right panel of Fig. 8.6).

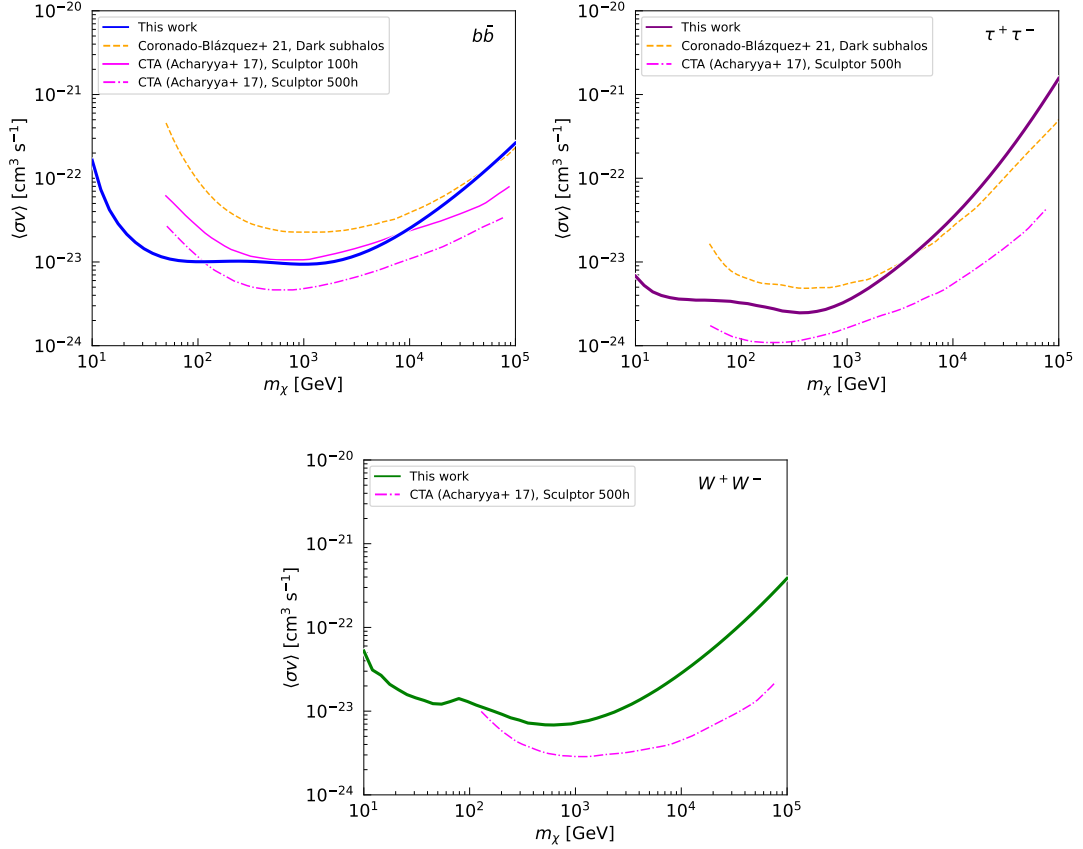


Fig. 8.27. Comparison of the predicted upper bounds on the dark matter annihilation cross-section $\langle\sigma v\rangle$, with existing constraints from the literature: bounds from non-observation of dark sub-halos (dashed yellow) [318], from the Sculptor dwarf galaxy [315] with 100 hours (solid magenta) and 500 hours (dot-dashed magenta) of observations with CTAO. Three channels are displayed: $b\bar{b}$ (upper left panel), $\tau^+\tau^-$ (upper right) and the W^+W^- (lower). The case for the $b\bar{b}$ channel is also shown in Fig. 8.5.

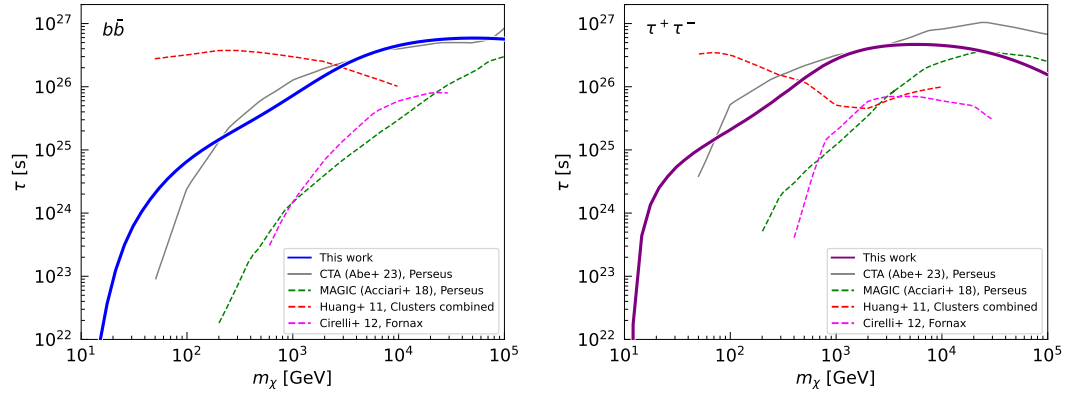


Fig. 8.28. Comparison of the predicted lower bounds on the DM particle lifetime τ for the $b\bar{b}$ and $\tau^+\tau^-$ decay channels, with existing constraints from the literature: limits from Perseus with CTAO [331] and MAGIC [332], from galaxy clusters [333], and from Fornax [334]. The case for the $b\bar{b}$ channel is also shown in Fig. 8.5.

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